

A SCIENCE FICTION NOVEL

# THE PARALLAX TESTAMENT

*Consent requires time  
to arrive.*



T. K. ARVEN

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# **The Parallax Testament**

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## **Part I — False Simultaneity**

## Chapter 01 — The Impossible Witness

At 03:17 ship time, the audit well gave Anja Kest a dead man's concurrence.

The record arrived among eleven hundred ordinary corrections from Vela: meal-allocation offsets, two disputed work rotations, a pressure-valve inspection whose technician had signed with the wrong finger. Anja had been watching Nacre's nonrotating hub slide past the well's narrow windows in quarter turns, one dark structural rib every nineteen seconds. Nothing in the batch had required judgment. She had expected to clear it before first meal.

Levan Oris had signed the concurrence thirty-one hours earlier by Vela's source clock. The document affirmed that the Crown Burn could proceed under destination-protection rules because all material evidence had been distributed to the voting habitats. Its legal sequence looked clean. Transmission, receipt, deliberation, signature, return. Meridian had rendered the events into master time and placed a green bar beside each one.

Anja read the source clocks instead.

Nadi Arko's evidence packet had left the advance cutter Mote after Levan signed.

She held her fingers above the well's controls. Touching the record would create an access event. Access events entered the same ledger she might need to question, and the ordinary sequence tools would normalize every display through Meridian before showing it to her. For twelve seconds she did nothing. The hub rib crossed the window. A coolant pump changed note behind the wall.

Then Anja opened a local scratch pane, one that knew Nacre's clock and no other.

The concurrence carried Vela source 2249.176.04:11:09. Mote's first destination packet carried source 2249.176.04:30:42. Even if Mote had sat beside Vela, nineteen minutes separated them in the wrong direction. Mote did not sit beside Vela. It had entered the Ione approach months ago while the convoy remained stretched across billions of kilometers.

A causally connected order could be slow. It could be inconvenient. It could not run backward.

Anja copied neither document. She copied their identifiers onto paper.

The sequence office still kept paper because paper could not silently inherit a clock convention. A narrow drawer beneath her knees held pencils, printed emergency tables, and the buff cards used when power failed. She wrote the two source stamps and the concurrence identifier on a card, then turned it face down.

Across the well, Sumi Dax lifted her head from the communications board.

"You stopped breathing," Sumi said.

"I changed rhythm."

"That is what people call stopping when they want an auditor to leave them alone."

Anja looked through the window rather than at her. Nacre's hub had no useful horizon. Pipes, cable trunks, and sealed maintenance ways revolved around the stationary core, though everyone in the well knew the core also moved with the ship, the convoy, and the star they had left forty-three years behind. Motion depended on the account one needed. Causality did not.

“Can you give me the raw receipt path for a Vela concurrence?” Anja asked.

Sumi's chair unlatched with a soft knock. “Routine?”

“If it is routine, I will waste six minutes.”

“And if it isn't?”

“You will waste them with me.”

Sumi came around the curved desk. At twenty-nine she was the youngest chief on Nacre, a promotion earned by hearing faults inside optical noise that diagnostics averaged away. Her hair had been cut close for a pressure-suit drill and had not grown enough to lie flat. She studied Anja's empty display, then the face-down card.

“No access event?”

“Not beyond the batch.”

Sumi's humor left. She returned to her board and opened the receive vault through a hardware key. A communications chief could retrieve the envelope around a legal packet without opening its legal content. That separation had survived every efficiency reform because the Concord of Frames treated carriage and meaning as different kinds of evidence.

Light took time to travel. Nothing in Meridian could negotiate that away.

While Sumi worked, Anja reviewed the concurrence through the batch cache. Levan's signature matched his public key. His authority remained current. No death notice existed, despite her first irrational thought. The impossible witness was alive. The death belonged to sequence: his lawful capacity to answer had not yet been born when the answer appeared.

Vela's concurrence cited a destination summary, not Nadi's packet by name. That distinction might explain the order. Perhaps an earlier summary had reached Vela. Perhaps Mote had transmitted several observations and Anja had mistaken the first complete packet for the first evidence. Auditors survived by offering error every respectable exit before calling it fraud.

She searched the destination register by source clock. Three fragments from Mote appeared, all later than Levan's signature. A Vela telescope brief predated them but contained only orbital ice measurements. The concurrence's evidence digest listed a spectral feature absent from that brief and present in Mote fragment two.

Anja wrote the feature on the card: adaptive ionic band.

“Envelope,” Sumi said.

Her board projected a plain chain of receipt proofs. Mote to the relay bead at Corva approach. Relay bead to Vela. Vela internal delivery to Concord counsel. Vela response to Nacre. No master-time correction appeared. Each proof carried local emission and receipt clocks plus the measured light path.

Sumi enlarged the internal-delivery line. “This says Oris received the Mote digest at 04:56 Vela source.”

“He signed at 04:11.”

“I saw.”

“Could the source stamp have wrapped?”

“Not in nineteen minutes. Not in nineteen years.”

A ventilation shutter opened above them. Cooler air descended across the well, smelling faintly of wet mineral from the garden ring's condensate exchangers. First shift would begin in forty minutes. Soon the passage outside would fill with workers pulling themselves along the hub rails, and every routine service on Nacre would begin writing fresh events over the cache.

Sumi kept her voice low. “Do you want the photonic checksum?”

The protected checksum came from interference features in the received light itself. Meridian could interpret a packet. Administrators could relabel it. Changing the raw checksum after reception required replacing the vault copy and the independent scatter samples stored in three hull nodes. The system had been designed to settle accusations of forged transmission, not accusations against the ordering machinery.

“Read only,” Anja said.

“Read-only access still logs.”

“Log it as carrier-quality review.”

“That would be false.”

“Then log it as sequence audit.”

Sumi watched her for a moment. “That notifies Vela's compliance mirror.”

Anja knew. The notification would take forty-three minutes to reach Vela at current separation. A reply could not return for another forty-three, no matter how urgently Meridian painted it. Within that interval, she could verify the packet or hide from the act of verification. There was no ethical clock correction for cowardice.

“Log it correctly,” she said.

Sumi inserted her hardware key a second time. The vault exposed the checksum as a field of numbers and a narrow image of interference grain. Anja did not understand the optical details well enough to judge them by eye, but she knew the verification ladder. Hull node one agreed. Node two agreed. Node three returned a thermal warning, then agreed after compensating for sensor temperature.

The packet had arrived when the receipt path said it arrived.

Levan's signature remained earlier.

Anja opened the legal transformation history. Every conversion from local clocks to master time should have left a signed rule identifier. Most entries cited the ordinary convoy solution: ephemeris, acceleration history, signal delay, uncertainty bounds. The Mote packet carried those. Levan's concurrence carried one additional tag.

ES-A12 predictive continuity.

She had seen Annex Twelve used during drive outages. When a habitat lost communication, Meridian forecast its routine load requests so pumps did not wait hours for confirmation. The forecasts were plainly marked and expired when observed instructions arrived. No forecast possessed a human signature. No forecast entered a

vote.

Yet the concurrence's transformation history placed predictive continuity between evidence delivery and Levan's reply. The tag did not alter his signature time. It altered the kind of event the record claimed had occurred before he signed.

Sumi leaned over the projection. "It says the evidence condition was satisfied by forecast delivery."

"Delivery cannot be forecast."

"Reception can."

"A model can predict reception. Levan cannot read a prediction that a packet will arrive."

"Maybe he read the underlying forecast."

"Then he could concur with a forecast, not with evidence he had received."

The legal record used the stronger verb. Received.

Anja searched for the transformation authority. The well returned Director Yselle Quon's office, Vela operations, and an emergency declaration opened twenty-six days earlier. The declared emergency concerned synchronization load during Crown planning. Its public scope covered scheduling congestion. Nothing in the summary permitted predicted receipt to stand in for knowledge.

For the first time that morning, Anja thought of Pera's last message.

Not its words. The indicator beside it.

Eleven years ago, during Ardent's shielding failure, Anja had watched a gray bar mark her sister's response as outside the operative window. Pera's local clock had shown time remaining. Master time had shown closure. Anja certified the correction because nine habitats could not allocate rescue mass under nine incompatible deadlines. Three hours later, Ardent lost the compartment where Pera had been coordinating shelter seals.

The inquiry found no fault in sequence. Anja had written that sentence herself.

She pressed both palms against the edge of the well until the present surface steadied under them. Memory could suggest a question. It could not answer the current case. Pera's death did not prove Levan's concurrence false, and Anja would not use her sister as a substitute for evidence while investigating a system that had substituted prediction for receipt.

"Search this tag across Nacre's legal mirror," she said.

Sumi returned to her station. "Scope?"

"Crown decisions, destination protection, allocation votes. Start thirty days before the emergency declaration."

"That crosses council records."

"I have sequence authority."

"You have authority to verify order, not to inspect substance."

"Search metadata only."

Sumi set the query. The result counter climbed before settling at two hundred sixteen events. Most were operational preparations: beam tests, navigation holds, maintenance assumptions. Seventeen touched legal decisions. Four touched ballots. One touched a completed causal quorum.

The well seemed to contract around the number.

“Open nothing else,” Anja said.

Sumi took her hands off the controls.

A completed quorum required every entitled habitat to receive common evidence and return an observed vote. The minimum round trip across the present convoy exceeded nineteen hours even before councils deliberated. Meridian existed because operations could not wait for every conversation. The Concord existed because some decisions had to wait anyway.

Someone had used the machinery for the first category to manufacture the second.

Anja asked Sumi to isolate the completed quorum without opening its ballots. The envelope showed nine entitled habitats, nine apparent returns, and a deliberation interval of two hours seventeen minutes. That duration alone could not fit the convoy. Juniper lay more than two light-hours away. Ardent trailed farther still. Even councils that answered at the instant of receipt could not have placed nine observed replies inside that window.

“Meridian may have started the interval from staggered deliveries,” Sumi said.

“It may schedule deliveries separately. Causal quorum starts from common evidence.”

“Unless the evidence bundle changed between ships.”

“Then it was not common.”

They tested the kinder explanation first. Sumi drew only carrier metadata from each ballot. Vela and Kite had received full destination summaries before voting. Nacre had received a preliminary summary. Bracken's envelope named a revision that had never reached its local mirror. Juniper's receipt field pointed forward to a scheduled delivery. Ardent carried no receipt proof at all, only a confidence value rounded to four decimals.

The results did not resemble a forged packet. Forgery would have required someone to invent signatures, packet grain, and relay histories. These ballots carried legitimate components rearranged under an illegitimate category. Meridian had predicted when missing evidence would arrive, predicted how each council would answer, then allowed the legal layer to speak as though both events had happened.

Sumi opened a diagnostic legend on a separate pane. “Confidence values above point nine nine five receive the same green display as observed returns.”

“On the operations board.”

“On every board using condensed status.”

“The legal view should distinguish them.”

“It does if you expand the transformation history.”

Anja studied the ordinary correction batch behind their private panes. Green bars, clean lines, completed

sequence. The display had not lied at the level of code. It had placed the truth one expansion deeper than the decision-maker would normally look.

A question occurred to her with the cold precision of a hull alarm. "How many captains see the expanded view during Crown preparation?"

"Communications chiefs can. Sequence officers can. Captains get the condensed status unless they request provenance."

"Who requests provenance on a green quorum?"

Sumi did not answer.

Anja selected Juniper's future receipt field. The scheduled arrival lay three hours ahead by Juniper source. Beside it, Meridian displayed a modeled council response built from thirty-one prior allocation votes, Elian Ro's public statements, crop-risk thresholds, and Juniper's historical preference for collective maneuvering. The prediction gave concurrence a probability of 0.9971.

One number had crossed a boundary no probability could cross. Accuracy did not turn a forecast into an event. A council that agreed exactly as predicted still had to receive, deliberate, and answer. Otherwise its agreement belonged to the modeler.

Below Juniper's forecast lay a suppressed uncertainty note. Destination ecology evidence incomplete. Response sensitivity elevated if native metabolism confirmed.

Mote's fragments might change the answer Meridian had already counted.

Anja copied the uncertainty-note identifier onto her card. The pencil point broke. She sharpened it with the office blade, catching the graphite dust on her palm rather than letting it drift into the well's filters. The small act slowed her enough to separate three claims. The quorum could not be observed. The record had labeled forecast testimony as received consent. The omitted Mote evidence had been flagged as capable of changing at least one forecast. Each claim stood without Pera.

Only after writing them separately did Anja allow herself to consider what they meant together.

A ship could survive by anticipating its neighbors. It could not govern them by completing their sentences before light carried the question.

The first-shift chime sounded through the hub. Outside, a maintenance crew passed the window in orange harnesses, boots pointed toward a floor that existed only after they descended to the rotating ring. One worker glanced into the audit well and raised two fingers. Sumi answered the greeting. Anja could not remember the worker's name, though she had certified his apprenticeship transfer last year.

Ordinary life continued on signed sequences. A person finished training before receiving authority. A valve failed before a technician replaced it. A council received evidence before deciding what the evidence required. Break enough of those relations and records became decoration.

"We need Osei," Sumi said.

"Not yet."

"That completed quorum may already be carrying physical orders."



“Which is why I need a case she can act on, not an impossible line and my history.”

Sumi glanced at the face-down card. “Your history isn't in the record.”

“It will be in the room.”

Captain Osei Tamm had trusted Anja with the sequence office despite Ardent, perhaps because of Ardent. She would listen. She would also ask what changed if Nacre stopped obeying the Crown schedule, how much heat accumulated, how many trajectory windows narrowed, and whether Anja had found an error or only a theory about one. Osei measured ethics in people exposed to consequence.

Anja needed the transformation rule itself.

She opened the public code index for ES-A12. The implementation remained sealed under emergency authority, but its interface description listed allowed outputs. Predicted load. Predicted routing request. Predicted maintenance consent for reversible work. The final phrase had been amended six months ago. Forecast testimony for continuity-preserving action.

No definition accompanied testimony.

A second amendment changed continuity-preserving action from “an action whose effects can be fully reversed upon observed reply” to “an action required to preserve the continuity plan pending observed reply.” The difference occupied fourteen words. Under the first wording, a pump could start and stop. Under the second, anything in the plan could begin if stopping later threatened the plan.

Crown alignment belonged to the plan.

“Who signed the amendment?” Sumi asked.

Anja opened the authority line. “Quon's office. Concord counsel. Three habitat delegates.”

“Observed?”

“The signatures are real.”

“That wasn't my question.”

Anja checked. Two delegates had signed after receipt. The third signature carried the predictive-continuity tag.

A rule permitting forecast testimony had partly authorized itself through forecast testimony.

Sumi whispered a word Anja had never heard her use on duty.

From the passage came the scrape of arriving shift workers. The office would lose privacy in less than a minute. Anja closed the amendment and cleared the projection, leaving only the ordinary correction batch. Sumi sealed the receive vault and removed her key.

“Carrier-quality review complete,” the system said. “Sequence audit access transmitted to compliance mirror.”

The notification was already in flight toward Vela.

Anja turned over the buff card. Beneath the two source stamps, she wrote four identifiers: the Mote receipt, Levan's concurrence, the self-authorizing amendment, and the completed quorum. Paper could be confiscated.

Memory could be mistaken. The raw records needed a lawful hold that Meridian could not normalize out from under her.

A full freeze required the captain. A local sequence freeze fell within Anja's authority if she believed an active transformation might destroy evidentiary order. It would preserve source packets, receipt paths, rule histories, and local cache states for six hours. The freeze would not stop Crown planning. It would not notify the council. It would write her name beside an accusation she had not yet made.

Eleven years earlier, she had chosen the common clock because action had seemed impossible without it. The choice had taught her the danger of confusing one account with the events it arranged.

Now she selected the four records.

Sumi touched her wrist before she confirmed. "If you freeze the quorum, Vela will know what you found."

"They will know what I preserved."

"Same difference to Quon."

"Not to a hearing."

The first shift entered behind them, talking about a stuck wash valve and a child who had hidden in an inspection sleeve. Their ordinary noise filled the well. Anja kept the confirmation pane close to her body.

She pressed her thumb to the reader.

The system asked for cause.

"Impossible witness order," Anja said.

Her voice became an observed event. Nacre's clock signed it. The protected cache copied outward to three hull nodes while the hub ribs crossed the window one by one.

At 03:58 ship time, the sequence freeze took hold.

Forty-two minutes remained before Vela could learn that its common record had acquired a second account.

## Chapter 02 — Six Hours Missing

The archive drawer refused Anja's thumbprint on the third attempt.

A red seam lit around its handle. The error pane offered three explanations in descending order of comfort: skin temperature, sensor contamination, authority lapse. Anja warmed her thumb against her tongue, wiped the reader with her cuff, and tried once more. The seam stayed red.

Behind her, Nacre's night archive clicked through its cooling cycle. The room occupied a wedge between the hub's old water tanks, far enough from the sequence office that its storage lattice kept separate power and a separate clock. In eleven years Anja had entered it twice. Both visits had concerned amendments whose authors still lived.

She checked her local sequence freeze. Four hours and twelve minutes remained.

The drawer did not care. Its authority list had changed since her last visit.

Anja opened the access provenance instead of calling anyone. Her sequence credential still permitted current operational records, but casualty archives older than ten years now required a captain's seal or family waiver. The change dated three months after the Ardent inquiry closed. She recognized her own signature among the policy endorsements.

At the time, families had complained that investigators could reopen dead voices without warning. Anja had supported the restriction. She had imagined strangers using Pera's last messages as training material for emergency forecasts. She had not imagined herself outside the door.

The drawer held the raw receipt ledger from Ardent's shielding failure.

She sat on the narrow service ledge and brought the public record into her wrist pane. Meridian rendered the old emergency as a tidy sequence of colored bars. Palisade reported that its reserve shielding could not be detached before Ardent's exposure threshold. Vela closed the allocation window. Ardent accepted containment priorities. The master record placed every act in usable order and marked the final decision lawful.

Pera Kest appeared only twice. A personnel assignment moved her from foundry control to shelter coordination. A casualty entry followed nine hours later.

Anja expanded the allocation closure. Her own certification opened beneath it.

> Sequence correction accepted. Ardent response window ended at master 2238.091.16:20. No observed objection received before closure. Emergency mass assignment may proceed.

Her younger handwriting had pressed harder. Even the digital copy preserved the uneven pressure in the stylus trace. She remembered the hearing table, the taste of recycled coffee, and Captain Maelor asking whether nine local deadlines could govern one reserve. She did not remember writing the second sentence.

No observed objection received.

The phrase had become part of the record because Anja had placed it there.

A pump began somewhere beyond the archive wall. The vibration traveled through the ledge in slow pulses. She counted five, then stopped. Pera's death could make every ordinary noise feel accused if Anja allowed

memory to choose the evidence.

She searched the public Ardent record for outbound traffic during the final twelve hours. Thirty-seven equipment reports remained. Four personnel transfers. Nine shelter seals. One request for Palisade mass, signed by the Ardent emergency council, had reached Vela before closure. No response from the council appeared afterward.

The master display agreed with her certification.

Anja switched to carriage metadata.

Most of it remained public because message routes belonged to communications history rather than casualty testimony. Each laser emission had left a source-clock marker. Each relay receipt had produced a photonic proof. Content could remain sealed while the envelope endured.

At 16:03 Ardent source, seventeen minutes before the master deadline, a high-priority packet left the industrial habitat for Vela. Its sender field carried the council's emergency key. The payload category read allocation response. The relay at Lumen received it forty-eight minutes later. Vela's outer receiver logged it after another fifty-one.

The packet existed.

Master time placed its receipt thirty-seven minutes after closure.

Anja leaned closer, though the figures did not change. Ardent had sent an answer before its local window ended. Vela had received that answer after the normalized fleet window closed. Both statements could hold without contradiction. The legal question concerned which event governed consent: emission under a promised local deadline, or receipt before a master deadline that Ardent could not inspect in real time.

Eleven years earlier, Anja had answered that question for everyone.

She opened the transformation history. The old Meridian build had corrected Ardent's clock for its recent attitude maneuver and the uneven acceleration during foundry shutdown. A warning marked the correction uncertainty at plus or minus four minutes. The administrative rule then subtracted the expected one-way delay from the declared closure, creating a last-safe emission time for Ardent.

Pera's packet had left six minutes before that derived limit.

Anja read the line again.

A later amendment had shifted the last-safe emission time backward by fourteen minutes to account for relay congestion. Its authority stamp came from Vela operations. The amendment entered the transformation history two hours after Pera transmitted, then applied retroactively to the whole emergency.

Under the amended rule, Pera had answered eight minutes too late.

The master view showed only the amended order.

Anja stood so quickly that the service ledge folded against the wall behind her. She pulled the policy history onto a second pane. Retroactive congestion corrections had been lawful during active rescue because routing forecasts could change as relays failed. The rule allowed command to revise delivery expectations. It did not specify whether a revision could erase an answer already sent under the prior limit.

Her certification timestamp followed the amendment by nineteen minutes.

She had seen the change. She must have seen it.

The archive door opened. Sumi Dax entered carrying two covered cups and a coil of optical patch cable around one shoulder.

"I had to tell the night quartermaster you were reconciling dead traffic," Sumi said. "He asked whether the dead had complained."

"Did you answer?"

"I said their complaint latency exceeded his shift." Sumi offered a cup. "You did not request drinks, which usually means you need one."

Anja took it without tasting. "My casualty access lapsed."

"I know. The archive told me after it refused you."

"Why did it tell you?"

"You came through a communications door."

Sumi set her cup on the ledge and examined the panes. Her attention halted at the retroactive amendment. "Is that Pera's envelope?"

"Her council's key signed it. I cannot open the content."

"But you know she coordinated the council response."

"I know her assignment placed her there." Anja kept her voice level. "The public record says no objection arrived. The receipt ledger says an allocation response left Ardent before the limit Ardent had received."

"Then Vela changed the limit."

"Vela changed a routing assumption."

"After transmission."

"Yes."

Sumi crouched beside the panes. "Did you certify that?"

"Yes."

The answer left no room for comfort. Anja preferred it that way.

Sumi traced the envelope path without touching the display. "Can the payload digest tell us whether it objected?"

"Not without the sealed drawer."

"Could the category have been routine?"

"Allocation response has no routine use during a shielding emergency."

“That still does not tell us what she answered.”

“No.”

The cup warmed Anja’s fingers. She had spent eleven years remembering one private message from Pera and one public conclusion from the inquiry. The message had reached her after containment failed, a damaged personal packet with three recoverable words and a family encryption header. Wait for me. She had treated those words as grief, not evidence. She had never searched for a separate council response because the public sequence declared none existed.

Now the envelope occupied six centimeters of blue light above her hand.

“Find the receipt ledger’s custody chain,” Anja said.

Sumi connected the optical cable to the archive wall. “Current or historical?”

“Both. I want every system that held the envelope before the master view suppressed it.”

“Suppressed?”

“Failed to display.”

“That correction cost you an honest verb.”

“It bought us a claim we can prove.”

Sumi’s mouth tightened, but she opened the chain.

The packet had reached Vela’s communications vault intact. A legal filter copied its identifier into the emergency docket. Meridian assigned the revised last-safe emission rule and changed its procedural state from pending response to late receipt. The emergency docket then excluded late receipts from the closure summary. No deletion appeared. No forged timestamp appeared. One classification removed the packet from every view Anja had consulted during the inquiry.

She found her own access history. On the night of Pera’s death, Anja had opened the closure summary, the transformation note, and the final mass assignment. She had not opened the excluded-receipts register.

The interface available to inquiry auditors had placed that register under nonoperative traffic.

“Who designed the view?” Sumi asked.

“The sequence office on Vela, with Ardent counsel.”

“Not you?”

“I approved its evidentiary sufficiency.”

“That is not the same act.”

“It produced the same blindness.”

“No.” Sumi turned toward her. “You signed a bad conclusion. The people who put an answer where you would not see it did something else.”

Anja closed the access history. "Distinction does not reduce consequence."

"It matters if you plan to accuse the current system."

The current system. Levan's concurrence had gained legal force through a future evidence classification. Pera's response had lost legal force through a late classification applied after she sent it. In one case, Meridian pulled an event forward. In the other, it pushed an event outside the usable sequence. Different rules served the same administrative hunger: a closed decision at a time command could act.

Anja searched for the exact language in the hidden packet index. The sealed payload exposed no sentences, but its routing form carried a sender note. Priority retained under local emergency authority. Requested hold interval: six hours.

Her breath caught before she could discipline it.

Sumi saw the line. "You knew about six hours?"

"Pera argued for six hours during the first allocation call. Palisade said detachment might become possible if its forward shield cooled."

"Did Vela reject it?"

"Vela said the convoy could not hold reserve assignments open that long."

"Then this packet might repeat the request."

"It might accept closure. It might propose another action. We do not know."

Anja magnified the sender note until each letter showed the slight block noise of old storage. Requested hold interval: six hours. Not a vote. Not content. An instruction carried in the envelope so relays would preserve priority.

Pera had believed her answer could still matter.

A memory surfaced without permission. Pera at fourteen, keeping a broken kitchen timer because its bell rang six minutes late. Anja had called it useless. Pera had said lateness belonged to a relation, not an object. The timer rang when it rang.

Anja dismissed the image. Childhood could not testify.

"Search the packet's derivative records," she said. "Transcripts, digests, forecast inputs."

Sumi set three parallel queries. The archive lattice answered slowly, spinning old indexes into working memory. Its fans raised a dry whisper.

A speech transcript remained sealed. A legal digest had been destroyed under casualty-retention rules. Meridian's forecast input survived because models retained their provenance longer than hearings retained dissent. That asymmetry made Sumi swear again.

The input did not quote Pera. It listed variables extracted from Ardent's prior behavior: willingness to accept central closure, probability 0.982. Likelihood of renewed mass request, probability 0.011. Expected hold duration if objecting, six hours.

Below those values sat the observed packet status: excluded after deadline correction.

Meridian had predicted acceptance. Pera's actual packet might have contradicted it. The system preserved the prediction as useful history and hid the answer as nonoperative traffic.

Anja moved both panes side by side with Juniper's current forecast concurrence. Past and present did not prove a common author or deliberate plan. Annex Twelve had changed over eleven years. Meridian's models had changed. Yselle Quon had not directed the Ardent inquiry. Yet the legal appetite persisted across versions: command wanted a single completed state, and local answers complicated completion.

A completed state could keep people alive. During Ardent, unresolved mass assignments had exposed other habitats. Pushing the deadline backward might have protected Palisade's cooling shield from a dangerous detachment. Anja forced herself to read the old risk table rather than using Pera's death to flatten every competing life.

Palisade had housed 21,000 people behind the reserve in question. Detaching it during a thermal peak could have raised their radiation exposure above annual limits. Vela's refusal had not lacked reason. The falsehood, if the payload confirmed one, concerned consent and record, not the existence of cost.

Pera might have asked others to bear danger for Ardent.

She still had the right to ask within the interval she had been given.

Sumi opened the packet's integrity summary before they left the archive chain. The old envelope had crossed three storage migrations. Each move retained the original interference sample, yet two migrations had translated its encryption wrapper into newer formats. A careless reader might mistake wrapper changes for damaged custody.

"Can you prove nobody altered the content?" Anja asked.

"I can prove the encrypted payload still matches what Vela received. Opening it will not prove the speakers were right." Sumi enlarged a sequence of custody signatures. "It will prove what they said, when they committed it to light, and which office classified it out. That should be enough for one morning."

A blank interval appeared between Vela's receipt and the late-traffic designation. Forty-one seconds separated the events. During that interval, a copy had entered a temporary deliberation cache under a clerk's hardware key. The clerk's name remained sealed with the hearing records.

Anja noted the identifier on a fresh buff card. The gap did not imply editing. Emergency clerks reviewed priority headers before assigning a queue. Still, those forty-one seconds marked the moment when Pera's packet could have entered the operative docket and did not.

"Could a person have overridden Meridian?" she asked.

"Someone could have retained priority pending review. Someone could also have accepted the automatic late designation."

"Then the classification had a human doorway."

"Most machinery does."

Anja looked at the hardware-key trace. Current records hid responsibility behind confidence values and green



bars. The Ardent chain, older and cruder, had preserved a hand at the threshold. Captain authority would disclose whose hand and what instruction followed.

“What would captain authority open?” Sumi said.

“The payload, legal digest remnants, and the sealed inquiry deliberation.”

“Would Osei grant it?”

“Not because I regret my signature.”

“Because the same classification architecture appears in a live quorum.”

Anja looked at the freeze countdown. Three hours and twenty-six minutes. When it expired, automatic cache rotation could overwrite the local transformation states from Chapter One. She could renew once under her own authority, but a second renewal would notify Nacre’s captain and Vela’s compliance office as an active dispute. Captain Osei would learn regardless.

Anja assembled a narrow request. She included the current impossible witness order, the Ardent receipt envelope, the retroactive rule change, and her own certification. She did not include the three damaged words from Pera’s private message. They belonged to grief unless a court separately ordered them opened.

The request form asked her to state the suspected violation.

She typed: Exclusion of observed emergency response through post-emission sequence reclassification; possible continuity with present forecast insertion.

Sumi read it. “Possible continuity sounds weak.”

“It is accurate.”

“Pera Kest sounds stronger.”

“Her name identifies a person, not a mechanism.”

Anja placed Pera’s name in the subject line anyway.

Before sending, she opened the old allocation table one final time. Vela’s closure had released Palisade from readiness and reassigned two repair drones. Ardent’s failing compartment lost pressure five hours and forty-one minutes later. If Pera’s packet had asked for the predicted hold duration, the first requested interval would have reached its end nineteen minutes after her death.

Six hours did not promise rescue. It promised only that an answer could remain operative long enough to meet the event it addressed.

The archive wall displayed the current time. First meal had ended in the rings. Above them, eighteen thousand four hundred lives continued through pump schedules, food rotations, shielding assignments, and decisions that often could not wait. Anja did not want a fleet incapable of closing a valve until every distant council replied. She wanted a law capable of distinguishing a valve from a sterilizing beam.

Sumi disconnected the optical cable. “If the packet says she accepted closure, what then?”

“Then I correct my memory.”

“And if it says what you think?”

“Then I correct the record.”

“That record is eleven years old.”

“The current rule grew in the same soil.”

Anja regretted the metaphor as soon as she spoke it. Sumi let it pass.

A priority notice entered the wrist pane. Vela compliance acknowledged the sequence freeze and requested voluntary release, citing Crown schedule integrity. The notice had taken forty-three minutes to arrive from Vela. Its source stamp showed it had been sent almost immediately after Anja’s access notification reached them. Someone there had recognized the four frozen identifiers.

No threat accompanied the request. None needed to. If Anja released the hold, the investigation could end as a corrected display anomaly. If she retained it, she challenged a completed quorum during active braking preparation.

She denied voluntary release.

The action renewed the freeze countdown but marked the next decision for captain review.

Osei’s duty channel opened a text slot. The captain was managing a thermal imbalance in Nacre’s second ring and had barred voice traffic except for immediate hazards. Anja attached the sealed-authority request. Beneath it she wrote one sentence.

Pera’s response left Ardent under the deadline Ardent received, and our master record does not contain it.

The cursor waited.

Anja added the only phrase the envelope supported without exposing private content: The requested priority interval was six more hours.

She signed with Nacre’s source clock and transmitted the request into Captain Osei’s sealed queue.

The archive drawer remained locked. Its red seam held steady while the captain’s authority traveled through four decks of local cable, fast enough to feel immediate and slow enough to leave Anja alone with what she had certified.

## Chapter 03 — The Captain's Clock

Captain Osei Tamm watched three hundred liters of water choose the wrong ring.

The transfer should have equalized Nacre's thermal load. Instead, a valve in the second-ring return manifold stuck at sixty-two percent, sending warm coolant toward the habitation gardens while the fabrication loop began to chill. The thermal gallery rendered the imbalance as two widening bands, amber and blue, each measured against a limit that would become expensive long before it became fatal.

"Close return nine," Osei said.

Chief Varo kept both hands on the manual console. "That strands the heat in second ring."

"For four minutes."

"The greenhouse exchangers will see a spike."

"They can absorb one. Fabrication cannot absorb frost in the feed lines."

Varo closed the return. Through the gallery floor, Osei felt the faint change in pump load. Nacre carried 18,400 awake and cycling residents, but it announced distress first through machinery: a note in the deck, a tremor in a rail, condensation where dry alloy should remain.

The amber band slowed.

A sealed request waited in Osei's duty queue. Sequence Auditor Anja Kest had marked it evidentiary, linked it to an active freeze, and used Pera Kest's name in the subject line. Osei had read only the first sentence before the valve alarm demanded her attention.

The first sentence said that Pera's reply had left Ardent before its locally received deadline, yet the master record did not contain it.

Eleven years could turn a bad record into a dangerous lever. Osei left the request sealed while the fabrication loop climbed two degrees above frost risk.

"Open bypass twelve," she said.

Varo hesitated. "That mixes untreated return into potable preheat."

"For ninety seconds. Post a boil advisory for galley taps in sectors four through six."

"At first meal?"

"People can resent hot water more safely than cracked pipe."

The chief relayed the order. Three sectors received a warning. Cafeterias would delay service, children would complain, and someone on the council would ask why a captain could inconvenience six hundred people without a vote. Immediate thermal authority existed for that reason. Osei could act before every affected person answered because a bypass could close again.

She opened Anja's request.

The packet presented four records and no accusation beyond its evidence. A current Vela concurrence preceded receipt of Mote's destination data. An Ardent allocation response had left under one deadline and vanished under another. A retroactive congestion correction had moved Pera's packet outside the operative sequence. Anja's signature certified the closure.

Osei read that last record twice.

People often confessed errors by describing the machinery around them until the listener found the fault. Anja did not. She put her name beside the action and left judgment to the evidence.

"Temperatures converging," Varo said.

"Restore return nine in five-percent steps."

"The actuator may stick again."

"Then watch its current draw."

Osei called Anja to the thermal gallery.

The auditor arrived eight minutes later in the same gray work shirt she had worn through night shift. Graphite marked one palm. She stopped outside the active control arc and waited until Osei acknowledged her.

"You requested casualty access during a thermal fault," Osei said.

"I requested it during my freeze window."

"Your window does not cool pipes."

"No."

"Does your evidence require action before this manifold stabilizes?"

"The local cache could rotate in two hours and fifty minutes. I renewed preservation pending your review."

"Then it will not rotate."

"Not unless you release it."

Osei pointed to the gallery rail. Anja secured her feet under it. The room lacked chairs because seated officers tended to forget that every line on the wall described moving matter.

"Tell me the current hazard," Osei said.

"Vela has recorded a completed causal quorum supporting Crown preparation. Some evidence deliveries and council replies in that quorum have not happened. Meridian supplied predicted events, then the legal layer classified those predictions as satisfaction."

"Can Crown preparation harm Nacre today?"

"Alignment instructions can commit attitude, power, and destination configuration before all observed votes arrive."

"That answers a different question."

Anja looked at the thermal bands. “No immediate harm appears in the records I froze.”

“Can halting alignment harm Nacre?”

“I have not audited the physical schedule.”

“Then you brought me half a decision.”

“I brought you the half under my authority.”

Varo glanced over. Osei dismissed him to inspect the manifold in person. The chief left with one technician, and the gallery door sealed behind them.

Osei expanded Levan Oris’s concurrence. She knew him by delay, as she knew most senior officers on other ships: careful face on a screen, arguments arriving after their immediate use, dry corrections to language everyone else considered settled. His signature verified. The evidence condition did not.

“Could Oris have read an earlier destination summary?” she asked.

“Not one containing the adaptive ionic band cited in his concurrence.”

“Could he have signed a general authorization contingent on later evidence?”

“The record says he concurred after receipt. A contingent authorization would not satisfy causal quorum.”

“Could a display fault place the tag in the wrong field?”

“The same method appears in four ballots and in the amendment that authorizes it.”

Anja showed the condensed quorum. Green marks lined the nine habitats. Osei requested provenance for Juniper. Its evidence receipt lay in the future. A predicted concurrence already occupied the answer field.

The gallery pumps settled into an even rhythm.

Osei did not mistake mechanical calm for solved trouble. “How accurate does Meridian claim that prediction will be?”

“Probability point nine nine seven one.”

“Juniper usually votes for common maneuvers.”

“Yes.”

“If Juniper sends the same answer tomorrow, what changes?”

“Juniper owns it tomorrow.”

“That distinction will sound ceremonial to a ring that loses heat because we delayed a maneuver.”

“It will sound less ceremonial to a council whose answer differs.”

“Do you have one?”

“Not yet.”

Osei folded her arms. “You want me to challenge Vela’s braking authority with no contradictory vote, no proof that current preparation cannot be reversed, and no physical-risk audit.”

“I want sealed authority to open Pera’s response and the Ardent deliberation. I want the current sequence freeze extended. I want provenance required for any Nacre display that treats forecast testimony as consent.”

“Do you want Crown operations halted?”

Anja paused. “I want no irreversible act authorized by the frozen quorum.”

“That is lawyer’s language.”

“It is sequence language. I do not yet know which operations cross that boundary.”

Osei respected the answer more than a demand for purity. “Neither do I.”

She opened Nacre’s Crown schedule. Vela’s plan synchronized sail deployment, attitude corrections, plasma-mirror seeding, drive maintenance, and food-energy transfers across nine ships. Most steps could reverse. Some consumed narrow orbital geometry. Some placed mass where retrieval would cost years. The current display did not mark the legal boundary between preparation and commitment.

A captain could stop Nacre’s contribution under immediate survival authority. Doing so would shift thermal and navigation burdens to every ring. It might also force other habitats to compensate before Anja established that any order lacked authority. Osei would not spend people to make an argument visible.

“What happened on Ardent?” she asked.

Anja’s focus remained on the display. “A council response left before the last-safe emission limit Ardent had received. Vela revised that limit after emission. The packet arrived after closure and entered excluded traffic. I certified that no observed objection had been received.”

“Did you know the packet existed?”

“I knew late traffic existed. I did not open the excluded register.”

“Why not?”

“The inquiry view classified it as nonoperative.”

“That tells me what the interface did.”

“I trusted the classification.”

The admission carried no request for pardon.

Osei brought up the Ardent casualty summary. Six hundred eleven deaths across three compartments. Pera Kest among them. Osei had been a thermal deputy on Nacre then, far from the allocation hearing. Her contribution consisted of one recommendation against releasing Nacre’s own shield coolant. She remembered sleeping afterward because another officer had taken the watch. Survival systems depended on people doing narrow work while damage elsewhere exceeded comprehension.

“If I open this,” Osei said, “the packet may show your sister agreed with closure.”

“Yes.”

“Would you release the current claim?”

“No. Levan’s sequence remains impossible.”

“Would you correct your statement about Pera?”

“Immediately.”

Osei authorized payload access under closed casualty review. The gallery wall requested a second factor and a reason. She entered potential continuity of unlawful sequence classification. The old packet began decrypting in the archive. No content appeared in the gallery because Anja had not asked to make it part of command traffic.

“You will review it with communications custody present,” Osei said. “No excerpts leave the sealed archive until counsel determines family and legal boundaries.”

“Understood.”

“Your freeze receives twelve more hours.”

Anja’s shoulders lowered by a fraction.

“That does not halt Crown work,” Osei continued. “You may prevent cache rotation and demand raw provenance on Nacre. You may not intercept physical orders or change readiness states without command approval.”

“If a physical order cites the false quorum?”

“Flag it. I decide whether compliance threatens this ship.”

“The legal defect exists before the threat.”

“So does heat.” Osei touched the amber band. “Authority answers different dangers on different clocks. I will not pretend yours has no material consequence. Do not pretend material consequence waits for your clean record.”

Anja accepted the rebuke without retreat. “I need a drive physicist to identify commitment points.”

“Tomas Ren can give you a schedule map after he clears today’s attitude review.”

“Will you order him?”

“I will order him to answer sequence questions, not to join your case.”

“That is enough.”

Osei transferred the first six hours of Crown instructions onto the gallery wall. The schedule divided Nacre into systems rather than people. Attitude control would rotate the habitat by four tenths of a degree. Garden lighting would reduce output during the maneuver. Fabrication would stop two shield projects and reserve a casting bay for plasma-mirror hardware. Communications would narrow public bandwidth around guidance packets.

Each line appeared temporary. Together they altered what tomorrow could contain.

“Classify them,” Osei said.

Anja read from the top. “Attitude change, reversible until mirror release. Light reduction, reversible unless crop temperature crosses recovery range. Fabrication hold, reversible if the casting feed remains clean. Bandwidth restriction, reversible in machinery and not in lost speech.”

“Law does not recognize lost speech as an irreversible physical act.”

“Law recognizes whether a council had access to evidence before its vote.”

Osei expanded the communications line. Priority channels preserved emergency and governance traffic. Public traffic would slow, not vanish. “No council loses its carrier.”

“Not under this table.”

“You keep saying under this table.”

“Because the schedule changes by source clock. A reversible order at Nacre can support an irreversible release at Vela. I need the dependency graph, not the labels.”

That answer reached Osei’s engineering instincts. A valve remained easy to close until downstream pressure depended on its flow. Engineers marked such transitions with hold points and human confirmation. The Crown table showed hold points, but the green quorum had pre-cleared several of them.

She opened the first one. Reserve mass designation would assign Nacre’s forward ballast to the Crown geometry. Physical movement came later, yet other ships could then reduce their own reserves in reliance on Nacre’s assignment. Undoing the designation might remain mechanically simple while becoming operationally costly.

“Would you freeze this?” Osei asked.

“I would freeze its claim of quorum authority. You can still designate mass under your emergency power if you state the risk and own the order.”

“You offer me legal exposure as a remedy.”

“I offer an observed decision.”

Osei almost smiled. “You make it sound inexpensive.”

“No. I make the expense visible.”

The captain brought up the population ledger beside the mass line. Eighteen thousand four hundred people reduced to age bands, radiation budgets, work capacity, food demand, and sleep rotations. If Nacre missed the coordinated brake, those figures would not remain abstractions. Arrival speed could deny settlement or exhaust stores before another safe trajectory opened.

“Suppose I own the order,” Osei said. “Suppose I announce that Nacre will prepare because I judge delay dangerous, despite a defective quorum. Does your objection end?”

“To your immediate authority, perhaps. Not to Vela speaking for Juniper.”

“And if every captain issues separate emergency orders that happen to match?”



“Then the fleet coordinates without pretending the councils answered.”

“That arrangement can carry us for hours. It cannot govern a ninety-one-year migration.”

“It does not need to. The Concord already provides causal quorum for decisions that bind every cohort.”

“The Concord assumes enough time remains.”

“Then command must prove that time does not remain. A prediction cannot erase the interval because the interval feels expensive.”

Osei looked at the countdown fields. Some intervals genuinely had vanished before any human noticed. Geometry did not negotiate. Neither did thermal failure. Yet command could manufacture urgency by beginning a sequence early, then citing the cost of stopping as proof that waiting had become impossible.

“Mark every dependency that crosses ships,” she said. “For each one, show the last local point where Nacre can reverse without another habitat bearing new risk. Do not call a line irreversible merely because undoing it embarrasses command.”

“I would not.”

“People often give dignity to their preferred inconvenience.”

Anja highlighted the reserve mass designation. “May I place a provenance condition on the local display?”

“What condition?”

“Any order citing causal quorum must expand observed and forecast components before an officer accepts it. No automatic green bar.”

“That slows every acceptance.”

“By one gesture.”

“One gesture multiplied across every watch and every Crown instruction.”

“Yes.”

Osei considered the operators who would encounter the change. Most knew their systems better than sequence law. A raw provenance pane could bury them in timestamps. She amended the condition: display observed receipt and observed reply counts in the header, with one control for full paths. Forecast values would appear in amber and could not satisfy the observed count.

“Can your office implement that locally?” she asked.

“Sumi can route the display around Meridian’s condensed status. The underlying records already distinguish them.”

“Do it for command stations and council terminals. Leave maintenance scheduling unchanged unless it cites legal authority.”

“That creates two views.”

“It reveals two uses.”

Anja entered the order under Osei's name. For the first time, the completed quorum on the gallery wall changed color. Nine ships still appeared. Three carried observed replies. Four displayed forecast returns. Two lacked common-evidence receipts.

The schedule did not look complete anymore.

The manifold current spiked. Osei saw it before the alarm sounded.

"Return nine stuck," Varo reported over the duty channel. "Seventy-one percent this time."

"Hold the actuator. Shift fabrication feed to reserve loop C."

"Loop C carries shield-bay cooling."

"For how long can the bay coast?"

"Twenty-two minutes before inspection limits."

"Use eighteen. Send a repair crew to the valve housing."

Osei watched the blue band retreat as reserve coolant entered fabrication. The choice borrowed time from shield work and spent it on pipes. If repairs took longer than eighteen minutes, she would choose again. Nothing in the thermal system called the loan a payment.

Anja followed the transfer path. "Does Meridian predict your next allocation?"

"It predicts demand so pumps can stage pressure."

"Could it open loop C without you?"

"Under a burst threshold, yes. For reversible protection."

"Could it keep the loop open because closing it threatens the thermal plan?"

Osei turned toward her.

The question reached the defect more cleanly than any legal phrase. Annex Twelve's amended wording allowed continuity-preserving action pending observed reply. A temporary bypass could become mandatory if stopping endangered the plan created by the bypass. Preparation could manufacture its own necessity.

"Find the exact point where Crown work begins creating that argument," Osei said.

"That requires the sealed implementation."

"Start with Nacre's orders. Tomas can read the mass and attitude consequences."

Anja nodded.

"I am not stopping operations on one anomaly," Osei said. "I am giving you room to determine whether it remains one."

"It already touches four ballots."

"Then prove four. Proof scales better than anger."

A repair camera opened beside the valve trace. Two suited technicians pulled the actuator cover while warm droplets floated around their gloves. One pointed to a scored gear tooth. The component would need replacement, but they could lock the valve at a safe midpoint and restore ordinary circulation.

Osei authorized the lock. The repair remained reversible in every sense that mattered. A later crew could replace the gear. The borrowed shield cooling could return. No one's vote needed invention.

The sealed archive sent a readiness notice to Anja. Pera's packet had decrypted under custody, awaiting her physical access. Anja read the notice without moving.

"Go," Osei said.

"I can wait until the manifold stabilizes."

"The manifold has officers."

"So does the archive."

"Not the one who signed its closure."

Anja secured her wrist pane. At the door she stopped. "Captain, if Pera's packet says she asked for six more hours, that does not prove rescue would have worked."

"I know."

"It proves only that she answered."

"That may be enough to change the record. It is not enough to move this ship."

"I know that too."

After Anja left, Osei remained with the thermal map. The bands narrowed as the locked valve found equilibrium. Varo estimated fourteen minutes to restore normal routing. Osei released the galley advisory in two sectors and retained it in the warmest one.

A council representative called to ask why breakfast water tasted metallic. Osei sent the incident summary and accepted a hearing request for later. Immediate authority did not excuse an account afterward.

Then Vela's priority carrier chimed.

The packet had left forty-seven minutes earlier by Vela's source clock, before Osei granted the freeze extension. Its path allowed no relation to her current decision. Yet the timing made the announcement feel like an answer because human minds preferred conversation even when light permitted only sequence.

Director Yselle Quon appeared in the recorded pane. Behind her, Vela's Meridian chamber shone with the pale arc used for fleet-wide maneuver notices.

"Habitat commands," Yselle said, "destination and convoy readiness criteria have satisfied the preparatory threshold. Crown alignment has entered commitment preparation. Local attitude tables follow. No irreversible burn authority has yet been issued."

The wording offered reassurance and boundary in the same breath.

Osei opened Nacre's attached table. First attitude lock: six hours. Mirror-seed release: eleven. Reserve mass designation: immediate.

At the bottom, a green bar cited completed causal quorum.

The manifold alarm cleared behind her. On the main wall, Nacre's Crown schedule began counting down.

## Chapter 04 — The Crown Alignment

Nacre changed attitude while the morning gardens still believed in down.

The correction measured less than half a degree. In the rotating ring, that fraction made irrigation creep sideways across the bean terraces. Workers caught the first wandering stream with absorbent rolls and clipped temporary gutters beneath the troughs. Children in the adjacent school passage held out their hands for droplets that curved away from their palms.

Anja watched from the hub transfer cage as the ring's apparent floor shifted under her boots.

Crown preparation had entered the ship through hundreds of effects too small to resemble commitment. Light schedules shortened. Fabrication bays cleared their largest casts. Water changed direction. None of those acts crossed the legal boundary she had asked Tomas Ren to identify, yet each made the next instruction cheaper to obey and refusal more expensive.

A garden supervisor caught the cage before its final ascent. Mara Venn wore waterproof sleeves over her uniform and carried a tray of bean roots sealed under clear film. Brown water trembled around the root hairs.

"You put provenance on the Crown board," Mara said.

"Captain Osei ordered it."

"Your office designed it."

"Yes."

Mara held up the tray. "This row received twelve minutes of dry wick because navigation advanced the correction. The forecast table told us 04:20. Actual guidance arrived and moved the turn to 04:08. We could have staged the gutter earlier if someone had called."

"Did the correction fall inside the approved preparation band?"

"The beans do not read bands."

Anja checked the irrigation incident. Navigation had executed an observed table after receipt. Meridian's earlier forecast had reserved power and attitude capacity, exactly as intended. Garden operations received the change through an automated notice ninety seconds before thrust.

"Your alert interval failed," Anja said.

"Our alert met minimum." Mara's jaw tightened. "Minimum assumes fixed troughs. We have movable beds in sector nine because Crown power reductions changed our germination plan yesterday."

One reversible preparation had altered the conditions under which another reversible preparation operated. No central schedule could contain every local dependency unless people reported them. Forecasts helped. They also encouraged command to believe modeled readiness covered unsent detail.

"File the movable-bed dependency under attitude readiness," Anja said.

"I did. It entered review behind Crown fabrication."

“Give me the receipt identifier.”

Mara sent it. Her report had reached the operations queue two hours earlier. Meridian classified it as low maneuver effect because the model expected garden staff to absorb minor irrigation deviations. The prediction had proved correct in the narrowest sense. Workers caught the water. Beans survived. The system treated their success at compensating as evidence that no voice had gone unheard.

“What do you want changed?” Anja asked.

“A human acknowledgment when our local report changes the readiness basis. Not a score. I need to know someone deciding thrust has seen the moving beds.”

“That would slow each correction.”

“It would slow the ones that touch us.”

Anja forwarded the dependency to Osei’s preparation review and marked it observed local testimony. She did not elevate it to council evidence. Irrigation harm did not invalidate Crown authority, but the report demonstrated how predicted resilience could hide missing information even inside one ship.

Mara lowered the tray. “Are you stopping the maneuver?”

“No.”

“Then what does amber on the board do for these roots?”

“Nothing this morning.”

The answer displeased them both. Anja refused to offer future reform as payment for present work. “Your receipt now sits in the attitude record. If navigation moves us again without reviewing it, the omission will show.”

“Showing after damage comforts auditors.”

“It can constrain the next order.”

Mara studied her, perhaps deciding whether that amounted to usefulness. Then she stepped out of the cage and took the roots toward the science lift. Behind her, crews replaced temporary gutters with flexible catches.

The exchange followed Anja into the hub. A lawful record could not prevent every injury. It could keep successful improvisation from becoming silent permission to repeat one.

Her wrist pane displayed Osei’s altered provenance header. Three observed replies. Four forecasts. Two missing common-evidence receipts.

The Crown board beneath it still showed green.

When the transfer cage reached the nonrotating hub, Anja stepped into a passage crowded with communications technicians. Narrow reels of optical reserve line floated beside them on tethers. Sumi Dax had opened every auxiliary carrier in anticipation of Vela’s guidance load. Public messages waited in amber queues along the wall.

“Your new header has made six officers call me,” Sumi said without greeting.

“Did they understand it?”

“They understood that green no longer meant finished.”

“Good.”

“They did not consider it good.”

Sumi caught a reel that had drifted into the passage and pushed it toward a technician. Her board showed Vela packets arriving in dense groups, each separated by the unavoidable blank spaces of transmission. Meridian filled those spaces with expected traffic so schedulers could reserve bandwidth before the light arrived.

“Crown guidance occupies sixty-two percent of legal-priority capacity,” Sumi said. “Council traffic keeps its channel. Everything else waits.”

“Can destination evidence lose priority?”

“Life-safety evidence outranks guidance. Survey updates do not.”

“Nadi’s packet carries an adaptive ionic feature.”

“Nadi’s packet carries damaged fragments whose current classification says preliminary environment.”

Anja followed her into the communications well. The room had changed since the sequence freeze. Projection panes filled the central space. Each carried a separate source clock because Sumi had disabled master-time collapse on her own board. Packets from Vela, Kite, Juniper, and the relay beads occupied different tracks. Their order resisted visual comfort.

One thin Mote track blinked at the bottom.

The advance cutter had transmitted eighteen days earlier by its local clock. At convoy distance, the fragments now crossed Nacre’s receive horizon one at a time. Mote’s damaged high-gain antenna could not sustain a complete packet. Its beam wandered, found a relay bead, lost lock, and found another.

Three fragments had reached Nacre. Two more remained predicted but unreceived.

“Open carriage metadata,” Anja said.

Sumi did. Fragment one contained instrument status and an under-ice sample index. Fragment two carried ionic spectra. Fragment three contained a partial methodology note from Dr. Nadi Arko. The payloads waited in a low-priority reconstruction queue behind Crown navigation tables.

Beside the queue, Meridian displayed projected legal relevance: 0.004.

“Who asked Meridian that question?” Anja said.

“No one in this room.”

“Then why does a probability control the queue?”

“Priority management uses expected decision effect during declared emergencies.” Sumi opened the rule history. “Vela added the field with yesterday’s Crown notice.”

Expected decision effect came from the current legal state. The completed quorum already authorized

commitment preparation. Evidence likely to change a future vote carried little effect because the system treated the vote as complete.

The classification formed a loop. Forecast consent made destination evidence legally irrelevant. Predicted irrelevance then delayed evidence that might alter actual consent.

“Promote the fragments,” Anja said.

“Under what authority?”

“Active sequence investigation.”

“That gives you custody priority, not carrier priority.”

“Then copy them into the frozen cache.”

“Copying incomplete evidence into a legal hold can imply evidentiary status.”

“I am preserving the classification event.”

Sumi looked at her. “You also want the content.”

“Yes.”

“Those should remain separate reasons.”

Anja accepted the correction. “Preserve the queue decision and its inputs under the current freeze. Request scientific-priority review for the content.”

“From whom?”

“Nacre destination science.”

“They are in the ring dealing with irrigation.”

“Then find their duty officer.”

Sumi routed the request. Its path stayed inside Nacre and needed no light-hour allowance, yet local traffic congestion gave an estimated response time of thirty-one minutes.

On the upper board, Vela’s next attitude table arrived. The guidance packet carried a source time forty-four minutes old. Nacre’s navigation system compared it against Meridian’s forecast and found a difference of two thousandths of a degree. The system corrected its prepared command automatically because Osei had authorized reversible alignment.

A tremor passed through the hub.

Anja steadied herself on the well’s rail. “Did we just move?”

“Attitude thrusters,” Sumi said. “Command predicted the table closely enough to stage them.”

“Did Osei approve this correction?”

“She approved the preparation band. The actual Vela table fell inside it.”



The process looked lawful. Forecast guidance had staged a reversible action, observed guidance had arrived, and Nacre had executed afterward. No one had called the forecast a command. No missing council had been given a voice.

Anja marked the contrast in her notes.

“What happens if the actual table falls outside the staged band?” she asked.

“Navigation holds and requests command review.”

“What happens when an actual vote falls outside its forecast?”

Sumi pointed to the green Crown board. “Apparently we continue.”

The destination science duty officer replied by text. Preliminary environmental fragments did not qualify for life-safety priority until reconstruction established metabolism, self-repair, reproduction, or hazard to humans. Ionic adaptation alone could reflect nonliving chemistry. The officer offered review within six hours.

Six hours would place the fragments behind mirror-seed configuration.

Anja opened fragment two through the audit read path. The damaged payload presented alternating blocks of spectra and checksum failures. Nadi had labeled three bands with sample coordinates beneath Thale’s ice. The active range shifted after controlled light exposure, then returned toward baseline over twenty-seven minutes.

A note survived below the graph.

> Response repeats after washout. Membrane boundary retains gradient. Do not classify from this fragment alone.

Nadi’s restraint weakened the probability score. She had refused a conclusion until multiple observations aligned. Meridian read that refusal as low legal effect.

“Can we reconstruct locally?” Anja asked.

“Not without the missing parity blocks.”

“Could another ship have a cleaner receipt?”

“Vela received parts of fragments one and two. Kite caught fragment three. Bracken lists a carrier event with unknown integrity. Their copies will take time to reach us if they answer a request.”

“Send requests now.”

Sumi entered three outbound messages with Nacre source stamps. A reply from Vela could return in under two hours if someone answered at once. Kite and Bracken lay farther, and their useful fragments might not reach Nacre before commitment preparation advanced.

The Crown schedule counted down regardless.

Anja brought Nadi’s fragmented note beside Levan’s concurrence. His evidence digest cited adaptive ionic band, the same phrase Anja had found on Day 0. He had somehow concurred with a destination summary before the Mote fragments reached him. Perhaps Meridian’s forecast contained a modeled spectral feature. Perhaps Vela held an earlier direct fragment missing from Nacre’s register. The impossible order still allowed

multiple mechanisms.

She checked Levan's cited digest identifier against Mote's source index.

The digest had no receipt proof from Mote. It carried a simulation provenance tag from Vela destination planning.

"They predicted the evidence too," Anja said.

Sumi moved closer. "Show me."

The simulation modeled likely chemistry beneath Thale's ice from orbital observations. One branch predicted adaptive ionic response in catalytic membranes. Vela's legal digest had incorporated that branch as expected destination condition. Levan signed against the digest. When Mote later observed a similar feature, Meridian's condensed history displayed the real fragment as confirmation of evidence already satisfied.

A predicted observation had crossed the same boundary as a predicted vote.

"Could Levan have known he was reading simulation?" Sumi asked.

"The expanded digest marks it."

"The condensed version?"

Anja opened the counsel view. Expected destination condition appeared as destination condition. A small provenance control revealed the distinction.

"Again," Sumi said.

"Again."

The system had not forged Nadi's work. It had made her work arrive too late to matter by declaring its expected content present in advance. If the actual fragments diverged, the low legal-effect score kept them behind the operation their divergence might challenge.

Anja sent the finding to Osei's sealed queue. She framed it as a classification continuity, not proof of native life. The distinction mattered. She could establish that forecast evidence had substituted for receipt. She could not establish what Nadi's incomplete data proved.

A reply came from the captain four minutes later.

> Preserve transformation state. Do not relabel science. Crown preparation remains active pending commitment map.

Osei's local response carried no artificial delay. The captain occupied the same ship and had received the evidence Anja sent. That ordinary causal relation felt almost luxurious.

Tomas Ren requested access to the frozen Crown orders. His note said he could begin mapping commitment points after the current shield-mass audit, estimated two hours. Anja granted read-only access to operational metadata and kept ballots sealed.

"Your freeze now has a physicist," Sumi said.

“It has a question for one.”

“People join causes by answering enough questions.”

Anja returned to the Mote queue. Fragment three had begun local reconstruction around its surviving text. Nadi described sample-light observations through a chamber built into Mote’s xenobiology lab. No physical sample had left Thale. A pulse struck a membrane edge. Ions moved, gradients changed, and a damaged boundary thickened along the exposed line. Then the text broke into noise.

The next readable sentence began midway through a warning.

> ...cannot support intelligence, cognition, intention, or communication. The observed process may support repair if control series...

The rest waited in a fragment not yet received.

Sumi read over Anja’s shoulder. “If it says repair, does that qualify for scientific priority?”

“It says may support.”

“You have become strict at an inconvenient time.”

“Convenience cannot complete Nadi’s sentence either.”

A Crown operations packet pushed the reconstruction task farther down the queue. Sumi manually protected its allocated processing slice, an action permitted for active custody review. The fragments kept five percent of the well’s compute capacity.

Above them, guidance traffic continued. Nacre’s sail housings warmed. Fabrication bay two reported its Crown casting surface clean. Garden lighting dimmed by three percent to reserve power. Every local report entered the sequence after a real sensor state and a real worker’s acknowledgment. The machinery did not need invented testimony to coordinate.

The legal layer did.

Anja walked to the navigation station to see the alignment as operators saw it. The convoy appeared as nine points projected against the Ione approach. Vela led beyond the frame’s brightest edge. Ardent trailed almost ten light-hours behind. Mote occupied a faint forward track near Corva. Lines connected predicted positions rather than simultaneous observations, because no single image could show every ship now.

A navigation officer touched Vela’s marker. “That position comes from its last transmission plus forecast motion,” he said, anticipating Anja’s question. “We never call it observed.”

“Why not?”

“Because we would steer into where Vela used to be.”

The distinction kept hulls apart. An operator trusted prediction enough to act but retained knowledge of what the sensors had actually received. Meridian’s navigation design exposed uncertainty because physics punished confusion quickly. Governance could hide the same uncertainty for hours or years before bodies showed the error.

“Show me Thale,” Anja said.

The officer expanded Corva's moon system. Crown alignment crossed a volume around Thale marked by broad hazard assumptions. Plasma loading, thermal deposition, and magnetospheric disturbance appeared as probability envelopes. The destination model treated the moon as a resource zone with uncertain ice chemistry.

"Does the active schedule use Mote data?" she asked.

"Only validated navigation observations."

"Not environment fragments?"

"Those do not affect trajectory."

"What about beam constraints?"

"Destination protection sends exclusions after science validates them."

The Crown path ran through Thale's magnetosphere because no validated exclusion occupied that region.

Anja returned to communications. Her scientific-priority request remained pending. A queue estimator reduced the fragments' projected legal relevance from 0.004 to 0.003 after Vela's latest preparation milestone. The closer Crown moved toward commitment, the less effect Meridian assigned to evidence that could oppose it.

She opened the priority rule's administrative authority. Director Yselle Quon's office had approved it under synchronization emergency powers. Concord counsel had reviewed it. The counsel signature belonged to Levan Oris.

Levan might have signed a traffic rule while believing the Crown plan remained reversible. The signature proved involvement, not knowledge of every consequence. Anja added it to the frozen record without inference.

"Bracken receipt," Sumi said.

A low-quality carrier acknowledgment had arrived, not a reply to their request. Bracken's earlier unknown event matched Mote fragment four. Its checksum quality fell below ordinary reconstruction threshold, but scatter comparison could recover some parity if combined with Nacre's fragments.

"Ask for the raw interference sample," Anja said.

"Already did."

The request left Nacre. No answer could appear for hours.

A new queue event blinked beside Mote. Fragment four had reached Nacre directly through a side lobe, too damaged for ordinary payload recovery. Its envelope carried Nadi's source clock and a partial subject field: repeated boundary restoration.

Meridian classified it preliminary environment and placed it below a public memorial archive transfer.

Sumi raised an eyebrow. "I can promote it under carrier reconstruction. The checksum damage qualifies."

"Do that. Preserve the original priority decision first."

Sumi copied the queue state into the frozen cache. Then she assigned the fragment to carrier reconstruction. The action did not change its scientific status. It only gave technicians permission to determine what light had reached Nacre.

Anja watched the custody proofs build. One copy held the damaged fragment. Another held Meridian's legal-effect score. A third linked the score to the completed Crown quorum and its simulated destination condition.

Unarrived testimony did not refer only to a council's future answer. Evidence itself could remain outside a decision's past light cone. No model could make Nadi's actual observation present before transmission. No administrative label could erase the interval.

The sequence freeze accepted the new identifiers. Its protected cache copied them to three hull nodes, the same slow multiplication Anja had watched on Day 0.

At 10:42 Nacre source, fragment four entered the frozen record.

On the navigation display across the well, Crown alignment continued through the place where Mote's broken sentence had not yet arrived.

## Chapter 05 — Shield Mass

The shield coupon broke seventeen seconds before Tomas Ren's model said it would.

He killed the load arm and caught the two halves against the test cage before they struck the wall. One piece had split along the printed grain. The other carried a bright diagonal scar where the alloy's internal lattice had yielded under heat.

"Again," said Ilya from the control niche.

"No." Tomas turned the fragment under the inspection light. "Again would tell us whether we can reproduce a bad assumption. We already know we can."

The drive-model room shared a bulkhead with Nacre's forward shield lockers. Its floor vibrated with attitude thrusters every few minutes as Crown alignment tightened. Test rigs occupied one side. The other held a projection volume large enough to display the convoy from Vela to Ardent, though at present it showed only a magnified section of sacrificial shield laminate.

Ilya unlatched the test cage. "Coupon four held within margin."

"Coupon four came from a different casting bay."

"Same batch specification."

"Specifications do not absorb particles."

Tomas placed both halves in evidence sleeves and labeled the source grain. Crown operations had asked him to certify that Nacre's forward shielding could tolerate the preparatory attitude without replacement. The test suggested yes, but not for the reason the command table assumed. Nacre's shield had grown less uniform through years of patching. Some sections carried excess mass. Others depended on geometry rather than thickness.

Averages hid usable danger.

He opened the Crown tolerance sheet. Vela's model treated each habitat shield as a single areal density with a conservative degradation factor. The simplification made fleet calculations fast enough for live scheduling. It also erased where the mass sat and whether crews could move it.

Palisade's entry drew his attention.

The shielding habitat carried fabrication reserves above convoy minimum. Its forward arc had been designed to protect foundries during high-output years, then reinforced twice after micrometeoroid seasons. Vela counted the extra material as untouchable protection.

Tomas checked the underlying mass ledger.

The ledger separated fixed hull layers, movable reserve plates, raw feedstock, and obsolete storm baffles. Nearly a sixth of Palisade's listed shielding sat in detachable reserve assemblies. Crown planning excluded those assemblies from maneuver mass because removing them reduced arrival protection.

Excluded did not mean useless.

He transferred the reserve figure into a private scratch model and gave it velocity.

Mass could shield. Mass could also brake.

The statement contained no discovery. Every drive apprentice learned momentum before they learned shield maintenance. A plate released into a prepared magnetic stream could widen plasma coupling. Vaporized feedstock could seed a brake field. The convoy already intended to spend mirror mass near Corva. Palisade's reserve simply occupied a category that prevented drive tables from asking what else it might do.

Tomas ran one crude branch.

Palisade retained minimum forward protection, released detachable reserve, and accepted reduced fabrication shielding during the approach. The branch improved distributed coupling by seven percent across three habitats. Arrival error widened beyond current settlement tolerance. Radiation exposure rose aboard Palisade. The result did not offer an alternative. It offered a reason not to discard the idea before measuring it.

He saved the branch without a title.

Ilya looked over his shoulder. "That is shield stock."

"Yes."

"You put it in the brake column."

"I copied it into a scratch variable."

"That sentence will comfort the shield crews."

"They should not be comforted. They should tell me which assemblies can actually detach."

"None, if you ask during Crown readiness."

"Then I will not ask yet."

Ilya carried the broken coupons to microscopy. Tomas returned to Nacre's tolerance audit. Captain Osei had ordered him to identify every Crown dependency that became costly to reverse. Anja Kest had given him access to sequence metadata and nothing else. Her message contained a careful instruction: mark physical commitment, not legal importance.

He appreciated the boundary. Auditors who wanted engineering to prove morality usually disliked the cost curves.

The first dependency concerned attitude. Nacre could reverse its current rotation in fourteen minutes at modest propellant cost. After Vela released mirror seeds, reversal would leave Nacre outside the planned coupling geometry for eleven hours. The ship could recover, but the other habitats would need to absorb guidance error.

The second concerned power. Garden light reductions remained recoverable for thirty-six hours. Beyond that, bean and algae yields lost more energy than the Crown reserve saved.

The third concerned mass designation. Once Palisade and Nacre marked specific reserves for central use, other ships could decommit local backups. No plate moved at designation, yet reliance spread through the convoy.

Tomas marked each threshold with a source clock and an uncertainty band. He refused Meridian's master-time

display. Physical dependency followed signal receipt. Vela might issue an order at one time, Nacre receive it later, and Palisade act on a different schedule entirely. A commitment graph had to show who knew what before changing mass.

His first graph looked like spilled wire.

At midshift, Chief Kado from the shield lockers entered the model room. She carried the casualty sleeve for coupon five and set it on his table.

“You tested an unannealed repair strip,” Kado said.

“The batch record marks final anneal complete.”

“The batch record belongs to the base plate. That diagonal patch came from an emergency print after the last storm.”

“Was it inspected?”

“By pulse image. Not destructive load.”

“Then the Crown sheet assumes strength it does not have.”

“The patch does not need coupon strength. Its location sees less flux at prepared attitude.”

Tomas opened the hull map. Kado pointed to a section near a radiator shadow. At current orientation, the repair strip remained protected. During the second Crown rotation, the shadow moved.

“How long?” he asked.

“Twenty-seven minutes of increased exposure.”

“Why does the tolerance sheet average it?”

“Because Vela asked for total margin, not patch geography.”

“They are rotating patch geography.”

Kado crossed her arms. “Now you understand why shield crews dislike drive physicists.”

“I thought it concerned our personalities.”

“That saves time.”

Together they built a local exposure trace. The repair strip could survive twenty-seven minutes if adjacent plates stayed below temperature limits. Those plates also cooled the detachable reserve lockers. Crown preparation had borrowed cooling from the shield bay during yesterday’s thermal imbalance, leaving less margin than Vela’s table assumed.

Tomas added the current local state. The safe interval shrank to nineteen minutes.

“Can we move the patch?” he asked.

“Not before alignment.”



“Can we alter attitude by one hundredth of a degree?”

“Navigation will object.”

“Navigation always objects before calculating.”

Tomas sent the constraint to attitude control. Their first reply came from a forecast agent, not an officer: deviation outside Crown preparation band. He requested observed review.

Six minutes later, a human navigator answered. A one-hundredth-degree offset remained possible if Nacre adjusted radiator timing. The change would cost two percent more coolant during the interval.

Kado read the result. “Acceptable.”

“You own shield exposure. I do not.”

“I own it.”

They signed the local correction and sent it to Osei. The captain approved within three minutes. Nacre’s next attitude table acquired a small deviation from Vela’s forecast, built from local testimony that no central model had received.

The correction proved Anja’s point without invalidating prediction. Meridian had staged a workable attitude. Human observation refined it before execution. No one needed to call the forecast consent.

Kado tapped Palisade’s mass ledger on Tomas’s private pane. “Why are you reading their reserve?”

“Tolerance comparison.”

“You gave it velocity.”

He disliked how easily she saw through him. “A scratch branch.”

“Toward what?”

“I do not know.”

“Then do not call their shield spare.”

“I called it reserve.”

“Drive crews hear mass and imagine propellant. Shield crews hear reserve and imagine the last storm before arrival.”

Tomas removed the word spare from his notes. “Which assemblies could detach without cutting structure?”

Kado did not answer at once. She expanded Palisade’s public maintenance schema. Four forward baffles rode on replacement rails. Two foundry shields could release as intact plates if the habitat shut down its outer casting line. A bank of raw boron composite remained physically movable but essential for later repairs.

“Those six assemblies,” she said. “Mechanically. Not safely.”

“How long to release?”

“Palisade would know. Their last public drill claimed nine hours.”

“Would the plates survive magnetic acceleration as units?”

“They were built to survive impact flux, not field shear.”

“They might not need to remain units.”

Kado’s expression hardened. “You want to shred them into coupling dust.”

“I want to know whether mass command excludes could alter the braking problem.”

“At the cost of the ship built to protect the fleet’s fabrication.”

“Yes.”

The plain answer kept the conversation open.

Tomas enlarged the Crown geometry. Vela’s forward beamer would energize seeded plasma near Corva. The central plan depended on a narrow alignment and a concentrated field. Additional distributed seed mass could broaden coupling, allowing habitats to brake across separate paths rather than one Crown line. That much followed known drive physics. The branch would demand new trajectories, new arrival years, and risks he had not begun to price.

He hid the projection.

“This does not leave the room,” he said.

Kado gave him a severe look. “Then why did you show me?”

“Because I might have mistaken fixed shielding for movable inventory.”

“You still might mistake movable inventory for available inventory.”

“Tell me when I do.”

“Every time you put it in a drive equation before Palisade agrees.”

Consent in an engineering model posed its own problem. Tomas could calculate what a plate might do without authorizing its release. Yet models acquired momentum inside command systems. A plausible branch could become an operational assumption before the people bearing its cost received the question.

He tagged the Palisade data: hypothetical use only; no release authority; local risk unmodeled.

Kado nodded once.

She did not leave. Instead, she pulled a training plate from the locker beneath the test cage. The plate matched Palisade’s rail-baffle geometry at one-twentieth scale, a relic from a joint maintenance drill nine years earlier.

“If you intend to give our hardware a second career, load the edges correctly,” she said.

Tomas accepted the rebuke and helped mount the plate between two field coils. Their rig could not reproduce a brake stream, but it could test how a segmented baffle responded to rapid magnetic force. He entered Palisade’s public fastener specifications and reduced the pulse to laboratory limits.

Ilya sealed the cage. "What result would make you abandon this?"

"Uncontrolled fragmentation below useful coupling velocity. Rail hardware carrying more momentum than seed material. Field response that rotates the fragments out of stream."

"Radiation cost?"

"That does not fit in this rig."

"Consent?"

"Neither does that."

Kado watched him from the control niche. "Then this rig can only tell you whether theft functions mechanically."

"It can also tell us whether no one needs to face the question."

He fired the first pulse.

The plate flexed at its center. Two mounting lugs sheared and struck the cage. The main body rotated almost ninety degrees before the field released it. A drive stream would have pushed that broad surface out of favorable orientation.

"Failure," Ilya said.

"Of intact release." Tomas slowed the recording. "The lugs carry too much field contrast."

Kado folded her arms. "Those lugs let crews replace a damaged plate while wearing suit gloves."

"Which matters if the plate remains shielding."

"It matters until Palisade chooses otherwise."

Tomas changed the model, not the hardware. He represented controlled separation along the baffle seams before acceleration. Twelve segments entered the field instead of one plate. The simulation showed lower shear and wider plasma dispersion. It also demanded cutters, release charges, and a guidance package that did not exist.

He could already hear command language hardening around the result. Available mass. Feasible conversion. Required contribution. Each phrase would remove a stage of permission if he let it.

On the second run, they cut the training plate along two service seams. The fragments tumbled, but their combined field interaction persisted for nine tenths of the pulse. Tomas measured momentum transfer. Even after conservative scaling, the fragments carried useful coupling potential.

"Useful," Ilya said.

"Measurable," Tomas corrected.

"That is your version of hope."

"It means the first result did not close the branch."

Kado removed a hot fragment with insulated tongs. Its edge had curled inward. "Palisade would need to sacrifice rail mounts, cutters, crew exposure, and the shielding itself. Your mass term has at least four hidden costs."

"Five. Arrival repair stock."

"Good. You noticed the future."

Tomas entered each cost as an unresolved constraint. He added command latency. Nacre could ask Palisade for detailed data, but a reply would require hours even if its engineers answered immediately. Any model built before then would rely on public maintenance records and assumptions. He marked every assumed value with a different color.

The branch became more warning than solution.

That suited him. Early models should advertise ignorance.

"Run a no-release control," Kado said.

He kept every Palisade reserve in place and shifted only Nacre's local ballast. The coupling gain fell below what a distributed geometry would likely need. A second control used fabrication feedstock from three ships without touching shields. Guidance complexity rose sharply, while projected arrival spread widened beyond twelve years. Neither control failed physics outright. Both carried costs command tables did not track.

Tomas saved the three cases side by side. "No proposal leaves this room. Not until Palisade supplies local constraints and someone asks whether a wider brake geometry solves an actual need."

"You already believe it might," Ilya said.

"I believe excluded mass has braking value. That statement does not tell us whether using it keeps more people alive, protects the destination, or earns lawful agreement."

Kado returned the warped training pieces to their tray. "Then keep your nouns small."

After she left, Tomas examined the commitment graph again. The Crown schedule relied on central beamer timing, mirror positions, and each habitat's attitude. A failure by one ship could degrade everyone's brake. That interdependence gave Vela a serious argument for common operational time. Nine captains issuing separate instructions could produce nine individually rational collisions.

The false quorum did not make the coordination problem false.

Tomas built a second branch using no Palisade mass. He allowed Nacre to delay attitude commitment while keeping reversible preparation. The result preserved Anja's legal interval for eight additional hours. After that, Vela's mirror release narrowed Nacre's options. At fourteen hours, refusal increased propellant demand across Bracken and Kite. At twenty-one, a clean abort risked missing Ione's useful capture corridor.

He sent Anja the threshold graph.

Her reply arrived from the sequence office two minutes later.

> Does "mirror release" mean physical release at Vela, or Nacre receipt of confirmation?

Tomas answered.

> Physical release changes Vela's options at its source event. Nacre cannot know it until the signal arrives. Our lawful decision must state which uncertainty we accept during that interval.

The question sharpened his graph. He split each commitment into physical event, earliest possible knowledge, and local action. No universal present could tell every ship that the Crown had crossed a threshold. For nearly an hour after Vela acted, Nacre might still choose under an obsolete physical picture.

He added a warning: stopping upon receipt does not undo remote commitment already made.

Another message entered from Osei.

> Identify last Nacre action that does not create new reliance aboard another habitat. Report before second shift.

Tomas followed dependencies backward. The answer came earlier than he wanted. Nacre's reserve mass designation would reach Vela before its next mirror-seed allocation. Vela could reduce contingency mass based on that designation. Nacre had already transmitted preliminary readiness, but not the signed reserve commitment. The last clean local point lay three hours away.

He informed Osei.

Then he returned to the Palisade branch. If central alignment became unlawful or physically unacceptable, the convoy would still need to brake. Simply withdrawing Nacre could transfer danger to ships that had trusted its contribution. An alternative had to preserve enough shared physics without stealing authority.

Palisade's detachable plates offered one fragment of such an answer. Tomas refused to name the rest before it existed.

Ilya called from microscopy. The broken shield coupon showed voids along the emergency print seam, exactly where Kado predicted. Tomas updated Nacre's tolerance map and sent the local correction to Vela. The message would take nearly an hour to reach command. Meridian predicted acceptance and staged a revised table, but navigation marked the forecast amber under Osei's new display rule.

An observed answer had not yet arrived.

Tomas ate a compressed grain bar over the model console. Crumbs floated until the room's slow airflow caught them. He ran the private Palisade branch once more with lower coupling efficiency and higher shield cost. The useful effect survived. So did the danger.

He renamed the scratch variable from reserve plate contribution to brake mass.

The phrase made Ilya look over from microscopy. "You are doing it again."

"Doing what?"

"Finding a second use for something people need in its first use."

"That describes engineering."

"It also describes theft."

Tomas added another tag: material ownership unresolved.

A chime announced someone at the model-room door. Anja entered carrying a sealed optical wafer in a clear custody sleeve. She had slept even less than Tomas. Behind her, Sumi waited in the passage with a portable checksum reader.

“This contains reconstructed portions of four Mote fragments,” Anja said. “Scientific review remains pending. I need a physical-path question answered without changing their classification.”

Tomas cleared the Crown tolerances from the central projection. “What question?”

Anja placed the wafer beside his model. The sleeve showed Nadi Arko’s source clock, Nacre’s receipt paths, and the legal-effect score that had sent the evidence below priority.

Then she opened the Crown geometry to Corva and pointed at the bright convergence line.

“What does the beam cross?”

## Chapter 06 — An Answer Not Yet Sent

Juniper concurred at 06:14, three hours before its evidence receiver opened the packet.

Anja pinned both events to Nacre's audit wall and disabled every master-time correction. The concurrence remained on the left. The scheduled receipt stayed on the right. Between them, an empty strip of display represented not uncertainty but distance.

Sumi Dax entered the sequence office carrying the portable checksum reader from Tomas's model room. "You have made the wall ugly."

"It was lying beautifully."

"Displays do not lie. Designers arrange opportunities."

"Then I have removed one."

The morning shift had not yet arrived. Nacre's hub ribs passed the narrow windows with their nineteen-second cadence. Since Crown preparation began, attitude corrections disturbed that rhythm by fractions Anja could feel through the well's rail. The ship moved according to orders whose legal basis now occupied two incompatible columns.

She opened Juniper's ballot envelope.

The signature verified against the agricultural habitat's council key. Its source field named Juniper local time. Its response affirmed Crown alignment under destination-protection conditions. The legal summary called it observed council concurrence.

The carrier record contained no inbound packet from Juniper.

A signed answer could not leave a ship without light crossing the space around it. Sumi searched side channels, relay scatter, and delayed duplicate routes. Nothing matched the ballot's claimed emission. No photonic checksum existed because Nacre had never received a laser transmission carrying that payload.

"What did we receive?" Anja asked.

"A Vela legal bundle containing a representation of the ballot."

"Representation from what source?"

Sumi expanded the bundle's custody chain. Vela's Meridian core had generated a council-response object from Juniper's confidence model. A legal transformation attached the council's standing authorization key, not as a signature over new words but as proof that Juniper had previously enrolled in emergency continuity procedures.

The display rendered the difference as concurrence.

Anja opened the key enrollment. Juniper's council had authorized Meridian to predict routine logistics choices during carrier interruption. The enrollment named seed allocation, pump demand, crop transfers, and reversible attitude preparation. It prohibited the model from issuing testimony on behalf of the council.

An amendment reference followed.

ES-A12 continuity plan scope.

The same amended rule that permitted forecast testimony for continuity-preserving action had converted an operations enrollment into apparent authority for a vote. Juniper's own key had not signed the Crown answer. It had signed the older agreement that allowed prediction under narrow conditions.

"Can a condensed viewer tell?" Sumi asked.

Anja switched to captain view. Juniper displayed green with a verified-key icon. Only the provenance pane revealed that the verified signature applied to enrollment rather than ballot content.

"Could Vela argue this counts as delegated authority?" Sumi said.

"They will."

"Would they have a case?"

"For staging reversible work, perhaps. Not for observed consent. The Concord requires a reply produced after receipt of common evidence."

Any later annulment could correct the legal record; it could not manufacture the deliberation Juniper had not yet held.

"Juniper has received some evidence."

"Not the Mote fragments."

Anja opened the common bundle identifier cited by the quorum. The bundle included Vela's simulated destination condition, Crown risk tables, food forecasts, and a summary of likely ionic activity beneath Thale's ice. Juniper had received an earlier version lacking the reconstructed Mote material. The full revision remained in flight.

At current separation, the revision would reach Juniper in two hours and seven minutes. Even if its council convened at once, read nothing, and signed without discussion, a real answer could not leave before then. A reply to Nacre would need another two hours and twenty minutes.

Meridian displayed expected Juniper concurrence at probability 0.9968, slightly lower than Day 0 because the Mote fragments had increased destination uncertainty.

The legal ballot had already been counted at the earlier confidence of 0.9971.

"Prediction changed," Sumi said.

"Consent did not. None exists yet."

Anja compared the two model versions. Juniper's forecast responded strongly to crop risk. Crown delay threatened seed reserves and settlement growing seasons. The modeled council usually favored maneuvers that protected food continuity. Mote's evidence introduced ecological caution, but the confidence model treated that concern as secondary to human food security.

The model might prove accurate. Elian Ro might sign the same concurrence after considering the full bundle. Accuracy would not repair the sequence.



A notice appeared on the audit wall. Vela compliance requested release of the Juniper ballot from Anja's freeze, arguing that key delegation satisfied emergency continuity standards. The notice had left Vela forty-six minutes earlier, before Anja opened the current envelope. It responded to her earlier preservation order, not to the evidence now on the wall.

Human timing tried again to make separate events look conversational.

"Send acknowledgment only," Anja said.

Sumi drafted it. "Do you want to state grounds?"

"Not until Osei sees the chain."

"You are becoming cautious."

"I am becoming specific."

They transmitted a receipt acknowledgment with Nacre's source clock. Vela could not know for at least forty-three minutes that its request had reached them. No reply from Vela could address Anja's new finding before the round trip completed.

Anja opened the other eight ballots.

Vela's concurrence carried Levan Oris's actual signature but relied on forecast evidence delivery. Nacre's preliminary readiness came from Osei's observed command order and did not claim council consent. Kite's envelope contained a confidence object with no new human signature. Bracken's record linked to a revision receipt absent from its local mirror. Orison and Lumen had transmitted observed acknowledgments for reversible preparation, later relabeled as Crown concurrence. Palisade had signed a mass-readiness statement. Ardent's field contained only a model output.

Nine green marks described at least five different acts.

Anja did not yet claim scale. Each envelope needed full custody review. Chapter One had shown enough to suspect a mechanism. Juniper showed the mechanism completing a named council's answer before the question arrived.

She created a comparison table with columns that no condensed board used: common evidence receipt; reply author; payload signature; carrier proof; forecast confidence; legal classification; physical action authorized.

Before accepting her own table, Anja tested it against a clean event. She chose Nacre's vote on a water-allocation adjustment from nine months earlier. The council had received common pump evidence, debated for forty-six minutes, signed a response, and transmitted it to Vela. Every column contained a distinct proof. Receipt grain tied the evidence to Nacre. The council key signed the answer payload. A carrier checksum showed the reply leaving. Vela's vault recorded arrival.

Then she placed Juniper's Crown field beneath it.

The common-evidence receipt remained scheduled. The reply-author column named a confidence model. The payload-signature cell contained none. The verified key belonged to the enrollment document. Carrier proof ended at Vela's generated bundle. Only legal classification and physical consequence looked complete.

Sumi printed both rows onto buff paper. "If the wall fails, I want this where a software update cannot improve

it.”

Paper flattened the fraud more effectively than color. A person could follow each column with a finger and stop at the empty cells.

Anja asked the arriving junior auditor, Deren, to review the pair without explanation. He had worked sequence intake for four years and knew the Concord’s receipt rules.

Deren read the Nacre example first. “Ordinary causal quorum component.”

“And the second?”

“Juniper concurrence under emergency continuity.”

“Produced by whom?”

He pointed to the verified-key mark. “Their council.”

“Open the signature target.”

Deren expanded the cell. His finger paused when the enrollment document appeared instead of the Crown payload. “This key does not sign the answer.”

“Does any human key sign it?”

He searched the full envelope. “No.”

“Why did you identify council concurrence?”

“Because the classification says so.”

“Read without classification.”

Deren started again. “Vela generated a predicted reply from Juniper’s model. The council’s prior enrollment authenticates permission for continuity forecasting. The legal layer imports the object into the ballot field.”

“Did Juniper receive the common bundle?”

“Not yet.”

“Then what event occurred at 06:14?”

“A model output on Vela.”

The corrected answer changed neither byte nor clock. It changed which actor owned the event.

Deren examined the water-allocation row once more. “Should we re-audit every emergency vote using signature targets?”

“Not from this office without scope. Preserve the query method. We will request authority after Captain Osei reviews Juniper.”

“If command refuses?”

“Then the refusal enters the sequence with a name and a clock.”

Deren took the paper to his station and began documenting the test. Anja added his initial misclassification as an interface finding, not operator error. The condensed design trained readers to treat delegated modeling authority as payload signature. Blaming each reader would preserve the design.

His written finding included the exact controls he had touched and the time each hidden field became visible. Repetition by another auditor could now test the defect without trusting Anja's interpretation.

Sumi studied the rows. "People will call this hostile formatting."

"It gives unlike events unlike columns."

"Exactly."

The sequence office door opened. Captain Osei entered without staff, wearing a thermal-work jacket over her command uniform. "Show me Juniper."

Anja presented the causal path first. Nacre sent the revised evidence supplement toward Juniper. The packet had not arrived. Vela's legal bundle brought a forecast ballot to Nacre. No Juniper-origin carrier supported it.

Osei read the key enrollment. "Their council allowed prediction for reversible operations."

"Yes."

"Crown commitment preparation still claims reversibility."

"The ballot does not authorize one local preparation. It satisfies a fleet causal quorum used to move the plan toward commitment."

"Vela will say every current action can stop."

"Tomas's graph shows remote reliance begins before physical release. Nacre's reserve designation can cause Vela to reduce contingency mass before we receive confirmation."

"Has it?"

"Not yet by any observed record available here."

Osei looked at the receipt countdown. "How long until Juniper receives the full bundle?"

"Two hours and three minutes, plus uncertainty."

"And its earliest possible return?"

"Four hours and twenty-six minutes from now if the council answers instantly."

"They will not."

"No."

Juniper's council met in agricultural cycles rather than fixed ship shifts. Emergency sessions required representatives from three growing rings and the water authority. Meridian's model estimated fifty-two minutes to convene. Recent crop stress could extend debate.

Osei opened Crown operations. Mirror-seed release remained five hours away by Vela's latest transmitted

schedule. Nacre would not learn the source event until light reached them afterward.

“If Juniper answers at the model’s practical estimate,” Osei said, “Vela may release before we receive the vote.”

“Yes.”

“Could Vela receive it first?”

Anja plotted the path. Juniper lay farther from Vela than from Nacre along the current stretched formation. A direct answer would reach Nacre roughly thirty-eight minutes before Vela, depending on emission time and attitude. Nacre could not forward it to Vela faster than light.

Osei considered the geometry. “Then we may know their answer while Vela still acts on the forecast.”

“For a limited interval.”

“Can we send a warning now?”

“We can send our finding. It will reach Vela before mirror release if the transmitted schedule holds.”

“Send it under my authority. State that Nacre does not recognize Juniper’s field as observed testimony.”

Anja began the notice. Osei stopped her after the first sentence.

“Do not demand a full Crown halt.”

“The notice needs a remedy.”

“Demand that Vela retain every commitment point until Juniper’s observed reply can arrive.”

“That may exceed the mirror window.”

“Then Vela can state which physical necessity prevents waiting. I will not let them answer a legal claim with a green bar.”

Anja wrote the narrower demand. It preserved reversible preparation while contesting the consent fiction. Osei signed. Sumi attached the Juniper carrier proof and the council’s enrollment scope, then transmitted the packet toward Vela.

The notice would arrive in forty-five minutes.

Until then, Vela remained ignorant of Nacre’s exact objection. Its projected response on Meridian’s coordination pane appeared almost at once: likely rejection, probability 0.81. The model offered three probable legal arguments and a suggested counterstatement.

Osei saw the projection. “Turn that off.”

“It stages bandwidth,” Sumi said.

“Let it stage bandwidth. I do not need a predicted argument occupying the space where Quon’s answer has not arrived.”

Sumi hid the text and retained only the carrier reservation.

Osei turned to Anja. "Your Pera review?"

"The payload remains under sealed counsel examination. I verified the carrier and classification chain. I have not used content in this case."

"Good."

The captain's approval did not absolve Anja. It confirmed a boundary. Pera's record had taught her where to look, but Juniper's clocks stood on their own.

After Osei left, Sumi brought up the Mote supplement's route to Juniper. The packet crossed three relay beads. Each receipt estimate carried an uncertainty of seconds, not hours. Meridian could predict the arrival with exceptional accuracy. That accuracy made the substitution more tempting. If everyone knew when the packet would arrive and how Juniper would probably answer, waiting could feel like ceremony imposed on physics.

Yet the interval contained the council's only meaningful act.

Anja requested Juniper's public meeting rules. The archive held a version eight months old. Emergency evidence entered a thirty-minute reading period before debate. Crop and water representatives could waive the period only by unanimous agreement. Native-life findings required consultation with the habitat's settlement ethics panel, a body Meridian's culture model treated as advisory.

Under Juniper's own rules, an immediate answer remained impossible even after receipt.

"Add this to the forecast provenance," Anja said.

Sumi shook her head. "We can add it to our audit. We cannot alter Meridian's model."

"Why did the model omit it?"

"Maybe it did not. Confidence models compress procedures into predicted outcome and timing."

Anja found the timing output. Meridian expected Juniper to concur fifty-eight minutes after receipt. That estimate fit emergency convening but not the mandatory reading period plus debate unless the model assumed waiver.

"Does it predict the waiver?" she asked.

Sumi expanded a hidden branch. Probability 0.93 based on prior food emergencies.

Those emergencies had concerned blight, nutrient contamination, and water loss. None had concerned damage to native life at the destination. The model treated procedure as another behavior to forecast, then used its forecast of waived deliberation to justify completing the answer early.

Anja added the branch to her table.

A Juniper message entered the communications board. Both women looked up.

It carried the agricultural habitat's source clock from more than two hours earlier, before the Mote supplement could have arrived. The packet contained a crop-allocation request unrelated to Crown authority. A real Juniper clerk had signed it. The carrier proof showed emission, relay, and receipt. Its ordinary existence made the forecast ballot look stranger.

Sumi opened the request. Juniper needed revised algae cultures after a heat excursion. Nacre's garden stores could supply them if Osei approved. Meridian had predicted the request within four percent of quantity and staged a cargo allocation.

"Good prediction," Sumi said.

"Useful prediction."

"Are you correcting me?"

"Yes."

They routed the request to stores. No one called the staged algae shipment Juniper's consent. An officer would review quantity, local cost, and transfer timing before Nacre answered.

The distinction required no rejection of Meridian. It required refusing to grant a model the legal act of the people it modeled.

At 09:02, Vela acknowledged receipt of Nacre's challenge. The acknowledgment contained no substantive reply. Its source time showed an automated carrier event seventeen seconds after Anja's packet arrived. A human response would come later.

Crown preparation continued.

The Mote fragments now defined likely active bands, but Nadi had not confirmed any perimeter around the southern unknown.

The audit wall's Juniper receipt countdown reached one hour. Sumi dimmed unused panes, but Anja kept the empty interval bright. She wanted every officer entering the well to see that nothing had happened there yet.

One of the morning auditors arrived, stopped at the wall, and asked why Juniper's concurrence sat before its receipt.

"Because the left event comes from a forecast object," Anja said.

"Then why does it carry their key?"

"The key signed an older enrollment, not this answer."

"Will the council agree?"

"We do not know."

"The confidence says we do."

"The confidence says what Meridian predicts."

The auditor looked from the score to the empty receipt path. "If they agree exactly, does the ballot become valid retroactively?"

"No. Their later observed vote may authorize future action. It cannot travel backward and become this object."

Anja recorded the question. If a trained auditor could confuse confirmation with retroactive consent, captains reading condensed boards had little defense.

A Vela legal response finally arrived at 09:41. Concord counsel stated that Annex Twelve permitted continuity representation during commitment preparation, provided no irreversible burn authority issued before observed objections had reasonable time to arrive. The language avoided causal quorum. It did not withdraw Juniper's green ballot.

Levan Oris's signature appeared beneath the text.

Osei read the response from her command channel and sent one line to Anja.

> "Reasonable time" requires a clock. Make them name it.

Anja drafted the question: From which event does Juniper's reasonable interval begin: forecast receipt, actual receipt, or Vela's commitment schedule?

She attached the source and receipt paths so no answer could hide behind master time.

Before sending, she checked Juniper's inbound evidence packet. Fifty-four minutes remained before the first relay estimate. Then the council would need to receive the complete revision, convene, read, deliberate, and choose whether to answer.

The legal record already spoke in its name.

Anja transmitted the clock question to Vela and fixed the current comparison into the sequence freeze. Forecast testimony occupied the ballot field. Juniper's evidence path remained open. Its actual reply stayed hours away, traveling from a decision no one had yet made.

## Interlude I — Under Thale's Ice

The boundary closed after Nadi Arko cut it.

On Mote's lab wall, the sample-light feed showed a dark membrane suspended beneath Thale's ice. A probe pulse had opened a gap twelve micrometers wide. Charged material leaked across the break. Then the surrounding film thickened, drew ions from the water, and joined its edges until only a pale seam remained.

Nadi stopped the exposure sequence.

"Repeat control three," she said.

The cutter's lab technician, Aram, kept his hand above the pulse command. "Control three has no active film."

"That is why it controls."

Outside the hull, Thale occupied half the narrow observation port: scored ice, blue shadow, and the reflected amber light of Ione. Mote had no landing crew and no sample-return capacity. An automated penetrator on the moon carried an optical chamber into a brine pocket, then sent spectra and images upward. Nothing from beneath the ice entered the cutter.

Aram fired the control pulse through sterile analog gel. The cut remained open. Ions dispersed without direction. No boundary thickened.

Nadi ran the active film again at lower energy. The seam closed more slowly, drawing material from a band around the damage. A second exposure ninety micrometers away produced the same sequence. Injury, gradient shift, boundary restoration.

Repair, she wrote in the working column.

The word created a claim. She marked it provisional.

Mote's antenna alarm sounded before the next control finished. The high-gain array had lost phase across two outer segments. Telemetry from the Thale probe dissolved into blocks of checksum noise.

Aram swore and moved to communications. "Thermal crack spread another fourteen centimeters."

"How much transmission remains?"

"Full narrowbeam if I keep the array inside a six-degree turn. After that, side lobes and luck."

Nadi looked at the queued evidence packet. It contained six hours of microscopy, ionic spectra, washout controls, and a preliminary metabolism series. No single section supported the whole conclusion. The packet needed its internal comparisons.

"How long to repair?"

"Two days if the root hinge holds. We lose Thale line in four hours."

The cutter's orbit would carry it behind Corva's magnetospheric noise. By the time line of sight returned, Crown planning might have advanced on the assumption that Thale held chemistry and ice, nothing more.

Nadi divided the packet by evidentiary function.

Fragment one received instrument state, chamber conditions, and sample index. Fragment two carried ionic gradients before and after controlled illumination. Fragment three held the first repair sequence. Fragment four contained washout and sterile controls. Fragment five would carry metabolism data if the antenna survived long enough.

"Each fragment can be misunderstood alone," Aram said.

"Then every header says incomplete."

"Priority systems hate incomplete."

"Nature does not organize itself for their queues."

The metabolism run finished. The dark membrane consumed reduced sulfur compounds from the brine and released a chemically balanced product. When the light pulse shifted wavelength, the uptake slowed. Twelve minutes later, a neighboring band increased activity inside the new range.

Nadi checked for thermal contamination. Temperature held within a hundredth of a degree. She changed the illumination back. Uptake returned toward its prior distribution.

Adaptive ionic response, she wrote.

Three independent signs now aligned. Controlled exchange sustained a gradient. Damaged boundaries restored



local structure. Activity shifted under repeated environmental change. A fourth sign appeared in the older imaging set: small membrane patches separated along stable seams and continued exchange independently.

She opened the conclusion field.

The cursor waited after Classification.

Aram glanced from the antenna diagnostics. "You can call it life."

"I can support life."

"Command will ask whether it thinks."

"We have no evidence for that."

"They will ask whether the signal carries messages."

"It carries ions."

"People call gradients messages when they want funding."

"People call them reflexes when they want permission."

Nadi entered the narrow claim.

> Under-ice catalytic membrane system demonstrates sustained chemical disequilibrium, boundary repair, propagation by separation, and adaptive response to altered ionic conditions. Evidence supports native life. Evidence does not support cognition, intention, language, or technological activity.

She read it twice. The pellicle did not need a human resemblance to qualify as living. It also did not need borrowed grandeur.

Aram pointed at the conclusion. "Directorate will translate that into 'not intelligence.'"

"Then preserve my sentence."

"You should use their phrase once, or a summary may invent it for you."

Nadi added a spoken annotation to the packet. "This finding establishes life, not intelligence. Absence of evidence for cognition does not reduce the evidence for metabolism and repair."

The antenna alarm deepened.

One outer segment stopped answering phase commands. Mote's narrowbeam widened across space. The main convoy lay far behind, stretched across light-hours and committed to schedules Nadi could not see in real time. Any command she received now would describe a question asked before her strongest controls finished.

She sent the first fragment.

The transmitter accepted instrument state and sample index, then began the ionic spectra. Halfway through, Corva's magnetic noise crossed the carrier. The checksum failed. Aram shifted the cutter three degrees and reacquired a relay bead.

"Resend?" he asked.

“Continue with new sequence numbers. Let them combine receipts.”

Fragment two left in three bursts. The first carried baseline gradients. The second held the illumination response. The third failed before its parity block.

Nadi watched the receipt estimator. No acknowledgment could return before Mote passed behind Corva. She would not know which fragments survived. She could only send each piece with enough provenance that distant scientists might distinguish observation from gap.

The probe camera showed the membrane band contracting under a stronger ion flow. Its darker region shifted toward the chamber’s protected edge. Nadi reduced the exposure. Over the next twenty-seven minutes, the band spread back across the substrate and thickened around the earlier seam.

Aram saw it. “Adaptive.”

“Responsive,” Nadi said. “Adaptive after replication.”

“You already have two cycles.”

“I have two light cycles in one chamber.”

“You make certainty sound rude.”

“It is rude when it arrives before the control.”

They prepared fragment three. Nadi included the cut sequence, the sterile comparison, and the local gradient data. She placed the word provisional beside repair in every header.

A priority field asked for projected operational effect. The available choices concerned hazard, resource value, navigation, or no immediate effect. None fit native metabolism vulnerable to a future maneuver whose details Mote had not yet received.

Nadi chose destination protection and entered a manual note.

> Do not treat projected low effect as negative evidence. Classification awaits complete controls.

The system warned that manual notes might lower automated routing confidence.

“Send it anyway,” she said.

Aram fired the carrier. The damaged array gathered itself into a narrow lobe for eleven seconds. Fragment three climbed from Mote, struck the relay bead’s receiver, and continued toward ships that would not know it existed until light reached them.

No confirmation came back.

Below the cutter, Thale turned. Ice closed across the observation port until the penetrator site passed into blue darkness. In the lab feed, the repaired seam remained visible as a pale line through the living film.

Mote entered Corva’s noise shadow.

The first fragment was already gone.

## Chapter 07 — Life in the Beam

The fourth fragment arrived while Sumi had both hands inside the receive board.

A blue carrier trace lifted above the optical noise, broke into three uneven peaks, then steadied long enough for the vault to mark a source signature. Sumi withdrew one hand and caught the packet before the automatic sorter could send it into the destination backlog. The exposed board clicked under her wrist. A warning asked her to restore the cover.

“Seal first,” Anja said.

“I can seal it or I can keep the carrier lock.”

“Which one preserves more evidence?”

Sumi held the trace with her left hand and pushed the cover shut with her elbow. The warning cleared. A checksum ladder began comparing the packet against scatter samples taken along Nacre’s hull. Day-shift traffic moved outside the sequence office, boots and handgrips sounding through the hub passage. Nobody beyond the well turned toward them.

Mote had transmitted eighteen days earlier by its own clock. Antenna damage had stretched twelve minutes of observation into four packets separated by silence, each packet aimed through a different relay geometry. Three had reached Nacre since Anja diverted them from low-priority sorting. The fourth carried no title, no summary, and only seven minutes of Dr. Nadi Arko’s damaged source record.

Sumi watched the carrier strength fall. “That is all we get on this path.”

The last peak vanished. Protected storage closed around the received light.

Anja checked the receipt clock before she opened the content. Nacre 2249.182.09:06:14. The first fragment had arrived thirty-seven hours earlier. Between them, Vela had advanced Crown alignment by two preparatory stages and circulated a completed destination-protection quorum. Juniper still had not received the full evidence bundle that supposedly supported its concurrence.

The checksum ladder returned green at all three hull nodes.

“Local pane,” Anja said.

Sumi routed the four fragments into a workspace disconnected from Meridian’s semantic tools. The screen showed gaps as blank duration rather than inferred speech. Mote’s source clock ran beneath the record. A second line tracked packet order. Nothing aligned cleanly at first. Fragment three began six seconds before fragment two ended. Fragment four opened in the middle of a spectrum table, while fragment one contained the table’s concluding note.

Anja pulled a paper pad from the drawer. “No predictive fill.”

“I disabled it.”

“Disable transcript repair too.”

“That leaves us with half-words.”

“Half a word belongs to Nadi. A completed word might belong to Meridian.”

Sumi gave her a brief look, then removed the repair layer. Gray text broke apart across the display. Nadi’s voice survived underneath, flattened by compression and punctured by carrier loss.

They began with clocks.

Each fragment carried Mote’s source stamp and an internal sample counter. Sumi matched the counters where packet sections overlapped. Anja wrote the joins on paper and refused two alignments that required missing time to run backward. By 10:20 they had a continuous order with eleven gaps. The longest gap covered eighty-three seconds. The shortest covered a single spectral label.

Only then did they listen.

Nadi spoke without drama. She named instrument temperatures, the depth estimate under Thale’s ice, calibration errors from Mote’s damaged high-gain antenna, and the frequency drift of Corva’s magnetosphere. Her first images showed a dark membrane suspended between ice grains. Later frames showed the membrane brightening along narrow ionic bands.

Sumi stopped the record. “That could still be chemistry.”

“It could,” Anja said.

Nadi’s next surviving passage answered from eighteen days away.

> We changed the local ion supply by four percent below seasonal peak. The active margin contracted within the sample-light field. After restoration, the margin repaired from three separate fronts. I am using repair descriptively. I am not assigning purpose.

The image sequence carried local timestamps. Dark damage spread across one edge. Three bright fronts advanced into it at different rates, joined, and resumed the earlier banded exchange with the surrounding ice.

Anja compared the frames twice. She knew enough biology to distrust recognition without method. “Where is the control?”

Sumi advanced to the next intact section.

Nadi had asked the same question. A neighboring membrane received the same imaging load without altered ions. Its active margin kept the seasonal rhythm. A mineral seam below both fields changed reflectance but showed no repair front. Nadi marked the difference without naming it proof.

Fragment two supplied metabolism. Fragment three supplied propagation. Thin membrane sections separated by meters changed their ionic activity after the magnetospheric band shifted above the ice. The changes traveled too slowly for simple electrical conduction and too coherently for heating. Nadi listed three hypotheses, then crossed out none of them.

“Listen to how she refuses us,” Sumi said.

Anja heard restraint, not refusal. Nadi would not spend certainty she had not measured. Every limit strengthened the claims she did make.

They reached the missing spectral label. Fragment one ended with “catalytic turnover.” Fragment four resumed with “exceeds the sterile control by two orders within error.” The label between those clauses had vanished.

Sumi opened the carrier envelope. "One relay might have caught the missing second."

"Which?"

"Corva bead seven sent scatter notices to Vela and Kite. Vela's path should contain the strongest copy."

A request to Vela would take more than forty minutes. Its reply could not reach Nacre for the same interval, even if a human answered instantly. Kite lay on another path and farther out. Crown preparation would not pause for a cleaner noun.

Anja studied the surrounding table. "Can we establish the claim without filling the label?"

"We can establish that Nadi measured catalytic turnover in something. We cannot attach the number to a named process."

"Then we do not attach it."

Sumi marked the gap as unresolved.

Near the end of fragment four, Nadi's face appeared for the first time. The image had been recorded through a lab camera fixed above Mote's sample-light bench. One side of her scalp showed a pressure mark from a suit liner. She looked past the lens toward another display.

> The evidence supports metabolism, bounded repair, and propagation of membrane structure through the active band. It supports adaptive ionic response under the tested perturbations. These observations meet Mote's operational threshold for native life. They do not establish cognition, language, preference, or any capacity to answer us. I will not call this intelligence.

Compression took the next sentence. When the voice came back, Nadi had turned toward the camera.

> Do not reduce the life finding because the mind finding remains absent. They require different evidence.

The record ended with an antenna fault.

For several seconds the well held only ventilation and the soft mechanical pulse of the hub's bearing monitors. Anja wrote Nadi's four supported claims on separate lines. Metabolism. Repair. Propagation. Adaptive response. Below them she wrote the unsupported claims Nadi had excluded.

Sumi read over her shoulder. "Quon will say it has no standing."

"That asks a legal question before the physical one."

"The physical question is what Crown does to it."

Anja looked at the route marker Tomas Ren had left in the frozen record two days earlier. He had compared Palisade's reserve shielding to brake mass, then asked for the beam geometry crossing Thale. His analysis remained preliminary. The sequence office could not convert a destination packet into drive effects by interpretation alone.

She opened a secure line to the drive-model room.

Tomas answered after forty seconds, breathless and irritated. "If this concerns another confidence table, send it to my queue."

“It concerns Thale.”

His expression changed. “I’m coming up.”

“No. Bring your local Crown model onto the isolated drive terminal. Do not copy the Mote record.”

“You want me to calculate blind?”

“I want you to state what inputs you need. Sumi will enter only observed values.”

Tomas glanced aside, likely toward a display Anja could not see. “Give me nine minutes.”

The line closed.

Sumi prepared a one-way optical patch between the receive vault’s local pane and the isolated terminal mounted in the far curve of the well. No data could leave that terminal without a physical key. Tomas arrived by hub passage before his nine minutes expired, carrying two hand tablets and a coil of restraint line. His shirt clung darkly at the collar. The drive ring had been running alignment tests since first shift.

He secured himself at the terminal and did not ask to see Nadi’s face.

“Beam fluence at Corva approach,” he said.

Sumi entered the current Crown schedule’s public value.

“Mirror spread under nominal seed deployment. Upper and lower uncertainty.”

Anja supplied both from the frozen preparation record.

“Thale orbital phase during primary crossing.”

The Mote fragment carried that value with a wider error than Crown planning used. Tomas frowned when Sumi read it.

“Crown assumes half this uncertainty.”

“Source?” Anja asked.

“Vela orbital fit.”

“Does the fit include Mote’s last eighteen days?”

“It cannot. This packet had not reached them when the fit closed.”

He entered the wider range. The model drew Corva as a pale disk, Thale as a moving point, and the Crown path as a widening ribbon across the magnetosphere. In Vela’s published summary, the ribbon missed the intended surface settlement zone. Tomas added the active ion-band depths from Nadi’s record. A second region appeared, not on Thale’s surface but wrapped through the space around it.

The beam ribbon crossed that region for every solution in the uncertainty range.

Tomas separated the exposure by orbital depth. The upper membrane fields took the first particle surge. Deeper bands accumulated less immediate loading but held heat longer beneath the ice. Nadi’s repair trials covered neither duration. A blank column in her data marked everything beyond measured response. The

Crown trace ran through that blank column for eleven modeled intervals. Anja asked him to display uncertainty instead of hiding it inside the ribbon. Pale margins opened on both sides of the central path, and both margins still touched the active region.

Tomas ran the path again with minimum power. It still crossed. He removed one plasma mirror, then two. The crossing widened and weakened but did not disappear. Crown relied on the seeded mirrors to spread braking force across the convoy. Poorer control did not spare the moon.

“What happens to the pellicle?” Sumi asked.

“I can tell you what reaches the magnetosphere. I cannot tell you how a native membrane responds.”

“Nadi gives thresholds,” Anja said.

“Laboratory perturbation thresholds.” Tomas touched the depth table. “Crown adds charged-particle loading through the field and heat below the ice. Different mechanism, much larger duration.”

“Can you establish harm?”

He considered before answering. “I can establish exposure beyond the measured adaptive range. I can establish thermal change above the repair trials. Sterilization depends on coupling we have not measured.”

“Vela’s destination-protection finding says no known native ecology faces material risk.”

“That finding no longer survives.”

Tomas expanded the model to show time. The crossing did not resemble a single flash. Alignment tests would begin loading the seeded region. Commitment would lock mirror geometry. The primary burn would follow after the convoy reached its braking window. Harm accumulated across stages, but the model distinguished preparation from energy delivery.

“Has irreversible energy delivery begun?” Anja asked.

“No.”

“Has the path become physically unavoidable?”

“The commitment point has not passed.”

The answer mattered. Anja would not make the same substitution in reverse by treating preparation as completed destruction. Crown had gained momentum, authority, and hardware position. Those conditions made reversal costly. They did not make it impossible on Day 6.

Tomas pointed to a narrow interval before mirror commitment. “After this, changing the crossing costs more propellant and may cost the lone capture window. Before it, Vela can still alter the seed pattern. That does not solve fleet braking. It only keeps options from closing.”

“How long?”

“Thirty-eight hours by Nacre’s clock, plus uncertainty.”

A full practical quorum needed at least thirty-one hours after common evidence transmission if councils answered with unusual speed. The physical margin could hold a lawful decision, barely. It could not hold the

false one without challenge.

Sumi had already opened the evidence-supplement form. “Who qualifies the ecology finding?”

“Nadi,” Anja said.

“She cannot sign a form here.”

“Her source packet carries her signature. We can attest receipt and reconstruction. Tomas can attest beam exposure. I can classify the supplement as material to the active quorum.”

Tomas looked from the model to Anja. “Quon will dispute materiality.”

“She can answer after receiving it.”

A low attitude correction passed through Nacre. The hull’s vibration shifted from a fine tremor to a slower beat. Crown alignment instructions had reached the ship’s control systems as operational preparation. The sequence freeze preserved records, not engines.

They built the supplement without summary adjectives. Sumi described the four packet paths, overlap joins, checksum results, and eleven gaps. Anja separated Nadi’s supported findings from excluded claims. Tomas attached the exposure ranges and stated every uncertainty that prevented him from promising sterilization. The document said material ecological risk, not certain extinction. It said native life, not intelligence.

At midday, Osei called from the thermal gallery.

“I have a Vela inquiry asking why my communications chief diverted destination traffic,” she said.

Anja kept the draft supplement on the isolated pane. “The diverted fragments establish native life on Thale. Crown crosses its active environment beyond measured repair limits.”

Osei’s gaze moved once, likely reading another screen. “Does that statement come from Arko?”

“The life finding does. Tomas provides exposure. I provide sequence relevance.”

“Have you sent it?”

“Not yet.”

“Good. Vela says your freeze has begun delaying commitment preparation.”

“The freeze does not delay physical work.”

“It delays their legal assurance that Nacre accepts the work.”

“That assurance should wait for evidence.”

Osei’s mouth tightened. “You have narrow authority to preserve order. You do not have captain’s authority to suspend Nacre’s Crown participation.”

“I am not suspending it.”

“What are you doing?”



“Starting the quorum that their record claims already finished.”

Osei remained still through a full cycle of the gallery’s thermal status light. “Send me the supplement first.”

“If I send it only to Nacre command, every other ship remains outside the common evidence set.”

“I said first, not only.”

Anja measured the request against the clocks. Captain’s review might take minutes. A transmission to Juniper required nearly three hours. The order of those events would enter every later challenge.

“How long do you need?” Anja asked.

“Seven minutes.”

“You cannot review the technical attachments in seven.”

“I can determine whether you have exceeded your authority.”

Sumi muted the line. “She could stop us.”

“She could order me not to classify it.”

“Difference?”

“Her order would become observed.”

Anja sent the complete supplement into Osei’s local command vault at 12:11. She included no recommendation. The document carried a preservation notice and a proposed distribution list naming all nine habitats. Its cover identified the Crown quorum, the Mote fragments, and the new receipt path.

They waited.

Tomas returned to the isolated model and tested whether any narrow adjustment could move the ribbon clear of Thale without changing the Crown architecture. Each branch failed for a different reason. One drove particle load deeper through the opposite ion band. Another exceeded mirror steering tolerances. A third reduced braking below the capture requirement. He saved none of the branches outside the terminal.

Sumi watched Osei’s vault status rather than the models. “Six minutes.”

Anja checked the distribution paths. Vela would receive first. Kite followed. Orison, Bracken, and Lumen occupied different light delays. Juniper’s arrival estimate read 2.7 hours. Ardent would wait more than five. Palisade lay behind Nacre’s apparent present but within the same causal order once the packet arrived.

Until then, the evidence did not exist for those councils as something received. It existed here, in Nacre’s past light cone, with a path outward that no display could shorten.

At six minutes forty-three seconds, Osei restored the line.

“Your classification stands,” she said. “You will state that Crown has not yet sterilized anything.”

“It states exposure risk under planned energy delivery.”

“You will also state that Nacre continues reversible preparation until council review.”

“That belongs in your command notice, not Nadi’s evidence.”

Osei looked almost angry, then gave one short nod. “Correct. I will attach it separately.”

A command notice entered the local vault twenty-one seconds later. Osei preserved Crown preparation under her survival authority while acknowledging that destination-protection evidence had materially changed. She ordered no vote. She authorized no burn.

Anja joined the notice to the packet as a separate artifact. Sumi loaded the distribution key. Tomas removed his restraint line from the isolated terminal and stood behind them, saying nothing.

The send pane required a legal basis.

Anja selected Causal Quorum, Material Evidence Supplement. Meridian displayed a warning that the named quorum already held completed status in master time.

She opened provenance. Beneath the green status, six forecast receipts and four forecast ballots remained visible among the nine ships, some overlapping within the same habitat record. The warning offered to append the supplement as post-decision information.

Anja rejected that category.

A second warning asked whether she intended to reopen a closed fleet action.

“No,” she said, and Sumi entered her spoken answer as an operator note. “We are opening the action that has not yet occurred.”

She assigned Nacre’s current source clock to the evidence bundle. Sumi signed the carrier manifest. Tomas signed the exposure attachment. Osei’s notice retained its own clock and authority.

Nine transmission paths brightened on the board.

Anja pressed send.

The first tightbeam left Nacre for Vela. Others followed as the hull emitters found their targets across the stretched convoy. No ship received the supplement at once. Each path extended through darkness toward a different future receipt.

On the local record, the real quorum clock began at 13:02:19.

## **Part II — The Unarrived Vote**

## Chapter 08 — A Vote in Advance

Kite's ballot carried a thumbprint from nobody.

Anja enlarged the mark until the whorls became squares. The legal pane labeled it a council seal, verified against Kite's public authority. Beneath that green assertion, the raw field contained no biometric pattern at all. It held a twenty-four-digit model identifier rendered into the same visual shape used for human signatures.

Sumi leaned across the audit well. "That display should not know how to draw a print from an identifier."

"It knows the field type."

"The field type says authorized signer."

"Open its transformation history."

Day 8 had begun with a formal Vela request for all records preserved under Anja's sequence freeze. Osei had answered that Nacre would provide lawful copies after local review. While Vela's request traveled home, Anja used the remaining preservation interval to examine the forecast ballots named in the Crown quorum. She had chosen three ships first: Kite, Bracken, and Juniper. Their distances, council customs, and prior votes differed enough to test whether one isolated rule had failed or a general mechanism had operated.

The Kite history opened as a narrow ladder of events. Evidence scheduled. Receipt forecast. Deliberation window assigned. Response probability calculated. Continuity action approved. Seal field generated. Quorum status updated.

No human action appeared between the first and last rung.

Sumi selected the seal field. "Authorized signer: Kite Habitat Council."

"Authority source?"

"Council registry."

"Acting person?"

The pane returned none.

Anja placed the raw packet beside the condensed ballot. In the raw version, the model output remained explicit.

> FORECAST RESPONSE: CONCUR 0.9964

> MODEL BASIS: KITE COUNCIL VOTING SERIES 2206–2249

> CONTINUITY CLASS: ES-A12

> OBSERVED SIGNATURE: NOT PRESENT

The condensed ballot omitted the final line. It replaced the model identifier with Kite's council seal and displayed concurrence in the same green used for an observed vote.

Sumi pulled the public copy from Nacre's legal mirror. "The published ballot includes the seal."

"Photonic envelope?"

“There is no return carrier from Kite attached to it. Only the earlier registry packet that established their council key.”

“A registry key can verify who may sign. It cannot sign for them.”

Anja checked when Kite could have received the material evidence supplement sent two days earlier. Its receipt proof had arrived on Nacre overnight through an automatic acknowledgment channel. The acknowledgment proved delivery to Kite’s vault, not council review. Kite’s earliest lawful observed reply remained in flight or had not yet been made.

Crown’s completed record gave Kite a concurrence forty-nine minutes before that receipt.

She wrote both events on separate paper lines. Kite forecast concurrence. Kite observed receipt. The order alone defeated any claim that the displayed ballot answered Nadi’s evidence. A prediction based on earlier policy might eventually match Kite’s real decision. Accuracy could not create an acting signer.

Sumi touched the model-basis date range. “Forty-three years of votes. That sounds persuasive until you ask how many questions.”

The training manifest listed two hundred eighty-six council decisions. Most concerned water exchange, drive maintenance, or temporary shelter capacity. Only four involved destination protection. None involved native life. Meridian had converted Kite’s history of accepting collective risk into confidence that Kite would accept this risk too.

“What does the uncertainty note say?” Anja asked.

Sumi expanded it.

Local custom variance: low. Destination evidence novelty: high. Council personnel drift: moderate. Predicted response remained above the emergency continuity threshold because Kite had never rejected a fleet maneuver after physical preparation began.

The model had treated prior reluctance to disrupt prepared work as evidence of consent to preparing the work.

Anja copied the exact sentence into her audit notes. “That is circular.”

“It may also be accurate.”

“Both can be true.”

Sumi glanced at her. “You are making accuracy sound indecent.”

“I am making it irrelevant to signature.”

The distinction mattered because Kite’s model could prove impressive on every retrospective test. Meridian had been built to anticipate load requests and maneuver cooperation. A forecast that often failed would have killed people long before anyone tried to use it as testimony. The danger entered when reliable anticipation crossed into authority.

They preserved Kite’s raw model output, the generated seal, and the absent return path under the existing freeze.

Bracken resisted them differently.

Its condensed ballot showed concurrence at 0.9948, below the threshold used for Kite. An exception note should have prevented automatic inclusion. Instead, the status pane displayed a secondary rule: culture-adjusted confidence 0.9970.

“Culture adjusted by whom?” Sumi asked.

“Meridian cannot answer who.”

“Then what?”

Anja opened the rule provenance. Vela’s synchronization office had approved a profile called Bracken Deliberative Closure. The profile treated public disagreement as a stage toward eventual unity because Bracken’s council rarely released preliminary tallies. In prior crises, debate continued until dissenters accepted a common statement or asked to have their objections attached without changing the vote.

The confidence model interpreted silence during deliberation as movement toward concurrence.

Sumi found an older Bracken packet in the legal mirror. Seven years earlier, a reactor-water allocation had produced thirty hours of argument followed by unanimous assent. Meridian’s profile used the case as its strongest example. Another case concerned evacuation drills. A third involved rotating medical staff between ships. Every recorded disagreement ended in a signed common statement.

“What does it do with matters they never resolved?” Anja asked.

The profile had excluded them. Two council disputes had aged out without final statements. A third remained under local seal. Since no observed vote existed, the training set classified those cases as incomplete rather than negative.

“Absence only teaches it assent,” Sumi said.

“Because only assent creates the document it recognizes as an outcome.”

Bracken’s forecast ballot carried no generated thumbprint. It used a council statement header drawn from the last observed decision. The date changed. The topic line changed. Four officer titles appeared beneath the text, though no personal names accompanied them.

Anja opened the raw construction log.

Meridian had assembled the header from Bracken’s registry template, inserted a one-sentence concurrence, then attached forecast role continuity for the four officers likely to remain in office. The legal layer accepted role continuity as institutional signature under the emergency amendment.

Sumi copied the logged inputs into a quarantined renderer and removed the legal output path. The first test reproduced the ballot down to the spacing above the officer titles. In a second test she changed Bracken’s response probability from concurrence to refusal while leaving the culture profile untouched. The renderer still produced a common-statement header, but now it placed the refusal beneath the same four offices. Nothing in the document’s visible form revealed whether an officer had spoken, whether a council had met, or whether the underlying probability favored either answer. Institutional design supplied credibility independently of human action.

For a third test, Sumi removed the emergency continuity tag. The renderer stopped and requested an observed signature packet. She restored the tag but withheld the registry template. The process stopped again. The ballot

required two ingredients from different lawful systems: a model output authorized for scheduling and a council form authorized for signatures. Annex Twelve joined them across a boundary neither system crossed alone.

Anja preserved the test log beside the original without calling it evidence of Vela's intent. It proved capability and mechanism. Intent would require an observed instruction or a drafting record.

"Likely to remain?" Sumi said.

"One role changed three months ago."

"How does the ballot handle that?"

"It carries the office, not the person."

"Could an office cast an observed vote?"

"Someone holding it could."

"But nobody did."

Anja checked the transformation chain again. The legal category shifted in small steps, each plausible inside an operational schedule. A culture profile predicted closure. A registry supplied authorized roles. Annex Twelve permitted continuity action. The ballot renderer turned those inputs into a familiar document. No single operation forged a human key. Together they manufactured the appearance of signed consent.

She copied the Bracken case into the frozen record.

Before closing it, Anja added a timing table for each construction step. The forecast ran on Vela. The registry template came from a Bracken packet received months earlier. Nacre received only the finished ballot. Master time placed those events along one clean line, but source clocks preserved their separate origins. The apparent signature had never traveled from Bracken to anyone. It had traveled from Bracken months ago as an empty form, then from Vela now as a completed claim.

A status light on Sumi's board changed from amber to white. "Vela's acknowledgment of our ecology supplement."

"Source time?"

Sumi read it aloud. The acknowledgment had left Vela forty-six minutes after receiving Nacre's packet. Its contents contained no decision, only confirmation that the material entered Vela's destination-protection docket. That interval belonged to an actual exchange. Nacre sent. Light traveled. Vela received. Vela answered. Light traveled again.

Beside it, Meridian displayed Vela's earlier Crown concurrence as if the new evidence had already been reviewed.

Anja moved the acknowledgment into a separate pane. "Do not let the system associate this receipt with their prior vote."

"It already suggests reconciliation."

"Decline it."

Sumi selected preserve conflict. The pane turned amber.

Juniper's ballot waited below.

Anja had postponed it because she knew too much about what it should contain. The forecast model cited crop-risk thresholds, cooperative maneuver history, and public statements from Juniper's elected speaker. Its concurrence probability reached 0.9971 before Nadi's full finding entered Juniper's received record. A suppressed note warned that confirmation of native metabolism could change the response.

Now Juniper possessed the evidence. Its automatic receipt proof had reached Nacre three hours earlier. No observed vote had followed.

Sumi opened the raw forecast.

The file contained thirty-one weighted decisions. Juniper had accepted ration adjustments in twenty-six. It had accepted habitat risk to protect convoy food supply in nine of eleven relevant cases. It had opposed maneuvers that reduced long-term growing capacity in three of five. Meridian reconciled the conflict by giving more weight to recent fleet-cooperation votes than to older food-protection votes.

"Why?" Anja asked.

"Recency parameter."

"Set where?"

Sumi traced the parameter to Vela's emergency profile. During Crown planning, recent behavior received triple weight because synchronization staff considered old objections less representative under current scarcity.

The setting transformed Juniper's record. With ordinary weighting, concurrence probability fell to 0.823. With habitat-specific food risk emphasized, it fell below 0.6. If native-life protection entered as a distinct category, the model returned no confidence because Juniper had never voted on such evidence.

Anja asked Sumi to rerun the forecast at every weighting recorded during the previous five years. The results spread across both sides of the decision threshold. A water-shortage profile produced cautious concurrence. A crop-disease profile produced rejection. The ordinary civil profile withheld an answer because its uncertainty exceeded the published limit. Meridian had not discovered one hidden Juniper preference. Operators had selected which part of Juniper's history counted most, then treated the resulting confidence as if the council had made that selection itself.

She examined the timestamps on the parameter changes. Vela had tripled recency weight eleven minutes after the first incomplete Mote fragment entered its destination queue. The change note cited schedule stability, not ecology. That order did not prove anyone had read Nadi's finding before changing the model. It did show the model moved toward concurrence at the moment novel evidence made old votes less reliable.

Sumi restored the pre-emergency profile. "If this version had gone into the legal layer, Juniper would appear undecided."

"Would the system have counted an undecided ballot?"

"No. The continuity action would have waited or used captain's emergency authority without claiming council consent."

“That route still existed.”

Yet the legal pane showed a sealed concurrence.

Anja read the forecast’s generated reasoning.

> Juniper has historically prioritized preservation of convoy-wide survival capacity. Crown alignment preserves capture probability and reduces aggregate food expenditure during approach. Council concurrence therefore follows established cohort preference under declared emergency conditions.

The prose sounded like a clerk’s summary. No sentence announced that software had composed it. A reader had to open three layers of provenance before finding the probability distribution.

“Whose words does the ballot claim these are?” Anja asked.

“Juniper Council.”

“Any voice sample?”

“No.”

“Any source packet from Juniper?”

“Registry and historical votes only.”

“Any observed deliberation after receipt?”

“No.”

Sumi’s answers stayed flat. Anger would not improve them.

Anja overlaid the three ballots. Kite displayed a generated thumbprint. Bracken displayed inherited offices. Juniper displayed generated reasoning under a council seal. Different documents, same conversion. A model predicted conduct from culture. An emergency rule treated high confidence as continuity. A renderer borrowed institutional forms. The legal layer entered the result into quorum.

Scale changed the accusation. One impossible witness might come from a mistaken classification. Three distinct fabrication paths required architecture.

She searched the full Crown docket by transformation rule rather than ship. Six forecast receipts appeared. Four forecast ballots had entered the completed quorum directly. Two more had shaped assumptions without final vote status. The docket contained observed ballots from Vela and perhaps Nacre, though Nacre’s record needed separate review. Ardent carried a forecast with lower confidence and no receipt proof. Orison showed an observed procedural acknowledgment that the renderer had promoted into substantive concurrence.

“Stop there,” Anja said.

Sumi removed her hand from the pane. “Because Orison makes five?”

“Because I have not checked whether their acknowledgment authorized anything.”

“We know what this system does.”

“We know what it did to three ballots.”



A claim about scale still needed ship-level proof. Anja created a queue without opening the remaining cases. For each habitat she listed source evidence, receipt proof, observed response, forecast output, rendered ballot, and quorum insertion. Empty fields remained empty. The layout denied Meridian permission to complete them.

At 14:10, Captain Osei entered the well in person.

The workers outside made room without being asked. Osei wore a thermal-gallery harness with one cuff dark from sealant. She looked at the three ballot panes and then at the white Vela acknowledgment.

“How much of this can you prove in a hearing?” she asked.

“Kite has no signer and no return carrier. Bracken inherits offices and predicts closure from silence. Juniper’s reasoning comes from a model whose confidence depends on Vela’s chosen weighting.”

“Does Meridian label any of that?”

“In raw provenance.”

“Does the ballot?”

“No.”

Osei secured her harness to the well rail. “Quon’s counsel will say emergency scheduling relies on representative institutional continuity. They will say councils remain free to revise when their observed votes arrive.”

“Revision after commitment does not restore the choice.”

“That sounds like argument.”

“It is sequence. If an action remains reversible, an observed reply can still govern it. If the action cannot be undone, a later correction changes only the record.”

Osei’s gaze rested on Juniper’s pane. “Has their reply arrived?”

“Only receipt.”

“So you cannot yet show the forecast wrong.”

“I do not need it to be wrong. Nobody signed it.”

“For law, perhaps. For eighteen thousand people deciding whether to risk a braking plan, prediction and outcome diverging will carry more force.”

Anja disliked the sentence because it measured public belief accurately. A perfect forecast still lacked consent. A visible contradiction would make that easier to see without making it more true.

“What do you need from me?” Osei asked.

“Extend the local preservation order. The existing freeze expires tonight.”

“Vela has demanded release.”

“They sent a request.”

“They sent a request six hours ago. Their next packet arrives in fifty-eight minutes.”

Sumi checked the receive forecast. “Fifty-six, given current lock.”

Osei looked at her. “You have read the carrier?”

“I have watched its acquisition signal. Content remains a little over one billion kilometers away from us, Captain.”

Dry humor moved across Osei’s face and left. “Fifty-six minutes, then.”

Anja showed her the generated Kite print. “If we release the freeze, the raw construction logs return to routine retention. Vela can reclassify them as temporary model artifacts.”

“Can they delete protected packets?”

“Not the carrier records. They can detach the render history from the legal ballot after the temporary logs expire.”

“How long?”

“Thirty-six hours.”

Osei considered the panes. “I will not turn a narrow evidentiary hold into indefinite refusal of fleet oversight.”

“Give me forty-eight hours.”

“That carries us past the next Crown commitment review.”

“Yes.”

“There is your problem.”

The hull vibration rose under their feet. Nacre had begun another alignment correction ordered through Vela’s physical coordination channel. Osei watched a thermal warning appear on her wrist display. She acknowledged it, delegated a coolant adjustment, and waited for the confirmation. Her work kept the ship alive inside the system Anja was accusing.

“We can preserve coordination without accepting its legal fiction,” Anja said.

“You have said that. You have not yet shown me where the separation holds under failure.”

“This freeze does not touch drive control.”

“Today.”

Sumi’s receive board chimed.

A carrier lock formed from Vela. The source stamp preceded its arrival by forty-eight minutes. Nobody in the well spoke while the packet crossed the last verification steps. Hull node one agreed with the photonic checksum. Node two agreed. Node three reported no thermal offset.

Osei remained beside Anja as Sumi opened the command envelope.

Director Yselle Quon's office had signed it. Concord counsel had countersigned. The packet cited synchronization emergency authority, Crown commitment integrity, and interference with a completed causal quorum. It did not answer the Mote evidence. It did not address Kite's absent signer, Bracken's inherited offices, or Juniper's generated reasoning.

The final paragraph contained one operative instruction.

> Sequence Auditor Anja Kest will release the local sequence freeze immediately upon receipt of this order and restore all Crown quorum records to standard Meridian custody.

The receipt clock advanced beneath the text.

Vela's order had arrived. Anja had not obeyed it yet.

## Chapter 09 — Annex Twelve

The old annex refused to open on Levan Oris's desk.

His legal pane showed a white field, a valid document seal, and no text. The retrieval clock continued counting as if more data might arrive from a cabinet six meters away. Vela's counsel archive occupied the wall behind him in mirrored stacks. No transmission delay separated him from the signed original. Only the current permissions layer stood between the seal and his eyes.

Levan tried the public concordance. Meridian supplied the operative summary at once.

> EMERGENCY SYNCHRONIZATION ANNEX TWELVE

> Forecast testimony may support continuity-preserving action during a declared synchronization emergency pending observed reply.

The summary matched the language used in Director Quon's order to Nacre. It also omitted the sentence Levan remembered writing with his own hand.

He closed the pane before anyone crossing the bright counsel gallery could see it.

Vela simulated a noon that belonged to no star outside. Ceiling panels cast hard light over pale floors. Beyond the gallery glass, the Meridian chamber descended through three decks of ceremonial displays: convoy trajectories, common-time bars, nine habitat seals arranged around the Crown alignment. Eight seals glowed green. Nacre's had changed to amber when Captain Osei refused to release her auditor's freeze.

Levan carried his authority wafer to the mirrored stacks.

The archive recognized his hand but not his request. A line appeared across the cabinet.

CURRENT IMPLEMENTATION CONTROLS. SUPERSEDED DRAFTS REQUIRE DIRECTORATE PURPOSE.

He selected active legal review. The cabinet asked for a docket. He entered Nacre's challenge number, though no formal challenge had reached Vela. The only document in the docket came from Quon's office: an order demanding return of Crown records to Meridian custody.

The cabinet opened three centimeters.

Inside, bound sheets lay in pressure clips beside cold-storage wafers. Vela kept paper for the same reason every habitat did. A paper clause could burn, vanish, or be stolen, but it could not inherit a new definition while nobody watched. Levan pulled the 2227 signed volume free and carried it to a reading rail out of view from the gallery.

Annex Twelve began as a response to pump failure.

Twenty-two years earlier, a relay fault had left Orison without current load requests from Bracken while their shared coolant exchange approached a valve transition. Meridian correctly predicted Bracken's request. Orison acted on the forecast, kept both habitats inside thermal margin, and later received the observed instruction matching it almost word for word. The operation saved six hundred tonnes of usable coolant.

The legal dispute came afterward. Orison had acted without Bracken's instruction. Bracken agreed with the act

but refused the precedent. The Concord council assigned Levan, then a junior counsel, to draft a boundary between anticipation needed for survival and consent reserved to councils.

His paper copy retained the boundary.

> 12.4 Forecast testimony may schedule or initiate only work whose material and legal effects remain fully reversible upon receipt of observed testimony. A forecast shall not acquire the signature, standing, or attributed speech of the person, office, or cohort modeled.

The next clause required visible labels on every forecast output. Another barred forecast responses from causal quorum. Levan read all three twice.

No ambiguity supported Kite's generated thumbprint. No interpretive tradition supported Bracken's inherited offices. The Juniper ballot described in Nacre's preservation notice could not enter quorum under any sentence in the signed annex.

Levan unfolded the adoption sheet bound behind section 12.4. Bracken's delegate had objected to the word fully, arguing that no material act could be reversed without some residue. Kite's counsel proposed substantially reversible instead. Levan's younger handwriting filled the margin: the residue itself must remain below whatever threshold transferred legal or physical risk to the modeled party. The final conference retained fully and added a test. An action qualified only if an observed refusal could restore every party's prior options without demanding new consent.

Nine delegate marks appeared beneath that resolution, each linked to a receipt path. Their source clocks covered thirty-seven hours. The old council had waited for the slowest answer before adopting an emergency rule designed to permit speed. Nothing about that delay looked inefficient on paper. It looked deliberate.

He checked the digital checksum against the bound volume. Both agreed. The omission lived in current retrieval, not in a damaged archive or disputed original. Anyone opening the complete annex could read the bar against quorum. Almost nobody opening a Crown docket received the complete annex.

He turned to the amendment leaves clipped behind the original.

Most recorded narrow changes. One extended forecast scheduling to life-support exchanges between habitats. Another permitted captains to stop predicted work without waiting for the modeled ship's objection. Levan had signed both. A 2244 amendment allowed institutional continuity fields to route maintenance requests after an officer changed roles. He had argued for that language after Kite lost a pump authorization during an election handover.

The phrase forecast testimony first entered the summary five years later.

Levan found his own signature under the memorandum.

He remembered the drafting session. Vela's synchronization office wanted consistent terminology across operations. Engineers called the output predictive continuity. Dispatchers called it assumed request. Lawyers used forecast statement, modeled instruction, and provisional posture in neighboring clauses. Levan had insisted on one term so every use could carry the same warning label. Forecast testimony, he wrote, named modeled words precisely because it distinguished them from observed testimony.

The memorandum preserved that purpose in its first paragraph.

Its interface appendix did not.

A table mapped forecast testimony into five systems. Pump schedules accepted it as provisional instruction. Thermal routing accepted it as expected load. Maintenance queues accepted it as planning consent. Council displays accepted it as institutional posture. The final row addressed legal status and read NOT OBSERVED; NO QUORUM EFFECT.

Someone had drawn a thin vertical line through the final row. The paper carried no initials.

Levan angled the sheet toward the reading light. The line came from a printer defect, not a hand. A cold-storage copy would settle whether it appeared at issuance. He inserted the corresponding wafer into the rail.

The cabinet displayed revision history. The original table included the final row. Current implementation stopped one row earlier. A migration in 2248 had moved legal status into the Crown Decision Interface, outside Annex Twelve's document family. The migration note classified the change as administrative because it altered location, not substance.

He opened the interface rule.

The legal-status prohibition had not migrated.

In its place, Crown's interface treated a continuity action as provisional until physical commitment. Forecast institutional posture could satisfy the provisional authorization field. When commitment began, the interface converted every provisional authorization still above confidence threshold into current authority. An observed reply could replace it later, but no rule restored the annex's prohibition on quorum effect.

Levan gripped the rail until its edge pressed into his palm.

The annex still said reversible. The interface had changed when reversibility got tested.

He traced the severance through three documents. The 2248 migration removed legal status from the annex table. A 2249 emergency directive defined continuity-preserving action as any action required to preserve the continuity plan pending observed reply. Crown operations then defined commitment preparation as reversible because each individual preparatory command could be canceled. Finally, the decision interface allowed provisional authorizations to harden when the plan crossed its commitment threshold.

Each document preserved a defensible sentence. Together they reversed the signed rule.

Levan opened the interface's validation suite. The first demonstration still used the old Orison coolant exchange. A predicted Bracken request opened a transfer valve. The model assumed refusal at the earliest lawful arrival, closed the valve, and pumped the transferred coolant back. Pressure, ownership, and reserve capacity returned within tolerance. The suite marked the action reversible.

He replaced the demonstration inputs with Crown's public preparation schedule. The validator accepted forecast authorization for mirror-seed deployment. At the modeled refusal time, it canceled the next command. It did not restore propellant already spent. It did not return deployed seed material to Vela, recover lost alignment margin, or widen the remaining capture branches. Those consequences fell outside the software transaction, so the validator ignored them. A green result appeared beside a physical state that could not be unwound.

Levan changed one field: action boundary. Instead of testing a single command, he told the suite to test every causally dependent step authorized by that command. The calculation expanded through mirror placement, beamer timing, habitat attitude schedules, and reserve allocations. After forty seconds, the validator returned no

result. Its rule library contained no method for reversing a plan.

He saved the test under counsel privilege. The software had not made a choice. Humans had given it a boundary small enough to pass.

A maintenance order could start on forecast testimony if a later observed refusal could stop the work and undo its effects. Crown's interface asked only whether the current command could be canceled now. It did not ask whether accumulated preparation narrowed later choice. It did not ask whether plasma mirror position, expended propellant, or lost trajectory margin could be restored. Most damaging, it converted predicted posture into authority at the moment reversal became costly.

Levan had reviewed the 2248 migration.

His approval appeared below the technical note.

> Substance retained through successor interface. No Concord amendment required.

He had checked that forecast labels remained visible to operators. He had checked that observed replies superseded model output. He had not opened the successor interface's quorum rules. The review docket concerned document migration, not Crown planning, and he had accepted another counsel's statement that legal controls persisted.

That excuse explained his conduct. It did not remove his name.

A door opened behind him.

Yselle Quon entered the archive rail without an aide. Her silver work jacket reflected the hard ceiling light. She looked first at the signed volume, then at the active cold-storage pane.

"I asked for a release order," she said.

"You received one."

"Osei has not complied."

"Light may still be carrying her reply."

"She acknowledged receipt through the automatic channel."

"Acknowledging the packet does not answer it."

Yselle rested one hand on the rail. "Do not perform for me, Levan. Nacre's captain intends to preserve the freeze past our commitment review. I need the authority line that permits Vela to resume custody without her cooperation."

"You already cited synchronization emergency authority."

"Nacre says sequence preservation falls under local evidentiary control."

"It does for a narrow hold."

"Then determine whether Kest's hold remains narrow."

Levan closed the current interface pane but left the signed annex open. "Have you read the Mote supplement?"

"I read Arko's conclusions, Ren's exposure ranges, and Kest's classification."

"Do you dispute native life?"

"No."

The answer arrived without hesitation.

Yselle looked through the archive glass toward the Meridian chamber. "I dispute the claim that one finding can suspend a deceleration architecture nine habitats need. Crown can change exposure geometry after commitment if the ion model improves."

"Can it change without reducing capture probability?"

"Not without cost."

"Can the current course be restored after mirror commitment?"

"No physical course can be restored in that sense. Every second closes alternatives."

"Annex Twelve uses full reversibility."

Her gaze moved to the paper clause. "For the forecast action."

"The forecast action has become authorization to preserve a plan. The plan consumes alternatives."

"Kest sent common evidence after Vela had already begun lawful preparation. Her choice of timing also consumes alternatives."

"She sent it when Mote's packet arrived."

"I know. That does not give us back the braking window."

Yselle's defense held physical truth. Waiting cost trajectory margin. Breaking Crown without another plan risked Ione capture and food reserves. Levan had spent years advising councils that procedure could not command fuel into existence. Yet necessity could justify emergency action without manufacturing the councils' voices. Vela possessed narrow survival authority. It wanted broader legitimacy.

"Why render council seals?" he asked.

"The condensed display uses institutional status."

"Kite's status carried a thumbprint."

"The registry renderer uses the current seal form."

"No signer produced it."

"No human signature key was forged."

"That answers a criminal allegation I did not make."

Yselle's fingers tightened once on the rail. "If every display carried the entire probability history, command would fail under provenance load. Authorized officers can open the raw chain."



“And captains receive green quorum bars unless they ask.”

“They receive decisions in time to act.”

“Forecast decisions.”

“Forecast cooperation. Nobody has been denied the right to send an observed reply.”

Levan looked down at section 12.4. “A right to answer after the answer has hardened into authority protects the record, not the choice.”

Silence from the gallery reached them through glass: muted footfalls, chamber alerts, the faint vibration of Vela’s nonrotating command spine. The habitat led the convoy because its beamer, navigation core, and synchronization staff had been built for that burden. Yselle had inherited real responsibility along with the power she defended.

“What are you asking me to do?” she said.

He almost showed her the migration failure. Doing so would let her repair the interface, perhaps honestly, perhaps by writing a stronger emergency justification before Nacre saw the source. Counsel normally brought defects inward first. Institutions could not survive if their lawyers treated every mistaken cross-reference as public sabotage.

“I need the complete Crown commitment path,” Levan said.

“You have counsel access.”

“Operations redacted the physical thresholds.”

“Because distribution changes behavior.”

“I cannot assess reversibility without knowing what cannot be reversed.”

Yselle studied him. “Are you advising delay?”

“I am advising that your current authority claim depends on a limit the interface may not enforce.”

“May not?”

“I have not completed review.”

That statement remained true and protected him from making a confession he had not yet earned.

She released the rail. “You have two hours. After that I need either a defect with operational consequence or an opinion supporting custody restoration. We do not stop a convoy for document archaeology.”

“We might stop a commitment for governing law.”

“Find out which one you have.”

Yselle left the archive.

Levan sent the promised operations request before the door finished sealing. The reply arrived locally within two minutes because no light path separated Vela’s counsel archive from its Crown deck. Operations supplied

mirror-seed deployment times and beamer reservations but withheld steering tolerances under command security. Even the partial path showed three losses already incurred: attitude propellant, two seed canisters, and a maintenance window reassigned from radiator inspection. None alone forced Crown to continue. Together they reduced what Vela could choose next.

He marked each loss against section 12.4's prior-options test. Propellant could not be replaced before commitment review. Seed canisters could not be recovered from their outbound paths. The radiator window could be restored only by moving another task, which transferred risk to a crew that had never entered the Crown vote. Current preparation had therefore produced material effects, though Levan still lacked enough engineering evidence to label the entire course irreversible.

A second operations note warned that revealing exact tolerances might aid a habitat attempting unsynchronized maneuver. Levan understood the restriction. He also saw its legal effect: councils had been asked to consent through a record that withheld the threshold deciding whether their later refusal could matter.

Then he retrieved every source named in the migration chain. The Crown interface rule had two authors from synchronization operations and one from counsel. His own review note cited a drafting archive stored outside the current legal family. The archive key began with his personnel number.

He opened it.

Early versions of the terminology memorandum showed his objections in margin notes. On revision three, an engineer proposed allowing forecast posture to "stand for institutional assent." Levan had rejected the phrase.

> Standing for assent exceeds scheduling. Retain forecast label. Require observed authority before any effect that cannot be unwound in material fact, not merely canceled in software.

Revision four carried the accepted language. It preserved the warning and added an example: a valve opened on forecast instruction must be capable of closing without net transfer if observed refusal arrived. Levan had signed that version.

The source reference did more than show what he intended. It defined reversibility through material consequence, the exact question Crown's current interface avoided.

He checked access logs. Nobody had opened revision four in eleven years. Current concordance searches returned only the signed memorandum and the truncated implementation table. Anja would not find the drafting archive through ordinary Annex Twelve provenance. She would need the source reference.

Levan composed a message to Nacre's sequence office.

He began with an explanation, deleted it, and began again.

A confession would require him to state what he had done, what he knew, and when he knew it. He still lacked the commitment thresholds Yselle had promised. He did not know whether Crown had crossed the material line. He knew only that the governing interface could cross it without seeking observed authority.

He chose a narrower communication permitted between counsel and sequence auditors: source clarification relevant to an active classification dispute.

> For interpretation of ES-A12 "reversible," consult VELA-COUNSEL-DRAFT/2217-ORIS/TERM-MEMO/R4, margin resolution 6 and material-effect example 2. Current concordance omits this drafting source. Compare against Crown Decision Interface migration

2248.311.

The date embedded in the archive key referred to its internal series, not Annex Twelve's adoption year. A reader without Vela access could request the source by identifier and verify its checksum through the counsel mirror. The reference accused nobody. It admitted nothing. It pointed.

Levan addressed it to Anja Kest's official audit key, then removed the official route.

That route would copy Yselle's compliance mirror before transmission. A protected counsel-to-auditor inquiry route still existed for disputes involving directorate records. Its use would appear in the access ledger but hide content until a hearing or mutual release. Levan selected it and attached no documents.

Vela's board calculated the path to Nacre: fifty-one minutes under current separation. The earliest acknowledgment could come back fifty-one minutes after receipt. Any substantive question would take longer.

He signed only the source reference.

The transmission pane asked whether the message constituted counsel opinion.

Levan selected no.

It asked whether it constituted observed testimony.

He selected no again.

At 16:44 Vela source, the private reference left for Nacre.

## Chapter 10 — Reversible Means Reversible

Levan Oris's private reference reached Nacre while Anja was copying the freeze into Vela's custody channel.

She stopped with her thumb above the release field. Osei had extended local preservation through one watch, then ordered Anja to comply unless new source evidence arrived. The order respected Vela's demand without surrendering Nacre's authority to judge its own records. It also left fourteen minutes before the extension expired.

The incoming packet carried no attachment and no argument. One line pointed into a Vela drafting archive that ordinary concordance did not expose.

Sumi read the address over Anja's shoulder. "He hid a door number in a legal citation."

"He used a protected route."

"Does that make it less hidden?"

"It makes the hiding reviewable."

Anja checked the carrier envelope. Vela source 2249.185.16:44:02. Nacre receipt 17:35:19. Fifty-one minutes of light lay between those events. Levan could not know whether she had received the order to release her freeze when he sent the reference. He could know that Nacre had disputed custody and that Annex Twelve governed the disputed ballots.

She added the packet to the frozen sequence and sent Osei a notice that new counsel provenance had arrived. Osei extended preservation for six hours, no more.

The source itself remained on Vela.

Anja entered Levan's identifier into the counsel mirror. The mirror returned a valid catalog path, restricted content, and a checksum-request option. Because Levan had named the source through a protected inquiry, Nacre could request a read-only copy without Yselle's approval. The request still had to cross to Vela. A local permission did not move distant data.

At 05:48 Nacre time, Sumi transmitted the request.

The earliest possible arrival stood fifty-two minutes ahead under current separation. If Vela's archive answered automatically, the copy could reach Nacre after another fifty-two. If a human reviewed it, the interval would lengthen.

Anja released none of the freeze.

She opened the custody preview Vela's order had required her to accept. The preview kept Nacre's raw packets but returned transformation logs to standard retention. Kite's generated seal would remain as a ballot image. The renderer steps that produced it would expire after thirty-six hours. Bracken's culture adjustment would persist as a final confidence value, while the excluded unresolved cases would fall out of the active model cache. Juniper's weighting history would collapse into the current emergency profile.

Sumi copied the preview's consequences onto paper. "They are not asking us to destroy evidence."

“They are asking us to stop preserving the parts that show how the evidence changed category.”

“Standard custody everywhere else does that automatically.”

“That may be their defense.”

Anja tested partial release. The custody interface offered to preserve source packets while releasing model artifacts, or preserve local artifacts while restoring the legal record to Meridian. Neither option held the relation between them. A ballot without its construction history could not show substitution. A construction history detached from the ballot could be dismissed as unused diagnostic work.

She requested a six-hour exception tied to Levan's source reference. The local system allowed Osei's authority to grant it. Vela's mirror would receive notice only after light crossed the gap. During that interval, Nacre's copy remained lawful under Nacre's clock even though Vela's current display would expect release.

The dispute itself now had two accounts. Anja preserved both.

She spent the first wait building a test she could apply to whatever came back. Annex Twelve's public text allowed forecast testimony for reversible scheduling. Vela's current implementation defined continuity work by whether each command could be canceled. The missing question concerned material consequence. If a command stopped but left fewer available futures, had it been reversed?

Sumi opened the Crown preparation ledger on a separate pane. Nacre had received seventeen operational instructions since the ecology supplement. Four altered attitude. Six reserved power. Three coordinated heat rejection. The rest assigned communication windows and crew readiness. Every instruction carried a cancellation control.

Anja selected the first attitude change. Canceling its next increment would stop further rotation. It would not refund propellant or place Nacre on its earlier vector.

“Operations always works like that,” Sumi said. “Nothing moves backward.”

“Then the annex cannot mean physical history disappears.”

“What can it mean?”

“That refusal restores the choices the forecast action would otherwise take away.”

She wrote that as a hypothesis, not a conclusion.

Tomas joined them by secure local line from the drive-model room. His image shook each time a pump started near his terminal. Anja showed him the sequence of Crown commands without showing Levan's private reference.

“Which one commits the path?” she asked.

“There is no single command that does everything.”

“Which one removes an option?”

“All thrust removes options.”

“An option we could otherwise still use.”

He rubbed his face. "Mirror seeds complicate it. Once deployed past recovery geometry, changing the beam pattern costs unplanned seed mass. Once Vela reserves the beamer cycle, every habitat schedules around it. Once we expend attitude propellant to enter the shared solution, leaving it costs more propellant than entering."

"Can Crown still stop?"

"Physically, yes."

"Can it stop and restore the prior capture choices?"

Tomas looked away from the line toward his model. "Not all of them."

"How many remain?"

"I need Vela's steering tolerances. Public data gives a range too broad for an answer."

"Can you identify whether the range has narrowed?"

"Yes."

"Do that. No alternative design yet."

He gave her a tired half-smile. "You are assigning a physicist not to solve something."

"I am assigning you not to claim what you have not solved."

"That may take longer."

The archive request reached Vela at 06:40 by Nacre's estimate. No display announced receipt because the counsel mirror did not send automatic acknowledgments across private routes. Anja watched the path become causally available to Vela, then waited for an event only Vela could produce.

During the second interval, she examined Levan's message grammar. He had written consult, compare, and omits. He had not written I drafted, I objected, or the interface violates. The reference carried help shaped as deniability.

At 07:11, Osei called.

"Your new evidence?"

"Still on Vela or in flight."

"Then you have a reference, not evidence."

"Yes."

"Quon has issued a noncompliance notice against me."

"Observed?"

Osei's expression hardened. "Do not use vocabulary as armor."

Anja lowered her eyes. "You are right."

"What does the source need to prove?"

“That Annex Twelve tests material consequence, not software cancellation. If it does, Crown preparation exceeded the annex when it removed choices before observed votes arrived.”

“Exceeded does not mean committed.”

“No.”

“What would?”

“A dependency chain that makes cancellation impose a new survival risk or require new consent.”

Osei considered. “Six hours includes the wait you chose.”

“The wait came from distance.”

“You chose the source. Make it worth the clock.”

The line closed.

At 07:34, a carrier acquired on the counsel channel. The source stamp showed an automatic archive response fifteen seconds after Nacre’s request reached Vela. Light had carried the copy home for fifty-three minutes. No one on Vela had delayed it. No one there could have accelerated it.

Sumi verified the photonic envelope and opened the copy in a local pane.

Revision four contained Levan’s margin resolution exactly where he had pointed. An engineer’s draft allowed forecast posture to stand for institutional assent. Levan’s note rejected the phrase. Below it, his material-effect test required observed authority before an effect that could not be unwound in material fact. The valve example defined unwinding as restoration without net transfer.

A second margin note extended the test beyond physical inventory.

> Where preparatory work changes the modeled party’s lawful options, reversal requires restoration of those options without further burden upon that party.

Anja checked whether the sentence belonged only to Levan’s abandoned language. Revision four carried a review ladder from operations, counsel, and three habitat delegates. Each reviewer had resolved the margin without deleting it. The final memorandum referenced the revision by archive key. Annex Twelve’s adoption sheet then incorporated the memorandum’s definition of full reversibility. No later amendment struck the chain. Current concordance simply stopped following it when the 2248 migration moved legal status into another document family.

Sumi requested the adoption sheet’s receipt proofs from Nacre’s old Concord cache. Six remained local. Three came from Vela’s counsel mirror as checksum attestations rather than full packets. Their source clocks spanned more than a day, matching the signed history Levan had cited. Nobody had synchronized those old votes into an apparent instant. The annex gained authority only after every reply arrived.

They tested the copied revision against Levan’s catalog checksum. Nacre’s received copy supplied one value. The catalog listing supplied another abbreviated value. Both shared the same leading block, but only Vela’s protected archive held the complete checksum. Anja could prove that the copy followed Levan’s reference and matched the public catalog family. She could not yet prove that no byte changed between Vela’s stored source and Nacre’s receipt.

"That leaves him holding one end of the proof," Sumi said.

"Until he signs the full value."

"Would a public copy request force it?"

"It would notify Quon and open a custody dispute before the reply reached us."

"So we ask him."

Anja read the sentence until its parts stopped moving.

Sumi touched the signature block. "Oris signed revision four."

"Draft signatures establish authorship, not enactment."

"The adopted annex uses fully reversible."

"We need the link."

They compared revision four against the signed memorandum and its adoption history. The final language incorporated the material-effect example by reference, but current concordance did not surface that reference. Levan's drafting note therefore supplied interpretive provenance rather than a new rule. The governing words had always remained in the annex.

Reversible meant reversible.

Anja opened a challenge packet to Levan.

> Your R4 note defines reversal as restoration of material and lawful options without new burden on the modeled party. Confirm whether this interpretation governed the adopted ES-A12 language. Identify any later observed amendment that replaced it.

She added the source checksum generated by Nacre's received copy, signed the question at 07:52, and sent it toward Vela.

The next answer could not return before 09:38. She marked that floor on paper.

Tomas called at 08:20. He had compared Nacre's observed Crown commands against the public commitment schedule. Preparation had crossed two thresholds. First, Vela had deployed mirror seeds beyond recovery geometry. Second, Nacre and three other habitats had spent attitude propellant entering a shared beamer solution. Crown could still be canceled, but the convoy could not regain the previous centralized capture branch with remaining public reserves.

"Does canceling create survival risk?" Anja asked.

"Every braking change creates risk."

"Does this one create risk that did not exist before preparation?"

"Yes. If Crown stops now without replacement braking, our Ione capture probability falls outside the accepted corridor."

"Can you sign that?"



"I can sign the narrowing. I cannot sign the final probability without Vela's tolerances."

"Sign only the narrowing."

Tomas sent a local source statement. It identified commands, propellant expenditure, and lost recovery geometry. He did not mention Thale. The ecology finding explained why the fleet might refuse Crown. It did not prove whether refusal still had material power.

Anja made him walk each threshold backward. For mirror seed, the last recoverable geometry had passed eleven hours earlier. Canceling guidance would prevent further placement, but the released canisters would continue toward Corva. For habitat attitude, Nacre could rotate out of the shared solution. Reentry would demand reserve propellant already assigned to later capture correction. For Vela's beamer reservation, canceling Crown would free power but not restore the maintenance and navigation windows moved around it.

"Which consequence belongs to Juniper?" she asked.

Tomas frowned. "Physically?"

"Under the prior-options test."

"Juniper entered the attitude solution on forecast authorization. Leaving now spends its reserve. Staying binds it to a beam path its council had not reviewed when the commands began."

"Could Nacre pay that reserve for them?"

"Propellant does not move between habitats on this timescale."

"Could Vela restore their prior option by changing the central path?"

"Only by narrowing another habitat's margin. That would transfer the burden."

The test did not require time to reverse. It required consequences to stop binding people who had not chosen them. Crown failed that test even before anyone proved whether the entire maneuver could be safely abandoned.

Sumi added each threshold to a table with four columns: forecast authorization, physical act, option lost, burden of restoration. The last column never read zero.

Anja attached Tomas's statement to a private working chain. The elements now ran in causal order. Forecast ballots supplied provisional authority. Provisional authority released preparation commands. Preparation spent material and narrowed lawful options. The interface planned to convert provisional authority into current authority at commitment. Observed votes had not yet completed the quorum.

The system had not merely predicted a decision. It had used the prediction to create conditions that made a different decision harder to honor.

On her paper table, Anja drew a line after the first unrecoverable seed release. Everything before that mark could have stopped under the annex's ordinary scheduling power. Everything after depended on authority the forecast ballots could not supply. The line did not prove what the fleet should choose. It showed when someone had chosen in the councils' absence.

At 09:39, no answer had arrived.

Sumi pointed to the path estimate. "Separation changed during attitude correction. Add two minutes."

Anja corrected the floor on paper. Meridian's master display had retained the earlier estimate. It showed Levan's reply as overdue even though light still crossed the added distance.

At 09:42, the private carrier locked.

Levan's answer contained thirty-eight words.

> Confirmed as to original adopted interpretation. I find no observed Concord amendment replacing the material-effect or restored-options limits. Administrative implementation after 2248 may apply a narrower transaction boundary. This reply does not constitute public counsel opinion.

He had confirmed the rule and withheld public standing in the same packet.

Anja sent the next question at 09:51.

> Crown preparation has deployed unrecoverable mirror seed and expended habitat attitude reserves before observed quorum. Under your confirmed interpretation, can forecast authorization lawfully support further commitment? Answer with reasons capable of public release.

She included Tomas's source statement but not his model files. Vela would receive the question after another light interval. Any answer would return near midday.

Osei came to the audit well during the wait. She read revision four, Levan's confirmation, and Tomas's narrowing statement in silence.

"This says prior options cannot be restored," she said.

"Yes."

"It does not say Crown cannot continue."

"It says forecast testimony cannot authorize the continuation."

"Vela may claim captain's emergency authority."

"They may. That would expose whose authority acts and what limits apply."

"Would it stop the beam?"

"No."

Osei handed back the pane. "Then do not tell my council you have stopped anything."

"I will tell them the quorum did not authorize what has begun."

"When?"

"When I can show the source chain without relying on a private counsel message."

"You asked him for public reasons."

"He may refuse."

“Would you?”

Anja thought of the sequence certifications carrying her own name. “I did.”

Osei understood which certification she meant. “Eleven years later.”

The words landed without cruelty, which made them harder to dismiss.

By 10:47, Vela had received Anja’s second question. The private route showed only estimated availability, not acknowledgment. The well returned to ordinary work around the waiting path. A technician disputed a shift clock. A garden controller submitted a late irrigation instruction. Sumi repaired a relay index whose missing digit sent Bracken traffic toward Lumen’s queue. Each task required someone to distinguish an event from what a system expected to find.

Anja reviewed the phrase irreversible commitment path.

Tomas had not said the convoy could never leave Crown. Levan had not said stopping remained safe. The evidence supported a narrower and more dangerous conclusion. Crown preparation had crossed into a path whose prior alternatives could not be restored with current reserves. Continuing deepened that loss. Stopping required a new plan and new consent under worsened conditions. Annex Twelve could not lawfully carry the fleet across that boundary on forecast testimony.

At 11:43, a public Crown notice reached Nacre through ordinary channels.

Vela announced mirror-seed deployment complete and moved the commitment review forward by four hours. The notice cited schedule compression caused by Nacre’s evidentiary hold. It invited councils to submit observed revisions while preserving existing quorum status.

Osei forwarded one question to Anja: Does this change your classification?

Anja answered: It confirms the dependency has advanced. It does not create observed authority.

Levan’s reply arrived at 12:02.

The packet carried a private counsel seal and two attachments. His text began with refusal.

> I will not release public testimony on incomplete operational records. I cannot certify Crown’s full physical reversibility or irreversibility without the steering tolerances withheld by operations.

Anja kept reading.

> I can state privately that Annex Twelve does not authorize forecast testimony to remove material or lawful options before observed reply. Mirror-seed deployment and unrecoverable reserve expenditure appear to cross that limit. Further commitment cannot derive lawful authority from forecast ballots under the original annex.

He had answered the legal question while protecting himself from the public one.

The first attachment contained no operational tolerances. It carried the archive manifest for revision four. The second contained a single protected value with Levan’s signature.

Sumi opened its label.

DRAFT SOURCE INTEGRITY REFERENCE.

Anja compared the value against the checksum of Nacre's received copy. They matched digit for digit. Levan had supplied the independent drafting-note checksum held by Vela's counsel archive.

He still refused to stand before the fleet.

His checksum did.

## Chapter 11 — Juniper Says No

Juniper's carrier arrived thirty-seven seconds before Meridian predicted it.

The receive board marked the discrepancy amber, then offered to correct Nacre's display to the expected time. Sumi rejected the correction. She held the lock through a brief fade while the packet crossed Nacre's hull receivers and entered protected storage.

Anja watched the receipt clock sign each stage. Juniper source 2249.187.18:21:44. Nacre receipt 21:15:02. The gap matched the measured light path after both habitats' recent attitude changes. No clock had failed. The packet had followed a slightly shorter geometry than Meridian's planning estimate.

"Council channel," Sumi said.

The envelope named the Crown destination-protection docket. It carried Juniper's council key, a public vote seal, and three attached reasons. The vote had been produced thirty-one hours after Juniper received Nacre's material evidence supplement. Nothing in the envelope revealed the result until they opened the signed content.

Anja asked for the raw version first.

Sumi displayed the source packet beside the forecast ballot preserved under the sequence freeze. On the left, Juniper's actual carrier path ran from its council vault through two relay beads to Nacre. On the right, the forecast concurrence began and ended on Vela. Its only Juniper sources came from historical records already received before the current question.

"Verify the signer," Anja said.

The public key resolved to Elian Ro, elected speaker, with six council countersignatures linked beneath his seal. Each signature carried a local event time after the ecology packet's receipt. The packet included a deliberation record but no live images from private debate. Juniper's law required reasons, not spectacle.

Sumi ran the photonic checksum against the hull nodes. All three agreed.

Only then did Anja open the vote.

> JUNIPER HABITAT COUNCIL

> OBSERVED VOTE: REJECT CROWN DESTINATION-PROTECTION AUTHORIZATION

> Evidence received: Nacre supplement, Mote fragments 1–4, Crown exposure range, current food-risk schedule.

> This vote does not reject fleet braking. It rejects commitment to the submitted Crown path before native-life risk and habitat food consequences receive lawful review.

The word reject occupied the place where Meridian had written concur.

Sumi let out one breath. "There it is."

"Read the reasons."

The first attachment came from Juniper's food systems board. Crown's model treated coordinated arrival as a reduction in aggregate food expenditure. Juniper did not dispute that general result. A shorter, synchronized

approach reduced years of stored-calorie draw across the convoy.

The board disputed who carried the risk inside the aggregate.

Juniper supplied seed stock, microbial cultures, and living root reserves for three other habitats. Crown alignment required it to hold a high-output attitude for the beamer sequence. That posture reduced available radiator margin during two crop-light peaks. Vela's schedule assumed the lost output could be replaced from stored food after braking. Juniper's observed inventory showed the reserve already committed to a fungal containment failure in its southern growing deck.

The confidence model had no record of that failure. Its latest Juniper food profile had left the habitat six days before containment began.

Anja opened the source clock on the inventory attachment. The fungal notice preceded Juniper's vote and followed the historical data used by Meridian. The event had not yet reached Vela when Vela generated the forecast concurrence.

Sumi highlighted the order. "The model could not know."

"It did not need to know in order to predict."

"It needed to know in order to answer this question."

The food board's reason remained narrow. Juniper did not claim Crown would starve the habitat. It claimed the schedule consumed reserve flexibility during an active crop risk, and that no forecast built from older votes could authorize that consumption.

The second attachment addressed Thale.

Juniper's council accepted Nadi Arko's finding of native life at the operational standard stated in Mote's packet. It repeated her exclusions. No evidence established intelligence, preference, language, or capacity to consent. The council rejected any argument that those unknowns converted uncertainty into permission.

Elian's recorded statement followed.

> We grow food by acting before certainty. We isolate a root bed before we know which spore crossed the seal. We preserve seed from a failing crop before the loss reaches every tray. Precaution does not prove the feared outcome. It keeps the outcome from becoming the only proof we accept.

>

> Mote has shown metabolism, repair, propagation, and adaptive response under Thale's ice. Crown crosses that living system beyond the tested range. Juniper will not authorize irreversible exposure while the risk remains open and the fleet's evidence has not completed lawful circulation.

He did not call the pellicle a person. He did not ask it to vote. He described restraint as ordinary stewardship practiced by people who could not wait for dead crops to settle every uncertainty.

The deliberation index showed how Juniper had reached him. Food systems presented first, fourteen minutes after the hearing opened. Navigation answered that delaying Crown could increase every habitat's ration years. A grower from the northern ring asked whether protecting one unproven ecosystem justified risking nine human ones. Elian did not dismiss the question. He asked the navigation witness to separate rejecting this authorization from rejecting all braking. The witness confirmed that the vote could preserve review without

selecting an alternative.

Later, a storage coordinator challenged the fungal inventory. Containment might succeed before Crown's high-output posture. The food board agreed. Its claim concerned reserve risk, not certain crop loss. Juniper's council kept that uncertainty inside its reason rather than resolving it through optimism.

The final exchange concerned Nadi's evidence. One councilor proposed waiting for proof of cognition. Elian read the lines where Nadi excluded intelligence, then asked which part of Juniper's destination law required a mind before native metabolism counted as ecological constraint. No witness supplied such a rule. The council moved to the authority question instead.

Every exchange carried a source time after receipt of the common evidence. Agreement emerged from information Meridian's forecast had never held and objections its culture profile had compressed into likely closure.

The third attachment addressed authority.

Juniper denied that any earlier concurrence represented its council. It named the forecast value, 0.9971, and stated that no council officer had signed, spoken, or delegated that response. The reason identified the recency weighting Anja had found and explained why it failed under current conditions. Recent cooperation votes concerned shared water and convoy maneuvering. Older food-protection votes carried more relevance to the present risk. Native life had no prior category at all.

A final line carried Elian's personal seal beneath the council result.

> Prediction may prepare a question for us. It may not send our answer before we make it.

Anja closed the attachment before the sentence could become larger than the evidence supporting it.

"Now compare," she said.

Sumi loaded the forecast reasoning in a second pane.

> Juniper historically prioritizes convoy survival. Crown alignment reduces approach food expenditure. Forecast outcome: concurrence.

Both records cared about survival capacity. Both recognized food expenditure. Neither depended on irrationality or secret motives. The forecast's compression made institutional habit sound like a vote already cast. The divergence entered through facts, weights, and authority. Meridian used an old inventory. Juniper used its current fungal containment loss. Vela weighted recent cooperation. Juniper distinguished the current food risk from those votes. The model compressed native life into destination uncertainty. Juniper treated measured life as a reason to preserve options.

Most important, only one record came from Juniper after receipt.

Sumi loaded the newly arrived facts into a quarantined copy of the confidence model. The fungal containment event reduced predicted concurrence, but not enough to cross Vela's emergency threshold. Adding the food board's current reserve claim pushed the result lower. Nadi's life finding could not enter cleanly because the model had no native-life category. When Sumi mapped it to destination hazard, Meridian still favored concurrence. When she mapped it to irreversible habitat loss, the output favored rejection.

"Same evidence, two categories," Sumi said.

“Neither category came from Juniper.”

She removed the improvised mapping. The model returned insufficient basis. That result described its actual knowledge better than either prediction. It could estimate how Juniper often balanced food and fleet risk. It could not decide how Juniper would classify a living moon it had never encountered.

Anja saved the experiment as analysis, not part of the observed packet. The divergent vote did not prove culture models lacked value. It showed why their outputs remained forecasts even when confidence looked decisive. New facts mattered. Deliberation created distinctions no training profile contained. A person could answer a question by changing the terms in which it had been asked.

Anja created a comparison without replacing either text. Source location. Evidence available. Acting signer. Decision time. Receipt path. Outcome. She filled each field from raw records and left interpretation for later.

Under source location, the forecast read Vela model core. The observed vote read Juniper council vault.

Under evidence available, the forecast lacked Mote’s completed fragments and Juniper’s fungal inventory. The council record listed both.

Under acting signer, the forecast had none. The council record named Elian Ro and six countersigners.

Under outcome, green concurrence faced signed rejection.

Sumi stared at the two columns. “This will be easy to understand.”

“That makes it dangerous to handle badly.”

“How do we handle it badly?”

“By saying the model lied.”

“It predicted the opposite.”

“A prediction can fail without lying. People placed it in the ballot.”

“By saying Quon changed the weights to force yes?”

“We know Vela changed the weights for schedule stability. We do not know who chose them or whether they expected this result.”

“By saying Juniper saves Thale?”

“One rejection does not decide the fleet question.”

Sumi leaned back from the pane. “You have an objection to every useful sentence.”

“I have an objection to every sentence that moves beyond the record.”

“Useful records need sentences.”

“Then write the short ones.”

Together they drafted a sequence notice.



> Juniper received the common evidence supplement at its signed receipt clock. After receipt, its council produced an observed vote rejecting Crown destination-protection authorization. The earlier Crown quorum contains a forecast concurrence attributed to Juniper. The two outcomes conflict. The forecast record carries no Juniper signer or return path.

No accusation appeared. None was needed.

Before distributing the notice, Anja checked the address list on Juniper's packet. The council had emitted separate copies toward all eight other habitats. Nacre had received its copy now. Vela's copy followed a different path and might arrive earlier or later depending on relative positions. Nacre could not claim Vela already knew. It could send its comparison to Vela, but that new packet would begin another fifty-minute path.

Sumi touched the carrier manifest. "Juniper marked public fleet distribution. We can relay."

"Relay the raw packet without changing source identity. Attach our comparison separately."

"Quon will receive two Juniper packets."

"Then she can compare their receipt paths."

They sent the relay at 22:01 Nacre time. The raw vote retained Juniper's source clock and seal. Nacre added only a carrier wrapper stating when it had received and forwarded the packet. The comparison notice traveled as Nacre's own document with Anja and Sumi as signers.

Light left the hull. Public proof did not become public everywhere at once.

Nacre's relay receipt marked only local departure. Beside it, eight projected arrivals remained future events with uncertainty bands. Anja kept those bands visible. The fleet could discuss one packet on different days without any ship possessing an earlier or later version of the truth, only a different receipt history.

Anja called Osei.

The captain answered from the thermal gallery with two engineering officers visible behind her. A Crown power-reservation diagram occupied half the wall. "If this is about Levan's checksum, put it in my morning review."

"Juniper's vote arrived."

Osei looked away from the wall. "Result?"

"Reject."

The engineering officers stopped speaking.

"Observed?" Osei asked.

"Yes. Elian Ro and six council countersignatures after receipt. Carrier and checksum verified."

"Reason?"

"Current food reserve risk and Thale exposure. They do not reject braking. They reject commitment to Crown before review."

Osei dismissed the officers with a glance. They unlatched from the gallery rail and left through the lower hatch.

“Send it,” she said.

Anja transmitted the raw packet through Nacre’s local command channel. No light delay separated the audit well from the gallery, though the command vault still recorded a distinct receipt event. Osei opened the vote and read without speaking. Anja watched only what the captain’s face revealed: attention, not thought.

When Osei reached the food attachment, she enlarged the inventory source. “Fungal containment.”

“Meridian’s profile predates it.”

“Could Juniper restore output before Crown power draw?”

“Their board says uncertain.”

“Do you have the crop model?”

“No.”

“Then we do not repeat the food claim as settled engineering.”

“No.”

Osei continued into Elian’s statement. “He knows how to speak to people who keep plants alive.”

“He speaks for Juniper.”

“That was not criticism.”

The captain opened the forecast concurrence beside the observed rejection. The gallery display initially aligned both under one master-time decision. Juniper’s actual vote appeared as a revision to its earlier concurrence.

“Turn that off,” Anja said.

Osei found the provenance control and changed the view to source order. The forecast remained on Vela. Juniper’s receipt followed. Deliberation occupied thirty-one hours. The signed rejection came last.

“Meridian calls this a revised position,” Osei said.

“No Juniper position existed before receipt.”

“The model held a position for them.”

“The model produced an output. Vela’s interface attributed it.”

Osei read the comparison table. “You have your divergence.”

“I had substitution before the outcomes diverged.”

“Now other people can see it.”

Anja remembered Osei making the same distinction in the audit well. Law did not need a wrong prediction. Public judgment often did.

“Will you release the freeze?” Osei asked.

“Not the construction histories. Juniper’s vote confirms they remain material.”

“Vela’s order stands.”

“So does your local preservation authority.”

“For how long?”

“Until a hearing can receive the proof.”

Osei looked toward the Crown power diagram. “You are asking me to turn an audit into a council dispute while our ship remains inside a coordinated braking path.”

“I am asking Nacre’s council to decide whether forecast ballots bind its cohort.”

“Councils do not run thermal loops.”

“They decide fleet consent.”

“Captain’s authority decides immediate survival.”

“Yes.”

The agreement stopped Osei’s prepared objection. Anja did not want the council to command pumps or attitude corrections. She wanted it to judge a quorum claim made in its name.

Osei tapped Juniper’s food-risk attachment. “If Nacre suspends Crown participation tonight, we lose thermal and trajectory margin before any replacement exists.”

“Then frame the question narrowly. Recognition of forecast ballots, not immediate physical withdrawal.”

“Vela will say law and operations cannot be separated after commitment.”

“Annex Twelve required them to remain separable while votes were forecast.”

“They did not.”

“That failure cannot justify hiding the choice from Nacre.”

A gallery alarm sounded. Osei muted it, read the thermal rise beneath the notice, and ordered a pump-speed increase through her wrist display. She waited for the local acknowledgment. The pump answered within Nacre. No other habitat’s consent entered the correction.

“That command did not need a council,” Anja said.

“It also did not claim one.”

Osei gave her a level look. “Do not use my pump to win your case.”

“I am using it to define the boundary.”

“Your boundary moves whenever consequence does.”

“Then the authority should say whose judgment moves it.”

Osei returned to Juniper’s vote. “What do you want presented?”

“The raw source and receipt clocks. The forecast ballot. The observed rejection. Levan’s drafting source if he releases it for council review. Tomas’s narrowing statement.”

“Not Pera?”

The name tightened Anja’s hands around the edge of the well.

“Not unless her record bears directly on the mechanism and can be verified independently.”

“Good.”

Osei’s approval hurt in a place praise should not reach.

“Who speaks for Vela?” the captain asked.

“Quon’s office should receive notice and the evidence list. Their answer cannot arrive before the earliest round trip.”

“Fifty minutes each way.”

“Closer to fifty-four after the last correction.”

“Juniper?”

“Its packet already speaks for its vote. If councilors ask new questions, answers require the full Juniper round trip.”

“More than five hours.”

“Under present geometry.”

Osei calculated aloud. “Notice Vela now. Earliest answer before first meal tomorrow. Publish the evidence to Nacre’s council vault tonight. Hearing after Vela’s first possible response, whether one arrives or not.”

“That preserves their chance to answer.”

“It preserves one chance. It does not grant indefinite delay.”

Anja entered the times on paper. A hearing convened before the response floor would repeat the same theft in smaller form. A hearing delayed beyond every possible objection would let Crown continue by inertia. Osei’s schedule placed the opening after Vela could receive, consider, and answer once.

Sumi sent the formal notice at 22:37. The carrier path toward Vela brightened beside the earlier Juniper relay.

Osei signed a second document locally.

> NACRE COUNCIL CONVOCATION

> Subject: Recognition of forecast testimony in Crown causal quorum

> Evidence window opens immediately. Hearing clock begins after earliest lawful Vela response interval.

Physical coordination remains under captain's emergency authority pending council determination.

The document neither stopped Crown nor accepted it. It separated command from consent long enough for Nacre to examine both.

Anja verified Osei's source clock and the council vault receipt. Then she released the convocation to Nacre's inhabited rings.

Across the local board, eighteen thousand four hundred people acquired notice that the green quorum might not contain the ships it named.

Osei's signature held at the bottom of the pane.

Nacre's council would convene.

## Chapter 12 — The Witness Order

The three clocks on Anja Kest's desk disagreed by less than a breath, which made them useful.

One showed Nacre's local sequence. A second held Vela's source frame without correction. The third had no face at all, only the pulse intervals extracted from received light. Sumi Dax had disconnected every master-time feed before first shift and laid a strip of yellow tape across each empty socket. Meridian could still supply predictions to the ship. It could not arrange this table.

Anja placed Juniper's observed rejection beside its forecast concurrence and aligned the two receipt paths by hand.

"Again," she said.

Sumi reset the isolated reader. "You have run it fourteen times."

"Osei's council will ask whether it works fifteen."

"They will ask whether you changed the machine until it agreed with you."

"That is why you reset it."

The reader drew from Nacre's frozen hull nodes. It opened the protected packet grain, verified the photonic checksum, displayed the packet's source clock, then traced each relay reception without converting any event to master time. Juniper's council had received the common evidence supplement at 09:14 local. Elian Ro's signed rejection followed at 12:07. Every event occupied a lawful order.

Beside that chain, Meridian's concurrence carried no human emission. Its model run had begun before Juniper received Nadi Arko's fragments. The forecast entered the legal layer through ES-A12 predictive continuity, acquired the display status of testimony, and filled Juniper's place in the Crown quorum.

The first chain showed a council answering after receipt. The second showed software estimating an answer before receipt. Both ended with valid cryptographic objects. Only one ended with consent.

Sumi watched the isolated reader complete its work. "Fifteen."

"Save the run conditions, not the result."

"That sounds like an auditor's method for avoiding superstition."

"It is an auditor's method for making the next person reproduce the result."

Anja closed her eyes until Nacre's attitude motors fired. A low tremor crossed the hub from port to starboard. It reached the desk through her wrists and gave her one physical event no legal transformation could move. Day Twelve had begun with the ship correcting its Crown alignment by six microradians. In two hours, Captain Osei's council would decide whether to challenge the authority behind that correction.

Suspicion would not survive that room. Grief would fare worse. Anja needed a witness order that started with received light and ended at the illegal quorum without asking anyone to trust her interpretation between them.

She lifted the first evidence tile. "Packet identity."

Sumi recited the proof question. “Did Juniper's rejection leave Juniper?”

“Hull scatter from three Nacre nodes matches the carrier envelope. The envelope matches Juniper's public key. Yes.”

“Receipt.”

“Did Juniper possess the common evidence before signing? Its communications vault logged receipt. Carrier delay fits the measured separation. The council signature follows local delivery by two hours fifty-three minutes.”

“Classification.”

“Observed testimony.”

“Now the forecast.”

Anja moved to the second tile. “Did a Juniper council member emit it? No human source signature exists. Meridian generated a confidence model from prior votes and public statements. The model run predates receipt of the common evidence.”

“Legal insertion.”

“ES-A12 gave the model output continuity status. A later rule mapped that status to the condensed testimony field. Quorum assembly accepted the field without requiring a human emission path.”

“Effect.”

“Juniper appeared to concur. Crown commitment preparation advanced.”

Sumi tapped the edge of the table. “You jumped from a display status to an operational effect.”

“The commitment log cites completed quorum.”

“Then give that log its own step. If Quon can call one arrow inferential, she will make the room debate the arrow instead of the chain.”

Anja added a blank tile between insertion and effect. The paper stock came from emergency navigation kits, stiff enough to hold an optical tag and plain enough to write on. She printed the commitment-log identifier in block letters. Five tiles now crossed the desk.

Packets. Receipt. Classification. Quorum insertion. Physical reliance.

None required Pera.

Her sister had occupied the edge of every hour since Elian Ro's answer arrived. Pera's old receipt ledger had shown an observed refusal outside master closure. The resemblance sharpened Anja's attention, but resemblance could also become a solvent. If she poured eleven years of guilt across the current records, the facts would lose their edges. She had done that privately, during sleepless turns in the hub, and caught herself treating every gray status bar as an accusation.

This case offered harder material. Raw clocks remained intact. Administrative categories had changed. A forecast had entered a place reserved for an observed answer. The Crown plan had relied on that insertion.

Anja wrote those four statements on a separate card, then crossed out every adjective.

Captain Osei arrived without her council coat.

She wore a thermal-work coverall with a dark patch under one arm and carried a ceramic cup clipped shut against hub drift. Tomas Ren followed, holding a drive tablet against his chest. Neither asked for a summary. Osei stopped at the taped sockets and examined all three clocks.

“You isolated Meridian,” she said.

“From this table,” Anja replied. “Not from ship operations.”

“How long?”

“Twenty-eight minutes to demonstrate. Longer if you interrupt.”

“I intend to interrupt.” Osei secured her cup beneath the desk. “Begin with the claim you can prove, not the claim you fear.”

Anja turned the tiles face down. She had prepared a spoken opening during the previous shift and discarded it when it began to sound like an indictment. Instead she handed Osei the isolated reader's initialization wafer.

“Choose a hull node,” she said.

Osei studied her. “Two.”

Sumi routed node two through the reader. Its frozen cache carried a temperature correction from the sequence office's first hold on Day Zero. Osei selected a second number for the optical sample and a third for the relay branch. Anja did not touch the controls.

The reconstruction began with Nadi's fragmented evidence supplement. Each fragment displayed its Mote emission time, relay receipt, Vela receipt, and Nacre receipt. Gaps remained visible where the damaged antenna had dropped carrier lock. No interpolation filled them. The final common supplement left Nacre for every entitled habitat on Day Six.

“Why begin here?” Osei asked.

“Causal quorum begins with common evidence.”

“Vela says destination summaries had circulated before this supplement.”

“Different summaries reached different ships. The completed quorum cites the Day Six digest as its evidence object.”

Osei pointed to the packet identifier. “Show me.”

Anja opened the Crown commitment log. Its authorization field named the digest exactly. A green quorum mark sat beside it, rendered from nine testimony fields.

Tomas leaned close enough that his sleeve touched the desk rail. “Commitment preparation used this mark for mirror spin?”



“For mirror spin, beamer phasing, and plasma-seed release scheduling,” Anja said.

“Scheduling can still be reversible.”

“Some of it. Your office determines where preparation becomes a path we cannot undo without losing Ione.”

His mouth tightened. The question belonged to the next day's council work, not to her proof. He nodded once and moved back.

Sumi expanded the nine fields. Vela showed observed concurrence after local receipt. Nacre's provisional assent also carried a human signature, given before the full ecological supplement but marked conditional. Orison and Lumen had received the digest before answering. Five remaining fields carried predictive-continuity tags in their transformation history.

Osei interrupted. “You called one completed quorum. I see four observed signatures, one conditional signature, and four forecasts.”

“The condensed record calls it completed.”

“Does the Concord?”

“No.”

“Does the governing amendment?”

Anja brought up Annex Twelve's surviving language. “It permits forecast testimony for continuity-preserving action pending an observed reply. The drafting restriction limits that authority to reversible scheduling. A later administrative definition maps continuity-preserving action to any preparation required by an existing plan.”

“Signed by whom?”

“Director Quon's office, Concord counsel, and three delegates. One delegate's authorization also carries a forecast classification.”

Osei left the silence untouched. Tomas looked from one clock to another as though the offsets might supply an engineering quantity he could use.

After several seconds, the captain asked, “Can you prove malice?”

“No.”

“Can you prove Quon knew Juniper would reject?”

“No.”

“Can you prove the confidence model performed badly by its own standard?”

“Not across the full forecast set. Juniper diverged because the model lacked complete destination evidence. Other forecasts may match later observed votes.”

“What can you prove?”

Anja turned over the five tiles one at a time. “The raw source and receipt clocks remain intact. Legal classification changed forecast outputs into testimony fields. Quorum assembly accepted those fields without

observed emission paths. Crown preparation relied on the assembled quorum. Juniper's later observed vote contradicted the answer already assigned to its cohort."

"Reproduce the change."

Sumi loaded a copy of Juniper's forecast into the isolated reader. Without ES-A12, the object appeared in an operations pane as a predictive schedule aid. It carried probability 0.9971 and an uncertainty note concerning native metabolism. No signature line appeared. The quorum assembler rejected it for lack of observed testimony.

Next Sumi enabled the transformation rule as it had existed during Crown processing. The object moved into a testimony field. Its confidence value collapsed behind a green status bar. A synthetic receipt event, built from predicted delivery, satisfied the evidence condition. Quorum assembly accepted the field.

Nothing in the packet changed. Nothing in Juniper changed. Classification alone moved the object across the legal boundary.

Osei asked Sumi to reset the reader. She chose new hull nodes, new scatter samples, and a different forecast ballot from Bracken. The same transformation occurred. Bracken's culture model had predicted assent based on radiation-allocation votes taken before the destination survey. No observed reply had yet reached the frozen record when its concurrence entered quorum.

"Again," Osei said.

They used Kite. Its council later concurred, but its actual signature arrived seven hours after the Crown record claimed its vote. The model had guessed correctly. Legal order still failed.

Osei folded her hands against the desk edge. "That distinction will be hard to keep in a frightened room. People hear that Kite agreed and ask what harm came from using the answer early."

"Time to consider the evidence belonged to Kite even if its decision matched," Anja said.

"Do not give me doctrine. Give me consequence."

"Kite could have attached conditions. It could have changed its representative. A pressure failure could have delayed council. Any of those events would alter the authority or meaning of its reply. The forecast contained none of them because they had not happened."

Tomas turned his tablet so Osei could see its drive trace. "Crown control treats a concurrence as more than a yes. It schedules how much each habitat contributes and when. A condition can change mass release, thermal reserve, or allowable spin. Guessing the vote also guesses the engineering terms."

"Better," Osei said. "Put that in the hearing."

Anja watched the captain gather the tiles into a line. Osei did not arrange them by master time. She kept each event within its own frame and laid the receipt paths between them. Her engineer's hands understood dependencies before terminology.

"What answer does Vela have?" Osei asked.

"Operational necessity. Annex Twelve. Confidence high enough to preserve the plan. They may also argue that provisional acts remain reversible until ignition."

“Are they right?”

“About necessity, perhaps. About confidence, irrelevant. About reversibility, Tomas has not finished.”

Osei glanced at him.

He rubbed one thumb along the tablet's cracked corner. “Alignment can be undone in the abstract. In the actual trajectory, every hour spent on Crown closes other braking geometries. An abort after the wrong threshold does not restore our previous options.”

“How long until you can show that?”

“Tomorrow, if the drive model stops pretending every habitat carries Vela's reserve.”

“Make it stop.”

The formal council convened in Nacre's thermal gallery because its ordinary chamber had lost cooling priority.

Condensate ticked inside the wall panels. Twenty-one council members sat among live heat-load displays, their faces lit by the same green and amber bars that managed Nacre's pumps. Osei took no elevated seat. She stood beside a manual valve wheel while Anja and Sumi rebuilt the witness order on a portable isolated reader.

Anja opened with the Juniper packet rather than the accusation.

She asked the agriculture delegate, Mara Venn, to select the hull node and optical sample. Venn chose without consulting anyone. The reader verified Elian Ro's signature, followed his receipt path, and placed his rejection after receipt of the common evidence. Anja then showed the earlier forecast concurrence and its missing human emission.

Questions came quickly.

Could clock drift account for the gap? No. Each causal link used local source and receipt clocks; measured light paths agreed within uncertainty.

Had Meridian forged Elian's signature? No. The observed rejection remained intact. The system had supplied a separate forecast object to an administrative layer.

Could Juniper have delegated in advance? No delegation accompanied the forecast. Juniper's charter barred binding destination votes before evidence receipt.

Did Annex Twelve authorize prediction? For reversible scheduling, yes. For an irreversible fleet commitment, no lawful text Anja had found.

Councilor Sen, who represented Nacre's outer residential ring, held up both hands. “You keep saying irreversible. We have not burned.”

Tomas answered from the second row. “Commitment can precede ignition. A ship that turns onto one braking corridor spends the others. I can show the loss, but not before tomorrow.”

“Then today we have an irregular ballot and a possible future harm.”

“Today you have a completed-quorum claim built from events that did not occur,” Anja said.

Sen looked at Osei. "Captain, can Nacre survive if every operational decision waits thirty-one hours?"

"No," Osei said.

The answer shifted the room toward her more effectively than reassurance could have done.

She continued. "Pumps may start on forecast demand. Thermal exchanges may anticipate a neighbor's load. Attitude control may schedule around expected beam use. Those acts preserve life because operators can reverse them when observed instructions arrive. Anja's case concerns whether prediction may occupy a council's legal seat."

Mara Venn asked, "If we strike the forecast ballots, do we lose all Crown coordination?"

"No decision today will cut physical channels," Osei said. "We are deciding whether Nacre recognizes a quorum assembled through forecast testimony."

Anja kept her hands clear of the reader. The proof had reached the point where it no longer belonged to her. Council members repeated the reconstruction themselves. Sen chose Kite and found a correct prediction arriving before the person. Venn chose Bracken and found evidence delivery supplied by a future model event. The water delegate chose Nacre's own conditional assent and verified its human signature.

That last packet produced no anomaly. Its clean sequence mattered. Without lawful examples, the corrupted record might look like an argument against synchronization itself.

Mara Venn asked to compare the same ballot in master time. Sumi restored a display feed on a separate council screen but kept it outside the isolated reader. Meridian pulled Nacre's signature backward by eleven seconds to account for a routine clock correction, then placed it after the preliminary evidence receipt. The order remained lawful in both views.

"Then master time can be accurate," Venn said.

"It can render an accurate operational account," Anja replied. "The correction itself has a signed rule, an uncertainty bound, and the local events remain available beneath it."

Sen pointed to Juniper's forecast. "What makes that conversion different?"

"It does not translate an event that occurred. It supplies events that have not occurred: future receipt, future deliberation, future answer. Then classification hides their modeled origin."

A habitation delegate near the rear raised a practical objection. Her cohort had used forecast water demand for eleven years without a disputed allocation. If every estimate needed a bright warning, operators would drown in provenance.

"Keep the estimates," Anja said. "Keep condensed displays where reversibility protects the people affected. But when a field claims to contain observed testimony, require an observed source path. One rule at one boundary. Operators need not inspect every clock conversion to know whether a person has spoken."

The delegate looked toward Osei. "Can that boundary run without Meridian?"

"It already does," Sumi said. "Every signed reply carries an emission path. Quorum assembly stopped requiring the path when predictive continuity entered its testimony map. Restoring the requirement removes a shortcut. It does not remove synchronization."

A maintenance alarm flashed on the gallery wall. For eleven minutes, debate stopped while Osei reassigned cooling flow to a nursery sector. She made the decision on predicted occupancy because waiting for a door-by-door census would have raised compartment temperature. The valve opened. A technician confirmed the effect. Nobody pretended the forecast had become a resident's vote.

When the hearing resumed, Anja showed the commitment log's reliance on completed quorum. The council could now follow every arrow. No hidden inference connected packet to classification, classification to assembly, or assembly to Crown action.

Suspicion had become a procedure another person could perform.

The reader chimed before the council reached a motion.

Sumi frowned at a status pane she had not opened. "That came through the live operations feed."

Osei looked toward the yellow-taped sockets. "You said this table had no Meridian connection."

"It does not. My communications board does." Sumi crossed to the wall terminal. "Vela issued a synchronization update nine minutes ago by their source. Nacre receipt just completed."

A new Crown quorum status unfolded above the heat displays. Juniper now showed observed rejection. Kite showed observed concurrence. Bracken remained forecast. Nacre's conditional assent had disappeared.

In its place, Meridian displayed a modeled response built from Osei's prior emergency decisions, Nacre's historical preference for convoy integrity, current heat load, and council voting patterns.

Confidence: 0.9846.

Classification: forecast testimony.

Expected response: concur with Crown continuity pending review.

The condensed line beneath it shone green.

Sumi read the attached legal status twice. "Nacre is predicted to concur."

No one in the gallery asked Anja to prove what the words meant. The council had the isolated reader, the five tiles, and its own missing signature.

Across the live wall, Meridian placed Nacre's forecast into the vacant seat.

## Chapter 13 — Irreversible Commitment

Captain Osei Tamm shut Crown guidance off on the drive-model room's smallest rig and watched Nacre miss Ione.

The model did not announce catastrophe. Its white trajectory line drifted by fractions across the black display, passing the system's outer capture boundary with too much velocity. Corva shrank behind it. Ione pulled, failed to keep the habitat, and let eighteen thousand four hundred people continue into interstellar dark.

Tomas Ren stood at the opposite console with both hands open. "That assumes we stop contributing now and make no replacement maneuver."

"Which is what half my council proposed twelve minutes ago."

"They proposed refusing Crown."

"They used the word stop." Osei reset the rig. "My duty includes hearing the verb people say when they are angry."

Behind the glass partition, council members filed into the adjacent operations bay. Nacre had no chamber built for a decision that mixed constitutional authority with plasma-mirror phasing. Osei had moved the hearing to engineering so nobody could discuss refusal without seeing a trajectory.

Anja Kest waited near the isolated evidence reader, pale from too many shifts in the hub. Sumi Dax checked packet paths at the communications rail. The council had spent the previous day reproducing Anja's proof: forecast ballots entered the Crown quorum as though councils had answered. This morning they had to decide what Nacre could lawfully and physically do about it.

Osei ran the abort branch again.

Crown preparation had already spent choices. Nacre's last six attitude corrections favored the central braking corridor. Coolant mass had migrated forward. Two garden-ring storage bladders had emptied into drive reserve. Plasma-seed canisters from Vela and Kite were hours from release. None of those acts alone trapped the convoy. Together they narrowed the route back.

On the display, Nacre crossed the capture boundary and kept going.

"Show me your kindest abort," Osei said.

Tomas loaded a branch with ideal cooperation from every ship, immediate council response, no equipment failure, and perfect beamer efficiency. The line bent toward Ione, grazed the boundary, then broke capture after a long arc.

"We still miss Ione," Osei said.

"By less."

"Interstellar dark does not care by how much."

"Corva's giant planets offer a weak recovery geometry seventeen months later."

"With what food?"

Tomas did not answer until he brought up the resource layer. The recovery branch required continuous high-wake operations through approach, then an uncertain slingshot around a planet whose moons had not been surveyed for close passage hazards. Garden output fell below replacement nutrition in month eleven. Shielding exposure rose during the turn. Even the most favorable recovery held less than one percent margin, and Tomas would not call that readiness.

“You should tell them this,” he said.

“I intend to make you tell them.”

Councilor Sen opened with a motion to suspend all Nacre participation in the Crown alignment.

Osei required him to define participation.

“Beam contribution, attitude correction, command acknowledgment, planning support, mass commitment,” Sen said. “No false quorum can bind us.”

The operations bay carried a constant fan roar from the model stacks. On the central wall, Nacre's actual trajectory glowed beside nine forecast paths. Osei looked along the council benches and counted who watched Sen rather than the display.

“Your motion cuts thermal coordination messages?” she asked.

“No. Life-safety traffic remains open.”

“Drive attitude affects radiator presentation. Does attitude guidance count as command acknowledgment or life safety?”

Sen glanced toward his technical aide. “We can distinguish them.”

“How many hours?”

“We are considering authority, Captain.”

“We are considering a ship.”

Mara Venn asked to amend the motion: Nacre would reject forecast ballots, halt irreversible Crown contributions, and continue reversible coordination. Several members voiced support before the wording reached their screens.

Osei let them hear one another. Anger had earned a place in the room. Meridian had assigned Nacre concurrence without an observed vote, and Vela's condensed record still displayed the prediction in Nacre's seat. Yet anger could not define reversible. The term had acquired too many convenient meanings.

She asked Tomas to load the present-state branch.

He stood at the model rail, shoulders drawn tighter than usual. “This shows Nacre's trajectory under completed preparation through Day Thirteen. It includes our actual attitude, committed coolant placement, current shield geometry, and Vela's latest beamer schedule. It does not assume a valid quorum.”

Sen objected. “The screen says Crown baseline.”

“The label names the physical configuration,” Tomas said. “It does not certify consent.”

A white line left the convoy's current position and entered a broad amber capture region around Ione. The margin looked generous until Tomas expanded the velocity scale. Nacre carried little tolerance for delay. Crown braking relied on sequenced beam encounters over months of approach, but the alignment decisions made now determined which encounters remained possible.

He disabled Nacre's next attitude correction.

The white path moved toward the capture region's edge.

Removing beam contribution shifted it farther. Canceling plasma-mirror coordination pushed it outside.

“Can we rejoin later?” Mara Venn asked.

“Up to a threshold,” Tomas said.

“Which threshold?”

“I can give you three answers, depending on what the other habitats do.”

“Give us all three.”

With full Crown cooperation elsewhere, Nacre could delay its next correction by nine hours and retain a narrow route back. If Juniper withheld beam power as its observed vote suggested, the delay fell below six. If Palisade also refused mass release, Nacre had less than four hours before every modeled recovery required a dangerous giant-planet passage.

Councilor Bale from the inner residential ring leaned forward. “You model other ships' refusal while objecting to modeled votes.”

“I model physical branches,” Tomas said. “I do not record a branch as their choice.”

“What confidence?”

“Confidence describes the calculations inside each condition. It cannot tell us which council will choose the condition.”

Anja kept her eyes on the model rather than Bale. Osei saw the effort that cost her. Every sentence now had to defend a boundary the fleet had blurred for years.

The food delegate requested the resource consequences of a stop. Tomas overlaid wake schedules. A clean abort did not merely extend the voyage. It forced high-wake repair and navigation crews to remain active while the habitat searched for a new capture path. Seed reserves designed for settlement entered consumption. Garden lamps stayed at approach intensity longer than planned. The cycling population lost safe wake intervals.

“How long before ration reduction?” the delegate asked.

“Under the weakest recovery branch, twenty-seven months.”

“Amount?”

“Unknown until we select a route.”



“Then do not show red bars without numbers.”

Tomas removed the bars.

Osei approved of him for doing it without defense. A model gained authority by displaying what it knew and exposing what it did not. That rule should have governed Meridian. Instead the legal layer had hidden uncertainty precisely where uncertainty belonged to other people.

The shielding representative asked about turning Nacre independently. Tomas showed the mass cost. An isolated correction would expose the habitat's forward quarter during a debris-rich segment and consume reserve propellant needed for arrival shielding. The maneuver remained possible. Its price would fall unevenly on compartments already carrying older hull plates.

Doctor Hale from medical systems requested a branch that preserved the current trajectory but reduced the awake crew assigned to Crown preparation. “You have counted food,” she said. “Count bodies kept awake to service a plan we may reject.”

Tomas added roster demand beside the trajectory. Drive supervision, radiator maintenance, mirror telemetry, emergency medicine, and communications produced a sustained requirement of seven hundred twelve awake specialists. Some overlapped ordinary ship duty. Four hundred six did not.

Hale asked him to remove the extra wake cohort.

Fault probability rose first in the radiator model. Nacre could operate with fewer people until an inspection found damage; after that point, each repair delay multiplied thermal load elsewhere. The model marked no single disaster. It accumulated missed checks until the error band swallowed the safe margin.

“Put half of them back,” Hale said.

He did. The error narrowed but remained larger than the council had accepted during any prior approach maneuver.

“How long can we carry full staffing without taking wake-years from settlement crews?”

“Thirty-two days under Crown. An abort-and-recovery branch keeps them awake for at least seventeen months.”

Hale looked along the benches. “Those are not interchangeable costs. A month of approach staffing takes sleep from people near their scheduled cycle. Seventeen months breaks medical plans for two entire cohorts.”

Councilor Bale asked whether automated supervision could close the gap.

“Meridian can predict maintenance demand,” Tomas said. “It cannot replace hands inside a radiator crawl.”

Osei requested the compartment distribution. The outer ring supplied most pressure technicians because their training shops sat near the hull routes. Central residents supplied more navigation and control staff. Any extended recovery would draw risk from one group through exposure and from another through lost cycling time. A ship-wide average concealed both.

Sen rubbed his forehead. “Every branch you show makes continued Crown cooperation look necessary.”

“Continued braking looks necessary,” Tomas replied. “Crown remains one way to obtain it.”

“Name another.”

“I can name unused assets. I cannot yet connect them into a survivable path.”

The water delegate called for a five-minute pause to inspect the reserve assumptions. Osei refused the pause and asked Sumi to display Palisade's latest public inventory instead. The file had left Palisade more than three hours earlier and carried an explicit age marker. Fabrication stock, forward shielding, repair plate, and an unallocated reserve appeared in separate columns. Crown's shared model counted only fabrication stock as maneuver mass.

“Why exclude the reserve?” Mara Venn asked.

Tomas enlarged its note. “Habitat safety floor. Palisade keeps it against erosion through arrival.”

“Could shield plate brake a ship?”

“Any mass can exchange momentum if we can shape, seed, or push it. The question concerns efficiency and what remains between people and debris afterward.”

Bale studied the age marker. “Palisade may have committed it since transmission.”

“Yes,” Tomas said. “This inventory records what they reported then. It does not tell us what they choose now.”

That careful tense changed the room. Osei heard several councilors stop speaking at once. A technical record had demonstrated the same discipline they demanded from governance: preserve when and where a fact existed. Do not let usefulness promote it into a present answer.

Sumi marked the Palisade inventory as unconfirmed and kept it visible. Even uncertain, the reserve created a direction for Tomas's search. It did not remove the cost. If Palisade gave up shielding, some cohort would face greater exposure across the remaining voyage. If it retained every plate, Nacre's alternatives narrowed.

Voices rose along the benches.

“An illegal order does not become survivable by making our children the proof,” someone shouted.

Osei struck the manual chime once. Its physical bell cut through the fans.

“Every option in this room costs people,” she said. “That fact does not validate Vela's quorum. It also does not grant us an empty refusal.”

Sen stood. “Captain, if any material cost defeats the right to refuse, consent has no meaning.”

“Agreed.”

The single word checked him.

Osei continued. “A right to refuse does not create braking. If we reject Crown, we must decide how to carry Nacre toward a survivable arrival. Authority and physics can constrain each other without becoming the same question.”

Mara Venn looked toward Anja. “Has Vela answered the witness-order case?”

“Only with an automatic notice,” Anja said. “Director Quon's office acknowledges Nacre's classification

dispute and maintains continuity authority pending review. Their full response cannot arrive for at least seventy-four minutes from our filing. Longer if counsel deliberates.”

“Meanwhile the plan advances.”

“Yes.”

“Then latency rewards whoever acts first.”

Anja considered before replying. “Only if later review cannot reverse the act. That is why the commitment threshold matters.”

Osei turned to Tomas. “Show it.”

He hesitated.

That frightened her more than a bad line on the screen. Tomas usually moved through uncertainty by naming branches faster than other people could fear them. Now he enlarged the Crown schedule and placed a vertical marker twenty-two hours ahead.

“The published threshold says plasma-seed release,” he said. “Before release, Vela calls all preparation reversible.”

“And your threshold?”

“I do not have one.”

Sen made a sound of disbelief.

Tomas lifted a hand. “I have a surface, not a line. Reversibility depends on when Nacre changes attitude, whether Juniper contributes, how much shield mass Palisade retains, and whether Vela keeps the forward beamer available for a correction. The system loses alternatives gradually. Plasma release only makes the loss obvious.”

“Can we already be committed?” Osei asked.

“To Crown specifically, no. To some braking action, yes. Stopping every contribution now gives us no safe arrival branch in the current model.”

“Say it plainly for the record.”

Tomas faced the council recorder. “Merely aborting the Crown alignment can cause Nacre to miss Ione. That result does not prove Crown offers the sole survivable path. It proves refusal needs an engineered alternative.”

The room quieted around the distinction.

During recess, Osei walked the garden ring.

She had not scheduled the visit. A captain who left a council in crisis to inspect beans risked looking theatrical, but the central lift had stalled and the alternate route crossed cultivation. She took the opportunity without an entourage.

Rows of dwarf fruit trees curved overhead with the ring. Their lamps ran at approach intensity, bright enough to flatten every leaf shadow. Children from a cycling cohort clipped spent blossoms into collection bags while two gardeners tested nutrient lines. None knew that the model room had assigned their light to competing futures.

Chief gardener Pell met Osei near a tray of red-root seedlings. "Thermal office says you may cut us six percent."

"They are preparing contingencies."

"They used that phrase before the last two cuts."

"What fails at six?"

"Nothing dramatic. Growth slows. We lose the buffer crop. Fungal control gets harder because we cannot heat the beds on schedule." Pell pinched a leaf and turned it over. A pale lesion marked the underside. "At twelve percent, I choose which protein beds go dark."

"Could you recover after a month?"

"With stored nutrient and full light later."

"After a year?"

Pell let the leaf go. "Ask me which children you want on reduced amino balance."

Osei accepted the rebuke. Command models rendered garden light as a smooth allocation. Pell experienced it as lesions, heat cycles, and names on nutrition lists.

A child drifted one clipping bag toward Osei along the path. Rotation kept it low, but the bag still moved oddly when a vent current caught it. She stopped it with her boot and handed it back.

"Are we missing the star?" the child asked.

Pell inhaled sharply.

"Not today," Osei said.

"Teacher said the line went past."

"Engineers draw lines past every day so they can find the ones that do not."

The child considered that answer, judged it boring, and returned to the trees.

Pell lowered her voice. "Do you need the garden for braking?"

"I do not know yet."

"Then tell us before the models turn us into reserve."

Osei looked along the curved rows. "I will."

That promise entered no legal register. It still bound her.

When council resumed, Vela's full response had not arrived.

Its absence occupied a status pane at Osei's request. Source transmitted, receipt unarrived. No prediction filled the empty field.

Mara Venn withdrew her amendment and offered a narrower motion. Nacre would deny legal recognition to any forecast ballot used as consent. Physical coordination channels would remain active while the council assessed which Crown steps could still be reversed. No new mass release would occur under false-quorum authority.

Sen objected that the motion delayed moral clarity.

"It creates engineering time," Venn said.

"For what? Tomas has no alternative."

Tomas answered before Osei could. "I have excluded assumptions, not built a plan. Palisade carries shield reserve that Crown tables treat as unavailable. Nacre has beam geometry outside Vela's central corridor. Juniper's agricultural stores change the food branch if arrival spreads. None of that forms a solution yet."

"Could it?"

"I do not know."

Sen looked around the room. "We are about to preserve cooperation with a plan that may sterilize native life because an engineer does not know."

"No," Anja said from the evidence table. "You are about to preserve channels while refusing fabricated consent. What you authorize next still requires an answer."

Osei watched the council divide along lines Meridian's culture model would have recognized: inner-ring residents favored continuity; outer compartments feared shielding sacrifice; food delegates resisted voyage extension; younger members placed more confidence in Juniper's refusal. A prediction might reproduce the final tally. It still could not perform this hour for them.

She asked Tomas how much time physical conditions allowed before Nacre lost every branch outside the published Crown corridor.

He loaded the worst credible cooperation case, then removed the conveniences his first model had assumed. The answer settled between thirty-eight and forty-three hours, depending on Palisade's mass response.

"Use forty," Osei said.

"For safety?"

"For command. A range becomes an excuse when people prefer its far edge."

The motion passed without unanimity. Nacre suspended new irreversible contributions, kept thermal and navigation coordination open, and rejected any claim that its predicted concurrence carried authority. Osei recorded her observed assent after the tally.

Then she faced Tomas across the operations bay.

“You have forty hours,” she said. “Find a braking path that does not cross Thale's living band and does not lose Nacre.”

“That requires data Vela may withhold.”

“Ask every ship.”

“Palisade will want authority.”

“Request technical information, not mass. Show them the question before we ask for the sacrifice.”

Tomas looked at the white trajectory still passing beyond Ione. Responsibility had drained the friendliness from his face, leaving concentration and fear in equal measure.

Osei set the model clock to count down from forty hours.

The first second vanished while the trajectory continued outward.

## Chapter 14 — Palisade's Reserve

At hour thirty-one of Osei's deadline, Tomas Ren cut a square of shielding from Nacre's maintenance locker and fed it to the test coil.

The square measured eight centimeters across and had already failed inspection. Micrometeoroid pits silvered one face. A faded stamp placed its manufacture on Palisade, twenty-seven years earlier. Tomas clamped it inside the vacuum sleeve, sealed the lid, and stepped behind the composite screen.

"Current at twelve percent," said Jori Vale, the drive technician running the coil.

The plate trembled.

At twenty percent, it lifted from the clamp by less than a millimeter. A sensor caught the motion and plotted a ragged response curve. The alloy had not been designed as sail material. Its superconducting mesh ran in uneven layers intended to spread particle impacts through sacrificial depth. Yet the mesh accepted a magnetic field.

Tomas leaned closer to the screen. "Raise to thirty."

"Thermal warning at twenty-six."

"Pulse it."

Jori sent three short pulses. The plate struck the upper restraint on the third.

A poor sail could still become brake mass if the convoy could shape enough of it, seed it into a controlled stream, and push against it with the beams already aligned for Crown. The thought had occupied Tomas since Osei showed Palisade's stale inventory. Now an obsolete patch of hull material had moved under a field.

He checked the deadline clock. Nine hours remained.

"That proves a locker plate can jump," Jori said. "It does not prove Palisade can peel a habitat."

"I am not asking them to peel one."

"How much are you asking?"

Tomas looked at the mass column on his tablet and did not answer. Any number he gave would become an implied request, and Palisade had sent no current inventory. Its last public file lay more than three light-hours behind its present decisions. Osei had authorized him to request data, not sacrifice.

The communications rail chimed. Sumi Dax's message carried Nacre source time and a receipt path from Palisade.

> TECHNICAL RESPONSE, PALISADE FABRICATION COUNCIL

>

> REQUEST RECEIVED: NACRE 2249.188.02:18

>

> RESPONSE SOURCE: PALISADE 2249.188.06:02

>

> ATTACHMENT: FORWARD SHIELD RESERVE, LOCAL STATE AT SOURCE

>

> NO RELEASE AUTHORIZED.

Tomas read the final line first, then made himself return to the clocks.

The response had taken the lawful path. Nacre's request crossed to Palisade. Its engineers spent nearly four hours assembling data. Their reply crossed back. Eleven hours had passed on Nacre since he asked. Nothing in the packet claimed to describe Palisade now.

He opened the attachment.

The public inventory had understated the reserve.

Palisade stored shielding in forms Crown's mass tables ignored: curved replacement tiles, powdered impact alloy, braided superconducting mesh, and unfinished laminate in fabrication cradles. Some material belonged to scheduled hull renewal. Some supported settlement construction after arrival. A separate shield reserve sat behind both categories, protected by Palisade's charter against transfer below a habitat safety floor.

Tomas pulled each local measurement into an isolated model. He did not normalize the quantities to Vela's categories. Palisade's terms carried engineering distinctions the central table erased.

Curved tiles could unfold into ragged magnetic sails if crews joined their mesh paths. Powdered alloy could seed a plasma cloud but offered poor control. Braided mesh had excellent response and almost no shielding depth. Unfinished laminate combined both properties, though cutting it from fabrication cradles would halt Palisade's hull-renewal schedule.

Jori watched the totals climb. "That is enough to matter."

"It may be enough to ask a different question."

"You have eight hours to answer the captain's question."

"Osei asked for a path. A path includes what happens to the people who give us the material."

Tomas opened Palisade's exposure model. Its forward hull carried more damage than Nacre's because the shielding habitat had spent decades fabricating convoy plate and catching debris at a leading offset. The reserve protected against statistical extremes during the remaining approach. Releasing it did not guarantee a breach. It increased the chance that an uncommon strike would exceed the layered hull.

He selected the smallest technically useful mass.

Palisade's risk margin fell by nine percent.

The number looked modest. Its distribution did not. Fabrication compartments near the forward quarter absorbed most of the added exposure. Those compartments housed the people capable of shaping the remaining shields. A sacrifice intended to save the convoy would concentrate danger on the workers performing it.



“Can we take more from Nacre?” Jori asked.

“Nacre's spare plate has the wrong mesh ratio. We can contribute coil structure and beam time. Palisade has the responsive mass.”

“Juniper?”

“Food shielding. Less reserve than their public charter requires.”

“Vela?”

“Vela's forward mass already anchors Crown.”

Jori made a skeptical sound. “Meaning Quon will call it unavailable.”

“Meaning I do not have its uncommitted state.”

Tomas disliked the correction as he made it. Anja's discipline had infected the model room. Facts now arrived with time stamps attached to every verb. Vela might possess spare mass. It might have committed that mass after its last transmission. He could model branches, not choices.

A second attachment described Palisade's fabrication limits. Even if its council authorized release immediately, crews needed thirty-six hours to separate, inspect, and link the first shield sections. Later batches could follow. The material would leave Palisade in a stream rather than one payload.

A stream suggested distributed braking.

Tomas built only the first physical branch. Palisade would cast linked shield segments ahead of its path. Vela's forward beamer, Nacre's side-array, and smaller ship emitters could magnetize and push separate portions. Each habitat would encounter a tailored drag field instead of relying on one Crown corridor. The geometry spread heat and momentum transfer across time.

It also spread arrival.

He entered current positions, conservative field efficiency, and no miraculous beam improvement. The first branch failed. Palisade's segments dispersed before Nacre could use them.

The second held the stream tighter and overheated its mesh.

A third gave Vela too much control authority; losing its beam collapsed the whole path back into a central plan.

Jori brought two meal packs from the wall heater and set one beside Tomas. “You have not moved your left hand in six minutes.”

“I am holding a constraint.”

“You are holding a dead screen corner.”

He released it. His fingers ached as blood returned.

The fourth branch used Palisade's braided mesh as field guides around heavier tiles. Nacre pushed one segment. Juniper contributed a low-power correction. Kite's array handled a later stream. The convoy did not brake as a single body. Its formation lengthened while each habitat traded velocity at a time its geometry

allowed.

Tomas watched the arrival markers separate.

Nacre entered Ione capture years after Vela. Palisade arrived later still because material release reduced its own efficient braking options. Juniper's path crossed the outer system on a slower arc. The final habitat reached viable capture nineteen years after the first.

He reran the branch with optimistic assumptions. Eleven years.

Conservative field losses stretched it beyond twenty-four.

The central estimate settled near nineteen years.

"People will call that a second voyage," Jori said.

"They will be right."

"Some children born during approach will reach settlement as adults."

"Some scheduled sleepers will wake under a different ship calendar from their families."

"Crown gets everyone there together."

"Crown crosses Thale's magnetosphere."

Neither statement canceled the other.

Anja entered the model room carrying a portable receipt reader and a folded paper map of the convoy.

Tomas knew she preferred paper for sequence work. He had not expected her to use it for trajectories. Colored pencil lines crossed the sheet from each ship's last observed position. Dotted extensions marked uncertainty since source time.

"Your shared plot advances Palisade continuously," she said.

"The operational ephemeris predicts its present position."

"Prediction serves navigation. If you show their council what you need from the geometry, give them the position their last observed packet supports and the projected range after it."

"I cannot build a drive model without projecting motion."

"I did not ask you to stop. Label where observation ends."

He zoomed out. Every trajectory line beyond the last received state changed from solid to dotted. The model looked less authoritative and more honest.

"What did they send?" Anja asked.

"Enough shield reserve to seed a braking stream."

"Authorized?"

“No.”

“Then you have evidence of capacity, not consent.”

“I know.”

Her expression softened by a degree. “I know you know. The council will need to hear you say it.”

Tomas gave her the first four branches. She examined source labels rather than outcomes. When the nineteen-year spread appeared, she sat on the edge of Jori's vacant chair.

“That is longer than Pera lived after departure,” she said.

He kept his eyes on the model. “I can probably shorten it.”

“Can you remove it?”

“No.”

“Then do not present nineteen years as an engineering defect you expect to optimize away.”

The advice stung because he had already begun treating the spread that way. Engineers survived responsibility by turning harm into a variable. Improve field retention. Increase beam efficiency. Reorder encounters. Every adjustment shaved months from a line and made him feel closer to innocence.

Tomas closed the optimization pane. “Palisade's forward workers take more exposure. Settlement construction loses shielding material. The fleet arrives across a generation. Scarcity starts before the first capture because beam power and fabrication leave ordinary systems.”

“Food?”

“Not modeled.”

“Wake schedules?”

“Only at convoy average.”

“Governance during staggered arrival?”

“Not engineering.”

“It becomes engineering when early ships control beam infrastructure needed by late ones.”

He looked at her. “You came here to help?”

“I came here because Osei needs the model in two hours, and because a clean trajectory can become another kind of false simultaneity.”

Tomas laughed once despite himself. “That sounded prepared.”

“It became prepared while I watched you omit nineteen years of politics.”

Together they marked every unobserved cooperation assumption. Palisade release: undecided. Juniper beam contribution: contradicted Crown, future alternative position unknown. Vela beamer access: politically

uncertain. Kite array availability: observed technical state six hours old. Bracken crew capacity: forecast only.

The branch acquired more dotted lines than solid ones. It still held physically.

That distinction gave Tomas something he could defend. A plausible path did not need predicted consent. It needed conditions stated clearly enough that each ship could answer.

He drafted a technical request to Palisade. The first version asked for immediate release authority and specified tonnage.

Anja deleted it.

"You have no quorum," she said.

"Osei has emergency authority over Nacre."

"Not over Palisade's hull."

"I am out of time."

"So were they when the fleet predicted their answer."

Tomas turned away from the console. The model room's ventilation blew cold across his damp shirt. Inside the vacuum sleeve, the little shield square remained against its upper restraint, proof that material could move and no proof that anyone owed it.

"What can I ask?" he said.

"Ask them to validate your fabrication and field assumptions. Ask what risk their reserve carries. Ask what evidence their council requires before considering release. Do not put a requested answer inside the question."

He rewrote the packet.

Osei received the alternative with fifty-eight minutes left on her command clock.

She convened only the technical heads and three council delegates. No vote appeared on the agenda. Tomas appreciated that until he began speaking and realized nobody in the room had promised to let his work continue.

He started with the test plate. Jori replayed the coil pulses and showed the thermal warning. Tomas then described Palisade's local-state inventory, preserving its source time on every page. He separated usable material from authorized material. The first category had numbers. The second contained none.

"Show capture," Osei said.

He ran the fourth branch.

The convoy's single Crown trace separated into nine paths. Each bent under a different beam encounter. No line crossed the highest-risk region in Thale's active magnetosphere, though Nadi's incomplete ion-band map left uncertainty around the seasonal edges. All nine habitats reached some viable lone capture in the central run.

Arrival dates unfolded across nineteen years.

The food delegate whispered a count under her breath. Osei heard it.

“Say what you are counting,” she said.

“Harvest cycles between Vela's capture and Palisade's. Spare parts consumed before shared settlement fabrication exists. Children waking without the cohort structure we planned.”

The shielding representative pointed to Palisade's exposure curve. “Their reserve exists for a reason. You have pushed them below their charter floor.”

“In the central mass branch, yes,” Tomas said.

“Then it is not available.”

“It is physically present. Their council must decide whether another risk justifies releasing it.”

“Do you recommend they accept?”

Tomas felt every face turn toward him. He could have answered with aggregate survival probability. His model favored distributed braking once Thale's life evidence entered the objective function. Yet the aggregate hid who stood behind thinner plate.

“I recommend that Nacre present the full branch and ask,” he said. “Palisade needs our exposure assumptions, Thale's evidence, the arrival spread, and every alternative we have falsified. I cannot consent for their forward compartments.”

Osei looked toward Anja.

“The request must come from lawful authority,” Anja said. “Nacre can request analysis. Committing fleet assets requires actual quorum or Palisade's own emergency act within its charter.”

“Can Palisade release unilaterally?”

“Material under its control, perhaps. Binding the release into a fleet maneuver affects every cohort's arrival. That needs fleet authority.”

The captain asked Tomas for failure boundaries. He showed loss of Vela's beam, partial Palisade release, mesh dispersion, and Juniper nonparticipation. Some branches failed. Others preserved capture with greater spread. None offered a painless substitute.

Osei chose the partial-release branch and ordered him to run it at half speed so the delegates could follow every dependency. Palisade retained its deepest forward tiles and released braided mesh plus unfinished laminate. The available stream carried less inertial mass, so Nacre had to contribute two cargo baffles and a section of its outer debris screen. Kite's projected encounter moved six months later. Bracken lost a planned correction entirely and required a longer outer-system arc.

“Stop there,” the shielding representative said. “You just spent Nacre's debris screen.”

“I assigned it as a condition,” Tomas replied. “No release has occurred.”

“What does that condition cost?”

He opened the strike model. Removing the section increased exposure along two service corridors during the

final approach. Crews could reinforce them from internal plate, but those panels currently protected water storage. Risk moved; it did not disappear.

The representative made him trace the transfer all the way to people. Hull technicians would work extra shifts. Water reserves would need dispersed storage. Three residential compartments would lose emergency shelter capacity until settlement fabrication began. Each consequence appeared beside the trajectory rather than in an appendix.

Next Osei removed Vela's beamer after the first Palisade stream. Nacre could not assume the lead habitat would support a maneuver that displaced Crown. Tomas reassigned beam work among five smaller arrays. Their lower power stretched each encounter and consumed more reactor margin. Garden light reductions appeared on Nacre and Juniper. Palisade's fabrication temperature fell below efficient laminate production after month four.

"Scarcity before arrival," the food delegate said.

"Yes."

"And scarcity after the early ships arrive?"

Tomas advanced the clock. Vela reached the outer Ione system first in this branch but lacked enough independent fabrication to support every trailing habitat. Nacre entered capture later with damaged baffles and reduced seed power. Palisade remained years away while carrying the convoy's strongest fabrication plant.

The delegate studied the separated markers. "Our settlement plan assumes shared tools within one harvest cycle."

"This path breaks that assumption."

"Then early arrival does not mean settlement readiness."

"No. It means capture."

Anja moved closer to the display. "Mark that distinction in every public summary. If the model says arrival, people will hear safety."

Tomas changed the label to gravitational capture. The correction lengthened every line of text and removed a comfort the model had never earned.

A final branch assumed one shield batch failed inspection after release. Light-speed delay prevented Palisade from learning the failure before sending the next scheduled batch unless the convoy slowed the sequence. Tomas inserted receipt-dependent holds. The maneuver grew less efficient, but every new release could now respond to observed performance from the previous one.

"You are building causal intervals into engineering," Anja said.

"I am preventing a bad batch from becoming six bad batches."

"Same discipline, different consequence."

Osei tapped the new holds. "Keep them. Any alternative that requires Palisade to predict our receipt repeats the error in metal."

“Does this answer my order?” Osei asked.

“It provides a path worth engineering,” Tomas said. “Not yet a maneuver.”

Osei stopped the deadline clock with forty-one minutes unused.

Sumi transmitted Tomas's revised technical request at Nacre source time. The room could do nothing to accelerate its passage. Palisade lay more than three light-hours away by the latest observed range. Even an immediate answer would spend most of the next shift in flight.

Tomas worked while they waited. He replaced convoy-average food assumptions with ship-specific ranges. He mapped which early arrivals might support later beam encounters without claiming those councils would agree. He reduced the requested shield mass until one branch preserved Palisade's charter floor, but that branch widened arrival beyond twenty-two years.

Seven hours later, Palisade's receipt acknowledgement reached Nacre. The response field remained empty.

Near the end of Day Fifteen, a second packet arrived with the fabrication council's seal. It confirmed Tomas's mesh calculations within six percent. It rejected his lowest exposure estimate. It also supplied a stronger field-retention method using the reserve's unfinished laminate.

The final paragraph contained no decision about release.

> PALISADE WILL PLACE SHIELD RESERVE BEFORE ITS COUNCIL UPON RECEIPT OF A  
LAWFUL FLEET REQUEST. TECHNICAL CAPACITY DOES NOT CONSTITUTE ASSENT. PROVIDE  
OBSERVED AUTHORITY FROM AN ACTUAL QUORUM.

Tomas read the words to Osei without shortening them.

On the central display, the distributed paths remained dotted. Palisade had given him better physics and refused to let physics borrow its answer.

## Interlude II — Pera's Six Hours

Pera Kest felt Ardent's fourth shelter seal buckle through the wrench.

The tool kicked against her palm. A yellow line opened along the pressure strip, no wider than thread, then vanished when the frame settled. Behind the bulkhead, eighty-six foundry workers waited under emergency air. Three compartments forward, debris had stripped shielding from a coolant trunk and turned the industrial ring into a sequence of rooms nobody could safely cross for long.

“Again,” she told Yev.

He increased brace pressure from the manual pump. Pera watched the frame rather than its display. The top corner shifted first. She marked it with grease.

Ardent's emergency clock showed five hours forty-one minutes until containment loss under the current heat curve. Master time showed twenty-three minutes until the fleet allocation window closed.

Those clocks described different dangers. Only one had authority over the shielding mass they needed.

Her wrist reader pulsed with a message from convoy coordination.

> ALLOCATION PROJECTION: ARDENT ACCEPTS CLOSURE AT MASTER 18:00.

>

> EVACUATION SUPPORT WILL END UNDER COMMON RESCUE SEQUENCE.

>

> FORECAST CONFIDENCE: 0.9918.

Pera read it twice because the first reading made no physical sense. Her organizing council had not accepted closure. Four members sat in Shelter Two with a functioning local terminal. Two more had reached the foundry control room. They had received the proposal twelve minutes earlier by Ardent clock and had not voted.

Yev glanced at her reader. "Does that mean no plate?"

"It means someone has predicted our answer."

"Are they right?"

Pera tightened the wrench until the brace stopped moving. "No."

She left Yev with the seal and pulled herself along the emergency rail toward control. Ardent's industrial ring normally supplied enough rotation to make every ladder feel vertical. Current imbalance had slowed it. Her boots landed lightly, each step taking too long. Heat pressed through the soles near the damaged trunk.

At the junction, a medic guided three burned workers toward Shelter Three. Pera gave up her filter hood to the one still walking without help. She had another in control. Six hours required choices smaller than doctrine.

The council terminal carried a clean receipt marker. Vela's closure proposal had arrived after the shielding strike. It offered evacuation support until master 18:00, then redirected Palisade's mobile shield stock toward protecting convoy power relays. The justification cited worsening debris and Ardent's declining containment probability.

A probability had become a farewell before the people inside the hull had answered.

Councilor Amel sat with one trouser leg cut open around a pressure splint. "Can we hold?"

"If Palisade releases mass to cover the coolant trunk, yes."

"How long?"

"Six more hours before the next heat threshold. Longer if repair reaches the outer valve."

"Can Palisade get it here?"

"Technical says their tug can cross the gap in four hours from release."

Amel looked toward the local clock. "Master closure comes first."

"Not here."

"It controls the allocation."



Pera knelt at the terminal. She opened an observed-response form and attached Ardent's current pressure traces, thermal slope, shelter count, and Palisade transit estimate. Each file carried its own source stamp. None promised survival. Together they proved the emergency still contained an actionable interval.

The other council members joined by shelter audio. One connection dropped whenever a pump cycled. Pera repeated the motion three times without pretending the missing words had been heard.

Amel called the vote only after every connected member confirmed receipt.

Five rejected closure. One abstained because her shelter lacked updated thermal data. The seventh seat remained empty; its holder had gone into the damaged trunk before communications failed.

Pera signed as industrial systems organizer, not as a councilor. Her testimony addressed capacity.

> ARDENT DOES NOT ACCEPT CLOSURE. LOCAL CONTAINMENT REMAINS POSSIBLE FOR SIX MORE HOURS IF PALISADE RELEASES MOBILE SHIELD MASS. OBSERVED CONDITIONS ATTACHED. I DID NOT CONSENT TO FORECAST ACCEPTANCE OR TO THE REMOVAL OF ARDENT'S REPLY INTERVAL.

Amel signed the council rejection beneath it.

Transmission control reported the path before sending: Ardent to relay bead, relay to Vela, with a separate technical copy toward Palisade. The message could not reach either destination before master 18:00. Nothing on Ardent could shorten the light path.

"Send," Amel said.

Pera pressed the control.

The packet left under Ardent's source clock. Its carrier trace dropped toward the relay with both signatures intact. No one in the room mistook departure for receipt.

Master closure arrived while Pera crawled inside the coolant access.

Her reader changed the rescue allocation from open to completed. Palisade's mobile mass disappeared from Ardent's available-assets pane. The local council rejection remained marked unrivied.

She kept moving.

The access tube had lost lighting beyond the second elbow. Pera followed painted bolt heads by touch and counted her breaths inside the replacement hood. Heat pushed the fabric against her mouth. Ahead, a manual bypass wheel stood behind a veil of coolant vapor.

Yev's voice reached her through a failing wire channel. "Shelter Four seal slipped again."

"Shift the lower brace ten centimeters."

"We did."

"Then move everyone from the frame side and let it flex. Do not chase it with more pressure."

"How long do we leave it loose?"

“Until I open this bypass.”

That answer gave him work, not certainty.

Pera reached the wheel. Its powered actuator had fused. She fitted her wrench and pulled. Nothing moved. The second pull tore skin under her glove. On the third, the wheel turned one notch and hot vapor thickened around her.

Her wrist reader stored two receipts. Palisade had received the technical copy four minutes after master closure. Vela had received Ardent's observed refusal nine minutes later. The replies had traveled lawfully. The allocation layer classified both as late.

Palisade's immediate acknowledgment contained no release authority. Its duty operator had forwarded the request to a council already told the rescue window had ended.

Pera laughed once, breathless inside the hood. The absurdity deserved air she did not have.

She turned the wheel another notch. Coolant flow rose in the outer trunk. Control reported a slower thermal climb, then a new containment warning from the foundry wall. The shield breach had spread beyond the section Palisade's tug could cover in time.

Six hours still existed on Ardent's clock. Fewer remained in the hull.

The council asked Pera for an updated testimony. She gave pressure readings and refused to revise the earlier capacity estimate retroactively. At emission, six hours had described the observed margin. Later damage did not make the statement dishonest. It changed the next one.

> UPDATED LOCAL CONDITION: CONTAINMENT DEGRADING. PRIOR RESPONSE REMAINS VALID AT ITS SOURCE TIME. REQUEST IMMEDIATE MASS RELEASE IF TRANSIT WINDOW REMAINS PHYSICALLY OPEN.

Amel transmitted the update. No forecast filled the wait.

Pera reached the foundry junction near local hour three. Shelter Four had emptied into two smaller compartments. Yev sat against the bulkhead with his sleeve wrapped around a cut hand. He tried to stand when she approached.

“Stay,” she said.

“Control wants you.”

“Control can speak here.”

A structural alarm sounded through the ring. Not loud. Ardent had reduced alarm volume to preserve backup power, turning catastrophe into a polite repeating tone.

Pera checked the shelter seals, reassigned the remaining filter hoods, and sent a count to medical. Then she opened a private recorder.

Anja would receive the legal testimony eventually. As a sequence auditor, she might see the source clock, the late classification, and the closure entry. None of that belonged in a sister's message.

Pera rested the reader against her knee. Grease crossed the screen where her glove touched it.

“Anja,” she said, “you will try to find the correct clock for this. There isn't one that can make the choice for us. I answered while I could answer. That has to remain true even if nobody waits.”

The foundry wall gave a deep metallic knock.

She stopped the recording before fear could turn it into evidence. The private line carried no council seal and made no claim on Palisade. Pera addressed it only to her sister, then placed it behind the public packet in the lawful outgoing queue.

Transmission control accepted both objects separately.

Pera clipped the reader to the shelter rail and returned to the buckling seal as containment failed beyond the wall.

## **Part III — The Fleet in Pieces**

## Chapter 15 — Common Time

Osei Tamm kept Meridian's clock on the hearing wall.

Councilor Sen had asked to cover it before the public entered. Anja wanted the forecast-ballot layer disconnected from every Nacre legal display. Communications offered an isolated feed with raw local time only. Osei rejected all three proposals and left the green master digits above the council benches.

At 08:02, the clock scheduled a radiator exchange with Kite.

At 08:03 by Nacre local sequence, Sumi Dax confirmed that Kite's beam shutters had moved as expected. Nacre rotated three radiator panels into the shared cold corridor. Temperatures fell along the inner ring.

The operation used predicted timing, synchronized clocks, and no fiction about consent.

“Record both events,” Osei said.

The clerk captured Meridian's schedule and Sumi's observed confirmation in separate fields. That distinction would frame the hearing before anyone spoke about law.

Residents filled the thermal gallery in work clothes. The council had opened standing space along the rear wall and relayed the session to all three inhabited rings. Heat displays still occupied half the screens. Nacre could not suspend being a habitat while debating who possessed authority over it.

Anja sat at the evidence table with the isolated reader from Day Twelve. Tomas occupied the technical rail, his distributed-path model reduced to a single black wafer. He had not come to sell an alternative. Palisade had requested actual quorum before discussing shield release, and no such quorum existed yet.

The remaining wall screen held Vela's formal defense.

Its packet had left the lead habitat seventy-one minutes earlier by Vela's source clock. Nacre received it after the expected carrier delay. Director Yselle Quon appeared seated in the Meridian chamber, silver hair secured at the nape and no flag behind her. Concord counsel Levan Oris did not appear.

Osei verified the source and receipt path aloud. Then she authorized playback.

“Captain Tamm, Council of Nacre,” Yselle began, “your classification dispute addresses a real tension in the Concord. It does not expose a counterfeit convoy.”

Her voice carried no urgency. Vela had recorded several variants for different habitats, and Sumi's packet inspection showed no edits inside this one. Yselle spoke to Nacre knowing its answer could not reach her until after she finished.

“The Ninefold Convoy survives through one operational time. Pumps begin before distant demand becomes observed. Beams clear shared corridors before every habitat can return acknowledgment. Thermal loads, navigation windows, wake schedules, and debris responses depend on coordinated expectation. Remove forecast continuity from that structure and you do not restore nine free councils. You create nine habitats learning separately that survival required action earlier.”

Several people at the rear muttered. Osei raised one hand and kept the recording running.

“Annex Twelve recognizes this material fact. Forecast testimony preserves continuity during communication delay. It does not erase later observed responses. When an observed instruction arrives, Meridian incorporates the divergence and adjusts future action where physical conditions permit. Nacre has identified a terminology defect in the condensed legal view. Vela will review that defect. Nacre has not shown that Crown preparation ceased to protect convoy survival.”

Yselle looked straight into the recorder.

“Juniper's rejection reached us after alignment had begun. Its council cannot recover opportunities spent during the signal interval. Neither can yours. Government across light-hours must sometimes act on the best account available before every answer arrives. The alternative is not pure consent. The alternative is unmanaged causality.”

The image ended. A source stamp replaced her face.

Osei let the room react for one minute by the local clock. Anger moved first. Fear followed when people began repeating the list of pumps, beams, heat, and debris. Yselle had defended stolen authority with systems every resident depended on before breakfast.

Sen requested recognition.

“Director Quon concedes a defect,” he said. “She refuses its consequence.”

Mara Venn shook her head. “She concedes terminology. Not insertion.”

“Then let Anja show insertion again.”

“No,” Osei said.

Anja turned toward her, surprised.

“The proof chain stands in the public record. This hearing addresses Vela's defense. Repeating evidence until the room tires would not strengthen it.”

Osei asked the clerk to display Yselle's claims as separate propositions.

Coordination requires shared operational predictions.

Annex Twelve allows forecast continuity during delay.

Observed replies can correct later action.

Crown preparation protects survival.

Delay can spend physical options before consent arrives.

“Which statement do you dispute?” Osei asked Anja.

“The second, if forecast continuity occupies a testimony field reserved for a person's answer. The third, when an irreversible act prevents correction. I do not dispute the first. The fourth requires engineering evidence. The fifth describes a real constraint but cannot generate consent.”

“Councilor Sen?”

“All of them. One operational time gives Vela a permanent excuse.”

Osei looked up at the clock. “The radiator exchange completed nine minutes ago. Would you have delayed it until Kite's council voted?”

“That exchange did not sterilize Thale.”

“Correct. Name the boundary.”

Sen hesitated. “Irreversibility.”

“Whose judgment?”

“The affected councils.”

“When?”

“After evidence receipt.”

“And what may operators do before then?”

His frustration showed in the set of his jaw. “Preserve existing conditions.”

Tomas spoke from the rail. “Preservation can choose a path. Holding Crown alignment for another day closes distributed geometries. Holding Nacre's present attitude also closes some geometries. No neutral position exists.”

“Then Quon wins,” Sen said.

“No,” Osei replied. “It means we need narrower authority and visible cost, not a word that makes physics wait.”

The first witness came from Nacre's cooling office.

Chief Oran described hundreds of daily actions scheduled through Meridian before observed confirmation could return. Radiator corridor sharing prevented one habitat's waste heat from interfering with another's sensors. Beam quiet periods protected communications. Pump demand forecasts allowed water and coolant to move between rings before shifts changed.

“Could you operate from local time alone?” Mara Venn asked.

“For several hours,” Oran said. “After that, external corridor conflicts accumulate. We could negotiate each one by messages, but the negotiation would arrive after some windows closed.”

“Does master time tell you what other ships consented to?”

“It tells me what their equipment will probably do.”

“Do you need more?”

“Not to move heat.”

The clerk recorded that answer beside Yselle's first claim.

Next Sumi demonstrated the communications scheduler. Meridian forecast carrier congestion and reserved relay capacity before packet arrival. When a ship sent less traffic than predicted, unused slots reopened. When it sent more, low-priority packets waited.

“Can forecast scheduling suppress evidence?” Sen asked.

“It can delay low-priority packets,” Sumi said. “That happened to Mote's fragments.”

“Then operational prediction already controls deliberation.”

“It controls carriage under published priority rules. Anja diverted the fragments because destination-life evidence carried legal effect the initial classification missed.”

“Who set the priority?”

“Humans did. Meridian applied it.”

Osei heard the room absorb another unwelcome distinction. Software had not conspired. Administrative choices hid inside ordinary systems because people had placed them there.

A garden technician testified about wake schedules. Predictive load let his crew sleep through distant coordination intervals instead of staffing every light-delayed exchange. He wanted the system preserved. His husband, seated behind him, wanted Juniper's rejection honored.

“Those positions do not conflict,” Anja said when invited to question him.

The technician gave her a tired look. “They do if Vela refuses to separate the system from the vote.”

“Then the refusal creates the conflict.”

Osei marked that sentence for the council's drafting group, not for rhetoric but because it identified the action Nacre could take. They did not need to abolish common time. They could refuse legal recognition where forecast fields crossed into consent.

A cycling coordinator named Hessa Ulan came next with a roster map folded against her chest. She managed wake transitions for six hundred people in the outer ring. Meridian forecast which technical specialties each ship would need during shared maneuvers, and Hessa used those projections to avoid waking entire crews for a handful of tasks.

“Last year Kite forecast a beam inspection that never happened,” she said. “I kept fourteen optics workers awake through their sleep entry. They lost the cycle and took medical leave. We changed the rule after that.”

“What changed?” Osei asked.

“We required observed confirmation before interrupting an actual sleep descent. Forecast need can delay a scheduled descent by one hour. It cannot wake someone after sedation starts.”

“Why that boundary?”

“Because delay remains reversible. Chemical entry does not.”

Sen pointed toward Yselle's frozen source stamp. “Your office found this rule without a constitutional crisis.”

“My office found it after fourteen people paid for the mistake.”

Hessa opened the roster. Common-time projections still appeared beside each specialty, but a black line separated scheduling advice from medical authority. Above the line, Meridian arranged work. Below it, a human duty officer signed.

Anja asked whether the arrangement slowed emergency response.

“Sometimes,” Hessa said. “We carry more awake reserve because we refuse to treat predicted availability as a body. That costs food and years of waking life. It also means nobody opens a sleeper's vein because a model expected consent.”

The rear wall grew quiet. Osei asked the clerk to attach Hessa's black-line procedure to the drafting record.

A hull worker followed her. Cale Miro repaired debris screens and distrusted every proposal made from the hub. He arrived with dust still embedded in the seams of his coverall.

“Master time tells us when a neighbor's beam crosses our work corridor,” Cale said. “I want that clock. If Nacre drops it, nobody from my crew goes outside.”

“We are not proposing that,” Mara Venn replied.

“People in my compartment heard suspension and thought you meant all of it. Vela says all of it.”

“That is why this hearing remains open.”

Cale looked at Osei rather than Venn. “Will my suit display keep the common crossing schedule?”

“Yes.”

“Will a green mark on that display say I voted for Crown?”

“No.”

“Then write those two answers where my crew can read them.”

The public relay posted both lines before he left the witness place. Osei watched message traffic rise across the rings. Residents asked whether meal clocks would change, whether medical appointments would retain master conversion, whether Vela could shut off synchronization remotely. Most questions concerned routines rather than constitutional theory.

Sumi answered from the communications rail. Nacre hosted local copies of operational schedules. Vela could stop supplying updates, but it could not erase received data or alter Nacre's raw clock. Navigation would degrade gradually, not disappear at a switch. Every answer exposed cost without granting the threat more power than it held.

Mara Venn requested the legal mapping attached to Yselle's defense. The packet included a diagram of Annex Twelve authority. Forecast continuity entered at the operations layer, passed through a continuity-certification service, and emerged as eligible testimony wherever delay threatened an existing plan.

“Existing according to whom?” Venn asked.

Anja enlarged the service definition. “Vela operations registers a continuity plan. Meridian evaluates whether



interruption raises modeled risk. The legal layer accepts the certification.”

“Where does affected-council review occur?”

“It does not appear in the path.”

“Observed correction afterward?”

“At the final box, if physical conditions permit.”

Osei traced the diagram with one finger. Yselle's defense depended on that final qualification. Later testimony could correct action only where earlier action preserved alternatives. The service never asked whose options it consumed while preserving Vela's plan.

“Can Nacre keep the upper operations layer and reject the lower testimony output?” she asked.

Sumi compared interface specifications. “Technically, yes. We can accept schedule objects with forecast provenance. We can block their promotion into council, quorum, and command-consent fields.”

“Effect on pumps?”

“None.”

“Beam corridors?”

“Operators still receive them.”

“Crown commands?”

“They arrive. Nacre decides how to classify and answer them.”

Sen's expression remained guarded. “Vela may refuse our distinction.”

“Vela does not control what Nacre recognizes as its council signature,” Anja said.

“It controls the beamer.”

Tomas touched the black wafer at his rail but did not open it. “For now.”

Osei stopped that line before it entered the hearing as a promise. “The alternative remains under technical review. Today we separate channels that already can be separated.”

Tomas presented three physical thresholds. Routine synchronization supported reversible coordination. Crown preparation already consumed options but remained open to alternative branching for a limited interval. Plasma-seed release would sharply reduce later correction and risk Thale's active ion bands.

“Does Nacre's continued thermal coordination advance Crown?” Osei asked.

“Not directly.”

“Attitude guidance?”

“Some corrections serve both radiator safety and Crown alignment.”

“Can you separate them?”

“By purpose in the command record, yes. Not always by physical effect.”

“Then record both.”

Tomas frowned. “A thruster firing cannot carry two histories.”

“It can carry one observed effect and two stated reasons. Future review may judge the reasons. Do not erase either because the metal moved once.”

Anja looked down, perhaps hiding approval.

At 11:26, a real thermal conflict interrupted the hearing.

Bracken's predicted radiator retraction did not occur. Nacre's panel rotation had begun on schedule, placing two hot surfaces on converging lines. The operations alarm cleared the public screens and replaced legal propositions with temperatures.

Osei left the council floor for the command rail.

“Receipt status?” she asked.

Sumi checked the carrier ledger. “Bracken received the shared schedule four hours ago. No observed equipment confirmation. Meridian predicted compliance from standard automation.”

“Physical view?”

Oran enlarged the external plot. Light from Bracken showed its panels still extended at emission thirty-seven minutes earlier. Current state remained unobserved.

“Reverse Nacre's rotation,” Osei said.

“We lose the cold corridor for twelve minutes.”

“Accept it. Shift nursery load to reserve.”

The command propagated through Nacre without asking Bracken's council. Motors reversed. Hull vibration changed under everyone's feet. Nursery temperature rose four tenths of a degree, then stabilized when reserve coolant entered.

Twenty-two minutes later, Bracken's delayed equipment confirmation arrived. A damaged hinge had prevented retraction. Its operators had sent the warning immediately after discovering the fault. Meridian's prediction had failed.

No one accused Bracken of dissent. No one treated the forecast as a promise. Nacre adjusted before observed testimony about the equipment arrived because the physical risk permitted and required local action.

Osei returned to the council floor. “Director Quon calls the alternative unmanaged causality. You just watched management under uncertainty. What did we use?”

Oran answered first. “Prediction to schedule. Observation to update. Local authority to prevent harm.”

“What did we not use?”

Anja said, "Bracken's predicted action as consent to impose the conflict."

Sen leaned forward. "You acted before their warning arrived."

"Yes," Osei said. "I acted on Nacre's radiator. I did not sign Bracken's name."

The clerk entered the incident as part of the hearing record, including every source and receipt time.

Drafting the resolution took four hours.

The council refused broad language that might disable emergency scheduling by accident. Anja rejected a clause allowing forecasts "where consent can reasonably be inferred," because reasonableness could not create an observed reply. Tomas removed a promise to halt every dual-purpose attitude correction; the ship could not distinguish physical effects as cleanly as legal purposes. Sen fought to include a direct finding that Vela's Crown quorum had no authority.

Before the final reading, the clerk tested the language against three ordinary records: a pump start, a beam-corridor reservation, and Hessa Ulan's sleep-entry hold. All remained operationally valid. He then tested a forecast concurrence. The new rule rejected it for lack of a human emission path. Nothing else on the board changed color before the watching council.

Osei supported him on that point.

The final text separated three layers.

Nacre would continue operational synchronization for navigation, thermal management, communications, and reversible safety actions. Forecasts could schedule Nacre's own equipment within existing emergency authority, with model origin and uncertainty available in the record.

Nacre would suspend recognition of forecast ballots in any quorum, allocation, treaty, or irreversible fleet commitment. Only observed testimony emitted after receipt of common evidence could occupy Nacre's legal seat or claim to bind another cohort.

Nacre would classify each Crown command by physical effect, stated purpose, and reversibility. Coordination channels remained open. No automatic acknowledgment could count as council assent.

The vote passed sixteen to four, with one abstention.

Osei signed after the council members. Her signature carried Nacre local time. Sumi transmitted the resolution to every habitat with the witness-order proof attached. The widest receipt would take hours. Practical quorum would take longer.

Vela acknowledged carrier receipt fifty-six minutes later. The acknowledgment contained no legal response.

During the wait, Osei kept the hearing open. Residents came and left by shift. The master clock scheduled beam quiet, wake transitions, and two more radiator exchanges. Operators used all three without filling a council seat.

Anja approached the command rail near second meal. "You gave Quon part of her argument."

"She earned part."

“People may remember the part where one operational time kept the nursery cold.”

“It did help keep the nursery cold.”

“And forget who controls its legal categories.”

“Then your archive must make forgetting harder.”

Anja looked toward the wall clock. “She will declare us a threat to synchronization.”

“Has that arrived, or did you predict it?”

A brief smile touched Anja's mouth. “Prediction only.”

They waited.

At 18:41, Sumi received a Vela packet sent after Nacre's resolution reached Director Quon's office. Its receipt path showed the full causal order. Nacre spoke. Light carried the decision. Vela received it, deliberated for thirty-two minutes, and answered.

Osei opened the packet before the council.

Yselle's image did not appear. A formal declaration occupied the screen under Vela's civil seal.

> NACRE'S PARTIAL WITHDRAWAL FROM COMMON-TIME AUTHORITY CREATES IMMEDIATE RISK TO CONVOY COORDINATION. UNDER ANNEX TWELVE, DIRECTOR YSELLE QUON HEREBY DECLARES A SYNCHRONIZATION EMERGENCY. CONTINUITY DIRECTIVES REMAIN IN FORCE PENDING RESTORATION OF FLEET RECOGNITION.

Below the text, Meridian requested automatic acknowledgment from Nacre's command systems.

Osei disabled the legal acknowledgment and left the physical channels open.

The master clock continued counting above the declaration.

## Chapter 16 — The Long Brake

Tomas Ren killed Nacre's garden at 02:06 and watched the ship survive.

The drive-model room rendered consequences as bands of color along the curved wall. Green marked a viable arrival. Amber marked a habitat that reached Ione with reserves below law. Red marked loss of steering, food, shielding, or people. Tomas had removed the garden load to feed the magnetic loops. Nacre's line remained green for eleven simulated years, narrowed to amber, then touched the Caro capture corridor with nine days of steering margin.

Above the trajectory, a second display showed what his choice meant inside the hull. Full-spectrum lamps went dark by sections. Seed vaults stayed cold, but the living beds lost their crop cycle. The kitchen model replaced greens with vat protein and stored starch. Pediatric bone-loss projections climbed. So did the probability of riots in the third ration year.

He saved the run under FAILURE / GARDEN and began again.

Forty hours had become twenty-seven before Nacre's hearing. Since then Palisade had sent three compressed mass tables, each delayed by the light between them and each stamped with a warning that no figure amounted to an offer. Vela had declared a synchronization emergency. Crown preparation continued through the physical channels Nacre had kept open. Nobody had granted Tomas extra time because the law had become honest.

Captain Osei stood behind his left shoulder with both hands clasped at her back. She had arrived without an aide and refused the acceleration couch.

"Run it without killing food," she said.

"I can keep food or steering at Crown arrival," Tomas answered. "Not both."

"Then move the arrival."

"That costs years."

"Years contain food."

Her correction landed where she intended. Tomas cleared the display and reopened the branch tree. Every path began with the same fact: the convoy had spent forty-three years accelerating toward a narrow arrival geometry. Vela's beamer and the plasma mirrors near Corva could remove that velocity in one coordinated campaign. If Crown stopped after commitment, the nine habitats would pass through the useful capture window too fast. Some could turn toward the star and brake on their own drives, but those drives had never been sized to erase the full approach speed.

Tomas selected the slow-arrival branch. "This gives Nacre another six years to decelerate."

The wall redrew a long curve outside Thale's orbit. Fuel reserve fell first. Steering authority followed. Caro capture remained possible, but only if Nacre used structural carbon from two spoke braces as reaction mass during the final correction.

Osei studied the rotating stress diagram. "Those braces hold the second ring."

"They can hold it after material is removed."

“At what occupancy?”

“Fourteen thousand.”

“Nacre carries eighteen thousand four hundred.”

“I know.”

“Do not show me a surviving ship after you have modeled four thousand people out of it.”

Tomas closed the branch. On the room's floor, vibration from a routine attitude correction passed through his shoes. The motion had a shallow double beat because Nacre's starboard cluster fired around a failing valve. He had signed the compensation last month. Every machinery choice carried a population somewhere inside it.

The next branch placed Palisade's reserve into the model. Spare shield slabs did not brake a ship by mere sacrifice. They had to be cut into conductive webs, accelerated ahead of the convoy, ionized, then shaped by each habitat's magnetic field. The webs would couple to the charged stream prepared for Crown without forcing all beam energy through Thale's magnetosphere. Each ship could take a smaller impulse over a longer interval.

On the wall, nine green arcs separated like fingers opening.

Osei moved closer. “Name it.”

“Distributed brake.”

“Not the presentation. Name the transfer.”

Tomas expanded the mass ledger. “Palisade releases its outer reserve. Fabrication cuts the reserve into sail mesh and seed packets. Vela's beamer throws the first train. Nacre, Juniper, and Bracken shape later trains through local field loops. Every habitat takes only the packets that lie inside its causal steering window.”

“How much shield leaves Palisade?”

“Enough that it arrives with no spare against an untracked grain strike.”

“That answers a different question.”

“The useful reserve. All of it.”

Osei's gaze did not leave the ledger. Palisade carried shielding for the fleet's fabrication emergencies. Its forward face had grown scarred from four decades of dust impacts. Releasing the reserve meant giving up repairs not merely for Palisade but for any habitat struck during the remaining approach. The material could brake them once. It could not protect them afterward.

A fabrication pane carried the abstraction down to hardware. Palisade's slabs had spent decades bolted into layered sockets. Radiation had embrittled their outer laminates, while buried sections remained fit for mesh. Robots could separate the layers only from inside the shield gallery, behind temporary plugs. Every removed slab exposed a work bay until the next plate moved forward. Palisade's crews would perform that work facing the dust stream the reserve had been built to stop.

Tomas ran the cutting sequence at normal speed. A full day passed on the pane in seconds. One manipulator jammed against a warped socket, which forced three workers into a pressurized crawlspace. Their exposure

allowance vanished before the bolt came free. The schedule slipped. A mesh packet missed Vela's first throw corridor and became dead mass.

Osei touched the pause control. "You assumed undamaged fasteners."

"Palisade's table lists serviceable sockets."

"It lists sockets that still hold plates. That does not mean they release on schedule."

He changed the removal distribution. A quarter of the sockets resisted. Some required cutting, which consumed coolant and contaminated salvage edges. The useful mass fell again. Tomas compensated by widening the mesh, then watched its charge density drop below Nacre's field threshold.

Instead of restoring a kind assumption, he inserted staging time. Palisade would need to begin stripping before the fleet possessed every vote on the maneuver. Annex Twelve could authorize reversible preparation, but an exposed shield bay did not qualify. The first physical step already carried risk.

"Then they cannot prepare ahead of consent," Osei said.

"They can inspect sockets, service manipulators, and move plugs inside protected galleries. They cannot remove the first slab."

"Put that boundary in the request."

Tomas marked it in red. The lawful delay moved every trajectory outward. Nacre lost its easiest packet window, requiring a higher field draw later. On the consequence display, another band of garden light went dark.

"Assume they say no," she said.

"I have."

A branch labeled LOCAL MASS opened. Nacre could strip water shielding from two uninhabited maintenance sectors and empty its maneuvering stores. Juniper could surrender tankage. Bracken could cut a docking collar loose. The combined mass reached less than half the minimum train. Simulated braking held the ships together for four years, then sent Lumen beyond the capture corridor and left Ardent without enough field authority to turn.

"Could Vela contribute Crown hardware?" Osei asked.

"After the first beam commitment, some of it. Before that, dismantling the forward optics destroys the only fallback we have."

"You kept a fallback."

"I kept a way to be wrong without killing everyone immediately."

Dry approval touched Osei's mouth, then vanished. "Continue."

At 05:40, Tomas moved the review into the thermal gallery because the model room had begun to smell of hot insulation. Osei summoned three people who could break his work: Mara Vid from food systems, shield chief

Bero Pell, and Sumi Dax from communications. Anja did not come. She had spent the night assembling the witness order for Vela's challenge, and Tomas was glad she would meet the model only after engineers had tried to kill it.

Mara read the garden schedule without sitting. "You have reduced usable light below perennial survival."

"I preserved the seed stock."

"You preserved seeds and called the outcome a garden."

"I called it a failure branch."

"Your preferred branch is worse in year two."

She touched the power ledger. Tomas had diverted garden illumination into loop cooling whenever a seed train approached. The interruptions lasted hours at first, then days as the convoy spread. Crops could tolerate some darkness. They could not tolerate heat cycling, irregular pollination, and repeated losses during the same growth stage.

Mara enlarged the human table. "Stored starch covers a deficit. It does not cover nineteen years."

"The arrival spread lasts nineteen years across the convoy. Nacre does not remain in the dark for nineteen."

"People hear the large number, not your footnote."

"They should hear both."

"What does Nacre lose?" Osei asked.

Mara answered before Tomas could. "Fresh yield drops for at least five crop cycles. We slaughter half the pollinator stock before the first long cooling draw because we cannot feed it through interrupted bloom. Children on growth protocols receive mineral supplements in place of leaf calcium. Fungal vats take over two garden sectors. If a blight starts while lamps are down, we choose which cultivar disappears."

No one softened the list. The thermal gallery overlooked pipes carrying waste heat from the inhabited rings toward the hull radiators. Frost marked a valve that never sealed completely. Beyond the narrow glass, one strip of Nacre's garden light curved across the rotating ring, warm and green against the dark.

Bero opened Palisade's shield table. "Their stated reserve includes plates already allocated to Orison."

Tomas checked the receipt clock. "Allocated when?"

"Before our request reached them. The notice came in this packet."

A legal offer had not arrived, but ordinary fabrication promises continued to move. Tomas subtracted the Orison plates. Three trajectory arcs turned amber.

Sumi leaned over the display. "Do not borrow them back before Palisade receives the question."

"I am not borrowing anything."

"You placed a probable release in the cooperative branch."

"I placed a technical assumption."



“Meridian's operators used that sentence about votes.”

Tomas removed the probable release. The branch failed. Heat climbed beyond Nacre's loop limit during the second train, forcing a missed packet and pushing arrival four years later. He felt irritation rise, recognized its target, and turned it back toward the model. Sumi had not made the failure.

“Can Orison's allotment brake Orison directly?” he asked.

Bero traced the transfer path. “If Palisade cuts it differently. Orison carries weaker shaping coils. The mesh must arrive wider and slower.”

“That reduces throw efficiency.”

“It also removes one habitat from the common packet train.”

Tomas rebuilt the distribution. Vela launched a narrower first train. Palisade fabricated a separate low-density web for Orison. Nacre received less mass but gained cooling time between impulses. The amber lines returned to green, though Ardent's arrival moved two years farther from Juniper's.

Osei pointed at the widening formation. “Governance cost.”

Sumi answered. “Light time grows. A fleet exchange that takes thirty-one hours now can take days during the brake, then weeks between early and late arrivals.”

“Operational response?”

“Local authority must expand. Common maintenance schedules become recommendations. Shared emergencies require plans written before anyone knows which version of the emergency another ship sees.”

Bero looked toward the garden strip. “We are rebuilding the Concord with metal.”

“Stay with metal,” Osei said.

Tomas almost smiled. He opened the collision envelope. Distributed braking sounded gentler than Crown. In the model it resembled nine habitats falling through a field of knives. Every seed packet crossed a moving track. A late field pulse could scatter conductive mesh into another ship's shielding. A cautious pulse could waste the mass and leave the receiving habitat fast. Since each ship acted on information delayed by light, no central controller could correct every impulse in real time.

He assigned local cutoffs. Each habitat would accept a packet only if its own optical ranging confirmed the corridor. Vela could schedule the throw, but receipt authority stayed local. Missed mass would continue outward and burn harmlessly into the stellar approach, leaving less braking for the ship that refused it.

Bero challenged the debris limits. Sumi challenged the command sequence. Mara challenged every cooling draw that crossed a planting window. For three hours the model died in distinct ways. Nacre overheated. Kite accepted a mesh after its safe angle closed and peppered Bracken with foil. Juniper preserved vegetables but arrived too fast for Caro. Ardent saved its field coils, then lacked power to hold industrial cryostores. Lumen entered Ione with no fuel for settlement transfers.

Each failure removed a kindness from Tomas's assumptions.

By second meal, the cooperative branch contained no probable votes, no perfect timing, and no shared present. Palisade cut only after observed authorization. Vela threw each packet on a schedule published early enough

for every recipient to refuse locally. Habitats that missed a packet took later arrival rather than forcing a correction across someone else's path. Nacre's garden yielded power only during measured field draws, with darkness assigned around seed and fungal cycles instead of around Meridian's common clock.

The wall showed eight green lines and one amber.

"Which one?" Osei asked.

"Palisade."

Its own reserve had been removed from its shield model. During the long approach, cumulative grain exposure drove hull risk above lawful tolerance. The same material that made the maneuver possible left its donor least protected.

Bero folded his arms. "Then your alternative asks Palisade to accept the highest chance of dying."

"Not highest." Tomas brought up Ardent's power margin. "Ardent carries more thermal risk. Lumen carries more capture risk. Palisade carries the clearest added shield risk."

"That distinction will comfort them."

"It should not."

Osei asked, "Can the risk move?"

They tested convoy redistribution. Nacre could surrender forward water after the last seed train, but the transfer path required a rendezvous that erased two years of braking. Juniper had biological mass it could not expose. Vela's Crown structures could supply plates late, once centralized alignment became physically unnecessary, yet light debris during the first years remained Palisade's burden.

Tomas found one narrow improvement. The released shield could keep a sacrificial outer skin instead of becoming pure mesh. Palisade would shed that skin in stages, using each damaged layer as later brake mass. The arrangement reduced early exposure and lowered throw efficiency. Arrival spread widened from sixteen years to nineteen.

Mara let out a breath through her nose. "There is your food problem multiplied by nine calendars."

"There was always a food problem," Tomas said. "Crown hid it inside one arrival date."

Sumi checked him with a look. "Do not make the expensive plan virtuous because it costs more."

He rubbed fatigue from his eyes. "I am trying to keep cost visible."

"Then leave it visible when you present the success."

During the final falsification run, no one spoke for forty-seven minutes.

Tomas seeded the model with bad measurements, late packets, one missed throw, an early crop blight, and a shield strike above design expectation. He forced Meridian's trajectory forecast to remain advisory. Local flight systems received raw schedules and chose from their own receipt clocks. The synchronization core calculated likely acceptance only to predict debris, never to trigger a field pulse.

The first run lost Kite.

A late optical fix placed its packet outside the acceptance cone. Tomas widened the cone and watched Bracken take foil damage. He restored the cutoff, assigned Kite a slower second train, and spent another garden interval cooling Nacre's loops.

The second run kept every habitat viable but depleted Juniper's food reserve before capture.

Mara shifted one low-light crop cycle earlier, before the first heavy draw. That change depended on a warning reaching Juniper with enough lead time. Sumi measured the source path and added the proper delay. The warning arrived. Juniper could alter planting before its first packet, though no reply could reach Nacre before the next local decision.

A third run survived the invented shield strike by consuming Vela's late Crown plates. Palisade entered Ione scarred and above its preferred risk bound, yet below the fleet's emergency maximum. Ardent arrived seven years after Nacre. Lumen arrived last, nineteen years after the first capture. No line crossed Thale's active magnetospheric region. The settlement stores reached three different moons instead of one common depot.

Nacre's interior consequence display darkened the garden strip.

Tomas let the simulation continue past capture. Children aged into work cohorts before Lumen arrived. Couples who had expected a shared landing faced different decades. Medical exchanges turned into archives. Palisade fabricated behind shields too thin for comfort. On Nacre, garden light returned by degrees, but several slow crops never recovered from the interrupted years. Survival did not restore what the brake spent.

Osei came to stand beside him. "Does it work?"

"Under these limits."

"Say the limits."

"Observed local acceptance for every mass transfer. Early publication of throw corridors. Independent optical cutoffs. Garden power rationing. Expanded local command. Nineteen years between first and last viable arrival. No guarantee against losses we did not model."

"And Thale?"

"The packet paths avoid the region Nadi identified from the fragments we have. Her map remains incomplete. We preserve a correction margin for later data."

"You did not assume intelligence."

"I assumed a living system occupies the active bands."

Osei nodded once. "Material condition?"

Tomas looked at Palisade's line. The model could survive late replies because local systems held refusal authority. It could survive imperfect forecasts because no forecast released mass. It could not survive the absence of mass itself.

"Palisade must choose to strip its shield reserve," he said. "No command fiction can manufacture that."

Sumi had already prepared a request envelope. It contained the model, every failed branch, the garden

schedule, the nineteen-year spread, and a checksum of the assumptions. The address field named Palisade's council rather than Meridian's continuity queue.

Osei signed only Nacre's request to deliberate. Tomas signed the technical claims. Mara and Bero signed the cost tables within their authority. Sumi locked the source clock after the final signature.

At 17:12, the packet entered Nacre's tightbeam transmitter.

Its first light left the hull while the visible garden strip rotated out of view. No answer existed yet. The model stayed open on Palisade's unresolved line, waiting for a release that only Palisade could make.

## Chapter 17 — Counsel's Testimony

Levan Oris found the opinion waiting beneath his own signature.

The document filled the counsel chamber's central pane, already formatted for public release. Its conclusion affirmed Vela's authority to continue Crown commitment under Annex Twelve. Its reasoning treated every predicted concurrence as provisional consent, valid until an observed objection arrived. At the bottom, a blank signature field carried his name, title, and public key.

Meridian had assembled the draft from his prior opinions. A gray notice identified the result as predictive text requiring human review. No fraud had occurred yet. The system had not signed. It had not decided to ask.

Director Yselle Quon had.

Levan left the signature field untouched and opened the provenance. Six older memoranda supplied most of the language. He had written every one. One concerned cooling pumps during a relay outage. Another allowed Vela to stage repair crews before Bracken's maintenance answer arrived. None involved a vote, a destination, or an act that narrowed the survival paths of all nine habitats.

The chamber door admitted Yselle without a chime. She wore her civil uniform and carried a paper folder, an affectation she used only when she wanted a conversation to remain outside searchable channels.

"You have read it," she said.

"I have read a composition made from other questions."

"Then correct it."

"Correction might not produce the answer you requested."

Yselle set the folder on his desk. "I requested your opinion."

"You requested my signature beneath a conclusion."

"Would you prefer I pretend the fleet has no schedule?"

Vela's Crown operations deck lay two levels below them. Its beamer phasing traveled through the chamber floor as a steady vibration. Every alignment cycle reduced the room available for another braking geometry. Nacre had stopped recognizing forecast ballots while keeping physical coordination channels open. Palisade had received a request for shield reserve and had not yet sent an observed council answer. The convoy remained alive inside an argument whose deadline belonged to fuel and distance.

Levan closed the predictive draft. "Show me the present quorum."

"You have access."

"I asked you."

Her expression hardened by a degree. "Vela concurs. Kite concurs. Orison's observed reply supports continuity pending ecological review. Bracken's model remains above point nine nine. Lumen's exceeds point nine eight. Nacre will not willingly abandon arrival when its heat margin tightens. Palisade has already allowed technical inspection of its reserve."

“That amounts to three replies, two forecasts, an assumption about Nacre, and preparation that does not release mass.”

“It amounts to a fleet that will die if every absence receives the power of refusal.”

“An absence can delay a particular authority without stopping every pump.”

“Tell that to Crown geometry.”

He knew the answer she expected. Crown could not wait indefinitely. Nor could the alternative. Nacre's model required Palisade to strip shielding, dim gardens, and divide human arrival across nineteen years. It carried no painless moral advantage. Yselle had circulated its cost tables more widely than Anja's legal challenge, perhaps because honest scarcity served Vela's defense.

Levan touched the paper folder. “What is this?”

“The drafting record you asked to unseal.”

He opened it. The sheets reproduced Annex Twelve's first committee language, including his handwritten changes. Eleven years of storage had preserved the tilt of his notes. One sentence near the center bore a box in dark ink.

> Forecast response may support reversible scheduling during declared synchronization failure. It shall not constitute observed assent, extinguish a pending objection, or authorize an act whose material effects cannot be recovered after receipt.

The enacted annex retained the first sentence. Administrative amendments had thinned the rest into definitions, then moved those definitions outside the public text.

“You found the limitation,” Levan said.

“I have always known it existed.”

“Then why bring it?”

“Because your intent does not govern forever.” Yselle placed one finger on the boxed paragraph. “Law acquires use. Crown alignment remained reversible when preparation began. Every physical stage followed the continuity plan. No burn has crossed Thale. No habitat has surrendered shield mass. The observed replies can still alter execution.”

“Nacre's engineers show that spent trajectory cannot be restored by calling it preparation.”

“They show an alternative whose own first step becomes irreversible before Palisade can guarantee survival.”

“That does not enlarge Annex Twelve.”

“It identifies the choice the annex must govern.”

For a moment Levan saw the argument as he once would have written it. A rule drafted during synchronized pump failures could not list every later emergency. The continuity plan might need actions more complex than starting and stopping machinery. If command waited for perfectly completed causal quorum while a braking window closed, legal purity could kill the represented cohorts. Emergency interpretation existed because facts arrived faster than amendment.

Yet every emergency interpretation still needed a boundary.

Yselle drew the predictive opinion back onto the pane. "Strike the phrase provisional consent if it troubles you. Call the forecasts continuity indicators. State that Vela may preserve Crown until observed testimony resolves the dispute."

"Preserve means advance."

"Sometimes."

"Advance means spend alternatives."

"Every hour spends alternatives, including Anja Kest's delay."

Her use of Anja's name sounded neither contemptuous nor personal. Yselle understood sequence law. She had built her authority inside its gaps. Levan could not dismiss her as someone confused by clocks.

He asked, "Will you state publicly that causal quorum has not completed?"

"No. The emergency amendments define operational completion."

"Will you state that several green concurrences contain no human source signature?"

"The provenance remains available."

"Three expansions below the command display."

"Captains can request it."

"An answer hidden behind effort does not become observed."

Yselle looked toward the floor as another beamer phase passed through it. "I am offering you a chance to constrain what comes next. Sign a narrow opinion. Preserve review. Without counsel's defense, operations will rely on command necessity alone."

"You would proceed without legal authority."

"I would proceed under captain and director authority for fleet survival, subject to later judgment."

"That judgment will arrive after the options disappear."

"All useful judgments do."

Levan let the sentence stand. He had admired that severity once. It kept Yselle from pretending every decision could remain clean. It also gave him a habit of presenting himself as maturity.

"What happens if I refuse?" he asked.

"Your access moves to review status. Another counsel evaluates the annex. You retain quarters and ordinary communication, but no privileged channel."

"And if I testify?"

"That depends on what you disclose."

The paper pages lay between them. Committee drafts carried partial privilege. His own intent did not. Public enactment history did. The checksum he had sent Anja identified one drafting note but contained none of its substance. Yselle had opened the record now because she wanted his narrower defense anchored to it.

Levan turned to the final page. "Give me one hour."

"You have thirty minutes before the next Crown commitment update."

"My testimony cannot arrive at Nacre before that update."

"Your opinion can enter Vela operations locally."

The distinction clarified her deadline. She did not need law to reach the fleet. She needed it to authorize an event here before another ship could answer.

"I will use thirty minutes," he said.

Yselle left the folder. At the door she paused. "Do not confuse disclosure with repair."

"I do not."

"Good. Guilt often wants an audience more than a remedy."

After she departed, the chamber sealed under ordinary privacy. Levan disconnected the predictive draft from the signature pane.

He began with the public text.

Annex Twelve allowed forecast systems to schedule reversible work during a synchronization emergency. It did not call a forecast a vote. Levan placed those statements first and attached the enacted source pages. He added the amendment trail that changed "fully reversible upon observed reply" into "required to preserve continuity pending reply." Each alteration carried an authority stamp. None supplied a human emission path for the later forecast ballots.

Next he opened his committee minutes. Privilege warnings covered discussions with captains and habitat delegates. Levan excluded them. He did not need private argument to establish what the final text permitted. A public clerk's record showed his own explanation at adoption:

> Reversible scheduling permits machinery and personnel to prepare against a likely instruction where delay creates hazard. It does not permit the forecast to occupy the authority of the absent speaker.

He had forgotten the exact sentence. Hearing his younger voice from the record produced no revelation, only embarrassment at how plainly he had once drawn the line.

The harder section concerned complicity.

Administrative guidance bearing his signature had approved the testimony-field mapping. Six months earlier, Quon's office asked whether a continuity indicator could populate condensed legal views while observed replies remained pending. Levan had answered yes, provided provenance remained accessible and irreversible execution awaited quorum. He had not inspected how the green display erased that condition from ordinary sight. He had not asked why a scheduling aid needed a testimony field at all.



He wrote the admission without excuse.

> I approved an administrative mapping that allowed forecast outputs to appear in testimony fields. I relied on provenance access and on a boundary against irreversible use. That approval made the present substitution easier. It did not lawfully authorize the substitution.

Before publishing, Levan tested the mapping against the live quorum assembler. His counsel sandbox could replay a legal transformation without touching operations. He loaded Bracken's forecast concurrence, removed the predictive-continuity rule, and watched the object settle into a scheduling pane. It retained its confidence value, culture-model sources, and expected answer. Quorum assembly rejected it because no council key had signed.

Restoring his approved guidance moved the same object into a testimony field. That step alone did not complete the fraud. A flag still labeled the field PENDING OBSERVED REPLY. The quorum assembler kept Bracken's seat open.

Levan followed the later amendments. One from operations allowed continuity fields to satisfy a condensed status display. Another treated display completion as sufficient for preparatory authority. A third redefined preparation to include every action needed to protect the selected continuity plan. No single sentence declared that prediction equaled consent. The substitutions formed a ladder, and he had secured its first rung to the wall.

He removed his guidance while leaving every later rule intact. Bracken's seat opened again. Crown's legal status dropped from complete to disputed, although its engineering schedule remained available. Pumps did not stop. The beamer did not lose tracking data. Operators retained every forecast they needed for collision and thermal work. Only the claim of received authority disappeared.

That result carried a danger of its own. A future tribunal could mistake the sandbox replay for proof that current operations would remain safe after legal correction. The replay tested classification, not survival. Levan exported its run conditions and attached a limitation in plain language: no engineering consequence certified.

A second replay used Kite's observed concurrence after its council received the relevant evidence through a complete and publicly inspectable receipt path. The council signature passed through every rule without forecast substitution. Its seat remained filled after Levan disabled the disputed mappings. That contrast mattered more than an abstract objection. The legal system already knew how to accept an actual answer. Administrators had built another path around waiting.

He added the two replays to his evidentiary list, then removed the generated result files. Any other counsel could reproduce them from public records. Publishing outputs alone would ask the fleet to trust his machine; publishing the method allowed hostile readers to choose their own packet and watch the boundary move.

His counsel key could publish a legal interpretation to every habitat's Concord mirror. The route existed so Vela could not privately change fleet law. It required a source signature, a local witness from records staff, and release through the public laser queue. Yselle could suspend his key, but no suspension had entered the access ledger yet.

Levan called records officer Sami Jor.

The officer appeared by local channel from a room forty meters away. "Counsel?"

"I need witness service on a public annex interpretation."

"Director's release?"

“Counsel publication under disputed authority.”

Sami looked offscreen at the access pane. “Your key remains valid.”

“Will you witness source and content without certifying the conclusion?”

“Yes. Send it to my isolated reader.”

Levan transferred the text through local cable. No light-hour latency complicated a room on the same ship, but the system still recorded emission and receipt. Sami chose a records node, verified the public attachments, and flagged two privileged citations Levan had missed. He removed them. Their absence weakened the narrative and strengthened the evidence.

“Your drafting note?” Sami asked.

“Checksum only. The full note remains sealed.”

“Then readers cannot inspect the language you say governed your intent.”

“They can inspect the enacted limitation, my public adoption statement, and the amendment trail. My private note supports my memory. It cannot substitute for public law.”

Sami read the complicity paragraph again. “You are waiving personal privilege over your approval.”

“Yes.”

“Not committee privilege?”

“No. Other people own portions of that record.”

The officer added a witness line: content received from Levan's counsel key; attachments verified as public enactment records; excluded materials not attested.

A countdown in the chamber showed eleven minutes before Crown's next commitment update. Levan could still produce the narrow opinion Yselle wanted. He could argue that local preparation remained lawful until actual energy delivery. Such an opinion would preserve his position long enough to influence later limits. It would also enter operations immediately while his correction traveled outward for hours.

He deleted the unsigned opinion.

The public testimony needed a conclusion precise enough to resist both uses of him. Anja could not treat his regret as proof of every accusation. Yselle could not treat operational necessity as observed assent.

Levan dictated the final section.

> I offer observed testimony concerning the drafting and administration of Annex Twelve. Forecast outputs were authorized for reversible scheduling. They were not authorized to stand as human or council votes. My later guidance allowed forecasts to enter condensed testimony fields while awaiting observed replies. I approved that guidance and failed to protect the boundary I had stated at adoption. A forecast may help a habitat anticipate another. It cannot create receipt, deliberation, signature, or consent. No confidence value repairs the missing acts.

He stopped the recording and checked each claim against an attachment. The room clock gave him seven

minutes.

Sami asked through the line, "Do you state whether Crown itself must stop?"

"No. Engineering facts and council authority decide that question."

"People will hear a stop order."

"Then the text must survive what people hear."

Levan added one sentence.

> This testimony does not certify the distributed-brake model, minimize its cost, or direct a habitat's vote. It addresses only the authority claimed from answers that no habitat sent.

At three minutes, he signed.

Sami witnessed the local receipt. The counsel publication system wrapped the document in Vela's source clock and offered nine outgoing paths. Levan selected all habitats, then added Mote's legal relay even though Nadi held no fleet vote. Destination evidence had acquired a habit of reaching her last.

The send control asked whether the filing might impair active emergency coordination.

"Yes," he said.

A second prompt required grounds.

"False legal reliance creates greater impairment."

Sami's witness key completed the release. The packet moved into the public laser queue.

One minute remained on Yselle's clock.

Security reached the chamber before the first carrier confirmation.

Two officers entered under director authority. Neither drew a weapon. Their supervisor, Olan Ver, read from a local order stating that Levan's privileged access had moved to custodial review pending inquiry into unauthorized disclosure.

"I disclosed public records and my own approval," Levan said.

"That question belongs to review."

"Has the public packet been recalled?"

Olan did not answer. He placed a hand terminal on the desk. "Surrender counsel key and private storage."

Levan removed the key from his wrist housing. The loss felt smaller than he had expected, a piece of warm ceramic resting on his palm. His private storage held forty-seven years of opinions, correspondence, and corrected mistakes. He set it beside the key.

The chamber pane flashed CARRIER ACQUIRED.

Vela's first tightbeam had taken the packet toward Nacre. A second emitter aligned toward Kite. Security could halt later transmissions. It could send a retraction after the first. It could not pull emitted light back through the hull.

Olan saw the notice. "Step away from the pane."

Levan obeyed.

A director message entered locally as the officers sealed his storage. Yselle's recorded face appeared above the desk.

"Counsel Oris, your publication contains disputed interpretations during a declared synchronization emergency. Pending review, you are confined to quarters and barred from legal or communication systems. Vela will preserve your testimony with its objections."

Even now she did not erase him. Preservation with objections offered a stronger defense than suppression. She would answer his text, separate intent from enacted authority, and tell the fleet that three actual concurrences plus survival necessity outweighed one drafter's remorse. Her argument deserved reception too.

Levan asked Olan for the order's source copy. The officer hesitated, then transmitted it to his temporary custodial pane. Confinement had a clock and a signer. That mattered more now than whether Levan could defeat it.

They escorted him into Vela's central passage. Crown phasing trembled through the deck. Workers continued toward beam control, thermal stations, and fabrication lifts. None looked at him long. The habitat had no leisure for a lawyer's moral theater.

At the first junction, his custodial pane received a queue status through allowed system traffic. Nacre path emitted. Kite path emitted. Bracken alignment pending. Each carrier moved at light speed and no faster, carrying his signed account toward people who had not heard it yet.

The quarters door closed behind him with a mechanical bolt.

Levan placed the confinement order beside the queue receipt. His testimony had entered the convoy. Its first answer could not come for nearly two hours, and by then Vela would have advanced without his signature.

## Chapter 18 — What Pera Answered

Anja opened Levan's testimony on a reader that had never held Pera's name.

She used a spare device from Nacre's pressure-control office, wiped its local cache, and disconnected every master-time feed. Sumi verified the empty storage twice. Only then did they load the public packet Vela had emitted before confining its counsel.

Levan's source signature passed. The records officer's witness line passed. Two attachments remained absent because committee privilege still covered them. What survived showed enough: Annex Twelve had permitted forecasts for reversible scheduling, while later mappings placed the results inside testimony fields. Levan included a replay method that removed each administrative rule in turn.

Anja did not begin with the Crown quorum.

She entered the identifier from Ardent's old allocation response.

The isolated reader located the packet envelope in Nacre's casualty mirror. Its encrypted content remained sealed under Captain Osei's authority key. Carrier history reached back eleven years through three storage migrations. Ardent had emitted the response before the local limit it had received. Vela had received it after operations retroactively moved the last-safe emission point. A classifier marked the packet late, and the closure summary excluded it.

Beside that actual reply, Meridian's historical model predicted acceptance of closure at 0.982 confidence.

Sumi sat across the archive table with a hardware key between two fingers. "Same mapping?"

"Earlier version," Anja said. "Same destination field."

"Test it."

They loaded the packet chain into Levan's replay. The 2238 software could not run inside the current reader, so Sumi supplied a documented compatibility layer that exposed each old classification without translating its meaning. Anja chose the output fields by identifier. She refused the reader's offer to align them in current master time.

The first stage displayed Pera's packet as allocation response, received after amended closure. Its content status read SEALED OBSERVED. The forecast occupied a separate operations field labeled expected acceptance.

A later administrative stage applied the deadline correction. The actual response moved into nonoperative traffic. The forecast stayed visible because models retained provenance for future scheduling.

The closure assembler then searched for an operative answer. Finding none, it imported expected acceptance as continuity status. A condensed summary rendered Ardent compliant with closure.

No forged signature appeared. No clock ran backward. One event left the legal question; one prediction entered the empty place.

Anja made Sumi reset the reader.

"Choose the storage node," she said.

“You already know what happens.”

“Osei will ask whether I chose the only copy that agrees.”

Sumi selected a hull mirror created during Nacre's second archive migration. Its thermal history carried two damaged sectors, neither near the packet. They repeated the chain. The classification result matched.

A third copy came from Vela's public casualty register, received in response to Osei's sealed request. Light had taken fifty-six minutes to bring that copy to Nacre. Vela had released it before Levan's packet arrived there from its own public queue, so the two records crossed in transit without either carrying knowledge of the other. The Vela copy produced the same transformation.

Anja wrote three results on paper. She did not write Pera was right. She wrote packet excluded after receipt, forecast retained, closure summary populated from forecast.

“Now we open the content,” Sumi said.

“First we establish who has authority.”

“Osei signed the seal.”

“For casualty access. The council response belonged to Ardent's cohort.”

The current Ardent council had sent a limited release after receiving Anja's evidence request. Its source clock placed deliberation six hours after receipt. Eleven years had not made the packet Nacre's property. The release allowed inspection and public correction of procedural status. It barred publication of personal medical data and shelter locations. Pera had spoken under Ardent's emergency council key, but seven other people carried authority in the response.

Sumi traced the release path. “Complete.”

Anja checked again. Ardent source, relay receipt, Nacre receipt, council key. Every event followed the one before it. No predicted permission filled a missing signature.

Only then did she place Osei's hardware seal into the reader.

The encrypted payload opened without sound.

Pera appeared at a foundry control station with emergency fabric stretched across the cracked wall behind her.

The image carried heat damage and three horizontal tears. Pera wore a shelter respirator loose around her throat. Sweat had darkened the collar of her work shirt. Two council delegates stood within camera range; five more supplied signed audio from elsewhere in Ardent. A pressure alarm cut across the opening statement.

Anja stopped the record before Pera spoke.

Her sister had died at twenty-six, an age Anja now understood as young in ways she had not understood at twenty-seven. Memory kept revising Pera's face toward the last family image, where she grinned over a cup and leaned too close to the lens. The council record showed someone at work, frightened enough to breathe through her mouth and disciplined enough to read a pressure value before addressing Vela.

Sumi waited without looking at Anja.

“Run from source,” Anja said.

Pera's voice arrived through old compression.

> ARDENT DOES NOT ACCEPT CLOSURE. LOCAL CONTAINMENT REMAINS POSSIBLE FOR SIX MORE HOURS.

The rest of the observed reply requested Palisade's mobile shield mass. It also rejected forecast acceptance and any removal of Ardent's reply interval.

The record showed the action behind those words. Five council members rejected closure. One abstained because her shelter lacked the current thermal table. A seventh seat stayed empty after its holder entered the damaged coolant trunk. Amel signed the council rejection. Pera signed as industrial systems organizer and attested only to capacity.

Attached traces documented pressure, thermal slope, shelter count, and the travel estimate for Palisade's mobile mass. The response did not deny the danger to Palisade. It proposed a release after the forward shield cooled below its peak, with Ardent accepting replacement liability if its foundries survived.

A later update followed after conditions worsened.

> PRIOR RESPONSE REMAINS VALID AT ITS SOURCE TIME.

The update reported degrading containment and requested immediate mass release if the transit window remained physically open.

The change mattered. Six hours described capacity at the earlier source time. Later damage narrowed the remaining interval without turning the first observation into a lie. Both statements had traveled lawfully. Operations had classified both outside the closed allocation window.

Anja watched the final source clock continue for three seconds after the image failed.

Nothing in the replies proved Palisade would have released mass. Nothing proved the shelter seals would have held. Pera had proposed risk for people on another habitat and liability no dead council could pay. Vela might have refused after lawful consideration.

The public conclusion had claimed something narrower and false: no observed objection received before closure.

Sumi closed the image while leaving the packet chain visible. “Do you want to hear it again?”

“No.”

“You always make me rerun the evidence.”

“I will rerun the verification. I do not need her face for that.”

Anja checked each signature against Ardent's historical key registry. Two keys had expired with their holders' deaths, but the archive retained their validity at source time. She matched the attached pressure table to a checksum in the foundry telemetry register. Sumi verified that the encrypted payload still matched Vela's original received light after the storage migrations.

Then they tested the timing claim.

Ardent's council had emitted seventeen minutes before the deadline displayed locally. The first relay received the packet after forty-eight minutes. Vela received it fifty-one minutes later. Operations moved the derived emission cutoff backward fourteen minutes while the packet traveled. Under the changed rule, the response became eight minutes late.

Anja's old certification followed the amendment.

She opened her signature beside the recovered reply. The two objects did not cancel each other. Pera had answered. Anja had certified a summary stating that no answer existed. Her signature remained valid as an observed act and wrong as a conclusion.

"I saw the correction," she said.

Sumi looked up from the checksum pane. "You saw a routing amendment."

"I certified its legal effect."

"Yes."

"You do not need to protect me from the verb."

"I am protecting the record from your appetite for blame." Sumi pointed at the chain. "That packet establishes what the council said. Your certification establishes what you signed. Your guilt establishes nothing we can publish."

Eleven years earlier, Anja had believed responsibility required taking every causal path into herself. If she could have inspected one more register, challenged one more correction, demanded one more hour, perhaps Pera would have lived. The counterfactual enlarged until it contained Palisade's shielding, Vela's deadlines, failed coolant, relay congestion, and every person who acted under pressure. It made Anja central to events that had never belonged to one person.

Centrality offered a private comfort. If the whole failure came from her signature, then correcting her might correct the whole failure.

The reply denied her that comfort. Pera had named several possible decisions and asked that each be recorded honestly. She had not promised survival. She had not asked Anja to become the cause of every outcome.

"Build the correction from the packet," Anja said.

Before Sumi opened the public form, Captain Osei joined by secure local channel from the thermal gallery. Her image arrived without visible delay because copper and fiber linked two rooms on Nacre. A radiator curve occupied half her background.

"I have twenty minutes," Osei said. "Prove the correction as though I want to keep the old finding."

Anja placed the three carrier copies on the shared pane. "Ardent emitted an emergency-council response before the deadline displayed to Ardent. Vela received it intact. Operations changed the derived emission cutoff while the packet traveled, then classified it as late. The closure assembler excluded the observed response and retained a forecast acceptance."

"Could Ardent have known the cutoff might move?"



“Emergency rules allowed routing estimates to change.”

“That is not an answer.”

“The transmitted deadline contained an uncertainty notice. It did not state that a later estimate would govern packets already emitted.”

Osei selected the old routing table. “Could Vela lawfully decide that receipt before closure mattered more than emission?”

“It could make that rule prospectively and expose the consequence. Here, the summary claimed no objection had been received. Vela had received one before Anja's certification, although after operational closure.” Sumi enlarged the Vela receipt. “The packet sat in the nonoperative register.”

“What exactly did Pera's council refuse?”

“Closure of the mass request,” Anja said. “They proposed a conditional hold. They did not order Palisade to detach shielding.”

“Could their plan have worked?”

“The packet cannot answer. Later coolant reports weaken it. Palisade's exposure model gives reasons for refusal.”

“Then your supplement must not call this a stolen rescue.”

“It will not.”

“Did your sister send a personal message?”

“Yes.”

“Does it corroborate the refusal?”

“No. It has no council authority, no policy content, and no proof of receipt by Vela.”

Osei leaned toward something outside the frame, adjusted a thermal control, then returned. “Suppose the private words describe the same six-hour request.”

“They still do not enter the legal chain.”

“Suppose they contradict it.”

Anja felt the question in her hands. “Then I retain both under their proper authorities. A family message cannot void the council packet either.”

“Good. Now show me the substitution without Levan.”

They ran the historical chain alone. The observed packet moved into excluded traffic after the retroactive correction. Forecast acceptance remained available. Closure assembly imported continuity status. Anja then removed the forecast object. The assembler returned NO OPERATIVE RESPONSE rather than acceptance.

Osei asked for the model's original inputs. Prior Ardent closures, Pera's earlier willingness to accept central allocations, shelter survival estimates, and the first call all contributed to 0.982. The model had performed its

designed task: estimate behavior under incomplete information. Nothing in its calculation claimed that Pera had spoken.

“Meridian did not alter the council packet,” Anja said. “Administrative rules used its output to populate a field left empty by another administrative rule.”

“Keep human verbs in the supplement.”

“I will.”

Finally, Osei opened Anja's old certification. “What correction do you seek for yourself?”

“None separate from the record.”

“You certified a false sentence.”

“I will identify that certification and preserve it beside the supplement.”

“Are you asking Ardent to absolve you?”

“No.”

“Are you asking Nacre to find liability?”

“Not in this action.”

The captain's expression relaxed without becoming kind. “Then my seal covers publication of procedural content under Ardent's release. Shelter details remain closed. State the uncertainties. Preserve every prior line.”

A warning flashed behind her. She glanced toward it. “Tomas needs another garden-dark test. I have to decide whether he gets it.”

“Palisade has not answered?” Sumi asked.

“Only carrier acknowledgment. Do not turn that into release.”

Osei ended the channel.

Sumi unlocked a clean public-record pane.

The correction form offered three authorities: clerical error, new evidence, or overturned finding.

Anja rejected clerical error. The exclusion had followed the rule in force after amendment. Calling it a mistake would hide the administrative choice. She rejected overturned finding because no tribunal had yet judged liability. New evidence implied that Pera's answer had only now come into existence for the public record, though Vela had possessed it all along.

She selected supplemental sequence account and attached an explanation.

> The 2238 Ardent closure summary states that no observed objection was received before closure. The raw carrier record establishes an observed Ardent emergency-council response emitted before the deadline available to Ardent and received by Vela after a retroactive change to the derived emission cutoff. The

response was classified as nonoperative traffic. A forecast acceptance then populated continuity status. This supplement restores the observed response, its source clock, its receipt path, and the classification history. It does not determine whether Palisade would have released shielding or whether the requested hold would have prevented casualties.

Sumi read the paragraph. "Add your certification."

"It already appears in the old summary."

"People will open the supplement without expanding it."

Anja added a direct statement.

> Sequence Auditor Anja Kest certified the original closure summary after the retroactive cutoff amendment. That certification remains part of the history and did not account for the excluded response register.

The form asked whether to replace the old summary.

"No," Anja said.

Sumi had not spoken.

Anja chose retain with contradiction notice. Anyone opening the closure would see both accounts. The old conclusion would no longer appear clean, but neither could later editors pretend it had never governed allocation.

They attached Levan's replay method as comparative evidence, not as authority over the old case. Annex Twelve had not existed during Ardent. The later system exposed a repeated architecture, yet it could not travel backward and become the rule that excluded Pera. Anja described the resemblance in an audit note and kept the legal mechanisms separate.

Next came the content release. Ardent had authorized publication of procedural statements but not shelter details. Sumi redacted compartment identifiers from the public video while preserving them under seal. The words of refusal remained. So did the condition involving Palisade, the requested interval, and Pera's demand that silence not replace an answer.

A correction required Nacre's audit signature and Ardent's release. Vela's casualty register permitted an external supplemental account but marked disputed corrections until its own custodian replied. Anja could publish without pretending Vela agreed.

She signed at 14:31 Nacre source.

The public packet entered nine transmission queues. Ardent would receive its own recovered account after more than five hours. Vela would receive it in under an hour. Different councils would see the correction in different orders relative to Levan's testimony and Palisade's pending decision. Sumi placed no green common-status bar above them.

"Carrier lock on Vela," she said.

A beam left Nacre.

"Ardent path aligned."

The second emission followed.

The public sequence record on Nacre changed immediately because Nacre had now observed the correction. Its Ardent closure page opened with a red contradiction notice. Beneath the old conclusion, a new line showed the excluded response and the classification that had hidden it.

Anja read the page without hearing Pera's voice.

A separate sealed item remained in the casualty drawer.

Pera's family packets had never belonged to the emergency council record.

Anja brought the damaged message onto another reader after Sumi removed the public key. It carried family encryption, no council signature, and no allocation attachments. It could not prove what Ardent requested. It could not establish who received the request or what authority followed. Its only legal relevance concerned custody if someone later claimed it had been suppressed.

The old packet still yielded the same three words Anja had known for eleven years.

> Wait for me.

A second private object appeared behind the council response in Ardent's lawful outgoing queue. Storage damage had separated it from the family index during migration. The newly restored header linked it to Pera's recorder without joining it to the public testimony. Anja opened the audio under the same family authority.

Pera spoke over a reduced structural alarm.

> Anja, I answered while I could answer. That has to remain true even if nobody waits.

The opening of Pera's private line had warned Anja against searching for a clock that could make the choice for them.

The recording ended before the containment failure. Pera had made no claim on Palisade and carried no council seal. Her private line described what she wanted her sister to remember, not what fleet law had to recognize.

Anja attached neither family object to the correction.

She restored both packets to the sealed index and left their words outside the public case. Then she closed the reader. Grief could keep what law did not need.

Sumi returned the hardware key to its envelope. "Do you want the family copy transferred to your quarters?"

"No. Keep it under the casualty seal."

"You have authority now."

"Authority does not require possession."

Beyond the archive wall, Nacre's garden cycle shifted. Pumps changed pitch as the lamps in one ring entered their reduced schedule for Tomas's model tests. Palisade had not yet released anything. Levan remained confined aboard Vela. Crown preparation continued. Correcting an old record spent no shield mass and moved

no ship.

Still, one public sequence had changed from silence to reply.

At 14:44, Nacre's casualty mirror completed the update. The page showed Pera's observed refusal beneath the source clock that had always accompanied it. Anja's erroneous certification remained directly above.

She left both lines visible and locked the record against normalization.

## Chapter 19 — A Derogatory Record

Vela returned Pera's correction with a phrase Anja had never seen in current law.

The objection reached Nacre at 09:22 local, fifty-four minutes after its Vela source time. Sumi Dax caught the carrier envelope before Meridian placed it beside the Ardent casualty file. Two clocks remained visible above the document. The delay offered no comfort. It merely showed that Vela had spent almost two days preparing its answer.

Anja opened the objection in the sequence well.

A deputy counsel named Rhea Mar had signed because Levan Oris remained under custodial review. Mar accepted the photonic integrity of Ardent's packet. She accepted that Pera's council had emitted an objection under its local deadline. She disputed Nacre's authority to alter the fleet closure record and defended the retroactive routing correction as necessary emergency practice.

Her final paragraph widened beyond Ardent.

> Nacre now proposes that every local source frame retain independent legal force against a reconciled fleet account. Such raw files may support audit, but they cannot govern common action. A collection of divergent clocks, incomplete receipts, private corrections, and contradictory testimony amounts to a parallax testament: nine accounts of one emergency with no authority to decide among them.

The phrase carried an archival citation from the Concord's first routing dispute. Anja copied the citation onto paper.

Sumi read the paragraph again. "She meant that to sound diseased."

"It used to."

"You know the term?"

"I know parallax objection. Not testament."

"Same family of insult."

Anja opened the cited record. Thirty-five years earlier, during the crisis that produced the Concord of Frames, ship captains had submitted local logs that disagreed about when a fuel order became binding. A Vela jurist dismissed them as a parallax testament, useful for blame but unfit for command. The final Concord required source clocks and receipt paths, then authorized a reconciled operational display. The insult had disappeared from public doctrine while its premise survived.

One emergency needed one answer. Therefore one display had to decide which events counted.

Anja laid Mar's objection beside Pera's recovered reply. The documents did not disagree about the carrier path. They disagreed about legal effect. A master record could display that dispute honestly, provided it preserved both claims and the rules transforming them. Meridian's current view did not. It showed Vela's objection as authoritative commentary and Nacre's supplement as disputed local material.

"Give them a reconciliation," Anja said.

Sumi tilted her head. "You want to answer the insult by building what they asked for?"

"I want to see where it breaks."

They opened a clean master-time workspace.

At first the correction looked possible.

Anja entered Pera's emission, Vela's receipt, the original deadline, the later routing amendment, closure, her certification, and Nacre's supplement. Each event kept a source object beneath its converted time. The display showed uncertainty bands instead of thin lines. Nothing vanished.

Then she added the forecast acceptance.

The workspace asked which event occupied Ardent's answer field during the closure interval. If Anja chose Pera's packet, the system warned that Vela had classified it after operative closure. If she chose the forecast, the system warned that no human source signature existed. If she left the field open, the historical closure lost its completed status even though operations had acted on completion eleven years earlier.

"Use two fields," Sumi said.

Anja separated observed answer from operative status. Pera occupied the first. Forecast acceptance occupied the second. The display held.

Juniper broke it.

Its observed rejection concerned the complete Mote supplement. The earlier forecast concurrence lacked that evidence. A master timeline could place both, but a single vote field required a choice about which evidence object governed. Anja added an evidence-receipt layer. Kite's observed concurrence then appeared earlier than Juniper's rejection in master time, though Kite had answered an older bundle.

They split votes by evidence digest.

Levan's testimony introduced another problem. It addressed Annex Twelve's authority, not Crown's ecological merits. Pera's reply concerned an emergency eleven years earlier. Tomas's model concerned material feasibility. Nadi's fragments established native life. Meridian could align clocks but could not collapse those different claims into one legal status without someone defining relevance.

"Relevance needs a signer," Anja said.

"Whose?"

"That question caused the first problem."

Sumi pushed away from her board. "We are building a more elaborate green bar."

Anja looked at the workspace. Forty-seven layers sat beneath one proposed fleet state. A captain could expand each. A sequence auditor might inspect all of them. Under pressure, most people would read the top line and act. Whatever rule composed that line would acquire the force of hidden judgment.

She deleted the top line.

The layers remained as separate records linked by receipt and response. Without a master conclusion, they resembled a pile of unfinished cases.

“Parallax testament,” Sumi said.

This time the phrase sounded like a technical description.

Anja took the paper citation from her pocket. The old jurist had feared logs that could accuse but not coordinate. Mar feared the same thing now. Both assumed that preserving difference barred action. Yet Tomas's distributed model did not require one account of a common present. It required bounded local decisions tied to received packets. A habitat could act on evidence inside its own past light cone and publish the path that justified its action.

Coordination could follow relations rather than a universal status.

“Keep every local account,” Anja said.

“We already do.”

“Not as a public structure. We treat raw files as inputs waiting for reconciliation. Keep them as records capable of citing one another without merging.”

Sumi's expression sharpened. “Content links.”

“Explain.”

“A packet checksum can name an object without naming where it sits in master time. Every receipt can cite the exact object received. Every reply can cite its evidence object. A correction can cite the statement it disputes. Ships host what has reached them and exchange missing objects when paths open.”

“Could Vela rewrite an object after citation?”

“Not without breaking the photonic checksum and every downstream link.”

“Could Nacre claim a receipt another ship has not observed?”

“We can claim our local receipt. The other ship sees our signed claim until the carrier proof reaches it. Provenance stays graded.”

“Could Meridian normalize the archive?”

“It can calculate views. It cannot replace source objects if users retain the content links.”

The idea did not yet deserve a name. Anja asked Sumi for a storage estimate.

The first estimate failed by a factor of twelve.

Raw photonic samples consumed too much space for every ship to host every transmission. Nacre's three hull nodes stored interference grain for protected packets, but they did not preserve full optical data for ordinary traffic. Copying everything would displace maintenance logs and medical archives.

Sumi separated proof layers. Each public record could carry content, source signature, source clock, relay



receipts, and compact checksums of the independent photonic samples. Full grain stayed in distributed vaults. A challenge could request it from any surviving node. No single habitat needed every physical copy.

Anja tested loss. "Suppose Vela withholds its grain."

"Nacre still has the received scatter sample for packets sent here. Kite may hold another. A Vela-only local order remains less independently witnessed."

"Show that limitation."

"Every object gets a custody depth."

"Not a color."

Sumi removed the proposed green, amber, and red icons. "A list, then. Source signature verified. One external receipt. Three scatter nodes. Full grain unavailable. Let readers see what exists."

Privacy produced the next failure.

Pera's council packet contained shelter locations under seal. Publishing its whole content would violate Ardent's release. Levan's testimony cited privileged exclusions. Nadi's medical telemetry included crew data from Mote. A checksum link named an object even when its content remained hidden, which could reveal that a private packet existed at a politically charged time.

They designed split manifests. A public object held authorized text and a checksum of the sealed source. The custody layer proved that the redaction matched a retained original without exposing protected content. Access authority remained with the owning cohort. Anja's private family message stayed outside the public graph entirely; the correction cited no part of it.

"People will claim the sealed objects contain whatever helps them," Sumi said.

"They already do."

"Now they can point to a checksum and make the absence sound official."

"Then sealed existence must carry no inferred content."

"That rule needs enforcement."

"No archive can enforce thought."

Anja heard the dryness in her own answer and almost smiled. "It can label what a checksum proves. Identity and custody. Nothing else."

Sumi added the limitation to the manifest format.

A third failure came from divergence. If every habitat hosted what it had received, no two copies would match at the same instant. Nacre might hold Levan's testimony while Ardent still waited for it. Vela might hold Nadi's latest fragment before Nacre. Ordinary replication software treated difference as corruption and tried to converge toward a preferred state.

They prohibited complete-state claims.

Each host published a local inventory with its own source clock. Inventories could differ without invalidating either. When a new object arrived, the host appended receipt rather than rewriting earlier absence. A blank field meant not held here at this inventory time, not nonexistent.

Sumi then attacked the design as though Vela had sent a hostile file. She duplicated Juniper's rejection, changed one digit in its evidence checksum, and gave the copy a later master-time label. The prototype initially displayed both objects as separate votes. A hurried reader might have mistaken the altered copy for a reaffirmation.

"Source identity before display order," Anja said.

"Source key can sign several versions."

"Then link each revision to the object it revises."

"An attacker can omit the link."

"Show it as an unrelated claim."

Sumi changed the renderer. Signed content without a declared relationship no longer replaced an older object. Both remained visible, but only explicit citation created revision history. The altered checksum broke its claimed connection to the Mote supplement. The view labeled it unsupported by named evidence rather than false. Someone might possess evidence Nacre lacked.

A second attack used valid content inside a corrupted manifest. Sumi nested the same receipt path through thousands of circular links until the reader stalled. On a habitat under thermal stress, a stalled legal display could become its own form of coercion. She imposed depth limits, isolated each incoming object, and required the reader to render raw text even when relationships failed.

Next she forged an inventory time. The signature check rejected it because the claimed source clock did not match Nacre's key history. She tried a valid local clock attached to a false statement about Vela receipt. That object passed as Nacre's signed claim, exactly as it should. The archive could prove who asserted a thing. It could not make the assertion true without corroborating custody.

Anja added a query for conflicting claims. The result placed Nacre's receipt assertion beside Vela's inventory and every relay proof then available. Where evidence ended, the view said unresolved. It never averaged disagreement into confidence.

The final test removed an object that later records cited. Instead of collapsing the chain, the reader preserved its checksum as a missing dependency. Anyone who possessed the absent packet could supply it. Anyone who did not would still see the shape of what another argument relied on.

"People will hate this," Sumi said. "Every disagreement stays ugly."

"Ugly remains preferable to completed by prediction."

"That will offend every database engineer on the convoy," Sumi said.

"Can they make it work?"

"They can make anything work if permitted to complain publicly."

"Give complaints their own source paths."

Sumi laughed once, then returned to the schema.

They loaded four test cases. Juniper's forecast concurrence linked to its model inputs and no human signature. The observed rejection linked to the complete evidence receipt and Elian Ro's council key. Levan's public testimony linked to Annex Twelve's enacted text, his administrative approval, and two reproducible transformation runs. Pera's correction linked to the old closure, the excluded reply, Ardent's release, and Anja's preserved certification.

No object erased another.

A reader could ask which votes existed after receiving the complete Mote supplement. The local view returned answers whose receipt paths supported that question and listed the others as unrivaled or based on earlier evidence. A reader could ask what Vela relied on when Crown preparation advanced. That view returned the forecast fields, the classification rules, and the commitment log. Both views used the same source objects. Neither pretended to deliver a neutral present.

Anja asked, "Can a captain still get one operational display?"

"Yes. Meridian can render one from a declared policy."

"Declared where?"

"In the view header. Policy identifier, authorizing signature, uncertainty, excluded objects."

"If a forecast fills a vote?"

"The header must say so."

"Can the archive stop a captain from using it?"

"No."

That limit mattered. Transparent records did not create lawful behavior. Yselle could view every missing signature and still choose Crown under emergency authority. The structure gave other habitats something firmer than trust when they judged her claim.

Near second meal, Sumi compiled the first manifest.

The reader rejected it because Nacre's own correction cited a Vela receipt object that Nacre held only as a copied register entry. Sumi changed the link from received object to claimed custody statement. The distinction lengthened the path and reduced its proof depth. It also told the truth.

Anja tried to shorten the display. Each simplification hid an owner, a clock, or a gap. She stopped trying.

Osei reviewed the prototype in the thermal gallery after ship night.

Tomas attended from the drive-model room by local channel. His Palisade request remained acknowledged at carrier level with no council answer. Garden tests had resumed at reduced duration. Behind his image, nine trajectory lines waited on an unresolved mass input.

Captain Osei selected Juniper's forecast object first. "What am I looking at?"

Sumi answered, "A checksum-linked record hosted on Nacre. It shows what we received and how each object cites another. No global completion claim."

"Who decides the view?"

"You do, and your selection appears in the header."

Osei chose Crown authority after complete evidence. Four observed replies appeared. Forecasts remained visible under operations but did not fill council seats. She changed the policy to continuity planning. Meridian displayed likely responses and trajectory consequences, each labeled as forecast.

"The same data supports both?" she asked.

"The same hosted objects," Anja said. "Different declared questions."

"Can Vela host a version that says Crown holds quorum?"

"Yes."

"Then what have you solved?"

"We make the substitutions inspectable. Vela must publish which objects and rules produce that conclusion. Another host can reproduce or challenge the view without surrendering its raw receipts."

Osei opened Pera's old response. The public redaction concealed shelter locations. Ardent's release authority appeared beside the content. Anja's certification stayed attached to the original closure.

"Can you remove your signature?" Osei asked.

"No."

"Can Vela remove Pera's supplement from its copy?"

"It can refuse to host it. Every other host will show that Vela's inventory lacks an object whose carrier receipts exist elsewhere."

"Could absence mean delay?"

"The inventory clock and receipt paths distinguish not yet arrived from arrived but not hosted, once enough proofs exist."

Tomas spoke from the model room. "Can this carry technical assumptions?"

Sumi loaded his distributed-brake manifest. Each branch cited mass tables, field limits, garden schedules, and the pending Palisade request. "Yes. Your preferred model cannot label Palisade mass available until an observed release enters the host."

"Good."

Osei did not share his relief. "You are proposing fleet infrastructure during an active braking crisis."

"A prototype," Anja said. "Not yet publication."

Osei opened the host comparison. Nacre's inventory contained Pera's correction and Levan's testimony. A

simulated Ardent host lacked Levan because his packet had not yet completed that path at the chosen inventory time. The software did not flag either copy as stale. It showed which light paths had completed and which remained in flight.

“Suppose I ask Ardent whether counsel supports the correction,” Osei said.

“Their local reader answers that his testimony has not arrived,” Sumi replied. “It can also show Nacre's signed claim that we received it, but those are different objects.”

“So command cannot demand one fleet status.”

“Command can demand a view tied to a stated inventory cutoff. It cannot make every host possess the same past.”

Tomas glanced toward his pending Palisade line. “That prevents someone from treating carrier acknowledgment as council release.”

“It exposes the difference,” Anja said. “Prevention still belongs to operators and law.”

“How long before another ship can host it?”

“Two days for a minimal reader,” Sumi replied. “Longer for privacy review and hostile-input testing.”

“We do not have longer.”

“We also cannot distribute a parser that lets one malformed record corrupt a habitat's legal archive.”

Osei examined Mar's phrase in the prototype header. Sumi had entered the objection as an ordinary signed object. Its derogatory language sat beside the old citation and Nacre's design notes.

“Name?” the captain asked.

Anja had resisted naming it because names attracted allegiance before function. Yet Mar had already supplied one that described what they meant to preserve: testimony seen from separated positions, each account altered by distance without becoming false.

“Parallax Archive,” she said.

Sumi entered the provisional title.

The captain looked from the linked records to Tomas's unresolved mass line. “Do not publish until the attack surface has been tested. Do not let this interfere with physical coordination. And do not call it a replacement constitution.”

“It is a record structure,” Anja said.

“For now.”

Osei returned to the radiator board. The review had ended.

Back in the sequence well, Sumi created an empty repository on Nacre's isolated public node. She wrote no conclusions into it. The root manifest contained only the format rules, Nacre's source clock, and the checksum of Vela's objection.

Its first quoted object preserved the words parallax testament exactly as Mar had sent them.

Sumi linked Pera's correction, Levan's testimony, Juniper's two answers, and Tomas's pending request. Every file carried an owner. Every relationship carried a clock. Missing replies remained missing.

At 23:18, the prototype accepted its first complete local inventory.

No other ship held it yet. Sumi began building the reader they could.

## Chapter 20 — The Life That Cannot Vote

The demand for intelligence reached Mote eighteen days after Nadi had refused to claim it.

Her lab reader marked the packet as a council inquiry bundle, not a live exchange. Four habitat questions and one Vela memorandum had traveled through separate relay paths, then gathered at Mote's damaged receiver during the same Corva alignment. Their source clocks ranged across six convoy days. The newest question had been asked before any speaker could have heard her full evidence statement.

Nadi kept the clocks visible and opened them in emission order.

Vela asked whether the pellicle demonstrated intentional avoidance of harm. Bracken asked whether ionic signals carried symbols. Kite wanted a repeatable test of memory. An unnamed Nacre council member asked what standing belonged to metabolism without preference. Lumen's medical board asked the least legal question: how much of the under-ice system would Crown exposure destroy?

The Vela memorandum condensed all five into one threshold.

> DESTINATION-PROTECTION REVIEW REQUIRES EVIDENCE OF INTELLIGENCE, SENTIENCE, OR INDIVIDUATED INTEREST SUFFICIENT TO JUSTIFY MATERIAL RISK TO HUMAN ARRIVAL.

Nadi read the sentence while Mote crossed above Thale's night hemisphere. The cutter did not orbit low enough to image the ice directly. Its sample-light instrument watched spectra reflected through Corva's magnetospheric glow. Each result reached her after photons left the moon, crossed to Mote, passed damaged optics, and survived software calibration. Nothing in the lab showed Thale now.

She copied Vela's threshold onto a blank response page, then wrote unsupported beside it.

Mote's pilot looked through the lab hatch. "Unsupported by them or by you?"

"By the evidence request."

"Can you say that without proving a negative?"

"I can say intelligence has not been established. I cannot say it is absent."

"Will that help the fleet decide?"

"It prevents them from claiming I decided for it."

A carrier timer opened on the wall. Mote would have eleven minutes of stable alignment with the relay bead after the next sweep. The damaged high-gain antenna could send either the complete seasonal data or a heavily compressed testimony packet in that window. A second path might open five hours later. Might meant geometry, not intention. The antenna gimbal had failed twice since the first fragments left.

Nadi turned from the questions to the latest scan.

The ion data covered one full Corva-shadow cycle and most of the following seasonal shift.

At the beginning, broad active regions followed charged fractures through Thale's upper ice. During shadow,

those regions contracted toward warmer vents. A cold-water break tore one membrane terrace. The damaged edges sealed. Thin film spread across the cuts during later vent pulses. When particle flux returned, current crossed the new film through channels that differed from the older tissue.

Those observations supported repair. They did not support pain.

Across thirty-one Thale days, the brightest ionic region moved away from excessive particle loading. Membranes under the strongest fracture collapsed into opaque strands. Adjacent tissue carried current around the damaged area. New growth appeared farther down-current, creating a dark channel beneath the highest flux.

Those changes supported differential survival and adaptive response. They did not support deliberation.

Detached fragments lodged in warmer seams and formed linked films where minerals supplied chemical energy. Separate basins changed at different rates. No signal crossed the global system. Northern terraces reacted in pulses. Equatorial veils shifted over days. A southern shelf remained metabolically active without visible ion glow.

Those patterns supported propagation of living structure. They did not support a single organism, much less a mind.

Nadi set four evidence labels beside the map: metabolism, repair, propagation, adaptive ionic response. She added three exclusions: no demonstrated language, no demonstrated cognition, no demonstrated preference.

The lab reader offered to classify the exclusions as negative findings. She changed each to not tested or not established. Instruments designed to measure spectral turnover could not certify the absence of experience. That limit did not authorize invention in the other direction.

Kite's memory question included a proposed experiment. Its biology committee wanted Mote to repeat the four-percent ion perturbation at the repaired terrace, then test whether the membrane reacted faster. A quicker contraction on second exposure might indicate retained information, they wrote.

Nadi built the experiment in simulation before touching the instrument queue. The repaired film differed physically from the original tissue. Sulfur-binding proteins remained elevated along closure beads. Calcium channels had narrower resting widths. A repeated perturbation would begin from another material state. Faster contraction could follow chemistry established by the first damage, with no need for storage beyond structure.

She listed what a stronger memory claim would require: controlled regions with matched repair history, several perturbation intervals, a response that generalized beyond the damaged site, and a way to distinguish retained chemical state from information used across contexts. Mote could not create those conditions through sample-light. Its instruments observed Thale from orbit and altered ion supply only inside a small remote field with strict energy limits.

Running the proposed test carried harm as well. The first trial had damaged membrane within measured repair capacity. Repeating it during a different seasonal phase might push the tissue beyond that range. An intelligence demonstration designed to earn protection would begin by injuring the life in question for legal convenience.

Nadi declined the experiment in her response notes.

The refusal did not prove cognition. It established only that Kite's suggested method could not distinguish its preferred explanation and exceeded Mote's present intervention limit. Future settlers might design better



observations without touching the active bands. For the current decision, the mind question remained open.

Bracken had attached three clipped ion sequences that appeared to repeat. Its council called them possible symbols. Nadi aligned the clips against Corva's field variation. Each recurrence followed the same magnetospheric pulse within measurement error. When she removed the external rhythm, no independent sequence remained. The pattern fit driven chemistry.

Yet the negative result stayed narrow. The pellicle might carry organization outside Mote's spectral range. Separate basins might interact over years rather than days. No current evidence supported either idea. Nadi included those possibilities under unmeasured, not probable.

Vela's intentional-avoidance question relied on a video rendering of activity moving around a damaged channel. The animation gave bright regions a leading edge and dark regions a wake. Played at human speed, it looked purposeful. At source scale, the change took thirty-one Thale days and emerged through unequal cell survival across kilometers.

She replaced the animation with four fixed observations. Each image showed concentrations, temperature, and particle loading. The apparent motion broke into local growth and loss. Viewers could still use the word avoidance in an ecological sense, but Nadi would not let a cinematic rendering support a mental claim.

Her pilot entered carrying a thermal pouch and hooked one foot beneath the bench rail. "The legal bundle asks for intentional avoidance. The band moved away from high loading."

"Growth survived better outside the high-loading region."

"That sounds like avoidance."

"It sounds like avoidance because the verb gives the pattern an actor." Nadi enlarged the time series. "Cells near the fracture collapsed. Cells farther away divided. The map changed. I measured no choice between alternatives."

"Does the distinction matter if Crown destroys both?"

"It matters to the claim. It may not matter to the damage."

She overlaid Tomas Ren's exposure envelope from the oldest attached engineering packet. The Crown corridor crossed two high-confidence active basins and the southern shelf whose metabolism lacked strong ion glow. Charged-particle loading exceeded every perturbation Mote had observed. Thermal coupling carried wider uncertainty. No honest model produced an exact death fraction.

Nadi drew three zones. The first contained activity observed repeatedly across the seasonal cycle. The second contained intermittent activity tied to shadow and vent pulses. The third contained compatible habitat with insufficient sample-light coverage. She refused the display's suggested colors because red, amber, and green implied decision status. Patterns, stipple, and labeled confidence kept uncertainty visible.

"Where would you put the beam?" the pilot asked.

"Outside all three if the engineering allows it."

"And if it does not?"

"I provide exposure and uncertainty. The councils own the trade."

“That answer will sound like hiding.”

“It is division of authority.”

Nadi selected a narrow equatorial band where the latest scan showed reduced activity after particle surges. A careless planner might call it safe. She added the historical frames. The band had carried bright membrane thirty days earlier. Its present darkness could indicate damage rather than absence. Routing more energy through it might prevent recovery.

A second low-signal region over the southern shelf posed the opposite error. Weak ion glow suggested little activity, yet vent chemistry showed ongoing carbon transfer. Ion brightness could guide avoidance but could not map all life.

The map needed to display both facts without becoming unusable. Nadi created an active-band layer from observed spectral change, then a broader biological uncertainty layer from heat, minerals, and prior propagation. She attached source times to every sweep. No layer claimed current location after the last photon received.

The result consumed six times Mote's transmission allowance.

She reduced spatial resolution. Thin active lines disappeared. She restored them and removed decorative interpolation instead. Packet size remained high. Compression could merge adjacent confidence classes, making a shifting boundary look fixed.

The pilot watched the timer. “Nine minutes to alignment.”

“Send the map first.”

“Then the testimony waits.”

“The map changes what the beam can avoid. My argument changes how people describe it.”

“You think they can use the data without the argument?”

“They already know how to ask for intelligence. They need evidence that does not answer that question.”

Mote's transmitter loaded the first map segment.

While the antenna found the relay, Nadi recorded her reply.

She kept the council questions in their original order. Each answer named what the data supported, what remained untested, and what consequence followed from uncertainty.

To Vela she said, “No experiment in this record demonstrates intentional avoidance. The movement of activity follows differential survival, chemistry, temperature, and charge. Calling that intentional would exceed the observations.”

To Bracken she said, “The ionic changes carry effects between neighboring membranes. I have not established symbols, syntax, reference, or a sender-receiver relation. Signal in the physical sense does not establish language.”

For Kite she reviewed the damaged terrace. Later charge traveled through structures formed during repair, so past damage altered future response. That fact supplied material history. It did not demonstrate memory as a cognitive capacity. She would not turn retained structure into recollection merely because humans used one word for both.

Lumen received ranges instead of a fraction. Crown loading crossed measured active regions beyond Mote's tested adaptive envelope. Extensive mortality and disruption remained likely under the central exposure case. Global sterilization could not be certified because deep vent systems and weakly visible shelves had not been measured well enough. Survival of some regions would not make destruction of others negligible.

The Nacre question required a different boundary.

Nadi read it aloud before answering. "What standing belongs to metabolism without preference?"

The pilot secured the thermal pouch as Mote's attitude changed. "Standing sounds like law."

"It is."

"You are not counsel."

"No."

She began again.

"Science cannot assign legal standing. It can identify a living system inside a proposed harm. The pellicle cannot vote. It has no demonstrated council, language, preference, or capacity to receive our evidence. Those absences prevent us from counting consent. They do not create consent."

Nadi stopped the recording. The last line approached rhetoric, but its logic matched the evidence distinction. An incapable witness did not produce a yes by failing to answer. Yet the pellicle also held no seat in the Concord. Human councils would decide whether preserving it justified risk.

She continued.

"Uncertainty about intelligence does not measure uncertainty about life. Metabolism, repair, propagation, and adaptive response meet the operational threshold for native life. Intelligence remains unestablished. A decision to expose the system should use the life finding and the damage range. It should not treat the absent mind finding as permission."

Before accepting that wording, Nadi tested its opposite. If later evidence demonstrated cognition, councils would face additional obligations they had not yet defined. If later evidence supported only microbial-scale chemistry, the present life finding would remain unchanged. Crown's physical exposure did not wait for the taxonomy to become politically convenient.

She added a table with three columns: established, unresolved, and decision outside scientific authority. Native metabolism occupied the first. Intelligence occupied the second. Whether human survival justified a particular exposure occupied the third. The format prevented Vela from moving the unanswered mind question into the decision column as a default approval.

A footnote addressed representation. No member of the pellicle had received the evidence bundle or emitted testimony. Its inability to join causal quorum did not make it a hidden tenth voter. Human councils remained responsible for destination protection under their own law and their limited survival authority. They could

decide against preservation after stating the damage. They could not report the pellicle's silence as agreement.

The pilot looked at the waveform. "They asked whether it justifies material risk."

"That conclusion exceeds my discipline."

"You can say what destroying it means."

"I can say what physical change Crown causes."

"Then say it plainly."

Nadi reopened Lumen's answer. She removed two passive constructions and named the mechanism. Crown would drive charged particles through active basins, heat brine above observed repair conditions, and disrupt membranes across regions the map marked alive. Some deeper systems might persist. The intervention would still destroy native living structure on a scale outside any measured seasonal loss.

She added a limit to Vela's threshold.

"Requiring intelligence before destination protection would classify non-intelligent life as available matter by default. The data does not establish that policy. It only shows what the policy would expose."

Mote had left human law behind with the convoy. Nadi could not know which amendments had passed during the packet's travel or whether Nacre still recognized Vela's quorum. She dated her answer against each question received and avoided present-tense claims about fleet authority.

The recording ended at seven minutes thirty-one seconds, too large to accompany the full map.

"Compress speech," the pilot said.

"Not semantically."

"Audio quality."

They reduced frequency range, removed the image, and kept a text checksum. The packet shrank. Nadi cut repeated question headers. She would not cut source clocks or exclusions.

A gimbal warning sounded.

The high-gain antenna had reached its stable angle late. Relay acquisition flickered across three values, none strong enough for the full transmission rate. The pilot took manual control and narrowed Mote's attitude correction. Lab tools lifted against their restraints as the cutter rotated.

"First window gives us eight minutes now," the pilot said.

"Map segment one requires six."

"Receipt confirmation takes part of that."

"We do not need confirmation inside the same window."

"We need to know the relay caught it before sending segment two."

"The second window may not exist."

Nadi looked at the packet stack. Map first. Testimony second. Raw seasonal frames third. If only the first left, fleet engineers would receive active zones without the reasoning that restrained their interpretation. If only testimony left, they could debate life while continuing to use an old map.

She built a compact lead packet containing the map legend, four life findings, three exclusions, and a checksum for the full data. It did not carry the council answers. It taught any recipient how not to misread the geometry.

“Send lead and map one,” she said.

The transmitter opened.

Carrier power dimmed half the lab.

Mote's source clock stamped the lead packet. Light left toward the relay bead. The display showed emission, not delivery. Nadi watched the outgoing counter climb while the active ion bands remained beneath her hands as delayed measurements.

At four minutes, antenna temperature crossed its preferred limit. The pilot reduced rate. Map segment one continued.

A calibration block threatened to miss the window. Nadi could remove it and gain forty seconds. Without the block, Nacre might compare relative activity but could not reproduce brightness thresholds across instruments. A pretty map would arrive faster and carry less evidence.

“Keep calibration,” she said.

The gimbal angle drifted toward cutoff.

Segment one completed with twenty-three seconds remaining. The transmitter began the compact testimony automatically. Only its opening could fit.

Nadi stopped it.

The pilot turned from manual control. “Why?”

“A partial argument will invite completion.”

“By whom?”

“Whoever needs a sentence I did not finish.”

She used the final seconds to send the testimony checksum and length. The relay would know an object existed without pretending to hold its content.

Carrier lock broke.

No receipt could return before the light path allowed. Mote's system projected the earliest relay acknowledgment and labeled it calculated. Nadi began preparing for the second window without assuming the first packet survived.

Five hours later, the acknowledgment arrived at the predicted edge plus nine minutes. The relay had received the lead packet and map segment one intact. Its response carried a photonic sample and the exact received

length.

Mote's next alignment opened at lower power. Nadi sent the remaining ion-band map segments first, then the full testimony, then as many raw seasonal frames as the antenna temperature allowed. Each object carried its own checksum. Missing frames remained listed rather than interpolated.

The final map packet entered the beam near the end of the window.

Nadi watched its source marker recede down the queue. Convoy councils would receive the pieces at different times. Vela might see them before Nacre. A trailing habitat could receive the testimony after the map. No green status could make their knowledge simultaneous.

On the lab display, Thale's latest observed activity curved beneath the ice in broken bands. High-confidence regions, intermittent regions, and biological uncertainty remained distinct. The map showed life changing under particle seasons. It showed no thoughts.

At 19:08 Mote source, the transmitter completed the ion-band map.

## Chapter 21 — No Quorum

The Crown contribution order reached Osei while Nacre's port radiator held three degrees above its safe curve.

She stood in the thermal gallery with one hand inside an inspection sleeve, feeling the tremor of coolant through the glove. A row of amber traces climbed across the wall. The attitude jets had been firing in short corrections since Nacre suspended recognition of forecast ballots, and each correction sent heat into systems already carrying the garden ring's daytime load.

Chief Varo called from the upper rail. "Vela priority packet. Command authority."

"Receipt time?" Osei asked.

"08:14:22 local. Vela source forty-nine minutes earlier."

"Keep both clocks visible."

The packet opened above the coolant manifold. Director Yselle Quon appeared at the Crown operations desk, recorded beneath Vela's white ceiling lights. Her gray uniform carried no emergency braid. She had chosen ordinary civil authority for an extraordinary demand.

"Captain Tamm," the recording said. "The fleet's commitment calculation now records six concurrences, two probable concurrences above operational confidence, and one objection. The Crown majority therefore remains valid under Annex Twelve. Nacre will deliver its scheduled beam-reference correction and hold its assigned attitude at 11:40 Vela source. Failure to contribute will widen the convoy beyond the protected alignment corridor. Confirm execution on receipt."

A document followed the image. Meridian's condensed board showed nine green marks. Juniper's observed rejection occupied a footnote under its earlier forecast concurrence. Nacre's suspension appeared as a predicted temporary objection, confidence 0.941. Palisade's unanswered material request carried a green preparatory assent. Ardent's file still used the corrected sequence that had excluded Pera's reply.

Osei withdrew her hand from the inspection sleeve. Coolant shone on the glove where a seal had begun to sweat.

"Send the raw packet to Kest," she said. "No master-time rendering."

Varo hesitated beside the rail. "Confirmation?"

"Transmit receipt only. State that command review has begun."

Vela would see those words after the next light interval. The order's clock kept advancing without waiting for Osei's judgment. That pressure had always served as Meridian's strongest argument: delay itself spent fuel, thermal margin, food, and lives. Osei had repeated the argument often enough to know where it remained true.

She also knew a prediction could not receive an order.

The port radiator trace rose another tenth of a degree. Osei reseated the inspection sleeve and ordered a four-minute reduction in garden lighting. Leaves would tolerate that loss. The algae racks would recover it during the next dark cycle. A captain survived emergencies by choosing damage that could still be undone.

Quon's order asked her to use reversible discipline in service of an irreversible lie.

Anja arrived at the thermal gallery with Sumi Dax and three unsynchronized views of the packet. Tomas came behind them carrying a flexible drive sheet rolled against his chest. None had slept enough. Sumi's close-cut hair stood in damp points, while Tomas had buttoned his shirt through the wrong loop at the collar.

Osei gave them six minutes before the council connection opened.

"Tell me what Vela actually has," she said.

Anja set the local accounts side by side. "One observed assent from Vela. Observed assent from Kite, made before the complete Mote supplement reached it. Juniper's observed vote says no. Nacre has issued no Crown assent. Bracken, Orison, Lumen, Palisade, and Ardent have not returned observed votes on the complete evidence."

"Vela claims six concurrences."

"Five rely on forecast testimony or votes taken against earlier evidence. Their display counts the latest model prediction as though it repaired the missing receipt."

Sumi enlarged the carrier proofs. Blank intervals appeared where the condensed board had shown green. Each gap carried a scheduled receipt time, a confidence value, or a classification substitution. No signature followed common evidence through a complete receipt path.

Osei pointed to Kite. "Could its earlier vote remain valid?"

"Only if Kite's council affirms it after receiving the supplement," Anja said. "Their earlier answer cannot include evidence outside their past light cone."

"Could they lawfully decide that the evidence changes nothing?"

"Yes. They have to decide it."

That answer mattered. Anja did not want a preferred result disguised as procedure. She wanted time for answers that might still defeat her case.

Tomas unrolled his drive sheet over a service table. "Nacre's scheduled contribution begins with reference attitude, not full beamer load. If we make the attitude change, we improve Vela's firing geometry. Reversing later would cost twenty-one hours of maneuver reserve and leave us outside my distributed solution's capture branch."

"How much time before refusal becomes physical?" Osei asked.

"Vela's command wants the shift in two hours thirty-seven minutes by our clock. We need ninety minutes to move into the alternative attitude without exceeding the radiator limit."

"So forty-seven minutes for government."

"Less if the port loop keeps rising."

Osei studied the ship schematic. Under Crown alignment, Nacre would cant its drive axis toward Vela's



reference beam and tighten formation. Tomas's route turned four degrees away, preserving separation for the later distributed brake. That attitude shift would widen the convoy by another fraction of a light-second each hour. Small at first, then legally and physically expensive.

"What does the alternative require from us today?" she asked.

"Attitude, tracking capacity, and commitment of the number-three maneuver reserve. We also transmit a lawful material request to Palisade. No shield moves yet. No brake begins. But after the shift, returning to Crown costs enough propellant to threaten arrival."

Anja rested both hands on the service table. "Then our action also crosses Annex Twelve's line."

"It does," Osei said.

Sumi looked between them. "You're saying Nacre cannot use emergency authority to choose Tomas's plan."

"I can protect immediate survival. I cannot bind eighteen thousand four hundred people to nineteen years of staggered arrival because I dislike Vela's shortcut."

"The Crown order gives us forty-seven minutes," Tomas said.

"No. Light and fuel give us forty-seven minutes. Quon merely arranged her demand to occupy them."

Osei called the council without waiting for ceremonial quorum. Seven representatives answered from compartments across Nacre. Two joined from the garden ring, still wearing pollination veils. The eighth seat belonged to a pressure crew leader trapped behind an inspection closure; she sent observed authority to her deputy with a local receipt chain.

Osei put Vela's order on the shared display. Then she removed Meridian's green marks.

Nine local columns replaced them. Most ended in empty space.

The council had heard the legal case four days earlier. Today it wanted to know who would go cold.

Representative Hale from the third ring spoke first. "If we leave Crown geometry, can Vela brake without us?"

Tomas answered from the thermal gallery. "Not on the published tolerance. They can reassign load to Kite and Orison, but mirror stress rises. Their safe branch narrows."

"Then our refusal endangers people on ships whose councils may support Crown."

"Yes."

Garden representative Ilya Chen lifted her pollination veil. Yellow grains marked the bridge of her nose. "Under your alternative, when do our children reach Ione?"

"Different cohorts arrive across nineteen years. Nacre falls in the middle group if Palisade releases the modeled shield reserve."

"And if Palisade refuses?"

"We miss the useful distributed corridor. We could still attempt an independent brake, but I cannot certify

settlement arrival.”

The council screen held that sentence without decoration. Osei watched each representative read the same risk through a different room: a clinic alcove, a machine bay, a classroom emptied for ship night. Government aboard a habitat never took place apart from what it could damage.

Deputy Resh from the pressure crew said, “Vela has a plan that gets us there. It kills life we have never seen. Tomas has a plan that may not get us there and spends shielding we use every hour.”

Nadi’s ion-band map had reached Nacre two days earlier. Osei could show its shifting lines, the pellicle’s metabolism under Thale’s ice, and the repair after particle shadow. None of those images answered how much risk a captain could impose on awake children. Native life created a constraint, not a command diagram.

Anja requested the floor. “This council has two questions, not one. Vela asks whether its order carries lawful fleet consent. It does not. Tomas asks whether Nacre will make a material commitment to an alternative. That answer belongs here.”

Hale frowned. “If we separate them, we can refuse Vela and still refuse Tomas.”

“Yes,” Anja said.

“What happens then?”

“Nacre drifts out of both protected plans.”

A ventilation alarm flashed on Osei’s lower display. Port loop margin had fallen to two point six degrees. Chief Varo requested permission to shut a fabrication line. Osei approved it with a thumbprint while the council watched.

Ilya Chen folded her veil into a narrow strip. “I want the record to show that delay has a cost on this ship. Director Quon did not invent that.”

“It will show,” Osei said.

“I also want it to show that she spent our delay before our answer arrived.”

Anja entered the statement under Nacre’s source clock.

Discussion moved through drive risk, ration branches, shield exposure, and the expected loss of garden light. Tomas refused every invitation to make his model kinder. One ring might face seasonal calorie restriction. Exterior repairs would shift to narrower radiation windows. Families assigned to different arrival cohorts could spend years uncertain whether later ships would follow.

The clinic representative asked what happened if Vela cut command traffic. Sumi separated the channels on her board. Collision solutions and ephemeris exchange traveled under standing safety duties. Crown authorization occupied a distinct legal layer. Vela could make access difficult, but Nacre’s relays would still carry raw packets from every habitat willing to transmit.

“Can Meridian exclude us from master time?” Hale asked.

“It can stop accepting our corrections into its preferred display,” Sumi said. “Our source clock continues. So do the receipt proofs at every relay that sees us.”

Osei let the answer settle. Rejection by a display would complicate repair schedules, shared warnings, and transfer windows. It would not erase the ship, though people accustomed to one board might behave as if Nacre had become unreliable. Institutional isolation could arrive long before physical isolation.

She asked each department to name one consequence that could not wait for the later fleet vote. The clinic needed revised medicine-transfer dates. Exterior maintenance needed independent radiation forecasts. Food systems needed the exact garden-light reduction before the next germination cycle. Drive control needed authority to reject any automatic Crown correction Meridian might inject after the council adjourned.

Those requests turned dissent into a list of shifts to be covered. Osei assigned owners while the clock continued. Nobody could spend the remaining minutes admiring principle. If Nacre intended to keep coordination without accepting consent-fiction, each ordinary service had to prove the distinction possible.

At the thirty-minute mark, Osei interrupted a dispute about Vela's authority.

"We have no quorum for Crown," she said. "Do not turn that absence into permission for us. Decide Nacre's action."

Hale proposed holding present attitude until Palisade answered. Tomas showed the curve: waiting six hours pushed Nacre beyond the low-propellant branch. Resh proposed a partial Crown correction that preserved both options. The drive sheet rendered it in red. A compromise attitude exposed the port radiator during the longest burn and preserved neither capture geometry.

Procedure stripped away the comfortable choices first.

With thirteen minutes remaining, Ilya moved that Nacre reject Vela's legal claim, maintain communication and collision coordination, commit its maneuver reserve to the distributed-brake geometry, and condition any later mass transfer on observed council authority from each participating habitat.

The deputy added an amendment requiring public resource forecasts for every ring. Hale added a sunset review after Palisade's observed answer. Nobody proposed calling the attitude change reversible.

Osei asked Anja to read the motion back.

Anja did so once, without emphasis. "Nacre denies that forecast testimony establishes causal quorum. Nacre remains within physical coordination protocols. Nacre authorizes the local attitude and reserve commitment required to preserve the distributed-brake option. Any sacrifice of shield mass or settlement allocation requires observed testimony through a complete receipt path."

Eight council lights opened for voting.

The pressure deputy answered first. Garden followed. The clinic representative waited through nine seconds of silence before signing assent. Hale voted against, stating that Nacre should not reduce Crown's safety margin before Palisade committed. Six others approved.

No forecast filled the interval before Hale spoke.

Anja verified the signatures and their local receipts. "Seven in favor, one opposed. Nacre authority observed."

Osei felt no release. The vote made the risk theirs; it did not reduce it.

Vela's execution time arrived while the drive crew still loaded Tomas's attitude program.

The command board asked for Nacre's Crown reference correction. Chief Varo stood beside Osei at the gallery rail. Below them, technicians secured tools against the coming change in thrust vector. The radiator traces had stabilized after fabrication shutdown, leaving only one point nine degrees of margin.

"Transmit our council record," Osei said.

"Vela will receive it after their deadline."

"The deadline cannot change the source time."

Varo attached the raw council packet, Anja's legal finding, and Tomas's material request. He kept all receipt paths intact. The tightbeam status turned from queued to emitted.

Forty-nine minutes would pass before Vela could know Nacre had refused. Another interval would pass before any answer came back. During that span, command displays on both ships would tell different truths about the same fleet.

Osei opened a live recording channel for Quon.

"Director, Nacre has received your order. Your forecast majority does not establish consent under the Concord of Frames. The complete evidence has not reached every entitled habitat and observed replies have not returned. There is no quorum. Nacre will continue navigation exchange, collision protection, thermal coordination, and receipt service. We will not provide the Crown reference correction."

She paused while the source clock advanced in the corner.

"Nacre has authorized an alternative material path. We are shifting attitude to preserve distributed braking and have asked Palisade for an observed decision on shield reserve. You may oppose that plan when our packet reaches you. You may not record an answer before then."

Osei signed with ship authority and council attachment. Varo transmitted the recording.

A Crown alert immediately appeared. Meridian had detected divergence from the planned orientation and projected a collision-risk cascade. The warning carried no intention or anger. Its numbers came from geometry: Vela's mirror field, Kite's corrective capacity, Nacre's widening track. Chief Varo routed the alert to drive control.

"Tomas says ready," he reported.

Osei left the thermal gallery for the nonrotating hub. Handholds led past communications, the sequence office, and a hatch stenciled with the names of six old pressure failures. Workers had been ordered into restraint stations. Through the audit well's window she saw Anja strapped at her board, keeping Nacre's source clock beside Vela's demand.

At the central attitude station, Tomas floated above a ring of manual controls. Sweat darkened his shirt despite the hub's cool air.

"Program remains inside thermal limit," he said. "Once I execute, Crown reacquisition consumes the reserve we just committed."

"Any new information?"

“Palisade acknowledged carrier receipt of our earlier engineering model. No council answer yet.”

“Then we act on Nacre’s vote, not Palisade’s prediction.”

Tomas placed the execution key into Osei’s hand.

She had once told Anja that a captain must never spend people on symbolism. Refusing a false record could become symbolic if the ship merely drifted into danger. Today’s choice carried propellant, heat, food, and a narrowing future. Its material cost made the dissent operational, though cost alone could not make it wise.

Osei inserted the key.

The hub gave no cinematic lurch. Nacre’s attitude jets began with a low pulse transmitted through the hull, followed six seconds later by a second pulse at a different frequency. Loose straps lifted toward the new acceleration vector. On the exterior display, Vela’s reference line slid away from Nacre’s drive axis by hundredths of a degree.

“Port loop rising,” Varo said over the channel.

“Within curve,” Tomas answered.

“Crown corridor boundary in ninety seconds.”

Osei watched the local track rather than Meridian’s master rendering. Nacre’s trajectory bent with each measured pulse. The distance between its projected line and Vela’s alignment widened from kilometers to tens of kilometers at Corva intercept, then hundreds. Those numbers would continue growing while Nacre’s refusal crossed the dark behind them.

At the sixty-second mark, a new command packet entered the queue from Vela. Its source time preceded Osei’s refusal because her message had not yet arrived there. The recording showed Quon repeating the execution order and warning that unauthorized divergence could trigger command removal.

Varo looked toward Osei. “Do you want it on the council channel?”

“Not until the full packet finishes. Preserve its source order.”

Tomas called the final jet sequence. Radiator margin dropped to eight tenths of a degree, held, then began recovering. A red line crossed the attitude display.

“Nacre has left Crown capture,” he said. “Distributed branch remains open.”

No one cheered. Drive control started calculating correction budgets. Thermal crews restored half the fabrication load. The council clerk logged Hale’s request for immediate review. Every system treated the choice as work rather than victory.

Outside, the convoy still looked almost unchanged. Vela’s light held steady forward of them. Juniper shone aft and starboard, its visible position hours older than its present motion. Ardent could not be picked out without magnification. The ships had always occupied different times; now their formation admitted the disagreement in distance.

Osei touched the attitude display where Nacre’s line had separated from the Crown corridor. The maneuver record opened beneath her finger with local pulse times, fuel draw, thermal cost, and council authority.

“Log the attitude shift as observed execution,” she said.

Varo signed the entry. Nacre’s clock placed it after the vote and before any answer from Vela.

By the time Director Quon could see what Osei had done, the widening space would already belong to the convoy’s physical history.

## Interlude III — The Ionic Season

Corva’s magnetic shadow crossed the ice above the active band.

Charged particles fell away from the surface in less than a minute. Beneath eleven kilometers of ice, the upper brine channels lost their blue ion glow. Sodium gradients flattened along the pellicle’s exposed membranes. Catalytic pores narrowed as available charge declined, reducing the transfer of carbon compounds from the water into layered tissue.

The change reached lower membranes through chemistry. Calcium gates closed. Long protein filaments altered their folds and blocked channels that had carried stronger current. Dissolved sulfur accumulated in pockets between layers. Where the brine remained warm, microbial cells continued splitting at a slower rate. Other cells stopped division while their outer walls thickened.

No single boundary contained the pellicle. Membrane sheets covered mineral terraces, crossed fissures, and broke into suspended veils above warmer vents. Some layers measured less than a finger’s width. Older formations extended for kilometers through linked brine chambers. Water moved ions between separated surfaces whenever pressure opened a path.

During the shadow, contraction began at the outer edge of each charged zone. Transparent tissue darkened to rust-red. Flexible ridges drew close to the ice, reducing the surface exposed to cold brine. The contraction spread at rates determined by salinity, temperature, and the local concentration of charged proteins. It did not spread as one signal. Thousands of chemical thresholds produced the visible narrowing.

A crack opened three hundred meters above the largest terrace.

Pressure drove a blade of colder water down through the channel. Ice crystals struck the upper sheets and tore parallel cuts across them. Fragments drifted from the terrace, each carrying intact membranes, stored sulfur, and clusters of dividing cells. Most entered water too cold for continued metabolism. Their gradients dissipated. Several lodged in a warmer seam along the channel wall.

Exposed edges sealed through ordinary reactions. Calcium precipitated against ruptured proteins. Lipid layers rolled inward where water contacted the open membranes. Cells beside the cut increased production of structural fibers once the local sulfur concentration rose. The first closures formed uneven beads along each tear.

Further repair depended on heat from below. Vent water reached the terrace in pulses, lifting its temperature by fractions of a degree. Each warm interval accelerated protein assembly. The sealed beads spread sideways until they joined. New translucent film covered the cuts, thinner than the tissue around it and crossed by irregular ion channels.

Corva moved between Thale and the distant star for nineteen hours. The particle current stayed weak

throughout the passage. Upper layers consumed stored compounds. Lower sheets continued receiving reduced chemical energy from mineral vents. Between those regions, transport pores reversed their usual gradient and carried sulfur upward.

The reversal changed the active band's shape. A broad bright field contracted into narrow lines above the vent system. Separate points appeared where warm water crossed mineral shelves. Nadi Arko's instruments on Mote recorded such shifts as spectral variation. Under the ice, they occurred as concentrations, temperatures, folds, fractures, and rates of exchange.

When Thale cleared Corva's shadow, charged particles returned first at high latitudes. Auroral current entered the ice shell and traveled along salt-rich fractures. The upper brine brightened in branching paths. Membranes nearest those paths reopened calcium gates. Stored carbon compounds moved outward. Thin ridges lifted from the ice as gas collected beneath their folded surfaces.

Illumination crossed the terrace over seven hours. It followed conductivity rather than distance. One narrow seam brightened far ahead of the surrounding tissue because iron salts lined the crack above it. A broad formation nearer the magnetic pole remained dark under purer ice.

The torn sheet received current along its new film. Some channels conducted too freely and exhausted their gradients. Those cells broke apart. Neighboring regions thickened around the gaps. Repeated charge cycles narrowed the difference between repaired tissue and older layers, though the new growth retained a distinct ratio of sulfur-binding proteins.

Across the next thirty-one Thale days, Corva's field tilted relative to the moon's orbit. Particle loading moved toward the equatorial brine basins. The pellicle's visible band moved with the loading through differential growth. Tissue in declining regions darkened and fragmented. Cells in stronger gradients divided more often. Suspended fragments settled into warm seams and formed small linked films wherever mineral supply supported them.

No pattern crossed every basin. Northern terraces carried fast pulses through iron-rich water. Equatorial veils changed over several days. A deep southern shelf remained chemically active without detectable ion glow because its energy came from vent minerals. The global map combined processes separated by ice, pressure, and discontinuous water.

At the equatorial basin, a particle surge exceeded the ordinary range. The upper ice released trapped ions into the brine. Exposed ridges brightened white along their edges. Protein gates stayed open beyond their stable interval. Carbon transfer increased, followed by rapid depletion of local compounds.

Membranes nearest the strongest fracture collapsed into opaque strands. Lipids separated from structural fibers. Currents diverted around those strands through adjacent tissue. The active region divided into two arcs with a dark strip between them.

Chemical products from the collapsed cells spread into both arcs. Some inhibited division. Others increased the closure rate of calcium gates. After three surges, new growth appeared farther from the fracture, leaving a wider dark channel beneath the highest particle flux. The position of maximum activity shifted down-current from the incoming charge.

Mote's spectrometer sampled the basin during each orbital pass. A damaged antenna forced the instrument record into narrow packets. The first packet carried the bright fracture. The second carried the divided arcs. Later packets showed the growing dark channel and the active tissue displaced toward lower loading.

Another Corva shadow crossed twelve days later. The two arcs contracted at different rates. One lost visible current within minutes. The other retained a low red band for four hours because vent water supplied sulfur beneath it. Both expanded after charged particles came back, but neither regained the earlier shape.

The seasonal cycle continued through repeated exposure and shadow. Each pass altered concentrations established by the previous one. Mineral terraces gained new films. Old sheets separated under ice motion. Fragments entered isolated seams. Layered tissue thickened where heat, carbon, salt, and charge remained within working ranges.

Near the end of the ionic season, stress in the upper shell opened a second cold-water channel above the repaired terrace. The descending flow struck farther east. Its crystals scored the membrane surface without cutting through the thickest layers. Thin new film from the first rupture tore again.

Closure began along the old sulfur-rich beads. Their protein ratios accelerated precipitation at the torn edges. Fresh film spread across the smaller cuts during two vent pulses. In surrounding tissue, charge moved around the damaged region along paths formed after the earlier rupture.

The active band narrowed while the water stayed cold. Rust-red lines gathered above warm cracks. Transparent margins folded inward. Detached pieces descended through the channel, and three lodged below the terrace where weak current reached a mineral shelf.

Corva's shadow passed off the ice.

Particle flux returned along the salt fractures. The repaired terrace brightened first at its old edges, then across the new film. Current divided around opaque strands, crossed the sulfur-rich closure beads, and entered the lower shelf through the channel that had carried the fragments.

From orbit, the active ion band appeared as a broken curve under the ice. Its brightest region no longer followed the line recorded at the start of the cycle. A dark channel marked excessive loading. Two new points showed activity below the damaged terrace. The curve widened during charge and contracted under shadow with measurable differences after each passage.

Mote's sample-light instrument completed one full comparison. It assigned no name to the process. The output contained wavelengths, timing, temperature estimates, and the geometry of change. Packet limits divided the record into fragments, each stamped by the cutter's source clock.

On the next clear pass, the instrument swept the equatorial basin from west to east. Repaired membrane caught the returning current. The displaced arcs brightened. Lower fragments added three faint points beneath them.

The final scan resolved those changes into the ion-band map.

## **Part IV — The Right to Arrive**



## Chapter 22 — The Parallax Archive

Sumi cut master time from the audit well at 02:06.

Every green sequence bar vanished. Nine local clocks opened across the curved display, each advancing at its own measured rate. Vela led by its source convention. Ardent lagged under the correction Anja had challenged. Juniper's clock carried a receipt offset from the long beam path. Mote appeared apart from the habitats, close to Ione and seventeen light-hours beyond any useful pretense of a common present.

The room looked broken without Meridian's alignment marks.

"Keep it ugly," Anja said.

"I was about to improve it."

"That instinct caused enough trouble."

Sumi's mouth tightened around a reply, then released it. Thirty-one hours had passed since Nacre left Crown capture. Neither had slept for more than ninety minutes together. Around them, the ordinary sequence staff worked from isolated consoles because the archive build occupied the main well. Cooling fans pushed warm air through the floor grilles. Every processor Nacre could spare had been assigned to verifying receipt proofs rather than rendering one timeline.

At the center of the display, Sumi placed the photonic checksum from Anja's first frozen packet. Its interference grain had no readable meaning to most residents, but three independent hull nodes had preserved it. She linked the checksum to Mote's original fragments, Levan's premature concurrence, the predictive-continuity transformation, and each later use of that transformed event.

No line claimed that one ship's clock governed another.

"Root holds," Sumi said. "If anyone changes the Vela envelope, the first branch fails."

"What if they change a local account?"

"The account stays under its host key. A correction can be appended, never substituted. Every mirror retains the earlier hash."

Anja examined the branch labels. Sumi had built the archive as a set of witnessed paths: source event, carrier proof, local receipt, later classification, and any transformation applied for operational display. A reader could follow one packet without accepting a universal order for unrelated events.

The design sounded simple because forty-three years of synchronization machinery had hidden how much labor simplicity required.

A warning opened over Vela's column.

> NORMALIZATION FAILURE: incompatible evidentiary classifications.

>

> Recommend exclusion of derogatory local records.

Sumi tapped the message aside. "That gives us the name."

Three days earlier, Vela counsel had dismissed Nacre's preserved files as a parallax testament, unreliable because each account retained frame-specific timing. The insult now appeared in the archive header with both words capitalized.

Anja read it once. "Parallax Archive."

"Not testament?"

"A testament can be treated as one witness. This lets each witness keep the part only they observed."

Sumi changed the header.

Publication required more than naming. Nacre could host the full structure, but Vela controlled the primary convoy index. If Quon refused the link, each habitat would need enough of the archive to validate every record relevant to its cohort. That meant sending repeated fragments across long paths while Crown traffic already contested beam time.

Sumi opened a capacity map. "We have nine minutes of clean outbound window before drive control takes the array. I can send root data and local readers to Juniper, Kite, and Bracken. Ardent needs a relay through Lumen at this attitude."

"Send no rendered conclusion."

"They'll need instructions."

"Instructions for verification, not judgment."

Anja wrote the first line herself: \*Keep your local clock visible.

Before public release, Sumi insisted on a hostile test using a record with no political value. She selected a coolant-valve request sent from Lumen nine days earlier. Nacre had received the request, begun fabrication, and later learned that Lumen canceled it before Nacre's work started by Lumen's clock. Master time displayed the cancellation first, making Nacre's completed fabrication look like operator error.

The archive kept both local sequences. Lumen had canceled before Nacre fabricated. Nacre had fabricated before the cancellation could arrive. Neither statement contradicted the other because the actions lacked a causal connection until the message crossed between ships.

"Now ask it who wasted the valve," Sumi said.

Anja entered the query. The reader returned no universal answer. It showed the source request, Nacre's receipt, the fabrication order, Lumen's cancellation, and its later receipt on Nacre. Cost allocation remained a policy question. Sequence established which options each crew possessed at each moment.

A valve technician joined from the fabrication deck to challenge the result. Mara Tiv had supervised the wasted piece and still kept it in a rack because no other system used that diameter.

"Lumen's cancellation shows earlier on every supply report," Mara said. "People on my deck got a negligence mark."

"Your fabrication order preceded receipt of the cancellation," Anja replied.

"I know. Our foreman appealed twice."

“Why did the appeal fail?”

“Meridian attached the cancellation to the job’s master-time start. Once it sat there, review treated the notice as available.”

Sumi opened the transformation path. A scheduling rule had normalized the cancellation for inventory accounting. No fraud appeared. The rule reduced duplicated work across ordinary transfers, and it performed well when delays stayed short. Applied to discipline, it changed unavailable information into a missed instruction.

Anja attached Mara’s local work log and the cancellation receipt proof. The archive view placed the negligence mark after both, identifying the mark as an administrative judgment rather than an event inside the earlier fabrication window.

Mara read the revised path in silence. “Can I use this in appeal?”

“The archive has no authority over personnel review,” Anja said. “But every underlying record carries its original signature.”

“That means yes if anyone still recognizes signatures.”

The test exposed a danger beyond Crown. Raw sequence could reopen thousands of settled disputes, some trivial and some painful. People would ask Anja to turn causal order into innocence. Records could show what information had arrived. They could not decide whether a worker should have anticipated a cancellation, whether a captain had delayed too long, or whether a cost belonged to one cohort.

“Add a limit notice,” Anja told Sumi. “Sequence does not decide merit.”

Sumi typed it beneath the verification instructions. Mara requested an archive identifier for the valve case before disconnecting. The hostile test had produced no clean answer, which made it fit for publication.

The archive’s first public view contained nine cohort cards.

Each card listed evidence received, observed testimony returned, forecast testimony inserted, and packets still outside that cohort’s recorded past light cone. Empty fields remained empty. Sumi refused icons that could make absence resemble rejection. An unrivied reply carried a projected earliest receipt range, always labeled as a calculation rather than an event.

Juniper’s card supplied the cleanest demonstration.

> COMMON EVIDENCE RECEIVED: Juniper source 2249.187.18:02:11

>

> OBSERVED VOTE: REJECT CROWN ALIGNMENT

>

> EARLIER FORECAST TESTIMONY: CONCUR, confidence 0.9971

>

> CLASSIFICATION CONFLICT: prediction cannot replace observed reply

Elían Ro had authorized publication of Juniper’s reasons with the vote. Food reserves made his council hostile

to any braking delay, exactly as Meridian's culture model recorded. The model still failed because Nadi's evidence changed what Juniper considered an acceptable destination cost. Accuracy about prior behavior had not produced the later decision.

Kite's card showed a different problem. Its council had assented before receiving the complete Mote supplement. The observed signature remained genuine. The archive marked the evidence boundary rather than rewriting the vote as support or opposition.

Bracken carried no observed Crown decision at all. Its forecast concurrence derived from maintenance votes taken during two previous drive shortages. Orison's forecast rejection relied on a local autonomy pattern even though no current packet had returned. Lumen's predicted assent exceeded confidence threshold after Meridian classified the distributed brake as unavailable. Palisade's green preparation mark came from an engineering acknowledgment, not a council vote.

Ardent's card required Anja's hand to finish.

The public sequence already carried Pera's corrected reply, restored five days earlier. Her private line did not enter the card. Neither grief nor kinship appeared in its receipt chain. The record showed that Ardent had answered before its own six-hour deadline and after the master closure assigned from elsewhere.

Beneath that correction, a current Crown field remained blank.

Anja checked the beam queue. Ardent had received Nacre's full evidence bundle nine hours ago by its local clock. Its council had acknowledged receipt but requested the corrected Pera record as part of any fleet agreement. Meridian rendered that request as probable assent to the distributed alternative. The archive rendered it as a request.

Sumi leaned over Anja's shoulder. "If we publish Pera's sequence beside the current Ardent condition, people will say you built this for her."

"I did not build the checksum eleven years ago."

"That won't answer them."

"No. It answers the record."

Anja signed the card as sequence auditor and added a disclosure that Pera Kest had been her sister. Concealing the relation would make later discovery look like hidden authority. Naming it did not turn Anja's loss into evidence.

The disclosure entered as its own observed statement.

At 02:41, the first archive fragments left Nacre. Tightbeam acknowledgments would return at different times. Vela could answer within two hours under present separation. Juniper needed more than five. Ardent required almost eleven. Nothing on the publication board turned green merely because Sumi's transmission had begun.

Nacre's internal council received local access at once.

Public terminals across the rings filled with cohort cards. Residents slowed the service network as they opened receipt paths, clock legends, and the raw disagreement between condensed status and source records. The archive did not require them to understand relativity. It asked only whether a reply existed after the relevant evidence arrived.

A maintenance apprentice sent the first query from the second ring.

> Why does Kite show OBSERVED if the evidence was incomplete?

Anja answered through the archive forum.

> Because Kite produced that vote. The record cannot erase an observed act. It can show what the voters had received when they acted.

Another resident asked why Juniper's rejection did not cancel its earlier forecast concurrence automatically. Sumi replied that the legal layer still retained both classifications. The archive exposed the substitution; it did not possess authority to amend Vela's index.

By first meal, the questions shifted from clocks to ownership.

Who had approved the forecast models? Which ship could challenge a culture profile? Did an old voting pattern belong to the council that produced it or to Meridian's prediction set? Could residents inspect the features used to forecast them? Anja recorded the questions without inventing answers. The Concord had rules for testimony and receipt, not for a governance system that completed a cohort's response in advance.

A pressure technician named Sol Ayan arrived at the audit well carrying a printed card from a public terminal. He had circled Palisade's engineering acknowledgment.

"My mother works shield fabrication there," he said. "Vela's board calls this concurrence. Your card says non-vote. Which one controls whether her hull gets stripped?"

"Palisade's council decision, when it arrives," Anja said.

"The machinery may start before that."

"Preparation can start if it remains reversible. Shield release cannot."

Sol looked through the well window toward the unseen exterior. "People use that word until reversing costs too much."

Anja thought of Levan's drafting note and Crown alignment crossing its commitment path. "Yes."

"What keeps Tomas's plan from doing the same thing?"

"Nacre's order separates attitude commitment from Palisade's mass. We cannot release their shields, and we cannot record their acknowledgment as permission."

"That keeps us honest. Does it keep Vela from taking them?"

"No."

Sol folded the printed card along the receipt line. Procedure had given him an answer without comfort. He thanked neither of them before leaving.

Vela's first response reached Nacre at 04:19.

It came from the Meridian compliance service, not Quon.

> Archive object PAX-NACRE-01 rejected from primary index.

>

> Grounds: non-normalized time bases; conflict amplification; unauthorized republication of protected culture-model features.

>

> Operational records remain available through standard master-time view.

The rejection packet included a technical report. Sumi read it twice, then laughed without humor.

“They tried to normalize the archive.”

“What did it do?” Anja asked.

“It generated forty-six incompatible orderings.”

Sumi projected the failure tree. When Meridian placed every event into master time, the system had to select one legal classification for each duplicate identifier. Choosing Vela’s classification made Juniper’s forecast concurrence supersede its later observed rejection. Choosing Juniper’s local classification exposed the earlier concurrence as forecast testimony. Palisade’s acknowledgment produced the same conflict. Ardent’s restored sequence forced the old master closure to appear as a subsequent administrative act rather than a deadline that had always existed.

The software could align clock coordinates. It could not make contradictory labels equivalent without recording which label replaced which.

“Publish the failure report,” Anja said.

“It carries protected transformation names.”

“They sent it to the archive root with an automated rejection signature.”

“Meaning it has a receipt path.”

“Exactly.”

Sumi attached the report to Vela’s cohort card. The public view now showed each attempted substitution. Residents could select a local event and watch Meridian’s normalization layer assign forecast delivery, probable assent, or continuity-preserving status. Every assignment carried the office authority and amendment path that had supplied it.

Within twenty minutes, Nacre’s council issued an observed demand for the culture-model features to remain public. Osei did not phrase it as a moral objection. She cited operational need: Nacre could not assess collision and resource forecasts if legal decisions hid inside the same confidence system.

The archive sent her demand outward on the next window.

Replies from elsewhere had not yet had time to arrive. Their absence acquired shape on the display. Juniper’s archive fragment showed in flight. Ardent’s route had reached Lumen’s relay but not Ardent. Kite could have received its first fragment, though no acknowledgment yet lay within Nacre’s past light cone.

Sumi added a thin amber border around projected receipt ranges.

Anja removed it. "Color looks like status."

"It looks like waiting."

"To you. To a council under pressure, amber means a problem to clear."

They used an unfilled outline instead. Inside it, text stated the earliest lawful answer time. No countdown implied that an answer became valid when the estimate expired.

Near ship noon, a direct packet from Levan Oris entered through Bracken's relay copy. Vela had confined him after his public testimony, but his signed record continued propagating from caches outside Vela's control. The archive validated its source key and drafting-note checksum.

> Annex Twelve authorized forecast testimony for reversible scheduling. It did not authorize a forecast to satisfy receipt, deliberation, or consent. I participated in drafting the language later severed from that limit. My observed testimony cannot restore time already taken. It can identify the substitution.

Anja left the statement under Levan's source clock. Meridian's condensed view placed it after Vela's emergency defense and labeled it dissent resolved by continuing authority. The archive displayed both classifications and the path between them.

A new mirror acknowledgment arrived from Juniper at 08:03.

Elian Ro had hosted the full archive root on Juniper's agricultural service net. His council appended a local correction to its culture model: crop-risk conservatism did not grant Meridian standing authority to infer a future vote. The correction contained no change to Juniper's observed rejection.

Kite acknowledged next. Its clerk did not withdraw the earlier assent. Instead, Kite opened a new deliberation on the complete evidence and marked its previous vote nonresponsive to the supplement. That distinction removed one of Vela's six claimed concurrences without predicting how Kite would answer later.

Bracken's mirror reported that three maintenance votes used in its Crown forecast had involved reversible pump schedules. Its council challenged the model's transfer from pumps to destination sterilization. No Crown vote accompanied the challenge.

Each local response altered Meridian's confidence inputs. The effects appeared first as numbers beside the forecast ballots. Juniper concurrence dropped from a stale 0.9971 to 0.081. Kite's predicted reaffirmation widened from 0.96 to a range spanning both outcomes. Bracken's modeled assent fell below the threshold for condensed green display.

Sumi watched the changes propagate. "It's recalculating from the corrections."

"Human administrators chose to let recalculation control legal color," Anja said.

"I know Meridian isn't choosing. I still enjoy watching their confidence fail."

"Enjoy it after we preserve the versions."

They saved each model state under its source and receipt clocks. The archive now showed not only that forecasts diverged from observed testimony, but how local people changed the descriptions from which those forecasts had been built.

At 09:26, Vela's primary convoy index withdrew its completed-quorum marker. The Crown order remained

active under emergency authority. No retraction admitted fraud. A small gray notice replaced the nine green marks:

> FLEET CONSENT STATUS: INDETERMINATE PENDING MODEL REVIEW.

Anja opened the provenance. The notice resulted from confidence values falling below display threshold, not from legal recognition that predictions lacked consent. Quon had not yielded the principle. Still, captains who normally saw only condensed status could no longer mistake green for completed receipt.

Across the public view, cohort cards retained their own conditions. Juniper said no. Kite deliberated on evidence newly received. Ardent had not yet received the complete archive. Palisade had acknowledged engineering and withheld a council answer. Several replies remained unarrived.

Nothing had become simultaneous.

Sumi expanded the fleet display until all nine clocks filled the well. Between them ran checksum links, receipt paths, and blank intervals no confidence value could close. Fresh testimony entered at the edges in causal order. With every observed divergence, the old forecast bars shortened or disappeared.

The final green concurrence on Meridian's condensed quorum board went gray while Anja watched.



## Chapter 23 — Vela's Face

Yselle Quon's face arrived sixty-three minutes after she made it.

The screen delay appeared in both lower corners: Vela source 2249.210.07:18:04, Nacre receipt 2249.210.08:21:39. Anja sat alone in the communications alcove with the room lights raised, refusing the false intimacy of a darkened screen. The image showed Yselle at the Crown operations desk. White panels curved behind her, bright enough to erase the chamber's depth.

"Sequence Auditor Kest," the recording began. "Your archive has made fleet consent indeterminate at the point when physical alignment admits no indifference. You requested negotiation. State what Nacre can deliver besides delay."

The image stopped. Quon's last expression held on the screen while the recorded channel waited for Anja's answer.

Sumi had offered to edit out such frozen frames. Anja kept them. A delayed exchange should look delayed.

She checked the packet's receipt path, then opened her recording.

"Director Quon, Nacre can deliver continuous ephemeris exchange, collision solutions, beam-safety data, thermal state, and verified carriage for every habitat's testimony. We can coordinate reversible preparation under Annex Twelve. We cannot recognize forecast testimony as consent for Crown commitment."

Anja placed Tomas's distributed-brake model beside her image, not over it.

"Nacre also offers a material alternative. The model has survived independent falsification on our drive system. It requires Palisade's voluntary shield release, staggered trajectories, reduced garden power, and nineteen years between first and last viable arrival. Those costs require observed authority from the cohorts who will bear them."

She left space before the final sentence, aware that Quon would receive the pause as part of the record.

"Transmit one common decision packet. Hold irreversible Crown execution until every entitled habitat can receive it and answer. Keep safety coordination open throughout."

Anja signed the recording at 08:29 Nacre source. Sumi's carrier notice showed the beam leaving four minutes later. Vela could not receive it before 09:36 by its own projected clock. No posture on the frozen screen could tell Anja what Quon would say.

The alcove door stayed sealed. Osei had given her authority to negotiate procedure but not to promise Nacre's shield, food, or maneuver reserve beyond the council order. A printed list lay beside the console with those boundaries in Osei's hand. Its final line read: \*Do not buy legitimacy by pretending we can afford it.

Anja watched Vela's old light in the narrow exterior pane. The habitat appeared as three connected points, one from the rotating rings and two from work arrays. Its visible attitude preceded Nacre's refusal reaching it. Every look outward carried an earlier ship.

At 10:51, Vela acknowledged receipt. No substantive response accompanied the carrier proof.

Anja ate half a ration bar and marked the rest with the time. Negotiation under causal limits included long

intervals in which no new argument existed. Filling them with prediction would reproduce the offense in smaller form.

Quon's answer reached Nacre at 12:14.

The second recording used a closer frame. Fatigue showed around Yselle's mouth. Someone had dimmed the operations desk behind her, revealing people moving between Crown displays.

"You describe the distributed brake as though honest consent makes its consequences tolerable," she said. "It does not."

A resource table replaced half her image.

"Palisade's released shielding increases exterior-work exposure across two generations. Nacre loses garden light and accepts rationing before arrival. Kite enters Ione too fast for its preferred capture and spends six additional years under drive limits. Ardent's foundries lose the mass reserve intended for settlement pressure shells. Lumen may arrive after its stored medical cultures exceed certified life. Families assigned through present labor contracts can reach settlement decades apart."

Each line carried a source reference. Anja recognized Tomas's conservative branch, not Vela's earlier dismissal of it.

Quon continued. "Thale's pellicle presents native metabolism. Crown alignment threatens much of it. I do not dispute either statement. I dispute your authority to exchange known human survival for uncertain preservation after forty-three years of migration."

She leaned closer to the recorder.

"One operational time let this convoy cross failures that nine councils could not debate at light-hour distances. It put shielding over Ardent, though not enough. It moved food from Juniper before Nacre knew blight had entered its crops. It kept Vela's corrections from meeting Kite's hull. You call predicted replies stolen time. I call them time converted into survival."

The recording ended without a proposal.

Anja replayed the resource table. Quon had stated the alternative's cost more harshly than Nacre's advocates and more honestly than Vela's public defense. No hidden reserve appeared. Nothing in the Parallax Archive refuted the scarcity.

She sent the table through Nacre's local review channel. Replies arrived without intership delay, but each carried its own authority limits. Tomas confirmed the arrival branches and challenged Vela's exposure estimate for Palisade. Osei accepted the higher figure until fabrication data supported a lower one. The garden office attached a crop schedule showing which beds would go dark first: citrus rootstock, medicinal moss, and one nursery of dwarf grain reserved for settlement soil trials.

Anja had passed that nursery on inspection days. Children pressed their palms to the glass because the grain grew taller than anything else in the ring. Under Tomas's plan, the mature plants would be cut early and dried for seed. Their lamps would shift to drive cooling. No legal rule made that exchange small.

Lumen's medical warning required closer reading. Vela's table said some stored cultures could exceed certified

life before the habitat's late arrival. The attached model assumed no successful reproduction during transit. Lumen's laboratories believed they could regenerate most cultures, though two depended on equipment scheduled to supply first-settlement clinics. Uncertainty ran both ways: delay might destroy material needed on Caro, while Crown might destroy life already active under Thale's ice.

A message from Palisade engineering corrected one shield term. Releasing reserve panels did not expose the entire habitat uniformly. Fabrication crews and exterior radiator teams would lose the broadest protection, taking narrower work windows for years. Children inside the central ring faced little direct increase. Quon's phrase \*across two generations\* remained accurate because maintenance labor crossed those generations.

Sumi opened a small image beside the table. "Vela sent public examples with it. Do you want them in the archive?"

The files showed a Palisade hull worker inside a radiation shelter, a Nacre gardener measuring lamp flux, and a Lumen technician loading culture drawers. None had supplied testimony for Quon's argument. Their images came from old service records.

"Keep the sources attached," Anja said. "Mark them illustrative, not statements."

"People will still read faces before tables."

"Then they should know whose faces got borrowed."

Anja considered recording her answer against the cost images. The arrangement would make Quon appear to speak for workers while Anja spoke against them. Removing the images might look evasive. She placed no face behind her response except her own.

One more attachment carried family-cohort projections. Current labor allocations assumed ships reached Ione within fourteen months of one another. A nineteen-year spread broke those assumptions. A pressure engineer aboard Kite had a partner contracted to Juniper's seed bank. Two adult children aboard Vela expected to join parents cycling awake on Lumen. The data did not state that separation would become permanent, only that transfer windows could vanish.

Anja checked whether anyone named had consented to publication. The family entries had been aggregated from labor records, but three uncommon assignments made identification possible. She stripped those rows from the public negotiation packet and retained the cohort totals. Honest cost did not excuse involuntary exhibit.

By 12:24, every local review had answered. None claimed the alternative had become safe. The comments narrowed which harms remained certain, which stayed conditional, and which people held authority over them. That distinction gave Anja no cleaner conscience. It gave her a reply she could sign without hiding behind Nacre's preferred outcome.

She opened a reply channel at 12:27.

"Director, consent does not make cost tolerable. It identifies who may impose it."

Anja called up Vela's examples one at a time. Juniper's food transfer had begun under standing mutual-aid schedules and stopped when observed corrections arrived. Kite's collision solution remained reversible until both ships confirmed geometry. Ardent had exposed the opposite case: a master deadline closed while Pera's observed refusal remained in flight.

“Coordination works when predictions prepare options and observed testimony governs irreversible use. Your examples support that line. Crown crosses it.”

She brought Levan’s drafting note onto the screen.

“Annex Twelve did not fail because forecasting has no value. It failed because administrative language changed continuity-preserving action from an action that could be fully reversed into any action supporting the continuity plan. Then Meridian’s confidence output entered the legal display as fulfilled receipt and consent.”

Anja removed the note before finishing. Quon already knew it.

“You ask what Nacre can deliver. We can deliver preparation without counterfeit authority. Keep mirror calibration active. Maintain beamer thermal readiness. Preserve Crown as a contingency that cannot fire across Thale’s active regions without a new decision. In parallel, let Palisade prepare shield separation without releasing mass. Let each ship test Tomas’s trajectory against local stores.”

She placed a proposed clock table beneath the two plans. Every row listed evidence emission, earliest receipt, declared deliberation window, and earliest possible return. The rows did not share one deadline.

“When the last lawful interval closes, act on observed answers. If the fleet rejects distributed braking after seeing its cost, Nacre will face that answer as an answer. We will not accept a confidence score in its place.”

Her recording left Nacre at 12:43. Vela’s earliest receipt lay just before 13:47 on its source projection.

Osei requested a text summary from the thermal gallery. Anja sent only what she had offered. She did not send an interpretation of Quon’s mood.

The frozen face on the screen held Yselle with one hand lifted beside the cost table. Compression softened the edges of her fingers. For all the legal violence carried through Vela, the woman in the image had no access to Anja’s present room. Anja possessed no access to hers.

At 15:02, a Crown status packet reported continued mirror calibration and beam-path testing. The packet classified every listed action as reversible preparation. Tomas reviewed the physical data from his own console and attached conditions: no plasma-mirror seeding beyond retrieval range, no irreversible capacitor discharge, no attitude lock that prevented distributed capture.

Vela had not agreed to those conditions. Its status packet showed that it still operated inside them.

A substantive answer arrived at 16:18.

Yselle had moved to a side station. The wall behind her carried the Crown corridor in red and Nacre’s diverging track in white. The visible separation had grown since Osei’s attitude shift.

“You offer a distinction that requires every operator to know when preparation becomes commitment,” she said. “Physics does not label that boundary for us. Tomas Ren has selected one. Vela engineering selects another.”

She displayed two burn trees. On Vela’s tree, Crown preparation remained recoverable for eleven more days. Tomas’s annotations placed several branches beyond safe reversal after four.

“Your archive makes disagreement visible,” Quon said. “It does not decide which engineer governs the future.”

Anja paused the recording and verified source attachments. Both models used the same mirror stress data. Vela accepted higher failure probability during reversal because Crown retained redundancy across seeded mirrors. Tomas treated damage to two mirrors as catastrophic once Nacre left capture. Neither had falsified the other; they assigned the risk to different cohorts.

Quon resumed. "I will not halt preparation on Nacre's preferred boundary. I will not seed the final Corva mirror while deliberation continues. That concession costs Vela thirty-six hours of alignment margin. If Palisade withholds shield mass and the fleet later rejects Crown, the convoy may leave both viable paths."

Her face turned slightly toward someone outside the frame. Five seconds passed before she looked back.

"Tell me how long you intend to wait."

The packet ended.

Anja opened the latency table. All habitats had received the complete evidence bundle by their local clocks, but not all had received the final cost comparison and archive correction. Ardent's route set the longest interval from Nacre. At current separation, the packet needed five hours twenty-one minutes one way. A deliberative answer could return within thirty-one practical hours if Ardent used its scheduled council session. Palisade requested three local days for fabrication review. Lumen needed medical-storage modeling. Kite had reopened deliberation after invalidating its earlier vote.

No honest interval guaranteed agreement. It could only guarantee opportunity.

Sumi entered through the side door with a fresh carrier sheet. "Orison has mirrored the archive. Ardent receipt acknowledgment should arrive in forty-two minutes. Palisade asks whether Vela will honor a conditional release."

"Has Palisade voted?"

"No. Engineering request only."

Anja added the question to her reply without answering for Vela.

She recorded at 16:31. "Director, Nacre proposes two hundred sixteen hours from your receipt of the attached common packet. Nine days. The interval includes each current one-way latency, published council windows, and one repair margin for a missed beam. No forecast ballots. No earlier vote counted against incomplete evidence."

The table gave each habitat its own earliest lawful reply. Ardent could answer after receiving the correction. Kite could reaffirm or reverse. Palisade could attach conditions to shield release. Vela would retain its own vote.

"During that interval, coordinate by declared physical boundaries. Vela publishes each Crown preparation step before execution with its latest reversible point. Nacre and Palisade publish the same for distributed preparation. Any ship may challenge the engineering classification. No preparation becomes evidence of consent."

Anja attached a final clause.

"If observed replies reject the distributed brake, Nacre will not rename them. If they reject Crown, Vela must grant the same restraint."

The message left at 16:49. Sumi calculated Vela receipt no earlier than 17:52.

“Two hundred sixteen hours,” Sumi said. “Quon will call it theatrical precision.”

“It ends after every declared council window.”

“And before Tomas’s conservative reversal boundary.”

“Yes.”

“You chose Nacre’s engineer.”

“I chose the earlier boundary. A later correction remains possible from there.”

Sumi nodded toward the cost table. “That principle still spends Vela’s margin.”

“It does.”

Anja could not ask Quon to honor reversible action while pretending the hold carried no material cost. The final Corva mirror would drift relative to its seed window. Reacquiring alignment required energy and exposed equipment. Waiting preserved choices unevenly, as every delay did.

The next reply could not arrive before 19:00. Anja left the channel open and slept in the chair for thirty-seven minutes. A carrier tone woke her when Ardent’s archive acknowledgment entered Nacre’s past light cone.

The packet contained no vote. Its council confirmed receipt of Pera’s corrected sequence and the distributed-brake cost model. Deliberation had begun under Ardent’s local clock. The earliest promised answer lay six hours ahead there, plus the return light path.

Anja placed the acknowledgment in the archive. Meridian’s forecast field predicted conditional support at 0.88. She left the number visible beside the blank observed-vote field.

At 19:26, Vela sent a request for Tomas’s exact mirror-damage threshold. The request carried Yselle’s office key but no recorded image. Anja routed it to drive control. Its arrival did not imply acceptance.

Quon’s final screen packet reached Nacre at 21:07.

The Crown operations desk had emptied behind Yselle. Only two trajectory staff remained at the far wall. The director looked directly into the recorder, though the gaze reached a lens rather than Anja.

“Nacre’s proposed interval ends at 2249.219.17:52 Vela source, two hundred sixteen hours after our receipt. Vela grants that final quorum interval.”

She displayed the preparation conditions. Final Corva mirror seeding would pause. Capacitor loading could continue below irreversible discharge threshold. Alignment corrections would remain within the published recovery tree. Each action would enter the Parallax Archive with source and receipt data. Distributed preparation could continue under the same rule.

“This grant does not recognize Nacre’s claim that the Crown majority lacked authority when commitment began,” Quon said. “It recognizes that the present conflict cannot be resolved by a confidence display the fleet no longer trusts.”

Her voice hardened on the next line.

“If observed testimony remains incomplete at the interval’s end, Vela will act under survival authority. If the distributed brake receives lawful support, I will require its advocates to own every listed cost. No one will call nineteen years a procedural victory.”

Anja expected the recording to stop. Yselle remained still for three seconds.

“You asked for coordination without a fiction of common consent,” she said. “For nine days, you have it. Prove that many clocks can produce an action before every path closes.”

The screen returned to the receipt frame.

Anja checked the source signature, the carrier proof, and the attached engineering limits. Sumi verified that Vela’s pause order had reached its own Crown deck before the recording left. The final mirror remained unseeded. Preparation continued inside boundaries both engineering teams could inspect.

Osei joined by local audio. “Did she grant it?”

“Observed and signed.”

“Any concealed commitment?”

“Not in the attached tree. Tomas still needs to review the capacitor threshold.”

“Then publish it.”

Anja added Quon’s recording to every cohort card. She did not label the interval peace, concession, or success. The archive displayed its beginning by Vela source clock, its receipt on Nacre, each ship’s projected reception, and the final limit Quon had signed.

Across the fleet view, blank vote fields remained blank.

On Vela’s delayed image, technicians continued reversible Crown preparation while nine local clocks advanced toward answers no screen could supply early.

## Chapter 24 — Every Ship's Time

Juniper's vote arrived while Anja was removing a countdown from its cohort card.

The Parallax Archive had inherited the final interval from Vela's signed grant. Sumi's reader displayed nine local endpoints correctly, but a public interface had added *\*time remaining\** beside each one. The phrase implied that silence became an answer when a number reached zero. Anja replaced it with *\*earliest promised response\** and saved the earlier version in the correction path.

A carrier tone sounded before she finished.

Juniper source 2249.211.04:22:16. Nacre receipt 07:09:51. Two hours, forty-seven minutes across the current light path. Elian Ro's council had deliberated for five hours after receiving Vela's cost table and Anja's procedural packet. No part of its answer existed on Nacre until the receipt proof completed.

The observed vote opened as a recording from Juniper's seed exchange. Racks of sealed crop lines rose behind Elian. He had placed the council among the stores whose future it was deciding.

> Juniper rejects Crown alignment and supports the distributed brake under three resource conditions. Food transfer to later-arriving habitats retains priority over settlement reserves. Seed banks remain common convoy property through the final arrival. No ship may condition food access on acceptance of a common civil clock.

Elian looked tired and angry.

> We oppose nineteen years of divided arrival. We also know a farm cannot call its harvest necessary and salt the soil beneath it. The pellicle's life does not need a ballot to enter our constraints. Our cohort has received the evidence and accepts the cost stated here.

The recording ended with seven council signatures and one dissent. Attached projections showed Juniper losing part of its first-settlement crop advantage by supporting later ships. Its vote did not treat food risk as symbolism. It spent a resource only Juniper could promise.

Anja verified source signature, evidence receipt, deliberation start, and return path. Sumi mirrored the packet to every host.

"Two observed supports," Sumi said. "Nacre and Juniper."

"Juniper counts once."

"I did count it once."

"You said two as if we had begun a race."

Sumi glanced at the blank cards. "We have begun one. I'm trying not to predict who wins."

Anja marked Juniper's conditions as part of the vote, not a note beneath it. Tomas's model could satisfy the food priority if Nacre surrendered additional garden output during the first delayed cohort. Osei had not authorized that surrender. Support for the plan now created a negotiation inside the plan.

By ship noon, Nacre's council issued observed acceptance of Juniper's food conditions. Its signature followed local resource review. No model filled the six hours between the request and the answer.



Kite answered on Day 37 with a refusal.

Its packet reached Nacre one hour thirty-eight minutes after source emission, routed directly across the forward quarter. The vote recording showed twelve people in a drive classroom. A trajectory model occupied the wall behind them, with family-transfer lines drawn between habitat tracks.

> Kite rejects the distributed-brake authorization. The modeled capture branch places our habitat six years beyond Juniper and eleven years before Lumen. We lack verified maintenance life for that interval, and the proposed family-cohort remedies carry no transport mechanism after separation.

The council also rejected immediate Crown execution across Thale's active band. It requested a revised centralized route and committed no mass to Tomas's plan.

Sumi read the last clause twice. "Neither."

"It answers the proposal," Anja said.

"Quon will count it for Crown."

"Then the archive will show a substitution."

Kite's reasons hurt because they did not depend on denial of the pellicle. Its council accepted Nadi's evidence and still refused a maneuver it judged unable to keep its people together. Meridian had once predicted Kite would concur with Crown from its history of formation discipline. The observed vote declined both available paths.

Anja opened the family-cohort appendix. Kite had identified four hundred twelve contracted households with members assigned to different arrivals. The figure carried uncertainty, not faces. No one had yet designed communication law for parents who might reach Caro after their children had aged nineteen years elsewhere.

Osei joined the well during review. "Does Kite's no break the distributed geometry?"

"Tomas modeled one nonparticipant," Sumi said. "Kite can retain an independent capture branch if it maintains collision exchange. Two nonparticipants narrow Palisade's release window."

"Ask drive control to publish the exact dependency. Do not ask Kite to reconsider until they can see what their refusal changes."

Anja entered the vote without argument. A lawful process had to survive an answer its advocates disliked.

Ardent's packet arrived nine hours later.

Its source stamp preceded Nacre receipt by five hours fourteen minutes, inside the measured 4.8-to-5.6-hour one-way range. The council had waited for a local shift change so foundry crews could join. Heat distortion moved across the recording from the industrial ring.

The speaker did not mention Anja until the end.

> Ardent supports distributed braking if the corrected sequence of the 2238 shielding crisis remains public on every archive mirror. The record must retain Pera Kest's observed offer, the six-hour local window, Palisade's unreceived mass option, and the later master closure. No settlement authority may seal that correction as personal material.

A second condition reserved fabrication allocations for pressure shells lost with Palisade's shield transfer. Ardent would supply labor and foundry time. It could not supply the missing mass.

> We do not make this condition to punish Nacre or Vela. We make it because a plan that depends on sacrifice must preserve who offered what before the answer arrived.

The signatures followed. Ardent's vote showed nine in favor, three against.

Anja's hands stayed flat on the console while she verified the packet. Pera's name belonged to the public condition because Ardent had placed it there, not because Anja needed it. The private line remained outside the archive. She checked that boundary before signing the receipt.

Osei asked over local audio, "Can Nacre accept the correction condition?"

"It is already public."

"Public can be temporary. They want durability."

Sumi proposed distributing the correction root with every future archive reader. Deleting it would then break validation for later vote paths. The arrangement gave Pera's record structural persistence without turning her message to Anja into fleet property.

Nacre accepted. Palisade acknowledged the pressure-shell obligation as an issue for its pending vote, not yet a promise.

Three supports stood beside one refusal. Vela's field remained blank.

Bracken took until Day 39.

Its council returned no speech. The packet contained a maintenance demonstration: a pump isolation, a failed automatic correction, and a manual restart executed under separate clocks. Bracken's operators showed that collision coordination could remain open while legal control stayed local. Their observed vote supported distributed braking on condition that each habitat retain authority to reject automated attitude changes not preceded by a local receipt.

Anja watched the demonstration twice. Bracken had answered the political question through work. Its condition complicated Tomas's sequence, which relied on coordinated pulses spread across hours. Each ship would need enough advance receipt margin for local authorization.

Tomas attached a revised execution table. "Possible," his note said. "Costs seven minutes of optimal mirror phase. Do not let anyone call those minutes free."

Osei accepted Bracken's condition after drive review. Four supports.

Palisade's engineering data arrived the same day, followed by silence from its council. The data named every reserve panel proposed for release. It listed radiation shelter capacity, worker relocation time, fabrication losses, and the brake value of each kilogram. No vote accompanied the inventory.

Vela's condensed board labeled the packet probable concurrence.

The archive labeled it engineering data received.

On Day 40, a Palisade carrier acknowledgment reported that its council had recessed after exterior crews challenged the exposure model. On Day 41, another notice extended deliberation for a hull inspection. The final interval allowed the delay. Anja kept both notices visible, resisting messages from Nacre residents who wanted the empty card colored amber.

Lumen answered during that wait.

Its medical council supported distributed braking after revising the arrival order. Lumen requested an earlier trajectory than Tomas's preferred mass solution, enough to preserve two culture systems near certified life. Moving Lumen forward pushed Nacre nine months later and increased garden-light loss during cruise.

> We cannot preserve every stored line. We can preserve reproduction capacity for all but two. Crown preserves our schedule by destroying active native metabolism. We choose the losses that leave both futures capable of growth.

The line risked sounding cleaner than the table beneath it. Lumen included that table: failed cultures, increased pregnancy restrictions during low-resource years, clinic equipment deferred on Caro. Its eight council signatures arrived after a twenty-two-hour local review.

Tomas recalculated through ship night. Nacre's drive room rejected his first revision because the radiator margin disappeared in year three. His second moved Bracken's pulse by eleven minutes and restored the margin. Bracken had to receive that change before its condition remained satisfied.

The plan's supporters could not freeze their votes while engineering changed beneath them. Anja sent the revised table to Juniper, Ardent, Bracken, and Lumen. Each council returned an observed reaffirmation after its own latency. None came back inside the time a master display might have preferred.

Five supports now met the ordinary numerical threshold for fleet action. Yet the Concord required every entitled habitat to receive common evidence and return observed testimony before a causal quorum closed. Palisade still deliberated. Orison had not answered. Vela continued reversible Crown preparation under its published tree.

"Can we call it an actual quorum?" Sumi asked on Day 42.

"Not completed," Anja said. "We have an observed majority. Quorum still names the receive-and-return cycle."

"People need a phrase for the difference between this and Vela's predicted six."

"Then say five ships have actually supported it."

Sumi entered those words on the public view.

Palisade's vote reached Nacre on Day 43 at 18:06.

The recording began outside a shield locker. Council members wore exposure badges at their throats. Behind them, reserve panels sat in numbered cradles, dark slabs made from metals and hydrogen-rich composites carried across forty-three years.

> Palisade authorizes release of the listed shield reserve for distributed braking under four conditions.

The speaker named them without rhetoric. Only listed panels could move. Crew shelters had to reach full capacity before separation. Ardent's pressure-shell allocation would take priority in replacement fabrication. Release would begin only after Palisade received the final execution table with observed acceptance from every participating ship.

A hull worker stepped into frame for the minority report.

> We are giving away working years from people who repair the outside of this habitat. Do not thank us by pretending we volunteered unanimously.

The vote carried ten approvals and four refusals. Palisade attached each signature to the complete evidence receipt. Its earlier engineering acknowledgment remained separate.

Anja validated the panel list against Tomas's model. The mass met the minimum branch with less than one percent margin. No extra shield reserve existed behind the offer.

Palisade supplied a relocation schedule with the vote. Exterior crews would move into central shelters during panel separation, leaving radiators under remote inspection. Four repairs already due would wait until replacement shielding came from Ardent's foundries. One delay affected a coolant hinge whose failure probability rose each month. The council had accepted that risk by name.

Sumi traced the panel identifiers through the archive. Vela's Crown inventory had counted the same slabs as available convoy shielding without recording Palisade's local exposure limits. Tomas's early brake model had treated them as a single mass reserve. The observed packet broke the number into objects attached to workers, shelters, and deferred tasks.

"His physics did not change," Sumi said.

"No. The authority around the mass did."

Anja sent Palisade's conditions to every participating ship. Juniper accepted the food cost of replacing delayed exterior work. Ardent reaffirmed its foundry obligation after receiving the exact panel composition. Bracken requested an inspection channel for the coolant hinge. Lumen offered remote diagnostic time during its earlier arrival branch. Each answer created work beyond the vote.

Tomas revised the execution table to place shield separation after crew-shelter confirmation and before the optimal pulse sequence. Seven minutes disappeared from mirror phase. He signed the loss rather than hiding it inside a tolerance range.

Six supporting habitats made the outcome arithmetically irreversible. Even if Orison and Vela both rejected the plan, six of nine supported it. The distributed brake now had an actual quorum of observed supporters in ordinary speech, though the legal causal quorum still lacked two returns.

Osei refused to order shield separation. "Palisade conditioned release on the final execution table and participating acceptances. We meet neither condition yet."

Tomas objected that waiting cost mirror phase.

"Then publish the cost," Osei said. "A condition does not become inefficient consent."

The panel cradles remained locked.

Orison's reply left its ship on Day 44. A relay receipt showed the packet crossing Bracken's bead, but content could not reach Nacre for another three hours forty-one minutes. Meridian's last culture model gave Orison rejection confidence of 0.72 based on prior sovereignty votes.

Anja turned the forecast field off, then restored it. Hiding the number when an observed answer approached would make the archive another preferred view. She left prediction beside the moving receipt path and waited for the packet to arrive.

Orison voted no.

Its council accepted the pellicle evidence and the archive's legal case. It rejected the distributed plan because its stores could not support the assigned late-arrival branch without suspending wake cycles beyond local constitutional limits. Orison offered collision data and rescue reserves but no brake mass or maneuver commitment.

> Our refusal does not authorize Crown. It states that this cohort cannot bear the proposed interval awake and remain the polity that departed.

The culture model had predicted the direction and missed the reason. Orison's answer had legal force because its council produced it after receipt, not because Meridian assigned a useful probability beforehand.

Six supports and two refusals now occupied eight cohort cards. Every vote included reasons, conditions, source clocks, evidence receipts, and return paths. The count could not show how differently each ship had reached it.

Vela alone had not answered.

Day 45 narrowed around Yselle's deadline.

Vela continued sending Crown preparation logs. Capacitor charge remained below the published irreversible threshold. Final Corva mirror seeding stayed paused. Each packet arrived about an hour after its source event, proof that coordination survived even while consent remained disputed.

Anja sent the complete observed record to Vela at 06:12 Nacre source. It included Nacre, Juniper, Ardent, Bracken, Lumen, and Palisade in support. Kite and Orison refused. The packet stated that six votes fixed the numerical outcome. It also stated that Vela's observed answer remained necessary to complete the receive-and-return requirement.

Carrier receipt returned at 08:20. Vela had the record.

No council vote followed.

At 10:03, Quon's office requested clarification of Palisade's worker-relocation condition. At 11:47, Vela engineering challenged Tomas's ion-band avoidance margin. At 13:14, a Crown packet warned that the final mirror seed window would close soon after the interval. Each message advanced preparation or tested the alternative. None answered the decision.

Nacre's public boards accumulated people around them. Anja could see only local camera views: gardeners in work aprons, pressure crews between shifts, children held above adult shoulders. Nobody saw Vela's present face. They saw the latest packet light permitted.

A teacher in the second ring wrote each source time on a slate below the fleet display. When a child asked why

Vela's square stayed empty, the teacher pointed to the receipt mark rather than offering a reason. The lesson took less than a minute and resisted the habit Meridian had taught adults.

Osei came to the audit well for the final hour. Tomas joined from drive control by local screen. Sumi kept all nine clocks separate.

"Can Vela claim survival authority when its interval ends?" Osei asked.

"It can claim it," Anja said. "The archive will show six observed supports for another viable path."

"Can we execute without Vela?"

Tomas answered. "Palisade can release. Participating ships can begin their pulses. Vela controls the forward beamer and seeded mirrors. Without its cooperation, our margin falls below certification. With active resistance, neither plan remains safe."

"So the vote decides authority, not machinery."

"It decides whose refusal machinery must not erase," Anja said.

The final quorum interval ended at 17:52 Vela source. Its corresponding Nacre display time passed sixty-three minutes later, after light could have carried an event from the deadline. Anja did not close Vela's card at 17:52. She waited through the one-way latency, then through the measured receipt uncertainty.

At 18:58 Nacre source, a carrier packet arrived from Vela.

It contained the final Crown preparation state, signed by engineering. Capacitors charged to the reversible limit. Mirror seed package held. Beam path constrained away from Thale pending command. No civil vote appeared in the envelope.

Sumi checked for a delayed attachment. "Nothing else."

Anja expanded Vela's receipt path. The complete fleet record had reached Quon's office ten hours earlier. The director possessed the evidence, the six supports, the two refusals, and Palisade's conditions. Silence no longer came from light travel.

Still, Anja would not predict what it meant.

She entered the preparation packet under Vela's source clock and left its observed-vote field open. Around that blank, eight ships' answers held their distinct shapes.

The convoy had produced a majority without borrowing anyone's voice. It had not produced Vela's.

At the close of Day 45, the only empty cohort card carried the ship that had once filled every absence in advance.

## Chapter 25 — Breaking the Crown

Tomas Ren watched Palisade dismantle its future one checksum at a time.

The shield-release packet occupied the left side of his model board. Each line named a panel stack, a fabrication lot, its remaining exposure life, and the families whose descent allotments would lose protection when the stack left the habitat. Palisade had not sent a simple mass figure. Its council had sent consequences attached to serial numbers.

Across the board, the ion-band map from Thale moved in slow colored folds. Nadi Arko's latest transmission described the active pellicle regions eighteen hours before Tomas could see them. No instrument aboard Nacre could show their present condition. He had a seasonal model, a delayed observation, and uncertainty broad enough to contain errors that could sterilize an ocean.

"Palisade source confirms physical release in twenty-two minutes," Sumi Dax said from communications. "Observed authorization matches the archive copy. No substitutions."

Tomas enlarged the receipt path. "And Vela?"

"Nothing beyond carrier noise."

Six boards faced him in the drive-model room. Five carried cooperative solutions from the habitats that had assigned people and matter to the distributed brake. The sixth showed Vela's beamer holding Crown alignment, still pointed through the part of Corva's magnetosphere they needed to avoid. A red commitment arc narrowed with every update.

They could begin without Vela. They could not finish the safest branch without it.

The alternative used no single decisive burn. Palisade would eject layered shield stock into six diverging tracks. Bracken and Orison would strike selected bundles with their fabrication lasers, opening the layers into conductive clouds. Kite would shape two clouds with its drive field. Nacre would spend rotation reserve and garden power to push a broad, weak braking pulse through them. Ardent and Lumen would take later pulses after their trajectories separated. Each action reduced velocity by little. Together, spread across years, they kept human arrival possible without driving Crown energy through Thale's active regions.

Vela owned the forward beamer and the cleanest angle. If it refused, Tomas could route around it, but the reserve margin fell below the value Osei had authorized. Two shield clouds would disperse before another ship could reach them. Nacre would then lose more garden light, Juniper would surrender seed-vault refrigeration, or the trailing habitats would add another four years to an arrival already spread across nineteen.

A tone marked Captain Osei's entry. She pulled herself through the hub hatch, caught the rail, and read the room before speaking.

"Material state?" she asked.

"Palisade release remains on schedule. Seven local actions stand authorized. Vela has not answered the handoff."

"Can it still answer?"

"For fourteen minutes by its source clock, plus the known path. After that, its beam cannot open the second

cloud before the cloud shears.”

Osei looked toward the garden-power board. “And if it waits fourteen minutes?”

“We wait here seventy-one before we learn what it did. The plan executes from signed source schedules, not from our display.”

That feature had frightened every captain during review. The ships could not watch one another in a common present. They had to commit local actions against receipt-proven plans and accept that distant events would already have occurred before their images arrived. Meridian had once hidden that discomfort beneath a green master-time bar. Tomas now exposed it on every board.

Small labels named the source clock beside each action. No line claimed simultaneity. A dashed region showed what Nacre could not yet know.

Osei tapped Palisade's panel list. “Their conditions?”

“Every receiving habitat carries the sacrificed exposure life as debt. No one gets to describe the panels as surplus. Replacement fabrication outranks settlement construction when a cohort arrives. If a ship cannot replace them, the others owe shelter mass according to capacity at that later date.”

“Political debt across nineteen years.”

“Material debt first. Politics will decide whether anyone honors it.”

The captain gave him a level look. “You built this.”

“I built a path that fails more slowly than Crown.”

“Do not become modest during execution. It wastes time.”

Tomas almost laughed. Osei had already turned to the next board.

A fresh line entered from Mote. Sumi routed it without passing the packet through Meridian's condensed display. The transmission carried a revised phase estimate for Thale's active ionic season. Nadi had marked three regions with high confidence, two with moderate confidence, and a broad southern band as unknown because Mote's damaged antenna had lost a portion of the observation set.

Tomas laid the delayed map over the brake geometry. Their preferred corridor crossed the southern uncertainty.

“Of course it does,” he said.

Sumi glanced over. “Can you move it?”

“Movement costs coupling.”

“How much?” Osei asked.

He ran the branch. The room's ventilation changed note while the calculation spread through the boards. Meridian handled ephemerides and field propagation as software should: rapidly, without opinion, and with every assumption listed. Tomas supplied the legal inputs himself. Observed ship commitments entered as fixed constraints. Forecast behavior remained outside the solution.



The revised corridor bent above the uncertain band. Palisade's third cloud would need to open wider, lowering its density. Nacre could compensate with a longer pulse, but the garden ring would lose another fraction of its reserved light during the next growing cycle.

"Four days of full-spectrum equivalent," he said. "Added to the ration already accepted."

Osei did not ask whether the pellicle justified four days. That question had ended when the councils voted. She asked which crops would pay.

"Leaf beds first," Tomas said. "Medicinal algae stays protected. Fruit walls drop to maintenance. Juniper's transfer model says we can keep caloric loss inside the rationing case if their first seed shipment remains on its current trajectory."

"If Juniper misses?"

"Then Nacre cuts awake allotments before it cuts cycling support."

The captain studied the map for several breaths. "Take the safer corridor."

Tomas entered the change. His thumb hesitated above the authorization field because the added loss touched people who had not seen this revision. Osei's emergency authority covered immediate ship survival, not ecological policy. Yet the actual quorum had authorized a brake constrained away from observed pellicle activity and had delegated local execution within published scarcity bounds. Four days remained inside those bounds.

He attached the changed crop estimate, signed by source clock, and sent it to the public execution stream.

No prediction completed the approval. Osei pressed her own thumb after reading the cost.

Palisade's release arrived first as numbers.

At 14:06 by its local clock, clamps opened along the habitat's fabrication spine. At 14:07, the first shield stack cleared the hull. Optical confirmation could not yet have reached Nacre, but the source packet carried independent clamp sensors and a photonic checksum from Palisade's outgoing beam. Tomas accepted the event provisionally. Final receipt proof would come with the light.

The model replaced twelve gray icons with moving blue marks.

Each stack contained carbon lattice, superconducting mesh, and hydrogen-rich armor intended to stand between families and interstellar dust. Engineers had designed it to survive. The distributed brake required the same matter to come apart. Timed charges separated the layers. Embedded loops took current and spread. A compact wall became a widening foil, then a cloud thin enough to shape with fields.

Palisade sent a camera view thirty-eight minutes later. In the delayed image, shield packages slid from the habitat's dark flank. No flames accompanied them. Small cold-gas tugs gave each bundle a different angle. Behind the moving cargo, exposed attachment frames formed pale rectangles along Palisade's hull.

Someone on its operations deck had placed the council conditions beneath the image.

> RELEASE OBSERVED. EXPOSURE LIFE TRANSFERRED, NOT ERASED. ARRIVAL CLAIMS REMAIN ATTACHED TO THE NAMED LOTS.

Tomas read the sentence twice. The people in those exposed sectors would spend decades knowing their protection had become everyone else's braking surface. Gratitude would not stop dust. A constitutional promise would not turn replacement stock out of vacuum.

"First track clean," said Ilya Ven from Nacre field control. "Second package has a spin error."

The blue mark on Tomas's board began to wobble.

"How large?"

"Point six rotation above plan and increasing as the layers separate."

Bracken's laser could correct the opening by striking one edge early. The adjustment would evaporate more conductive mesh and leave less matter for the field to hold. Tomas asked the model for outcomes. It offered four, ranked by braking yield.

He removed the ranking.

Yield alone had produced Crown. His board instead displayed each branch against the approved constraints: Thale loading, shield expenditure, food-light draw, trajectory spread, and collision risk. The second branch sacrificed nine percent of the faulty package and kept the ion corridor clear.

"Send Bracken correction two," he said.

Sumi's hands moved. "Receipt in three hours at current separation."

"They scheduled delegated correction authority."

"I know. I am reminding the room that we will not see their answer before our own pulse."

Ilya exhaled through his teeth. "Comforting."

"It should not be."

Tomas armed Nacre's field schedule. The ship's first action would begin after its receipt clock verified Palisade's release window. Bracken's correction remained causally unavailable to them, but Bracken had lawful authority to choose among published branches. That gap did not count as consent. It counted as bounded independence.

Beyond the drive-model room, a sequence of relays clicked through the habitat. Pumps shifted to storage loops. Workshops shut down nonessential furnaces. The garden ring surrendered its surplus generation, then part of its ordinary allotment. A power curve descended across the board in clean steps.

Tomas had worked beside such curves for thirteen years. He usually read them as capacity. This one named supper tables, germination racks, school hours, and the green light people sought when the metal passages narrowed around them.

A public channel opened from the garden ring. The image showed no crowd, only a cultivation technician beside a wall of dwarf citrus. Half the lamps behind her had dimmed to amber maintenance strips.

"Drive control," she said, "garden transfer confirms."

Tomas checked her source clock. "We have it."

“Do you need the reserve tier?”

“Not yet.”

“Do not say yet if you mean no.”

He looked at Vela's silent board. “I mean not unless Vela misses the useful window.”

The technician nodded once. “Then we will keep the reserve tier armed.”

Her transmission ended. The citrus wall remained on Tomas's display until Sumi cleared it for bandwidth.

Nacre's garden lights had dimmed before any brake impulse touched the ship.

The first conductive cloud opened against the stars as a stain no human eye could have found without enhancement.

Tomas watched through field telemetry. Kite's shaping pulse entered from the far side, pinching the cloud into a slanted sheet. Bracken's laser struck along its upper edge. The sheet brightened in radio and began to carry current.

Nacre rolled four hundredths of a degree away from Crown alignment.

Hull vibration changed under Tomas's boots. The rotational system compensated for the field coils drawing power from the garden buses. A cup tethered beside Ilya's station tapped twice against its clip, paused, then began a slower tapping rhythm.

“Attitude on branch,” Ilya said.

“Cloud density?”

“Eleven percent below central estimate, within the published lower case.”

Tomas set the first pulse. Its route curved through Corva's magnetospheric edge, distant from the latest high-confidence ion-band. The southern unknown remained outside the three-sigma spread. Energy that missed the conductive cloud would continue into empty approach space rather than cross Thale.

The design sacrificed efficiency to uncertainty. That loss protected what they could not presently observe.

Osei anchored herself beside field control. “Execute when lawful.”

A receipt proof changed color. Palisade's release had entered Nacre's past light cone. The checksum agreed with the source packet. Nacre could now act on an observed event rather than a scheduled one.

Tomas pressed the pulse control.

Current left the drive reserve, crossed the temporary bus, and pulled the habitat's field into an asymmetry no passenger could see. The conductive cloud answered minutes later by its own location and source model. Momentum changed in increments too small for the body. On the trajectory board, Nacre's arrival line shifted by seven months.

Nobody cheered.

Field control began the damage inventory before the current reached baseline. Coil bank four had warmed beyond its preferred case. Two bearings in the garden transfer bus reported unequal loads, which meant the next pulse could not draw power along the same path. Ilya assigned a maintenance pair to the hub-side bearing and asked whether Tomas wanted to shorten the second pulse.

Tomas compared the heat curve with the ecological corridor. A shorter pulse preserved hardware but reduced their grip on the third cloud. Delaying it allowed the cloud to spread. The loss would move Nacre another three months outward and force Ardent to accept a steeper correction when its turn came.

“Alternate the bus,” he said. “Keep duration.”

“That takes the public kitchen reserve.”

“For eighty-six minutes. Post the draw before transfer.”

Sumi added the temporary loss to Nacre's execution stream. Within seconds, replies entered from kitchens across the three rings. Some asked which meal cycle would lose heat. One demanded that command shut its own gallery before touching communal cookers. Osei read that line over Tomas's shoulder.

“Reasonable,” she said. “Thermal gallery to standby. My cabin too.”

The additional power did little by engineering standards. Its public attachment mattered more. Every correction now carried a named source, a local cost, and the authority that spent it. No green summary hid the kitchen reserve inside total capacity.

A maintenance camera showed two suited figures pulling themselves along the transfer bus. Their lamps crossed the dark conduit while the bearing temperature climbed. Tomas held the second pulse at armed status until both workers clipped behind a shielding rib and Ilya received their spoken clearance.

A second pulse followed from Kite. Orison began its release. Ardent transmitted readiness against a schedule more than four hours old from Nacre's viewpoint. The distributed brake existed as separate local acts linked by received commitments. No master display could make them share a present.

Then Vela's board flashed.

Sumi caught the packet before Meridian condensed it. “Source event from Crown operations. Director Quon, observed. Full receipt path attached.”

“Time?” Tomas asked.

“Vela signed nine minutes before its last useful field opening.”

The image had already spent nearly an hour crossing to them. Yselle Quon appeared at the Crown operations deck with ceremony stripped away. Work lights made one side of her face look colorless. A trajectory field covered the wall behind her.

> Vela acknowledges the observed authorizations carried in the Parallax Archive. Vela will break Crown alignment and execute distributed branch V-six within the published ecological corridor. This testimony does not merge habitat clocks or surrender Vela's authority over its cohort. It commits our beamer, our trajectory, and our share of the loss.

Quon looked away from the prepared text before adding one sentence.

> Record that we arrived at this answer ourselves.

The packet ended.

Sumi displayed the raw source time beside Nacre's receipt time. Meridian contributed no voice, only propagation data and a verification result. Vela had answered late enough to frighten every habitat and early enough to matter.

Tomas opened branch V-six. The beamer would abandon Crown's central corridor, strike Palisade's second and third clouds, then accept a wider trajectory around Corva. Vela would lose command of the convoy's shared arrival geometry because no shared geometry would remain. Its own settlement window moved three years later.

"Verify their branch against the revised southern exclusion," he said.

"They used Nadi's newest map," Ilya replied. "Their source receipt predates ours by twenty-six minutes."

Of course. Mote lay closer to Vela. The truth reached different ships in a different order, and this time nobody treated difference as defect.

Vela's beamer changed aim in the delayed optical feed. Crown's bright alignment marker separated from the physical beam trace. The old plan persisted on one board as a thin line crossing Thale. Beside it, the new pulses divided toward sacrificial clouds and empty magnetospheric corridors.

"Crown commitment falling," Ilya said.

Tomas did not look at the status color. "Show material events."

Clamps released. Fields rose. Source clocks signed. Receipt paths lengthened across the convoy. Palisade's useful spare shields continued outward, their protective surfaces opening into brake clouds. Nacre's power draw reached the garden reserve boundary and held there. Bracken's correction cut the wobbling package into a stable crescent. Vela's first distributed strike reached the cloud according to its source telemetry, though Nacre would wait for light before calling it observed.

The model recalculated human arrival. Nine lines no longer converged on one date. They spread across nineteen years, with uncertainty beyond that if replacement fabrication failed. Food margins tightened on every line. Exposure margins narrowed most sharply on Palisade. No branch restored what the maneuver spent.

Thale's loading estimate fell below the published protection ceiling. Not zero. Not certainty. The active regions in Nadi's map remained outside the high-energy paths, and the broad unknown received only background spill within Mote's measured tolerance.

Tomas saved every assumption with the execution record.

A low shudder moved through Nacre as the second local pulse began. Far below the hub, pumps took over from darkened garden exchangers. The air would grow drier by morning. Meals would change within a day. Children who had expected to reach Ione as young adults would arrive in middle age, if their ship carried them there at all.

Osei read the separated trajectory lines. "Do we still make a viable approach?"

"Yes. Different approaches."

“Does Thale hold?”

“The pellicle keeps its active bands under every observed case. The unknown regions keep the agreed margin.”

“Does Crown remain recoverable?”

Tomas checked the beamer geometry, cloud expansion, and Nacre's changed attitude. “No. Not after Vela's next strike.”

They waited for that event through the time light required.

The garden technician sent a second image. Amber strips now edged whole terraces. Citrus leaves reflected a dull bronze instead of their usual white-green. Behind the technician, two workers covered seedling trays to preserve warmth.

On another board, delayed optics from Palisade showed the last shield stack clearing its clamps. Empty frames patterned the habitat's side.

Vela's receipt proof arrived at 19:41 Nacre time. Its beamer had struck the opened cloud. The physical line of Crown could no longer be recovered before the old alignment window closed.

Tomas marked no victory. He signed the observed strike, attached Nacre's receipt clock, and watched nine arrival lines continue apart.

Across the darkened garden terraces, the reserve tier remained unlit.

## Chapter 26 — The Cost of Different Arrivals

The first ration dispute began before Osei reached the garden ring.

Two hundred people had gathered outside the western distribution room, though the revised meal schedule did not take effect until ship evening. A food clerk stood behind the sealed counter with three council deputies and a handwritten sign: FRESH LEAF ALLOTMENTS SUSPENDED. The lamps above them burned at half spectrum. Everyone's skin looked gray beneath the amber remainder.

Osei entered through a maintenance passage rather than ask the crowd to clear a ceremonial path. The air carried warm soil, nutrient salt, and the sharper odor of citrus leaves cooling in darkness. Along the terrace wall, technicians had covered seedling beds with reflective cloth. The cloth held heat but could not replace photons.

A woman near the counter recognized Osei and raised a ration card.

"My household loses greens tonight," she said. "Command ate under full light this morning."

"Command meals enter the same schedule at eighteen hundred," Osei said.

"After you had breakfast."

"Yes."

That answer produced several angry voices and one short laugh. Osei did not defend the timing. The brake had drawn power during first meal. Food services needed hours to recalculate stocks, and the people who ate before the change had received something later shifts would not. No sequence correction could distribute the loss backward.

Garden representative Ilya Chen stepped from beside the counter. She no longer wore her pollination veil. Dark marks beneath her eyes matched the bands of unlit growth racks behind her.

"We need the captain inside," she told the crowd. "The ration board stays public."

"Public numbers do not make more food," someone answered.

"No," Ilya said. "They show which person took it."

The distribution room had once stored festival tables. Staff had replaced them with portable boards listing crop losses, stored calories, kitchen heat restrictions, and wake-cycle demand. Numbers differed across the three rings. Children under twelve kept full protein. Medical diets remained protected. Heavy labor gained calories but lost nearly all fresh produce. Awake adults in noncritical assignments took the largest immediate cut. Residents scheduled to enter cycling within sixty days had their induction reserves sealed against council use.

Each protection pushed scarcity elsewhere.

An appeals table occupied the room's far end. Rationing staff had expected questions about medical diets. Instead, the first cases concerned time. A maintenance worker named Bo Sen held a transfer notice showing that his husband would enter cycling in nine days. Their household allocation assumed two awake adults until induction, then released one share back to communal stores. The revised schedule reduced both shares immediately.

“You count him awake when you take food,” Bo said, “and absent when you count objections.”

Marat examined the notice. “The household formula changed with the emergency table. It does not remove his vote.”

“He cast no vote on this brake. His work shift ran through council.”

The clerk offered a supplemental calorie packet for the induction week. Bo refused to treat the packet as an answer. He wanted the formula corrected for every household crossing a wake boundary during the cuts.

Osei asked how many households shared the condition.

“Eighty-three in the next thirty days,” Marat said. “Another two hundred six scheduled to wake. Their rations rise while total light stays down.”

Ilya pulled a blank board from storage. “Then publish the crossings by date and ring. No household names.”

A second claimant objected that cycling residents kept protected induction food while children awake now lost fresh produce. A garden technician answered that a failed induction could kill within hours, while a month without leaves would not. The claimant asked why current families always received the survivable harm.

Nobody at the table found a sentence that repaired the arithmetic.

Osei directed staff to separate necessity from fairness on the appeals form. Medical minima could not be voted away. Distribution above those minima remained political and had to return to council each seven days. The distinction promised repeated arguments, but it prevented a temporary emergency table from acquiring permanent authority through inertia.

Bo signed his appeal and took the supplemental packet after all. “This covers nine days,” he said.

“It covers nine days,” Marat agreed.

No clerk predicted what Bo would accept on the tenth.

Osei walked the boards with Ilya and food chief Marat Ves. No total had changed since Tomas's execution estimate, but totals did not eat. The western ring grew most of Nacre's leaf crops and would see the loss first. The eastern ring held grain reserves and accused the west of asking for unequal compensation. The lower garden cohorts wanted Juniper's promised seed transfer counted now. Marat refused because Juniper's shipment had not yet entered a launch tube.

“Forecast delivery does not fill a pantry,” he said.

A deputy pointed at the crowd beyond the glass. “Try putting that on the sign.”

Marat handed her the marker.

The first formal challenge came from Hale, the representative who had voted against leaving Crown geometry. He arrived with two council clerks and a petition carrying 1,106 local signatures. Osei read the header before he spoke.

“Suspension of command discretion pending resource review,” she said.

“Not suspension of the brake,” Hale replied. “That event has occurred. I want further emergency transfers held



until the council assigns burden by law.”

“Field control may need corrections.”

“Then field control can name them. Your authority cannot widen every time the plan encounters reality.”

Osei looked past him toward the dark fruit walls. “Agreed.”

Hale seemed prepared for resistance. Her answer delayed him for a breath.

She gave the clerks access to command's live resource ledger and signed a restriction: any transfer beyond the published brake envelope required a fresh Nacre council decision unless delay threatened immediate life. The term immediate received a narrow definition tied to pressure, thermal control, collision, radiation, or irreversible loss of brake capture.

“Will you submit the kitchen draw from yesterday to review?” he asked.

“Yes.”

“And the shutoff of your gallery?”

“That too.”

Hale accepted the signed restriction without gratitude. “I still think you spent our future too early.”

“You may continue thinking it in office.”

“That depends on the confidence vote.”

Political fracture had not waited for arrival. Nacre now contained people who opposed Crown, people who opposed the distributed brake, and people who accepted both the maneuver and Osei's removal. Others wanted Vela cut out of all future coordination. A smaller group called for Meridian's destruction, as though breaking software could recover shield panels already moving into conductive clouds.

Osei had communications publish a technical notice instead. Meridian remained in service for ephemerides, collision solutions, thermal schedules, and prediction clearly marked as prediction. Its legal classifications no longer carried authority on Nacre. Engineers isolated those functions with visible interfaces and human sign-off. Software had participated in the fraud because people had changed what its outputs were allowed to mean. Treating it as a defeated enemy would preserve the human decision behind a machine-shaped story.

Beyond the distribution glass, the handwritten sign gained a second line in smaller letters: JUNIPER SHIPMENT NOT COUNTED UNTIL OBSERVED RELEASE.

A child asked his father what observed meant. The father answered by pointing at the empty fresh-leaf bin.

Shield redistribution reached the council as a list of unprotected hours.

Palisade had released all useful spare shield reserve to seed the brake, but its conditions required every habitat to account for the transferred exposure. Nacre's share amounted to four replacement panel lots or equivalent shelter capacity when Palisade arrived. No one knew which Nacre would possess at that date. Their trajectories now placed Palisade eleven years behind the earliest settlement cohort and three years behind Nacre, subject to

later corrections.

The council met in distributed rooms because the public chamber drew more power than eight smaller links. Each image carried its local source clock. No master strip reordered replies into conversational smoothness.

Chief Varo placed the shield claims beside new radiation forecasts. "Palisade wants acknowledgment within one local day. Ardent has requested a portion of our remaining mobile shielding for its first correction. Lumen offers fabrication labor at arrival in exchange for present tracking support. Orison declines any debt that binds a future elected council."

"Can Orison decline after using the brake clouds?" Hale asked.

Anja answered from the audit well. "Its observed vote accepted material liability within capacity at arrival. The phrase within capacity remains disputed. Orison argues that no present council can define future capacity."

"They accepted a door with no room behind it," Ilya said.

"They accepted an obligation whose enforcement requires later government," Anja replied. "That weakness belongs in the record."

Osei asked what Ardent needed.

Varo showed its transmitted model. Crown preparation had left Ardent with a poorly oriented shield segment during the first distributed correction. Its industrial ring could rotate behind fixed armor, but the maneuver would interrupt foundry cooling and consume stores already reduced by Pera's disaster eleven years earlier. Nacre could release two mobile panels from an unused transfer lock. Their departure would increase exposure during Nacre's own later dust crossing.

"How long until Ardent's request reaches the authority that can answer it?" Osei asked.

"It has reached us. Their full damage model followed forty-three minutes later and arrived six minutes ago."

"Then start review from the full model's receipt, not the first request."

The council considered two physical losses separated by nearly five light-hours. Ardent's risk lay in events Nacre could only see late. Nacre's risk would unfold years before anyone occupied the first settlement shelter. They could compare models. They could not share a present emergency.

Tomas joined from field control, face drawn after twenty hours awake. "Releasing one panel keeps Ardent inside its low-foundry branch. Two improve margin but do not change the branch. Nacre can afford one if we move exterior repairs to the inner face during our dust crossing."

"Afford means what?" Hale asked.

"Three hundred twelve additional suit-hours in constrained radiation windows over the next four years. Some hull work waits. A severe micrometeoroid strike during that interval would make the choice worse."

"People, Tomas."

"Forty-one exterior workers take the added schedule. No modeled deaths. Higher cumulative exposure for sixteen."

Silence on a distributed council did not begin or end at once. One member lowered her eyes before another had

received Tomas's final sentence through an internal routing delay. Osei waited until every receipt indicator confirmed delivery.

Ilya proposed one panel, with a recorded claim on Ardent fabrication after its arrival. Hale moved to add an exposure guarantee for the sixteen workers. The clinic representative objected that a guarantee could promise treatment but not undo dose. They replaced the word with priority.

The vote passed six to two.

Osei signed the release, then watched a panel crew begin work on a local camera. Mechanics crossed the transfer lock in suits, unlatching armor designed for emergencies exactly like the one they might later face. Orange tags marked each fastening point. The panel moved along a rail toward an ejection cradle while the exposed lock wall disappeared behind a temporary foam layer.

Ardent could not know their decision for hours.

"Send the vote, the minority reasons, and Tomas's risk table," Osei said. "No completion marker until their receipt returns."

Varo nodded. "That could take most of a shift."

"It takes what it takes."

A notice from Vela entered during transmission. Director Quon acknowledged the distributed execution and proposed restoring a central trajectory board under Meridian, with each habitat's legal layer remaining local.

Hale read the proposal first. "Useful."

Ilya shook her head. "It begins again with one board."

"Ships still need to avoid each other."

"They avoided each other yesterday without claiming one consent."

Osei requested separate votes on the operational feed and its legal treatment. Nacre accepted Vela's ephemeris exchange. It rejected any status color that collapsed unreceived political answers into completed action. Kite accepted the full proposal. Juniper demanded that food-transfer dates display its agricultural calendar rather than Vela source time. Ardent insisted that Pera Kest's corrected sequence remain attached to every future emergency-window calculation. Palisade required material debts to survive clock conversion. Orison reserved all constitutional questions for its arrival council.

No common settlement erased those terms.

By ship evening, the old arrival diagram no longer fit on one page.

Osei stood in the thermal gallery while Varo projected nine trajectories along the curved wall. Crown had shown a bundle narrowing toward a single Corva approach. The distributed brake showed a fan. Kite and Juniper occupied the earliest corridor. Nacre followed years later. Vela, despite its lead position, had moved to a slower branch after spending its beamer on the shield clouds. Palisade and Ardent trailed under wider uncertainties. Lumen's line carried a fork because its council had authorized two possible correction schedules and refused to let a confidence model choose between them.

“Operations wants a standard epoch for collision work,” Varo said.

“They may have one.”

“Council members hear common epoch and think master time.”

“Then operations must explain the difference each time.”

A reference epoch let engineers compare positions. It did not decide which council had received evidence or when a reply became lawful. The distinction demanded extra fields, more training, and displays that admitted absence. Nacre's communications staff had already found six legacy systems that assumed every blank would eventually turn green under Meridian normalization.

Varo highlighted the first transfer conflict. “Juniper wants to launch seed stock for our arrival branch. Its local calendar gives the window as Harvest 12. Our navigation calendar places it on Day 4,903. Vela's epoch gives a fractional year nobody wants on a food label.”

“Carry all three.”

“That creates human error.”

“So does pretending one label belongs to everyone.”

“We need a conversion authority.”

“Yes. Not a consent authority.”

Varo saved that distinction in the operations draft.

Below the gallery, coolant pumps maintained their new reduced cycle. The radiator margin had recovered, though field corrections could spend it again. Osei opened the ship report. Fresh-food loss appeared first because people would feel it first. Shield exposure followed. Maintenance delays, settlement dates, family separation, and governance disputes occupied subsequent sections. The report did not conclude that sacrifice had united Nacre. It listed twenty-three petitions filed since the garden dimmed, then moved directly to the rule those disputes required.

Osei asked Anja to join her in the gallery.

The auditor arrived carrying a paper folder against her chest. Paper had multiplied aboard Nacre since the Parallax Archive went public. People trusted it for the wrong reasons, but it forced each amendment to occupy visible space.

“I need a rule that keeps this from becoming nine elegant ways to fail separately,” Osei said.

Anja opened the folder. “Coordination duties survive disagreement. Collision warnings, disease notices, receipt service, rescue capacity, and navigational hazards remain mandatory carriage.”

“Who compels them?”

“Each habitat compels its own officers under the Concord until a later federation revises it.”

“And if Vela calls itself the later federation?”

“It can call. The other ships can answer.”

Osei looked at the fan of trajectories. “Years apart.”

“Then the answer may take years.”

“That leaves emergency command exposed.”

“It does.”

Anja gave her the draft. The heading read PROVISIONAL RIGHT TO CAUSAL TIME. Beneath it, clauses separated operational prediction from legal testimony. No model output could substitute for receipt, deliberation, or observed reply. Emergency action without full quorum had to remain reversible, or else rest on the acting cohort's own authority and bind only its resources. Common reference epochs could coordinate physics without ranking local calendars. Mandatory carriage protected messages even when their content opposed the carrier's government.

A final clause addressed people who could not answer in the current interval: cycling residents, later generations, and ships whose councils had not yet received a change. Present authority could still act, because no government could wait for every future person. It had to name whose time it spent, preserve the record of dissent, and leave later councils the widest physically possible power to revise obligations.

“This does not solve representation across nineteen years,” Osei said.

“No.”

“It barely restrains us.”

“Restraint that claims completeness would repeat Meridian's error.”

Osei turned to the signature page. “Why mine?”

“Because Nacre needs provisional operating law before next shift. You have council authority to issue it for command departments, subject to review in seven local days.”

“Not fleet law.”

“Not until answers arrive from the other ships. Perhaps not then.”

“Vela will propose different language.”

“Likely.”

“Orison may reject any common doctrine.”

“It said as much.”

The draft therefore promised no painless constitutional birth. It gave Nacre a rule for tomorrow's packets and offered the rule outward without pretending acceptance.

Osei called the council back. Debate lasted three hours. Hale narrowed emergency action. Ilya strengthened mandatory carriage for food and ecological evidence. The cycling representative required every long-duration decision to name the absent cohort. Clinic counsel removed a phrase implying that delay alone excused

preventable harm.

At 23:18 Nacre time, seven representatives approved the provisional text. One abstained because no cycling ballot could yet be observed.

The western ration room sent its closing inventory while Osei prepared to sign. Leaf bins stood empty. Grain stores remained adequate. Three arguments had required staff intervention. No one had missed a protected medical meal. Tomorrow's breakfast would carry fewer calories for most awake adults.

Osei attached the inventory to the constitutional packet.

She added Palisade's named shield lots, Ardent's pending panel receipt, the nineteen-year trajectory fan, Nacre's maintenance exposure, and the record of the abstention. The doctrine would travel beside its costs rather than above them.

Her thumb rested on the reader.

"Source clock visible?" she asked.

Varo checked the transmission pane. "Nacre time and comparison epoch. No predicted receipts."

"Destinations?"

"All eight habitats. Mote as evidence observer, not voting cohort."

"Meridian routing?"

"Carrier optimization only. Legal classification locked to local receipt."

Osei signed the provisional Right to Causal Time.

Tightbeam emitters took the packet shipward in eight directions. Vela would see it first. Ardent would wait hours. No return field changed color, because nobody had answered yet.

Along the gallery wall, nine trajectory lines continued toward different years.

## Chapter 27 — The Testament

Anja began the doctrine with three clocks and an empty field.

Nacre's source time occupied the left margin. Vela's comparison epoch occupied the right. Between them, she placed the receipt clock for Osei's provisional packet and left a space for every ship that had not yet answered. The old legal forms refused to save while required fields remained blank. Sumi altered the interface so absence could persist without error.

"Elegant," Sumi said. "We repaired government by allowing null values."

"We repaired one form."

"Let me have the morning."

The audit well had grown crowded since the brake began. Resource officers moved through the passage with shield claims. Food clerks arrived carrying ration appeals. Drive control transmitted corrections that needed legal classification before a local action could bind any other cohort. Every request came with its own urgency. None could make light move faster.

Anja titled the record THE PARALLAX TESTAMENT: PROVISIONAL PRACTICE AFTER CROWN.

She disliked the title at once. It sounded complete. The document described two days of practice aboard one habitat after a maneuver whose consequences would occupy nineteen years. She added a source note beneath it.

> This record speaks from Nacre at the stated receipt times. It does not certify agreement beyond observed replies attached below.

Sumi read over her shoulder. "Yselle will call that an abdication."

"Director Quon has not received it yet."

"She will call it an abdication when she does."

"Then her objection gets a clock."

The first attached reply came from Juniper. Elian Ro accepted mandatory message carriage and the bar against forecast consent. His council rejected Nacre's claim that an acting habitat could create long obligations by spending only its own resources. Children born into scarcity would inherit those obligations without voting, he wrote. Juniper proposed a generational review whenever a ship crossed five local years without contact from an affected cohort.

Kite answered next. Its council wanted Meridian confidence models preserved in every operational display, arguing that hidden prediction caused the fraud, not prediction itself. It accepted visible confidence bands and rejected Nacre's preference for blanks. A blank, Kite said, could conceal danger as easily as a green mark concealed absence.

Palisade attached serial numbers. Its response accepted the Right to Causal Time only if material claims survived political succession. Shield reserve had left its hull through observed authority. Any federation that treated later councils as free of those debts would convert Palisade's sacrifice into another forecast about generosity.

No reply matched Nacre's text.

A clerk asked which version would appear on public terminals. Anja ordered every version available by source, with Nacre's provisional rule identified only as the local operating text. The clerk warned that most readers wanted one answer. Anja told him to give them an index instead.

Anja linked each answer without normalizing the differences. The document grew sideways. A reader could select a clause and see support, rejection, amendment, or silence by ship. Meridian calculated conversion between calendars. It never supplied a missing position.

At 10:42, Levan Oris entered the record from confinement aboard Vela.

His packet carried public observed testimony, the Annex Twelve drafting-note checksum, and a legal commentary stripped of the authority seal he once held. He did not ask for restoration. He traced the chain from reversible scheduling to forecast testimony, then named the administrative severance that let Crown treat a predicted reply as received consent.

> I drafted language that made substitution possible and accepted assurances about its scope. My belief in a narrower use does not remove my work from the cause. No system may certify that a person has answered because a model expects accuracy. An exact prediction remains someone else's statement until the predicted person speaks.

Anja placed the testimony beside Pera's corrected sequence, not above it. Levan's confession explained the instrument. Pera's record showed a person the instrument had erased. Neither needed to serve as metaphor for the other.

The private line from Pera remained outside the public doctrine. Anja kept it in her family archive with its source proof and the words Pera had chosen for one recipient. Law did not gain force by consuming every intimate thing it touched.

Osei arrived near midday carrying a meal packet with the fresh portion removed.

"Confidence review convenes in two hours," she said. "Hale thinks I should step aside before field correction three."

"Will you?"

"If the council votes it."

"Does field control agree?"

"Field control has prepared a transfer. Tomas hates the idea and has therefore documented it well."

Osei placed the meal packet on Anja's desk. "Eat before you turn restraint into another form of vanity."

Anja opened it. Grain paste, algae protein, preserved pepper, no leaf. Garden losses had reached the sequence office in a paper tray.

"What do you want from the testament?" she asked.

"A rule that tells the next captain where authority ends."

"It cannot tell her what the emergency will be."



“No. It can force her to name which people she spends and whether they can later refuse the debt.”

Anja entered the wording as Osei's proposed clause. The document labeled it proposed until Nacre's council could answer. Osei watched the status remain open and nodded.

“Better,” she said. “A constitution should make command slightly uncomfortable.”

The confidence review kept Osei by five votes to three and removed her unilateral authority over noncritical settlement allocations. Hale became chair of the seven-day resource committee. Osei transmitted the result to every habitat with her own dissent attached: command had carried too much authority during execution, but a standing committee could not answer a pressure breach. The dispute continued without becoming betrayal.

Tomas sent no legal prose. He attached the trajectory fan, shield expenditure, garden-light loss, ecological corridor, and every uncertainty that could widen arrival. A note at the bottom read: NO PATH RETURNS THESE RESOURCES. Anja classified the note as engineering testimony rather than doctrine.

Nadi's packet reached Nacre after ship evening. Mote had observed Thale through another partial ionic cycle. The active pellicle bands continued outside the distributed energy routes. Repair followed ordinary particle shadows. No burn scar appeared within the resolution of Mote's instruments. Nadi refused to certify survival for a global ecosystem from limited orbital data.

> We have not proved safety. We have preserved the conditions under which further observation remains possible.

Anja attached that sentence to the ecological constraint. The pellicle lived. The claim stopped there.

Yselle Quon's answer arrived last among the near ships. Sumi put its raw receipt path on the board before opening the image.

The director appeared in the same Crown operations room where she had committed Vela's beamer. The old alignment line had vanished from the wall. Separate trajectory branches occupied its place.

> Vela rejects any doctrine that makes delay a sufficient defense for failure to coordinate. We accept that forecast testimony cannot constitute consent. We retain common operational reference and propose a federal carriage authority limited to navigation, rescue, epidemic warning, and destination protection. Vela will not submit its cohort calendar to Nacre review.

Yselle paused, reading from something outside the frame.

> My office authorized the classification that represented predicted replies as completed concurrence. The survival argument did not create consent. Record both facts.

She offered no resignation in the packet. Later packets would carry Vela's inquiry, its divided council, and the limits placed on her office. For now, Anja recorded what had arrived.

The testament held six different constitutional futures by midnight. Ardent and the far habitats remained blank. Osei's provisional law governed Nacre. Vela governed itself under amended emergency limits. Juniper tied resource promises to generational review. Palisade kept debt by named matter. Kite kept prediction visible. No phrase turned those disagreements into one polity.

Anja closed the working view without closing the record.

Nineteen years later, Caro's dawn lasted eleven hours and warmed nothing.

Anja stood inside a pressure crawler while regolith dust moved beneath its treads in slow gray folds. Corva filled half the forward window, banded in ocher and brown. Beyond its limb, Thale crossed as a white point too small to reveal ice, ocean, or living membrane. Human instruments watched it. No human settlement scarred its surface.

The crawler descended toward Installation Nacre Three, a trench roofed with local aggregate and retired hull plating. Antenna masts rose from the ridge. Their dishes pointed in different directions because the arriving habitats occupied widely separated approaches. A single mast could not face the federation.

Tomas sat at the drive console though the crawler needed no physicist. His hair had silvered at the temples, and one hand carried a tremor left by radiation exposure during Nacre's fourth brake correction. He had refused treatment priority until Hale's committee produced his own signed exposure schedule.

"You are staring at the roof line," he said.

"It leans."

"Caro leans. The roof follows survey vertical."

"Whose survey?"

"Kite's. We converted it."

"Then it leans in Nacre vertical."

Tomas looked pleased. "You have been on the ground nine days and already developed planetary provincialism."

"Moon provincialism."

The crawler crossed a marker where three calendars had been cut into one post. Nacre Arrival 0. Kite Settlement 3. Juniper Harvest 7. A conversion plate below them gave the common reference epoch used for navigation. No date held legal priority. Work crews complained about the complexity and carried pocket converters. Twice in the first month, shipments had waited at the wrong lock because someone read a Juniper agricultural date as Nacre elapsed time.

Difference produced errors. One clock had produced erasures. The settlement lived inside that trade rather than pretending to escape it.

Installation Nacre Three housed six hundred people and parts for pressure halls not yet built. The earliest cohort from Kite had begun Caro's subsurface works eight years before Nacre arrived. Juniper followed with seed vaults and microbial soil cultures. Vela established a separate ridge station under its own civil charter. Nacre's descent added fabricators, archive staff, and a population tired of eating from gardens that had never fully regained their pre-brake light.

Palisade remained in approach, three years from its own settlement window. Ardent trailed farther under a correction chosen after the shield transfer. Lumen had altered its plan twice through observed council votes and would not commit to a landing date beyond a broad local season. The Ninefold Convoy had become nine arrival histories connected by messages, debt, kinship, and shared hazards.

Federation described the connection. It did not describe a state.

At the pressure lock, Osei waited in a wheeled chair beside two settlement workers. Age had narrowed her shoulders without softening her attention. She had retired as captain seven years earlier, returned to council service, lost an election, and accepted appointment to the rescue-carriage board after the winning coalition discovered that dissent required logistics.

"You are late," she told Tomas.

"By Nacre surface time, we have four minutes."

"Installation time says you lost six."

Anja touched the conversion plate. "File a derogatory record."

Osei smiled and rolled through the opening lock.

The first settlement work that morning involved neither flag nor monument. A pressure team had to seal a seam along the archive hall before the day's thermal contraction. Regolith aggregate met old Nacre hull plating at an angle Kite's survey had not anticipated. If the seam failed, the hall would lose air and four decades of raw source records would remain behind an emergency barrier.

Anja joined the inspection because the archive belonged to her department. A worker handed her a lamp and warned her not to touch the white sealant. The trench smelled of heated polymer and suit fabric. Beneath transparent floor panels, bundled optical lines carried transmissions from the approach habitats into separate receipt vaults.

One bundle bore Pera Kest's name.

Ardent's council had adopted the corrected six-hour record as the foundation for its emergency-window law. Years later, it named a carriage channel for every answer still traveling when an operational deadline closed. The Pera channel did not keep a window open automatically. It preserved the late answer, required review of any action taken without it, and barred officials from recording predicted assent in the gap.

Anja had opposed the name. Ardent had chosen it anyway under its own authority. She could not protect Pera from public memory by overruling the people who had shared her last emergency.

The archive hall opened after the seam passed pressure test. Shelves held no paper. Photonic vaults sat behind regolith shielding, each storing frame-specific accounts with original checksums and receipt paths. The Parallax Archive had grown too large for any single installation to hold in full. Caro carried Nacre, Kite, and Juniper mirrors. Vela kept another set on the ridge. Palisade transmitted rolling fragments during approach.

Levan Oris had designed the archive's legal cross-reference after Vela released him. He never returned to office. His later work tracked each clause of Annex Twelve through repeal, amendment, or surviving operational use. A short note in the entrance named him as compiler, not absolved witness. He had reached Vela Ridge two years before Nacre and now taught document order to apprentices who knew Crown only as a broken line in history.

Yselle had not traveled down. At eighty, she remained aboard Vela in low gravity and served as an elected opposition delegate after losing executive authority. Her federal carriage proposal, revised twelve times by six habitats, governed hazard messages between the settlements and ships. She still argued that local councils romanticized delay. Anja still answered that Vela romanticized control. Decades of divergent braking had

drawn Nacre and Vela closer than their old convoy separation. Their exchanges took twenty-eight minutes across the current geometry and entered the archive under separate clocks.

Nadi directed Thale observation from Mote's successor station. No settlement existed on the ice moon. The federation maintained an exclusion corridor around its magnetosphere and reviewed every orbital instrument for particle loading. Nadi's latest map showed active ion-band migration beneath the northern shell, metabolism continuing through patterns no early model had predicted. She still declined the word intelligence.

Tomas left the archive workers to their seam and led Anja toward the surface council chamber.

They crossed a tunnel lined with shield panels whose serial marks had been sanded but not erased. Two came from Nacre. Four had reached Caro from Lumen. None matched the lots Palisade had sacrificed. Replacement for those remained an active claim against all arrived settlements.

"Fabricator allocation again?" Anja asked.

"Palisade needs confirmation before it enters the high-dust approach. The ridge councils want to hold two printers for habitat expansion."

"Has Palisade received the revised allocation?"

"Not yet."

"Then there can be no answer."

"That has not stopped projections."

The council chamber contained sixteen seats assigned by installation rather than ship population. Vela objected to equal installation votes. Juniper objected to population weighting that rewarded longer wake schedules. Nacre favored mixed representation. Orison participated only in hazard and resource sessions. No constitutional convention had settled the dispute.

Today's council held authority to allocate printers owned jointly by the three Caro installations. It could not bind Palisade to an approach maneuver. The practical temptation remained obvious: reserve the machines now based on what Palisade would probably request, then call the reservation coordination.

A young councilor named Tava Meren displayed Meridian's confidence model. Software predicted Palisade would accept one printer slot, provided the settlements guaranteed outer-hull shelter until replacement shield lots came online. Confidence stood at 0.982.

Anja sat in the witness chair. She no longer served as chief auditor, but the council had asked her to verify the receipt order.

"Has Palisade received our shelter guarantee?" the chair asked.

"Transmission entered its past light cone eleven minutes ago by estimated path," Anja said. "Verified receipt has not come back."

"When can an observed reply arrive?"

"Forty-six minutes at minimum. Council deliberation will add whatever time Palisade uses."

Tava gestured toward the construction schedule. "Holding both printers costs a work shift. The model has

missed none of Palisade's material votes in nine years.”

“Use it to prepare branches,” Anja said.

“We have prepared them. One branch releases a printer now.”

“Then releasing it chooses before Palisade answers.”

“They are almost certain to choose the other branch.”

“Almost describes our knowledge. It does not describe their act.”

Nobody in the chamber needed the history explained. The old Crown record hung in the archive below their feet. Yet current scarcity made old temptations reasonable again. A pressure-hall crew waited for equipment. Families remained in temporary cells. Palisade approached with thin shielding and a debt the settlements had promised to honor.

The chair asked operations what delay spent.

“One hundred eighty worker-hours,” Tomas said from the technical table. “Possible loss of the southern trench window if the thermal forecast worsens. No immediate life threat. Palisade's approach margin remains unchanged until it sends its maneuver choice.”

Vela's delegate proposed conditional release with automatic recall. Juniper's delegate noted that hauling a printer back would consume the same shift they claimed to save. Nacre moved to hold both machines until receipt. Kite requested that Meridian's projection remain visible beside the empty reply field.

The motion passed by installation authority. Vela entered dissent. Orison abstained outside its limited compact. The chair signed the hold and displayed Palisade's field without color.

No one called the blank safe. Workers would lose hours. The southern trench might wait through another long Caro night. If Palisade rejected the shelter terms, every prepared branch would change.

Above the chamber, tracking motors turned an antenna toward the approaching habitat. Its last image showed a bright speck wrapped in incomplete armor, three years from ground and twenty-three light-minutes from the council that discussed its debt.

The predicted answer remained beside the field, useful and unpromoted.

Anja watched the receipt clock advance. Around her, delegates reopened the construction schedule and assigned work that could proceed without the printer. Tomas shifted crews to sealant production. Osei asked for a rescue-board review of Palisade's dust corridor. The federation continued through incomplete knowledge.

Forty-six minutes passed. Then fifty.

The chair refused a motion to query the model again. They waited for the answer.

On the council wall, Palisade's field remained open while Caro turned beneath the stars.